



National Comprehensive HIV Prevention, Care and Treatment Training for Healthcare Providers

Participant Manual



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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!

Revised on August, 2021

(DRAFT)

Foreword

The Ethiopian Health Policy takes decentralization as a principle that guides the efforts to improve quality and coverage of preventive and curative health services. In this effort, the role of health workers assumes a central position. It is well known that a large segment of the Ethiopian population gets primary care from these health workers. In addition to decentralization maintaining and improving the quality of care provided at all levels of the health care tier is critical through capacity building.

Therefore, this comprehensive HIV prevention, care and treatment training manual is primarily intended to build the capacity of health workers at all levels with recent updates like Index case testing, PrEP, and third-line ARVs so that they can be confident enough to solicit and test high risks, provide combination prevention or initiate ART and follow patients on chronic HIV/AIDS care. It is our hope that health workers, managers and trainers will benefit from this updated manual.

I would like also to take this opportunity to thank all contributors who have involved in the development of the training material.

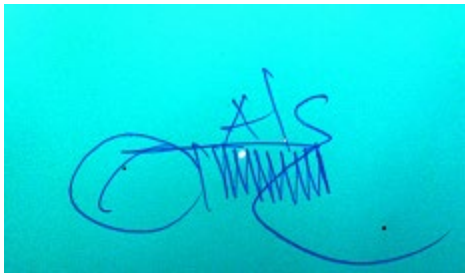


Hiwot Solomon (BSc, MPH)
Director, Disease Prevention and Control Directorate
Ministry of Health- Ethiopia

APPROVAL STATEMENT OF THE MINISTRY

The Federal Ministry of health of Ethiopia has been working towards standardization and institutionalization of In-Service Trainings (IST) at national level. As part of this initiative the ministry developed a national in-service training directive and implementation guide for the health sector. The directive requires all in-service training materials fulfill the standards set in the implementation Guide to ensure the quality of in-service training materials. Accordingly, the ministry reviews and approves existing training materials based on the IST standardization checklist annexed on the IST implementation guide.

As part of the national IST quality control process, this National Comprehensive HIV Prevention, Care and Treatment Training for Healthcare Providers package has been reviewed based on the standardization checklist and approved by the ministry in March, 2020.

A handwritten signature in blue ink, appearing to be 'Assegid Samuel Cheru Human', is written over a solid black rectangular background.

Assegid Samuel Cheru Human

Resource Development Directorate

A/ Director

Ministry of Health- Ethiopia

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Mirte Getachew	MOH-Ethiopia
Dr. Zelalem Tadesse	ANECCA
Dr Ambachew Tefera	USAID/ Project HOPE
Dr. Eshetu Gezehagn	USAID/PHSP
Dr Ghion Tirsite	WHO
Dr. Miftah Kemal	ICAP/E
Dr. Beyan Jeylan	ICAP/E
Yalemsew Derib	GHSC/PSM
Samrawit Ejigu	PSI/E
Ismeal Ahmed	CDC/E
Wakgari Adu	ADU Consulting
Dr.Yoseph Gutema	ICAP/E
Dr.Tolossa Eticha	AHMC
Seid Yusuf	Arsi University
Belete Ayalneh	GHSC/ PSM
Arega Aweke	Hawassa University
Mulu Legesse	MOH /PMED/PSM
Dr. Fethia Keder	PSI/E
Asmamaw Silesh	CHAI
Getu Assefa	MOH/FANTA
Getnet Hailu	EPHI
Chane Temsgen	CDC/E
Seble Mamo	MOH-Ethiopia
Zenabua Yalemwdeg	EPHA
Yetimork Tekle	MOH-Ethiopia

Dinku Worku	NEP+
Kidist Zealiyas	EPHI
Melaku Bekele	Private
Wintana Hailu	Private
Dr.Petros Mitiku	MOH/ ICAP
Tsion Tsegaye	EPSA
Dr Zerihun Hika	MOH/ICAP
Tigist Kassahun	EPSA
Teferi Flatie	FHAPCO
Dr. Leul Endale	MOH-Ethiopia
Alemtsehay Abebe	MOH-Ethiopia
Dr. Mesfin Girma	ICAP/E
Bayisa Chala	NEP+
Abraham Kebede	NNPWE
Dr. Andinet Asmamaw	ICAP/E
Martha Sheferaw	Mekedem Ethiopia

Acronyms and Abbreviations

3TC	Lamivudine
ABC	Abacavir
AFB	Acid fast bacilli
AIDS	Acquired Immune Deficiency Syndrome
ANC	Antenatal Care
ARV	Antiretroviral
ART	Antiretroviral Therapy
AZT/ZDV	Zidovudine
CBC	Complete Blood Count
CD4 cells	Type of T-lymphocyte, white blood cells
CMV	Cytomegalovirus
CPT	Cotrimoxazole Preventive Therapy
DBS	Dried Blood Spot
DRV/r	Ritonavir boosted Darunavir
d4T	Stavudine
ddI	Didanosine
DHS	Demographic and Health Survey
DNA	Deoxyribonucleic acid
DOTS	Directly Observed Therapy Short Course
DTG	Dolutegravir
EFV	Efavirenz
FBOs	Faith-based organizations
FDC	Fixed dose combination
FHAPCO	Federal HIV/AIDS Prevention and Control Office
FMOH	Federal Ministry of Health
HAART	Highly active antiretroviral therapy
HBV	Hepatitis B Virus
HCV	Hepatitis C Virus
HIV	Human Immunodeficiency Virus

IDV	Indinavir
INSTIs	Integrase strand transfer inhibitors
IP	Infection Prevention
IPT	INH Preventive Therapy
IRIS or IRS	Immune Reconstitution Inflammatory Syndrome also called Immune Reconstitution Syndrome (IRS)
LFT	Liver Function Test
LPV	Lopinavir
MTCT	Mother-To-Child Transmission (of HIV)
MD	Medical Doctor
NFV	Nelfinavir
NGO	Non-governmental Organization
NNRTI	Non-nucleoside reverse transcriptase inhibitor
NRTI	Nucleoside Analogue Reverse Transcriptase Inhibitor
NVP	Nevirapine
OIs	Opportunistic Infections
PCR	Polymerase chain reaction
PEP	Post-exposure prophylaxis
PI	Protease Inhibitor
PITC	Provider Initiated Testing and Counselling
PLHIV	People living with HIV
PMTCT	Prevention of mother-to-child transmission (of HIV)
RAL	Raltegravir
RNA	Ribonucleic acid
RTV/ r	Ritonavir
PI/r	Ritonavir boosted Protease Inhibitor
RFT	Renal function test
RT	Reverse transcriptase
STI	Sexually Transmitted Illnesses
TAT	Turn around time
TB	Tuberculosis

TLC	Total Lymphocyte Count
TLD	Tenofovir/ Lamivudine/Daltegravir FDC
TPT	TB Preventive Therapy
U/A	Urine analysis
UNAIDS	The Joint United Nations Program on HIV/AIDS
UP	Universal Precautions
WHO	World Health Organization
YMDD	(tyrosine-methionine-aspartate -aspartate) locus of the HBV reverse transcriptase gene for 3TC resistance
ZDV	Zidovudine (also abbreviated as AZT)

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INTRODUCTION TO THE MANUAL

Building the human resource capacity is one of the milestones for catering the multifaceted interventions related to HIV/AIDS prevention, care, support, and treatment. The need for coordinated and standardized comprehensive HIV/AIDS training tailored to the capacity and intervention needs are well articulated in both the Ethiopian Strategic Plan for Intensifying Multi-sectoral HIV/AIDS Response and the Road Map for Accelerated Access to HIV/AIDS Prevention, Care, Support, and Treatment Services in Ethiopia.

The expansion of comprehensive HIV/AIDS programs throughout the country is coupled with the capacity building of health care providers and awareness creation to the community to ensure access to HIV prevention, care, support, and treatment services at all levels.

Based on these principles the guideline revision panel has prepared this comprehensive HIV prevention, care and treatment training material to be used in conjunction with the accompanying facilitator guide, photo booklet and expert patient training materials.

Core competencies for trainees

Competence is the ability to effectively and efficiently deliver a specified professional service. This implies that the health worker is able to practice at proficiency (mastery of learning) in accordance with local conditions to meet local needs.

The core competences expected of trainees from this course are:

- Organize setup and run HIV Care/ART service delivery
- Perform targeted PITC (adults and pediatrics)
- Effectively identify, inform, elicit, test and link high risk partners and children of index cases,
- Do clinical evaluation of patients coming to the HIV Care/ART clinic
- Screen, diagnose and treat OIs
- Provide prophylaxis in particular CPT and INH
- Treat STIs using syndromic approach
- Manage exposed and Infected infants/ children
- Request and interpret laboratory tests
- Prepare patients and initiate ART for adults and children, including pregnant women
- Conduct adherence counseling, patient tracking and out-reach services.

- Practice appointment spacing for eligible clients
- Evaluate appropriate clients for Oral Pre-Exposure prophylaxis (PrEP), initiate PrEP and follow,
- Identify clients who need PEP, prescribe and follow up PEPDiagnose treatment failure and treat for possible second line regimen
- Provide comprehensive Palliative care
- Provide age appropriate HIV disclosure to children and adolescents
- Practice universal infection prevention protocols
- Document and use data at the facility level and report based on the national reporting requirements

COURSE SYLLABUS

Course description:

This twelve days training course is developed for health care professionals to deliver comprehensive HIV prevention, care and treatment services at any level of the health care tire system in the country.

Course Goal

To provide standard comprehensive HIV/AIDs care and treatment services at any level of the health care tier system in the country.

Participant Learning Objectives

- Describes the HIV epidemics at global, regional and national level
- List target groups for routine HTS through PITC
- Describe Partner and family index testing Services
- Do clinical evaluation of patients coming to the HIV Care/ART clinic
- Describes opportunistic infections (OIs), their diagnosis and managements
- Provide prophylaxis in particular CPT and INH
- Discuss the management of STI
- List the components of care of HIV exposed infant
- Describe the timeframe for follow- up visits of HIV exposed infants

- Perform DBS sample collection, drying, packaging, transportation and use on results for decision making
- Describes the chronic care for HIV
- Discuss patient preparation and initiation of ART,
- Define Adherence and discuss the use of 5 A's to prepare patients for ART adherence;
- Describe the criteria for implementation of appointment spacing model,
- Describe the principle and application of HIV pre-exposure prophylaxis (PrEP)
- Describe the principles and the management steps of HIV post-exposure prophylaxis (PEP),
- Discuss how to suspect/identify, work up and manage treatment failure cases
- Define palliative care and its role in the management of HIV.
- Explain psychosocial needs of children and adolescent to provide age appropriate HIV disclosure.
- Discuss universal infection prevention protocols
- Describe the use of data at the facility level and the national reporting requirements

Training Method

- Interactive presentation & discussion
- Demonstration
- Roll play
- Case studies
- Group work
- Buzz group discussion
- Brain Storming
- Guided Clinical Practices
- Recap

Training Materials

- Participant manual
- Facilitator guide
- Standard PowerPoint slides and videos
- Expert patient trainer guide

- Role play scenarios
- Job aids & observation check lists
- Course evaluation formats
- Knowledge assessment quaternaries
- Penile Model
- Condom
- Antiretroviral Drugs
- Flipchart & Markers
- LCD Projection & Laptop computer

Participant selection criteria

For basic training, health care providers with at least a diploma or first degree in Nursing, a Health officer or General practitioner who is willing to work in the ART clinic will be selected for this training.

For training of trainers (TOT), participants will be selected from health facilities, training and higher institutions, HIV program managers at different level of the health system that who already has the basic training on ART. Participants need to be directly involved in the HIV service delivery or HIV program management and with proven experience and facilitation skill.

Trainer selection criteria

Trainers should be health professionals at least with first degree in health or MPH/ MSc and who are certified with training of trainers (TOT) on comprehensive HIV prevention, care and treatment. Clinical experience in providing ART care and treatment is a plus.

The trainers must have experience using the master learning approach to provide the training. It is also recommended that at least two clinical trainers are required for a class of trainees and each practical attachment shall be led by a trainer.

The trainers can divide the course roles and responsibilities according to their expertise.

Method of course evaluation

Participant evaluation:

- Pre-course assessment-At the beginning of the training participants will take precourse standard pretest.The result will be used by facilitator and trainers to assess the baseline knowledge of participants,
- Participants will use standard checklist for Practical exercises
- Post course knowledge assessment,
- Skill assessment of observed practice during role plays, practicum and skill stations,
- Attendance, 100%

Course evaluation

- **Daily course evaluation**-Participants are expected to complete daily evaluation form to reflect the training process,
- **End course evaluation**-participants will complete the end course evaluation at the end of the course to see their overall reflection on the training (trainers, methodology, venue and materials.)

Criteria for certification

- Participants need to attend all classroom and practical sessions, 100% attendance,
- For basic training, Participants need to score more than >70%and for TOT more than 80% in post course knowledge assessment.

Course duration

The total duration of this training is twelve days which includes two days of practical sessions.

Training Venue selection & Suggested class size

Comprehensive HIV prevention, care and treatment need to be delivered in ISTC,CPD or other designated center and assuming role play & other group activities, therecommendnumber ofparticipant need to be 20-25 participants per class.

HIV PREVENTION, CARE AND TREATMENT (Basic ART) TRAINING SCHEDULE

Date	Time	Activities	Facilitators
Day 1 Monday	8:30 – 9:00AM	Registration	
	9:00-9:15AM	Opening Speech	
	9:15-9:25 AM	Introduction, Ground Rules and Expectations	
	9:25-9:30 AM	Introduction to the comprehensive HIV/AIDS Prevention, Care and Treatment Training Course for health care providers	
	9:30-10:30 AM	Pretest	
	10:30-10:45AM	TEA BREAK	
	10:45-11:15AM	Chapter 1. Overview of HIV/ AIDS in Ethiopia	
	11:15–12:30PM	Chapter 2. Introduction to HIV/AIDS Pathogenesis and Immunology	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:30PM	Chapter 3- Integrating prevention with treatment	
	3:30-3:45PM	TEA BREAK	
	3:45 - 4:25PM	Chapter 3- Integrating prevention with treatment, contd.	
	4:25 – 5:20PM	Chapter 4- HIV diagnosis in adults and children 4.1 HIV Diagnosis in adults and adolescents	
	5:25-5:30PM	Daily evaluation	
Day2 Tuesday	8:30-8:45AM	Recap of Day1	
	8:45-10:30 AM	4.2 HIV Diagnosis in infants, children and adolescent	
	10:30-10:45AM	TEA BREAK	
	10:45-11:55PM	Chapter 5- Care of HIV exposed infant: HEI follow up form	
	12:30-2:00PM	LUNCH BREAK	
	2:00-4:00PM	Practical exercise on HIV rapid testing	
	4:00-4:20PM	TEA BREAK	
	4:20-5:25PM	Chapter 6-Introduction to Chronic HIV care	
	5:25-5:30PM	Daily evaluation	
Day3 Wednesday	8:30-8:45AM	Recap of Day2	
	8:45-9:30AM	Chapter 6-Introduction to Chronic HIV care.... Role play on the use of 5A's.	
	9:30-10:30AM	Chapter 7: Clinical Evaluation and WHO clinical staging	
	10:30-10:45AM	TEA BREAK	
	10:45-11:45AM	Chapter 7: Clinical Evaluation and WHO clinical staging... Clinical review form	

Date	Time	Activities	Facilitators
	11:45 -12:40PM	Chapter 8 : Approach to growth and development assessment	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:15PM	Chapter 9: Introduction to Ethiopian National HIV care/ART patient, follow up form and intake form	
	3:15-3:30PM	TEA BREAK	
	3:30-5:00PM	Chapter 9: Introduction to Ethiopian National HIV care/ART patient, follow up form and intake form.....	
	5:00-5:25PM	Chapter 10: Adherence and Adherence preparation	
	5:25-5:30PM	Daily evaluation	
Day 4 Thursday	8:30-8:45AM	Recap of Day 3	
	8:45-10:00AM	Chapter 10: Adherence and Adherence preparation (with role play).....	
	10:00-10:30AM	Chapter 11: Prevention, screening and management of common OIs, ...CPT	
	10:30-10:50AM	TEA BREAK	
	10:50-12:30PM	<i>Skill station I on clinical evaluation and adherence preparation</i>	
	12:30PM-2:00PM	LUNCH BREAK	
	2:00-3:30PM	Chapter 11.2 Management of Opportunistic Diseases of the respiratory system	
	3:30-3:50PM	TEA BREAK	
	3:50-4:50PM	Chapter 11.2 Management of Opportunistic Diseases of the respiratory system	
	4:50-5:25PM	Chapter 11.3 Management of Opportunistic Diseases of the GI system	
	5:25-5:30PM	Daily evaluation	
Day 5 Friday	8:30-8:45AM	Recap of Day 4	
	8:45-9:40AM	Chapter 11.3 Management of Opportunistic Diseases of the GI system....	
	9:40-10:30AM	Chapter 11.4 Management of Opportunistic Diseases of the nervous system	
	10:30-10:50AM	TEA BREAK	
	10:50-11:10AM	Chapter 11.4 Management of Opportunistic Diseases of the nervous system	
	11:10-11:50AM	Chapter 11.5 Cutaneous manifestations in HIV positive individuals	
	11:50-12:30PM	Chapter 11.6 Prevention and management of STI	

Date	Time	Activities	Facilitators
	12:30PM-2:00PM	LUNCH BREAK	
	2:00-2:30PM	Chapter 11.6 Prevention and management of STI.....	
	2:30-3:00PM	Chapter 11.7 Visceral Leishmaniasis and HIV co-infection	
	3:00 -3:30PM	Chapter 11.8 Screening for co-morbidities, cervical cancer screening, Mental illness and advanced HIV diseases.	
	3:30-3:50PM	TEA BREAK	
	3:50-4:20PM	Chapter 12 Antiretroviral Therapy 12.1 Introduction – Introduce Retesting register	
	4:20-4:40PM	Chapter 12.2 When to start ART	
	4:40-5:25PM	Chapter 12.3 What ART regimen to start	
	5:25-5:30PM	Daily evaluation	
Day 6 Saturday	8:30-8:45AM	Recap of Day 5	
	8:45-10:30AM	Chapter 12.4 Adherence Support for ART initiation and monitoring(including EPT role play)	
	10:30-10:50AM	TEA BREAK	
	10:50-12:30PM	<i>Skill station II Adherence preparation and Initiation</i>	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:30PM	Chapter 12.5 Monitoring response to ART	
	3:30-3:50PM	TEA BREAK	
	3:50-4:10PM	Chapter 12.5 Monitoring response to ART... Introduce appointment spacing register	
	4:10-5:25PM	Chapter 12.6 Monitoring and management of ARV drug toxicity	
	5:25-5:30PM	Daily evaluation	
Sunday – Day off			
Day 7 Monday	8:30-8:45AM	Recap of Day 5	
	8:45-10:00AM	Chapter 12.7 Diagnosis and management of treatment failure Introduce VL testing register and EAC checklist	
	10:00 -10:30AM	Chapter 12.8 Drug interaction and Adverse drug reporting Chapter 12.9 Pharmaceutical supply chain management of HIV	
	10:30-10:50AM	TEA BREAK	
	10:50-12:30PM	<i>Skill station III-ART initiation and Monitoring</i>	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:30PM	Chapter 13 Psychosocial support and pediatric HIV disclosure	
	3:30-3:50PM	TEA BREAK	
	3:50-4:30PM	Chapter 13 Psychosocial support and pediatric HIV disclosure	

Date	Time	Activities	Facilitators
		With role play....	
	4:30 -5:25PM	Chapter 14: Nutrition and HIV/AIDs	
	5:25-5:30PM	Daily evaluation	
Day 8 Tuesday	8:30-12:30PM	Practicum 1 at Health Facility	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:00PM	Discussion of practicum experience	
	3:00-3:30PM	Chapter 14 Nutrition and HIV/AIDS	
	3:30-3:50PM	TEA BREAK	
	3:50-4:30PM	Chapter 14 Nutrition and HIV/AIDS	
	4:30-5:25PM	Chapter 15 Palliative care	
	5:25-5:30PM	Daily evaluation	
Day 9 Wednesday	8:30-12:30PM	Practicum II at Health Facility	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:00PM	Discussion of practicum experience	
	3:00-3:30PM	HIV Care /ART Patient Monitoring tools: Pretest and Introduction to HIV M&E	
	3:30-3:50PM	TEA BREAK	
	3:50-5:25PM	Continue M and E Intake form and Follow up form exercise	
	5:25-5:30PM	Daily evaluation	
Day 10 Thursday	8:30AM-12:30PM	Practicum III at Health Facility	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:00PM	Discussion of practicum experience	
	3:00-3:30PM	Pre-ART register, ART register, monthly and cohort introduction	
	3:30-3:50PM	TEA BREAK	
	3:50-5:25PM	Pre-ART register, ART register, monthly and cohort introduction.....	
	5:25-5:30PM	Daily evaluation	
Day 11 Friday	8:30AM-12:30PM	Practicum IV at Health Facility	
	12:30-2:00PM	LUNCH BREAK	
	2:00-2:30PM	Discussion of practicum experience	
	2:30-3:30PM	Pre-ART register, ART register, monthly and cohort exercise	

Date	Time	Activities	Facilitators
	3:30-3:50PM	TEA BREAK	
	3:50-5:10PM	Pre-ART register, ART register, monthly & cohort exercise ...	
	5:10-5:25PM	HIV M&E Posttest	
	5:25-5:30PM	Daily evaluation	
Day 12 Saturday	8:30-8:45AM	Recap of Day 11	
	8:45-10:30AM	Chapter 16 Standard precautions, PrEP and PEP	
	10:30-10:50AM	TEA BREAK	
	10:50-12:30PM	Chapter 17 Clinical Team building and HIV service delivery	
	12:30-2:00PM	LUNCH BREAK	
	2:00-3:00PM	Presentation clinical Team building and HIV Service delivery	
	3:00-3:30PM	Post-test	
	3:30-3:50PM	TEA BREAK	
	3:50-4:10PM	End of course evaluation and reflection on the training	
	4:10-4:40PM	Certification and Closing Remark	

Chapter 1: Overview of HIV/AIDS Situation in Ethiopia

Chapter Description:

This chapter describes the HIV epidemics at global, regional and national level in line with the responses made and achievements done so far. The national strategies and objectives are also detailed in this section.

Enabling objectives:

By the end of this chapter participants will be able to:

- Describe the global and national magnitude and trends of HIV epidemic.
- Explain the national response for HIV/AIDS including policy and strategic directions.
- Describe the current status of the Ethiopian ART program.

Chapter outline:

- Introduction
- Epidemiology of HIV/AIDS
- The national response to HIV/AIDS Epidemic
- ART program in Ethiopia

Time Allocated:40 minutes

Learning activities

- Power point presentation,
- Discussion and Reflection

1.1 Introduction

In Ethiopia, the first confirmed cases of HIV were detected in 1984 and the first hospitalized AIDS patients were reported in 1986, suggesting that the HIV epidemic had probably began in the late 1970's or early 1980's. The first sero-survey conducted in 1984-85 among military recruits showed a prevalence of 0.07%. Initially, the epidemic was localized in urban areas, along the major commercial routes and among certain occupational groups. By 1988, high rate of HIV prevalence were detected among commercial sex workers (17%) residing along the main trading roads and long-distance truck drivers (13%). In some urban areas, prevalence rate as high as 38% were recorded among sex workers. In Addis Ababa, HIV prevalence rate in female commercial sex workers rose rapidly, from 24.7% in 1988 to 54.3% in 1990.

Since HIV incidence data generating systems were almost non-existent, at the beginning of the epidemic, sero-prevalence data from specific populations have been the only option to estimate the magnitude and monitor the trend of the HIV epidemic in the country. Like in other African countries, the main source of HIV prevalence data in Ethiopia has been from sentinel surveillances on pregnant women attending antenatal clinics (ANC). ANC-based HIV prevalence monitoring in urban Ethiopia was initiated as early as 1989, but expansion to rural areas has occurred only since 2001.

1.2 Epidemiology of HIV/AIDS

1.2.1. Global

According to the Joint United Nations Program on AIDS (UNAIDS) Gap Report 2018, globally, there were 36.9 million people living with HIV and 21.7 million people accessing antiretroviral therapy at the end of 2017. This number is rising as more people are living longer because of antiretroviral therapy, alongside the number of new HIV infections—which, although declining, is still very high. An estimated 0.8% of adults aged 15–49 years worldwide are living with HIV, although the burden of the epidemic continues to vary considerably between regions and countries. There are 1.8 million children younger than 15 years living with HIV.

The number of people who are newly infected with HIV is continuing to decline in most parts of the world. There were 1.8 million new HIV infections in 2017 – a decline of 47% from 1996. In 2017, 180,000 children were newly infected with HIV. This is 68.9% lower than in 2002, the year with the highest number, when 580,000 children became newly infected with HIV. Providing access to antiretroviral medicines for pregnant women living with HIV has averted more than 900,000 new HIV infections among children since 2009.

In 2017, there were 940,000 AIDS-related deaths. AIDS-related deaths have fallen by more than 51% since 2004, when the highest number of deaths were recorded. Almost 21.7 million people were receiving antiretroviral therapy globally at the end of 2017. The percentage of people living with HIV who are not receiving antiretroviral therapy has been reducing from 90% in 2006 to 41% in 2017.

1.2.2. Sub Saharan African regional

The Eastern and Southern Africa remains the region that is most severely affected. In 2017, there were an estimated 19.6 million people living with HIV in this region, nearly 53.1% of the global total. Ten countries – Ethiopia, Kenya, Malawi, Mozambique, Nigeria, South Africa, Uganda, United Republic of Tanzania, Zambia and Zimbabwe – account for 81% of all people living with HIV in the region and half of those are in only two countries – Nigeria and South Africa. In sub-Saharan Africa, three in four new infections are among girls aged 15–19 years and young women aged 15–24 years are twice more likely to be living with HIV than men.

The primary contributor to the scale of the epidemic in this region is heterosexual transmission and the increased vulnerability to and risk of HIV infection among adolescent girls and young women.

AIDS-related mortality declined by 42% from 2010 to 2017 in eastern and southern Africa, reflecting the rapid pace of treatment scale-up in the region. A significant decline (48%) was seen in South Africa and other countries that recorded major declines in AIDS-related deaths include Rwanda (76%), Eritrea (67%), Ethiopia (63%), Kenya (60%), Botswana (58%), Burkina Faso (58%), Zimbabwe (57%), Malawi (51%) and the United Republic of Tanzania (44%). This success is directly due to the rapid increase in the number of people on antiretroviral therapy. The region has witnessed an expansion in the coverage of HIV treatment to record numbers of people for the past three years.

There were 800,000 million new HIV infections in Eastern and Southern Africa. The reduction in new HIV infections between 2010 and 2017 were strongest in sub-Saharan Africa due to sharp reductions with 30% decline. Women continue to account for a disproportionate percentage of new HIV infections among adults (aged 15 and older) in sub-Saharan Africa: they represented 59% of the 980,000 new infections in the region. Despite gains in preventing new HIV infections, sub-Saharan Africa remains the region most severely affected, with nearly 1 in every 25 adults (4.4%) living with HIV.

1.2.3. National

Currently the data sources for HIV prevalence estimations in Ethiopia are ANC sentinel surveillances and national Demographic and Health Surveys (DHS). From these data sources, estimated prevalence and other indicators of HIV/AIDS of the country are synthesized. According to the 2016 Ethiopia Demographic and Health Survey, the national HIV prevalence

among adult population aged 15-49 years was 0.9%. The prevalence greatly varies by gender, age, and residence. By gender, the prevalence among women was 1.2% and among men 0.6%, with more women in urban areas having the highest risk of infection. HIV prevalence among young people was low for both sexes, but young women have a threefold higher HIV prevalence (0.3%) than young men (0.1%). Among women, HIV prevalence peaked earlier at 30-34 years reaching 3.7% while in men the peak was 3% at 35-39 years.

According to EPHI HIV related estimation and projections for Ethiopia for 2018, the national HIV prevalence is 0.96%, with a total of 610,335 people living with HIV, of which 62% are female. Besides, there were an estimated 13,488 people newly infected during 2018 (61.5% females) and an annual AIDS deaths of 13,556.

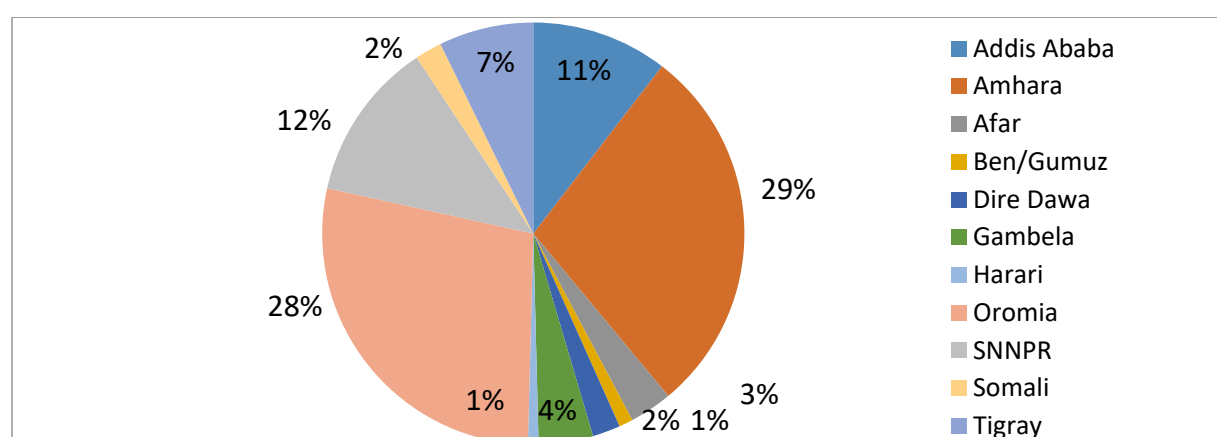


Figure 1.1. Annual new infection proportion by region, 2018

The EPHI HIV related estimation and projections for 2018 shows that there are differences in prevalence among regions and city administrations with the highest prevalence being in Gambella region, followed by Addis Ababa city administration, and Dire Dawa, where the HIV prevalence is 4.32%, 3.58% and 3.03%, respectively. The lowest is in SNNPR and Somali regional states with HIV prevalence of 0.42% and 0.16%, respectively.

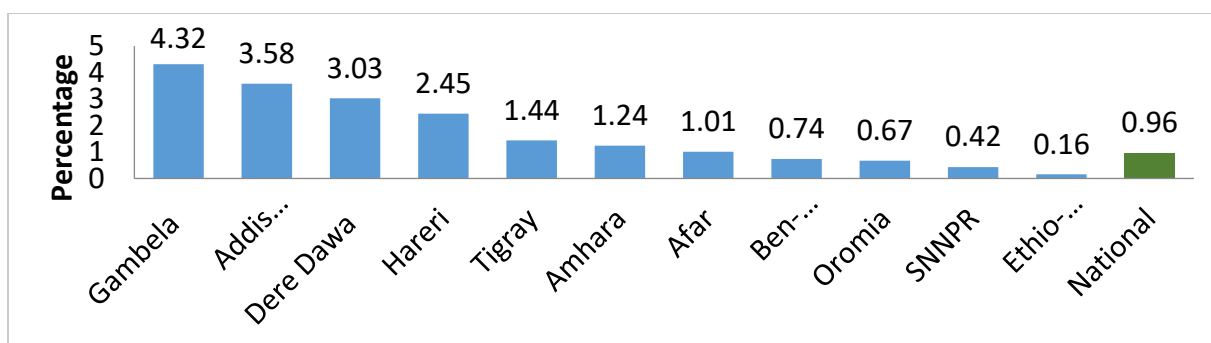


Figure 1.2. Adult HIV prevalence in Ethiopia per region, 2018

a. Urban versus rural trends

Throughout the epidemic period the estimations at hand show the urban prevalence of HIV in Ethiopia surpass significantly those of rural areas. The reason can be due to the differences in socio-economic and structure that make urban dwellers more vulnerable for HIV infections. The trends show that there is significant declining of HIV prevalence in urban areas. According to national estimates, HIV prevalence declined from 7.9 % in 2004 to 2.9% in 2018 in urban areas of the country. The data on rural areas show no significant decline, rather, they show stabilization. For instance, the prevalence in rural areas was 1.0 % in 2004 and stabilized on 0.4% from 2012 to 2018. According to EDHS 2016, the current HIV prevalence is seven times higher in urban areas than in rural areas, (2.9%versus 0.4%, respectively).

b. The national response to HIV/AIDS Epidemic

Ethiopia responded to the HIV epidemic promptly by establishing a taskforce in 1985 soon after the report of the first confirmed case of HIV. Two years later, in 1987, the national taskforce was upgraded to a department level under Federal Ministry of Health (FMOH). The department had the responsibility of coordinating the national prevention and control program. Subsequently, short-and medium-term plans were prepared and implemented in collaboration with national and international partners. However, the National AIDS Policy was issued only a decade later (1998). Further, in 1999, the Strategic Framework for the National Response against HIV was prepared. Both documents served as the basis for the expanded and multi-sectoral response against the epidemic.

Despite the prompt initial response, the national response was evaluated to be slow and frequently interrupted. As a result, further restructuring in the response mechanism was required. In April 2000, the National AIDS Council (NAC) to be chaired by FDRE president was established with secretariat offices from federal down to kebele levels. This further evolved into an office, the HIV/AIDS Prevention and Control Office (HAPCO), in 2002.

The 1998 Ethiopian HIV Policy and Strategic Plan for Management (SPM) of 2010 – 2014 have clearly articulated the guiding principles:

- **Multi-sectoralism, shared sense of urgency, ownership** and active involvement of the community, - **leadership commitment, partnership,**
- **Gender sensitivity, public health approach,** promotion and protection of **human rights,**
- Meaningful/ Greater **involvement of PLHIV,** and
- Best use of resources, **equitable** and **universal access,** sustainability and coordination.

Ethiopia has achieved exemplary successes in terms of HIV service expansion and uptake, which impacted to a 95% decline of new HIV infection from 1994 to 2012 and 73 % reduction of AIDS deaths compared to the periods 2006 to 2016 respectively. Ethiopia has developed HIV/AIDS prevention care and treatment strategic plan in an investment case approach which is being implemented from 2015-2020. This strategic plan aims to pave the path for ending AIDS by 2030 through averting 70,000- 80,000 new HIV infections and saving about half a million lives till 2020. The targets set in this investment case are in line with the three 90's (90-90-90) targets set by UNAIDS to help end the AIDS epidemic.

c. ART program in Ethiopian

As was elsewhere, Ethiopia's initial response to the epidemic had primary focus on prevention, with little attention to treatment. With the introduction of highly active antiretroviral therapy (HAART) in resource-limited settings in early 2000's, Ethiopia was among the first few African countries to introduce ART in 2003 in selected health facilities following the issuance of the national antiretroviral drugs (ARVs) supply and use Policy in 2002. The first adult treatment guideline was issued in 2003, and it has been regularly updated (in 2005, 2007, 2014 and 2018). Although, a pediatrics treatment guideline was developed in 2007. The current guideline, revised in 2018, is comprehensive that consolidates HIV prevention, care and treatment interventions both for adults and children.

Since Ethiopia started its free ART program in early 2005, a lot of lives have been saved due to the concerted efforts of the government and its partners. However, until recently, like many other countries, Ethiopia, had provided ART only for patients who reached a specific threshold in HIV disease progression. Starting ART used to impose long waits and multiple clinic visits which could lead many patients to drop out of care before they even begin treatment. Ethiopia addressed this by implementing ‘Test & Start’ strategy, which enables all people diagnosed with HIV to start ART regardless of their clinical staging and CD4 count status.

By June 2018, a total of 431,939 adults and 22,624 children under the age of 15 are taking ARV. Based on the spectrum estimate the 2018 ART need is 553,820 for adults and 56,514 for children under 15 years of age. Recently ART service is being available in more than 1,500 Health facilities. According to the new spectrum estimate for 2018, ART coverage for adults (age 15+) has reached 78% but the coverage remains low for children (age <15) living with HIV (40%).

Chapter summary

- There were approximately 36.9 million people living with HIV (PLHIV) (4.8% being children <15 years) at the end of 2017 with 1.8 million people becoming newly infected with HIV globally.
- The Eastern and Southern Africa is the most affected region, with 19.6 million people living with HIV in this region, nearly 53.1% of the global total.
- According to EPHI estimates and projections for Ethiopia for 2018, the national HIV prevalence is 0.96%.
- Ethiopia has developed HIV/AIDS prevention care and treatment strategic plan in an investment case approach which is being implemented from 2015-2020.

Chapter 2: Introduction to HIV/AIDS Pathogenesis and Immunology

Chapter description:

This chapter describes the HIV pathogenesis and our body immunology in response to HIV. The life cycle of HIV in human body and the function of normal immune system will be described in this section.

Purpose: To describe the natural history of HIV/AIDS in human body

Enabling objectives

By the end of this chapter the participants will be able to:

- Describe the function of normal immune system
- Describe the pathophysiology of HIV
- Discuss the life cycle of HIV
- Explain how HIV infection evolves in patients and the association between opportunistic infection(OI) occurrences and CD4 cells decline

Chapter outline:

- Understanding the Immune system
- HIV and the immune system

Allocated time: 50 minutes

Learning activities

- Reading
- Video show,
- Discussion,

2.1. Understanding the Immune System

One has to understand the anatomy and function of the immune system in order to understand the pathophysiology of HIV infection. This is because HIV is by and in large infection of the immune organs. The function of the immune system is to defend our body from invasion by microorganisms; this is because our body is continuously exposed to potential infectious agents through respiration, food intake or the skin that is exposed to the environment.

The immune system is organized in such a way that the body can defend itself immediately. This function is called innate immunity where by the body reacts immediately against infection or injury. Examples of innate immunity include the skin that defends the internal organs by acting like a mechanical barrier. For instance, patients who sustain burn injury develop infection easily because the skin loses its capacity to defend the body. Similarly, the acidity in the stomach kills pathogens that are swallowed with food. However, innate immunity is short lived and this defense mechanism lacks ability to recall previous infections.

The body has another type of immune response called adaptive immunity; this response is a slow but progressive defense against foreign bodies like infections and cancer cells. In addition, the response is long lived because the body learns from past experience by way of immunologic memory. An example for this type of immunity is vaccination. The body learns from the vaccine to produce specific proteins called antibodies. These antibodies are produced by cells called B-lymphocytes and the response is known as humeral immunity. Vaccine will stimulate T-lymphocytes and this process is called cellular immunity. Immune cells are produced or stationed in the thymus, bone marrow, lymph nodes, the spleen and liver. In addition, immune cells are found along the gastrointestinal tract like the tonsils or the respiratory tract collectively called mucosal-associated lymphoid organs.

2.1.1. How HIV Infects Cells?

HIV stands for Human Immunodeficiency Virus. Viruses are simple in structure and cannot replicate alone and thus they require the components of other cells to replicate. HIV, like all viruses, must therefore enter other cells if they are to replicate and survive. HIV is from a special family of viruses known as retroviruses.

HIV can be passed from one person to another through:

- Sexual contact (vaginal, anal and oral intercourse), vaginal secretions, semen
- Blood and body fluids
- Mother to child (during pregnancy, birth and breastfeeding)
- Percutaneous exposure

Viruses need a receptor on the cell surface in order to attach themselves and get inside. The CD4 receptor molecules present in a cell makes it susceptible for infection by HIV. These receptors occur on T-cells and other cells in the monocyte-macrophage cell lines. Thus, HIV

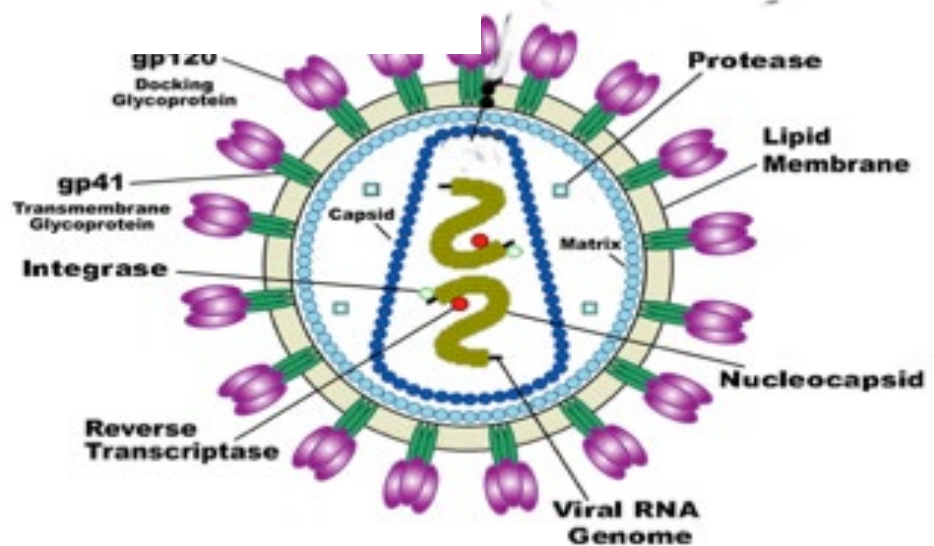
infects these cells and uses them for its multiplications. In the process, the infected cells got killed and consequently the body becomes defenseless.

2.2. HIV and the Immune System

The white blood cells play an important role to defend the body against all kinds of diseases. Lymphocytes are a type of white blood cells. The CD4 cell is a special type of lymphocyte with a marker on its surface called CD4. The process of HIV replication and immune response are summarized as follows.

1. HIV infects and destroys the CD4 cells, which are critical to guarantee body defense. These CD4 cells can be compared like army general/commander. When an army commander is destroyed during a war the whole army loses its power to defend against the enemy. Similarly, HIV destroys the CD4 cells and consequently, the body is exposed to conditions called opportunistic infections.
2. Familiarity with the sixstages of the HIV life cycle is essential for understanding the effects of ARVs on HIV.
 - HIV **attaches** to the CD4 cell and releases RNA and enzymes.
 - The reverse transcriptase enzyme makes a **DNA copy** of the viral RNA.
 - New viral DNA **integrates into the CD4** cell nucleus using the enzyme integrase.
 - **New viral components** are then produced, using the cell's "machinery"
 - The new viral components are, then, **assembled together** using the enzyme protease
 - Then **released** as new viruses.

Figure 2.1 HIV virus cell structure



*Watch a video show on HIV life cycle attached.

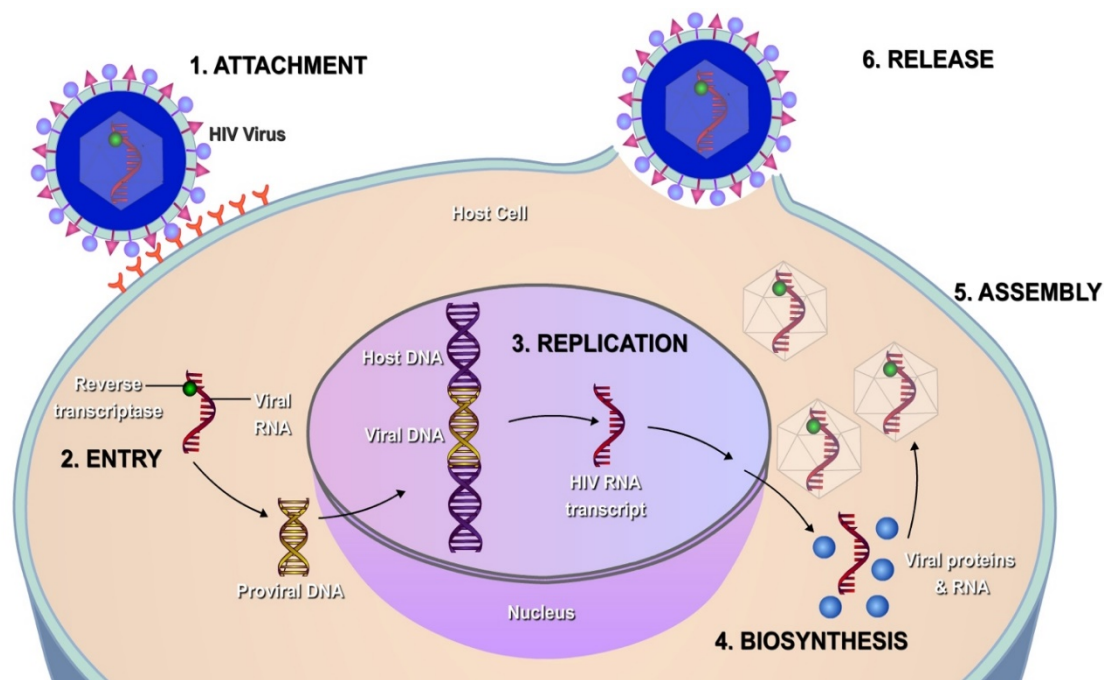


Figure 2.2. Life cycle of HIV

3. HIV uses the CD4 cell like a factory. It needs the machinery inside the factory (CD4 cell) to replicate. So, HIV enters the factory and starts replicating, using the CD4 cell's machinery. Millions of new viruses are released from the factory (CD4 cell). These new viruses then move on to infect other CD4 cells which also become factories for HIV.
4. The extent of immune damage inflicted by HIV is assessed by the CD4 count and the number can be counted in a small sample of blood. This is called the CD4 count and tells us how strong the immune system is. A CD4 count between 450-1500 cells/mm³ indicates the immune system is coping well and managing to remain high in spite of HIV. However, over time, the CD4 cells are progressively destroyed and the CD4 count falls. A low CD4 count tells us the immune system is weak.

5. The Viral Load (VL) indicates the activity of HIV infection and hence viral load is another useful blood test to tell us how much HIV is in the blood. Over time, the viral load increases as more and more viruses are produced resulting in rapid progression to AIDS.
6. The **clinical course** of HIV infection has three stages.
 - i. Primary HIV infection
 - ii. Asymptomatic stage/latent infection
 - iii. AIDS

2.2.1.Primary HIV infection

The initial stage following HIV infection is called primary HIV infection. Patients develop nonspecific 'flu-like' symptoms, which do not lead directly to the diagnosis of HIV infection because the symptoms are nonspecific. These symptoms include;

- Fever,
- Fatigue,
- Pharyngitis,
- Lymphadenopathy,
- Rash etc.

During primary infection, the patient could be negative for HIV specific antibodies despite the presence of infection. This stage is called window period and is characterized by very high viral load signifying extreme infectiousness in spite of false negative antibody result. The switch from antibody negative to antibody positive is called sero-conversion and most patients sero-convert within 3 months after exposure to the virus. Clinical management includes symptomatic treatment, counseling for risk reduction and repeat antibody testing. Patients are most likely to transmit HIV during the early stage of infection.

2.2.2. Asymptomatic stage/latent infection

When a person gets infected with the HIV, the virus will start to attack their immune system. During the first years of infection, the immune system-although weakened a bit by the HIV virus - still functions quite well. The infected person will have no symptoms, or only minor symptoms like skin diseases, a little loss of weight, or repeated sinusitis. A lot of people do not

know they are HIV+ at this stage. The clinical latency stage lasts an average of 10 years, but some people may progress through this stage faster.

2.2.3. Acquired Immune Deficiency Syndrome (AIDS)

After several years, the person's immune system will be very weak; they are vulnerable to diseases that they could normally fight off. These diseases are called opportunistic infections (OIs) because they take advantage of a weakened immune system to cause disease.

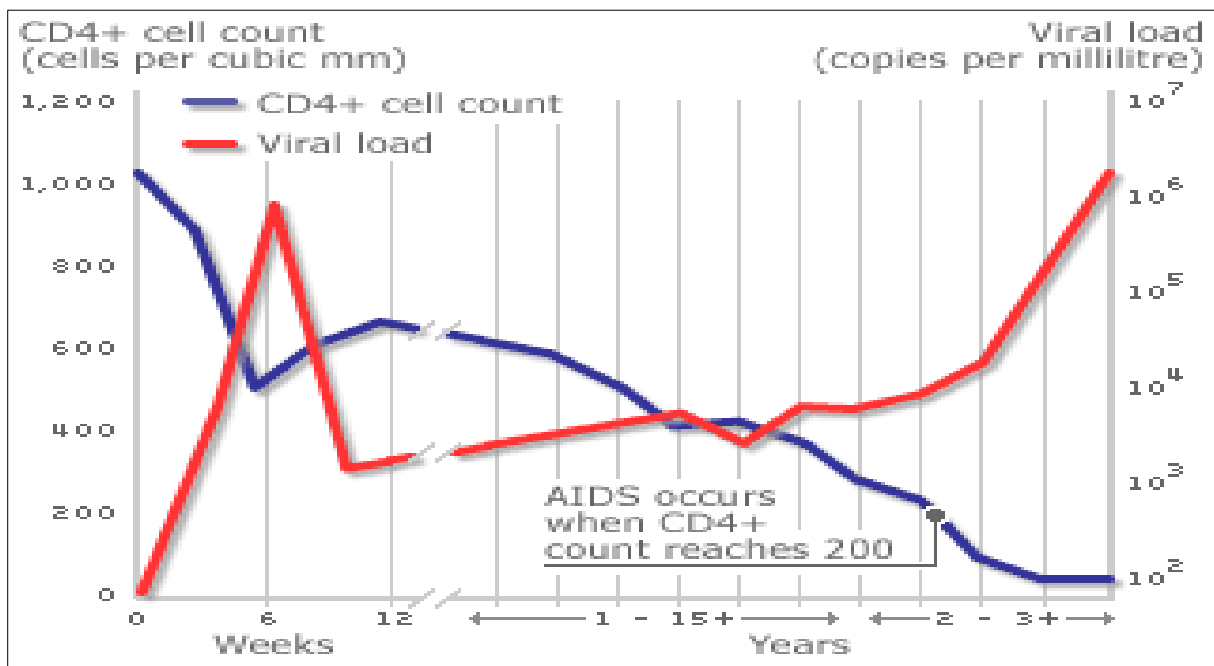


Figure 2.3 Disease progression pattern among adults

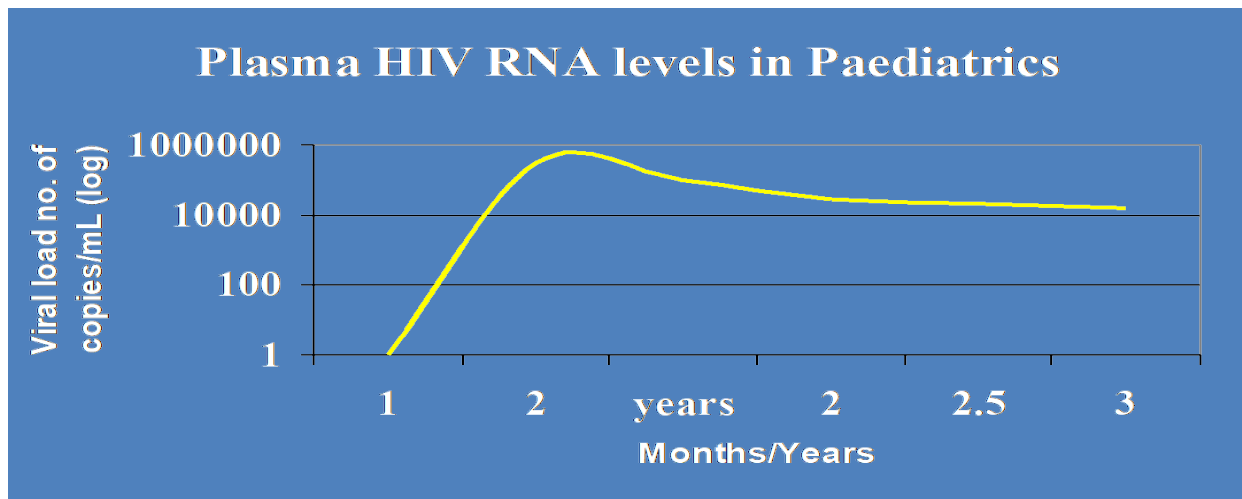


Figure 2.4. Viral replication pattern among infants and children

NB: HIV RNA levels in perinatally infected infants are generally low at birth (<10,000 copies/ml), increase to high values by age 2 months, and then decrease slowly after the first year.

Patterns of HIV disease progression varies among patients because of the viral set point (serum viral load level at 6 months to one year after acquisition of HIV infection) following seroconversion. Accordingly, three types of progression are noted for adult patients:

1. **Typical progressors** account for 90% of individuals who can stay for 8-10 years before developing symptoms. The viral set point is medium in this group.
2. **Rapid progressors** account for 5% of individuals who develop AIDS within 3 years. The viral set point is high in this group.
3. **Long term non-progressors** account for 10% individuals will have stable CD4 count for more than 8 years. The viral set point is remarkably low In this group

For Pediatric HIV infected populations, the following patterns are usually observed:

- **Category 1 (25–30%):** Rapid progressors, who die by the age of one and who are thought to have acquired the infection *in utero* or during the early perinatal period.
- **Category 2 (50–60%):** Children who develop symptoms early in life, followed by a downhill course and death by the age of three to five.
- **Category 3 (5–25%):** Long-term survivors, who live beyond the age of eight.

Ways of explaining the evolution of HIV infection to patients:

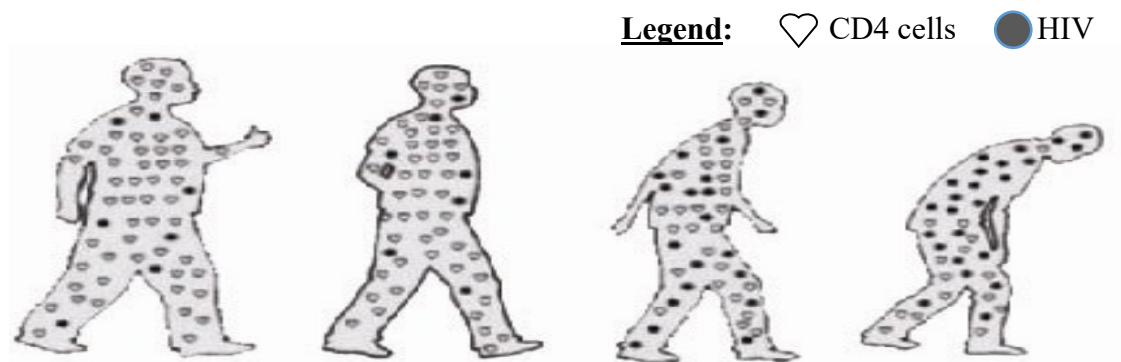


Figure 2.5. Natural course of HIV infection

Beginning: skin diseases, minor loss of weight...	After 5-10 years: chronic diarrhoea, brain problems, other opportunistic infections.
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As the CD4 level declines, the risk of getting opportunistic infections increases.

People with good immune systems have CD4 counts between **450 and 1500 cells/mm³**. In general, we can say:

- Some opportunistic infections develop when the CD4 count decreased below 450 cells/mm³,
- Very serious opportunistic infections start manifesting when the CD4 count decreased below 200 cells/mm³.

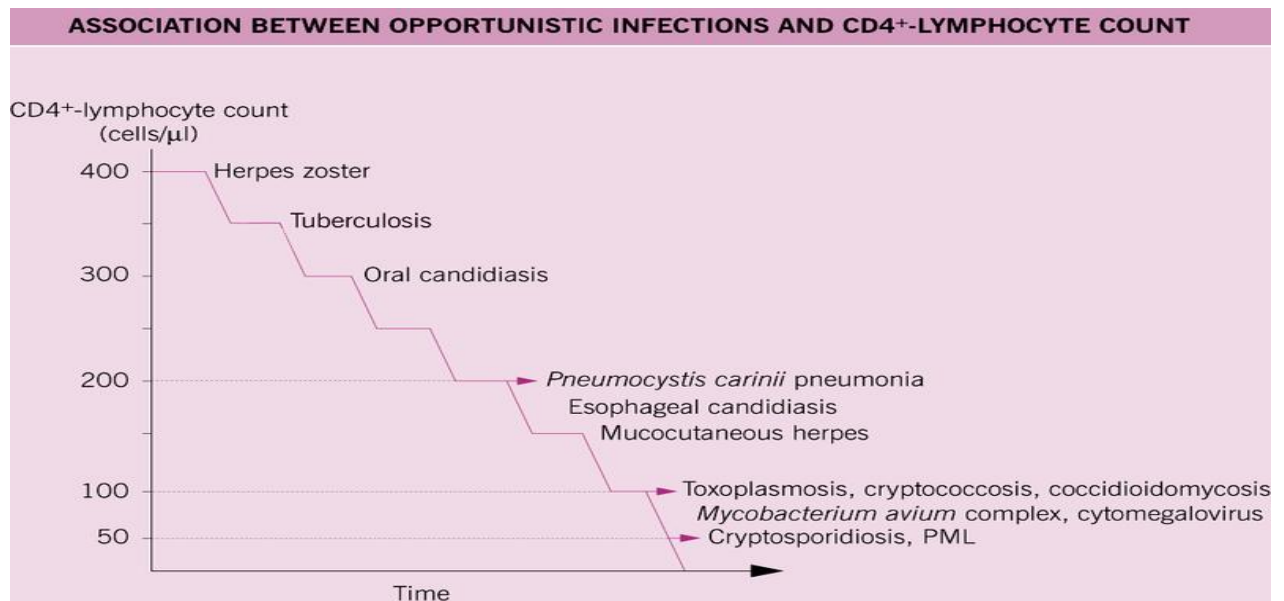


Figure 2.6. Association between OIs and CD4⁺ lymphocyte count

Chapter summary

- The body is protected from infection by the activity of the immune system
- The immune system has innate and adaptive components
- HIV infection is a chronic disease that affects the immune system of the body
- Progressively CD4 cell count decline with advancing disease, As CD4 cell count declines below critical levels, OIs occur,
- Clinical course of HIV infection varies from asymptomatic early stage to advanced disease
- HIV can only be suppressed by ART, there is no cure!

Chapter 3: Integrating Prevention with Treatment

Chapter Description:

This chapter describes the HIV prevention strategies, clearly understanding the different ways HIV infection can happen and NOT happen. The national HIV prevention roadmap with the 6 pillars are also detailed in this section.

Purpose: Understand combination prevention in HIV

Enabling objectives:

By the end of this chapter participants will be able to:

- List the different ways of getting infected with HIV
- Explain the most common ways of preventing HIV transmission;
- Explain the 6 pillars of combination prevention of the national HIV prevention roadmap
- Demonstrate proper use of male and female condom
- Counsel clients on prevention during every treatment encounter.

Chapter outline:

- Introduction
- Modes of transmission
- HIV prevention- combination strategy
- Condom use

Allocated time: 120 minutes

Learning activities

- Brainstorming,
- Case discussion,
- Reading
- Demonstration,
- Discussion,

BRAINSTORMING STORY

Read the story below carefully. Try to find as much information as possible about HIV/AIDS in this story. You can write down your ideas in the section that follows the summary of the story.

The story of Aselef and Alemayehu

1

- The story starts in 1993
- Aselef was a beautiful girl living in Adama Town
- She had just finished University and was to get married to Alemayehu
- The family of Aselef really liked Alemayehu

Alemayehu, one year ago

2

- Alemayehu owns a shop in Nazareth
- Last year, he went on a business trip
- During the trip, one night, he drunk a couple of beers, and met a sweet lady in the bar
- He spent the night with her

Alemayehu

3

- Alemayehu did not know it, but during that night, he got infected with HIV

March 1993

4

- Aselef and Alemayehu got married
- They had a beautiful party and good food with a lot of people invited

April 1993

5

- One day Aselef felt sick
- It was not very serious, it looked like flu with a sore throat and fever

Aselef got infected

6

- Aselef did not know it.
- Alemayehu also did not know it, but he transmitted the virus to his wife

Beginning of 1994

7

- Aselef is pregnant
- She had been very happy
- The first child of the family was born, a baby boy
- Asegid was not infected by HIV

1995, Dagim was born

8

- The family felt very happy, their business is going well and Aselef gave birth to another baby, Dagim
- Dagim got HIV through MTCT

1996, Dagim

9

- Aselef started to worry. Dagim had fever, diarrhea and had not been gaining weight.
- She went to the doctor

The Doctor talked about HIV/AIDS

10

- The doctor said that probably the baby had HIV
- The doctor wanted to do HIV test but she did not accept
- She returned home and did not talk about this with any one

End of the year 1996

11

- One night, Dagim died
- The whole family became very sad

Alemayehu, 1997

12

- Alemayehu started to be sick very often
- Alemayehu went to buy different treatments
- After one treatment, he got a bit better but got relapsed

1999, Alemayehu died

13

- The last month had been very hard for Aselef
- Aselef had been in a very bad mood due to the disease
- She also had to manage the business alone and take care of her sick husband

Aselef thinks...

14

- Aselef felt very depressed
- She did not felt sick yet, but she understands that this terrible disease had come in to her family,
- She is afraid for herself and her child, what will happen to Asegid, if she dies?

- 1.How was Alemayehu infected with HIV? _____
- 2.How was Aselef infected with HIV? _____
- 3.Do you think Dagim died of AIDS? _____
- 4.How was Dagim infected with HIV? _____
- 5.Why didn't Asegid get HIV? _____
- 6.How much time was there between Alemayehu's possible infection and the year he started to have serious symptoms? _____
- 7.How much time passed between the time Alemayehu got infected and the time he died? _____
- 8.Why do you think Aselef was sick shortly after marriage? _____

EXERCISE 3-2: List the modes of HIV transmission and ways of prevention.

3.1. Modes of HIV Transmission

Getting infected with HIV does not happen as easily as getting measles or influenza. Viruses like measles and influenza are transmitted by air, typically during social contact with infected people.

HIV needs 'transport' to get into the body of a person. This 'transport' can be blood, semen, vaginal fluid, or breast milk.

HIV can be transmitted through:

- **Unprotected Sexual practices** with an infected person.
- **Transfusion** with contaminated blood or blood products.
- The use of **needles, syringes** and **cutting** or **perforating objects** contaminated by HIV-infected blood, Sharing contaminated sharps for certain traditional practices (tooth extraction, uvulectomy, infibulation/ female genital mutilation, circumcision, tattooing/ scaring)
- **MTCT** during pregnancy, labour and delivery and breastfeeding from HIV infected women.
- Organ **transplantation** from HIV infected donor.
- Other contaminated **body fluids** like ascites, pleural effusion, pus and other discharges from an infectious lesion.

Box 3.1: Ways HIV can be transmitted through

a. Transmission through Sexual Relations:

HIV is primarily considered a sexually transmitted infection. **Unsafe sex** with an infected person is contributor for the majority of HIV infections. However, different types of sexual relations have different degrees of risk for transmitting the HIV infection. According to severity:

Anal sex: Among sexual practices, anal sex represents the biggest risk of infection if one of the partners is HIV infected. The anal sex practice represents the **biggest** risk because the anal mucosa does not produce a natural lubrication, is fragile, and wounds and bleeds very easily. The penis can have micro-lesions, which permit the entrance of the virus into the bloodstream.

Vaginal sex: The HIV virus can be found in large quantities in semen, and to a lesser amount in vaginal secretions of infected persons. The risk of infection is still high, but less than with anal sex, because the vagina produces a natural lubrication, and is more elastic. Vaginal sex, nevertheless, represents a serious risk of HIV infection because the vaginal mucosa can still have micro-lesions during penetration, which permits the entry of the virus. The HIV virus in vaginal secretions can penetrate the penis through micro-lesions. The soft tissue of the foreskin in **uncircumcised** men seems to be especially vulnerable to the entry of the HIV virus (increased risk).

Oral sex: Compared to anal and vaginal sex, oral sex represents the lesser risk for infection. However, very small wounds in the mouth can increase the risk of infection.

Important

The presence of sexually transmitted infections (discharge, ulcers) increases the risk of acquiring and transmitting HIV. This is because people with a sexually-transmitted infection have a higher concentration of HIV in the genital mucosa, and/or because transmission of the virus is facilitated due to the presence of lesions in the mucosa.

b. Transmission through blood and blood products:

The blood is one of the agents to transmit HIV. HIV can be transmitted through:

- Transfusion of infected blood or contaminated products made from infected blood.
- Needles and syringes with infected blood (even small quantities).

- Perforating or cutting instruments contaminated with infected blood. Transplantation of organ donated by HIV infected person.

c. Transmission from mother to child

The term Mother-to-Child Transmission (MTCT) of HIV is used in this document because in majority of cases (>90%) the source of the child's HIV infection is the mother. Use of the term mother-to-child transmission does not imply blame, whether or not a woman is aware of her own infection status.

MTCT is the transmission of HIV from an infected mother to her baby. It can occur during:

- Pregnancy
- Labor and delivery
- Breast feeding

MTCT is also called “Vertical transmission” or “Perinatal transmission”. It is the main cause of HIV infection in children. Note that not all HIV-positive women will transmit the virus to their child. A summary of rates of transmission without intervention is provided in Figure below.

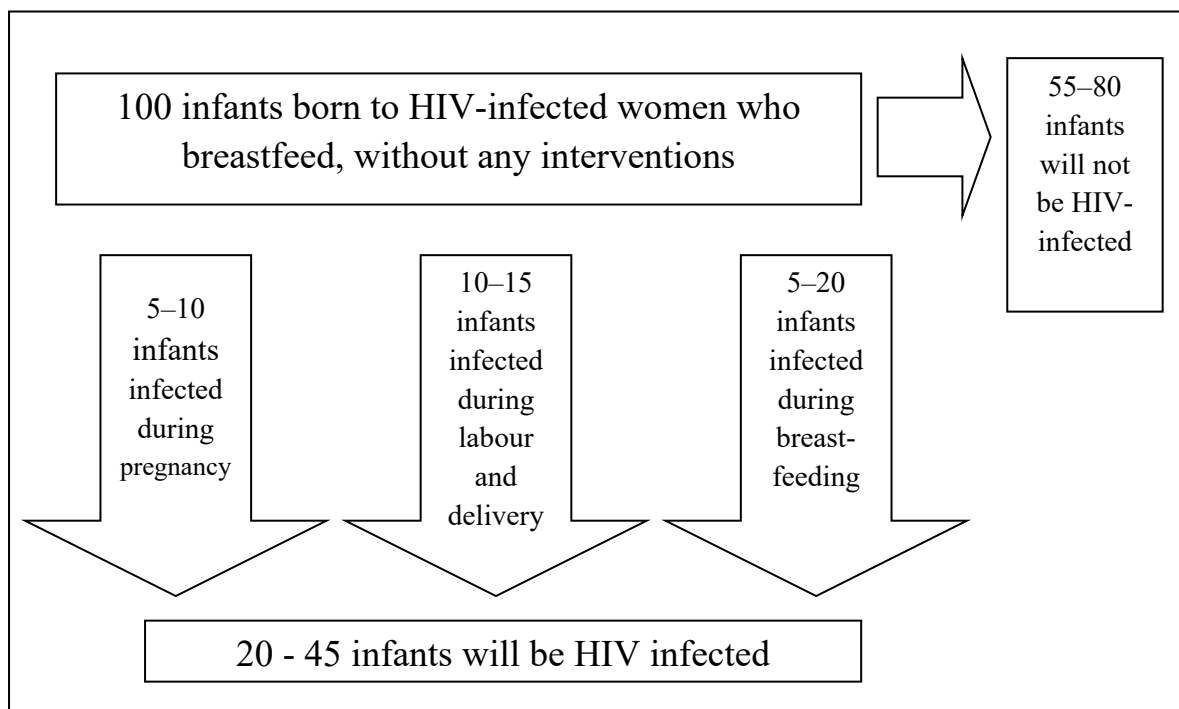


Figure 3.1. HIV outcomes of infants born to HIV positive women without intervention

The risk of transmission during pregnancy is low, as the developing baby is protected by the placenta. During labour and delivery, the risk is increased through sucking, absorbing or aspirating blood or cervical fluid. Several risk factors influence MTCT of HIV. Some of the major factors are:

Maternal factors:	Infant factor:
• High viral load • Low CD4 count with advanced disease • Prolonged rupture of membrane • HIV infection during pregnancy/breast feeding • Mixed feeding • Crackled nipples and breast abscess	• Prematurity • Oral thrush and ulcer • Birth order (first twin) in twin pregnancies • Invasive fetal monitoring during labor and delivery • Instrumental delivery

d. Other ways in which children can get HIV are:

- Sexual abuse
- Unsafe injections/injection by local healer
- Transfusion of blood donated by HIV infected persons
- Wet nursing by untested woman
- Manipulation by local healer (uvula cutting, milk teeth extraction, FGM)
- Feeding children by chewed food by the mother/adult care taker
- Using sharp object contaminated with HIV infected blood

Ways HIV NOT transmitted:

- Other body fluids like tears, saliva, sweat and urine
- Personal contacts: kisses on the mouth, hugging, handshakes
- Social contacts: during the work, in school, cinema, restaurant, and sauna
- Air or water: sneezing, coughing, swimming pool, swimming in the sea
- Contact with common items: pens, toilets, towels, sheets, soap
- Insects: mosquito bites or other insects.

3.2. HIV PREVENTION

Ethiopia implements Combination HIV prevention approach that leads to achieve maximum impact for preventing new HIV infections by combining biomedical, socio-behavioral, and structural interventions that are human-rights based and evidence informed, in the context of a well-studied and understood local epidemic. According to the National HIV prevention Roadmap 2020, the combination HIV prevention package has six prevention pillars, which includes Combination prevention for **adolescent girls, young women & their male partners**; Combination Prevention for **Key & Priority Population**; **Promotion & distribution of Condom**; Voluntary Male Medical circumcision (VMMC); Pre-Exposure Prophylaxis (PrEP) and STI prevention and control. The socio- behavioural and structural interventions of the combination prevention include provision of the socio- behavioural change and communication (SBCC) service, economic empowerment, prevention of gender based violence, and reduction of stigma and discrimination on PLHIV.

3.2.1. Ways to prevent HIV infection (Primary Prevention)

Interventions that help to prevent HIV infection at individual level are:

- ABCD (Abstain, Be faithful*, correct and consistent use of Condom and Dialog or Discussion on HIV risk, need for periodic testing and concerns)
- Implementation of Test and Start
- Prevention of Mother to Child transmission (PMTCT)
- Positive Health Dignity and Prevention (PHDP)
- Voluntary medical Male circumcision (VMMC)
- Infection Prevention (IP)
- Post Exposure Prophylaxis (PEP)
- Pre-Exposure Prevention (PrEP)

*“If you and your partner do not have HIV, it is highly advised that two of you should be in a faithful relationship and not have sex with any other partners. This will protect both of you from getting HIV.”

The following points make prevention of HIV transmission very important:

- The existing effective treatments control HIV, but doesn't cure

- HIV treatment is for life, and might have annoying side effects
- Some viruses have become drug-resistant
- There is still no preventative vaccine for HIV

3.2.2. HIV prevention among People Living with HIV

Also called, Positive Health Dignity and Prevention (PHDP) refers to interventions that help HIV positive people to take part in safer sex behaviors so that they can protect their partners and/or future children from getting HIV and preventing themselves from acquiring new strains of HIV (superinfection) or other Sexually Transmitted Infections (STIs).

a) **Why Prevention with Positives is important?** Providing prevention service for HIV positives:

- It prevents HIV transmission in discordant couples
- It decreases the potential transmission of ARV-resistant HIV strains
- Enables PLHIV protect themselves from getting sick with STIs
- Promotes adherence to their medication
- Improves quality of life for PLHIV
- Encourages testing and treatment for their partners
- Promotes PMTCT, protecting their babies
- Prevents unwanted pregnancy
- Reduces HIV transmission in the community.

b) **The Five HIV Prevention Steps for PLHIV:**

1. Provide **prevention messages** on: **risk reduction, disclosure, partner testing** (encourage their sex partners and children to be tested), **prevention of STI, condom use&FP** to your HIV positive clients at every visit
2. Assess client's **adherence** to ARVs and other medications at every visit
3. Assess client for signs and symptoms of **STIs** at every visit
4. Assess **pregnancy status** and intentions of client's partner at every visit
5. Provide **condoms** for dual protection

In providing family planning counseling, providers should:

- Respect the **right of all women**, regardless of HIV status, to decide the number and timing of children
- Encourage **dual protection** (using two forms of contraception where one should be a condom).
- Provide **condoms** wherever possible and refer clients to a convenient affordable source
- Provide integrated FP/HIV and STI services at all levels of care
- Offer information about prevention and HIV counselling and testing
- Counsel men and women who know they are positive, assisting them to make well-informed decisions

Family planning services shall be integrated within HIV care settings to avoid unintended pregnancy among women living with HIV (WLHIV). WHO recommends **routine offering of family planning** services to all adults and adolescent clients in HIV care services. For women living with HIV, special emphasis and support shall be provided on selecting modern contraceptive because some ARVs reduce the **potency (lifetime) of hormonal contraceptives**. This applies for WLHIV already using or new to hormonal contraceptives. It is also important that WLHIV shall be well aware that FP methods only prevent unintended pregnancy and do not prevent HIV and other STIs.

Use of **condom** with chosen method of FP for dual prevention of pregnancy and STIs/HIV shall be promoted whenever FP is considered. As a provider, please ensure HIV positive Women of child-bearing potential have access to and utilize effective FP methods with Condom at every visit.

Note:

For all PLHIV, counsel on safer sex and reducing risk of transmission

- Even on ART, you can still **transmit HIV**
- Educate on **symptoms of STIs**, and necessity to consult for prompt treatment
- Educate on **myths** associated with HIV infection, transmission, and treatment
- Emphasize **not** to have sex with casual sex partner,
- Emphasize that **Condom** prevents both **HIV/ STI** and **unintended pregnancy**,

c) Protecting the partners of PLHIV:

It is very important to know whether the patient's spouse or partner(s) are already infected or not with HIV. This means careful counselling on disclosure and active support for partner testing is important.

Many HIV-positive people assume that their partners are already infected which is not always the case. It is common to get couples with “discordant” HIV test results i.e., one is positive and the other is still negative. An HIV-negative partner in a couple is at a **very high risk of infection**. It is important for couples to develop to protect their sexual partners from infection and they should be supported in their planning efforts including ART therapy for the HIV infected partner.

While encouraging disclosure and testing of partner(s), it is important to discuss barriers to disclosure and explore benefits of disclosure.

- If the patient is ready, develop strategy and provide appropriate support (role play, refer to support group, etc.).
- Encourage and facilitate partner notification services and partner testing.

Adapt the information you provide to the person and the different situations, using the prevention section of the patient education material (job-aid). This contains special considerations for adolescents, women, men, pregnant women, and couples, both for discordant (one partner is positive, one partner is negative) and concordant (both are positive).

d) Counsel the patients on consistent and correct use of condoms during every sexual encounter.

- Emphasize that it is essential to use condoms consistently especially when already infected with HIV.
- Educate and demonstrate on how to use both male and female condoms.
- Provide condoms or explain how to obtain them.
- Discuss the potential barriers to consistent and correct use of condoms and explore how to overcome these barriers (providing techniques/skills for negotiating on condom use; role play with the patient).

e) Important considerations when using male condom

Condom demonstration and re-demonstration Steps

Emphasize that latex and vinyl are the only materials from which condoms are made that are proven to be effective barriers to HIV. You will need a **penis model** for the demonstration.

1. Show how to inspect the condom by checking the condom package to make sure it is not punctured and not expired. (If the condom package is punctured or expired, throw the condom away and repeat inspection with a new condom.)
2. Carefully open the condom package with your fingers. (Stress that you should never use a sharp object because it may puncture the condom.)
3. Find the tip of the condom with the forefinger and hold it so that the ring hangs down like a little hat.
4. Hold the tip with the forefinger and thumb as you place the condom on the penis model, ring on the outside. For uncircumcised person, the foreskin shall be retracted before applying the condom.
5. Roll the condom down to the base of the penis with the other hand.
6. Tell the client that after sex, hold the condom at the base and pull the hard penis away from the partner. Do not spill any liquid on the partner.
7. Slide the condom off.
8. Dispose immediately in proper waste container.

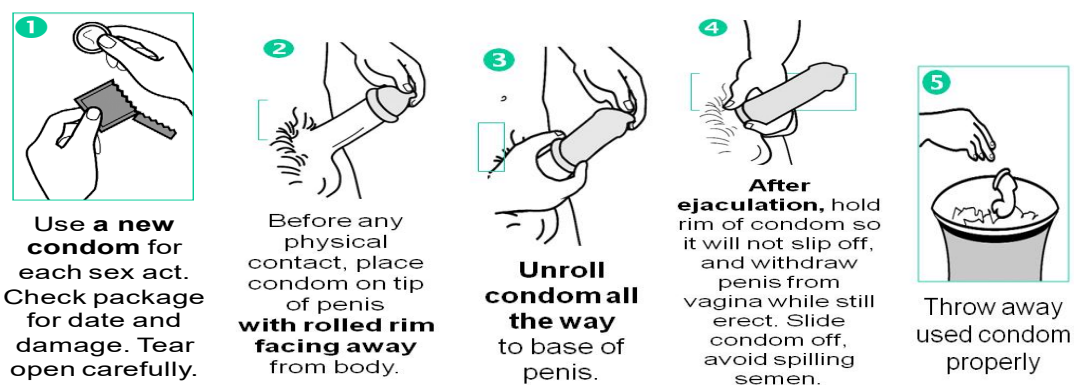


Figure 3.2. Demonstration steps on correct use of male condoms

f) Important consideration when using female condom (show female condom):

- Look at the expiry date.
- Check that the package is not damaged.

- The condom should be inserted **before** penis touches vagina (can be up to 4 hours before sex).
- After ejaculation, the woman must move away from partner and take care not to spill semen on vaginal opening.
- Dispose of used condom properly.

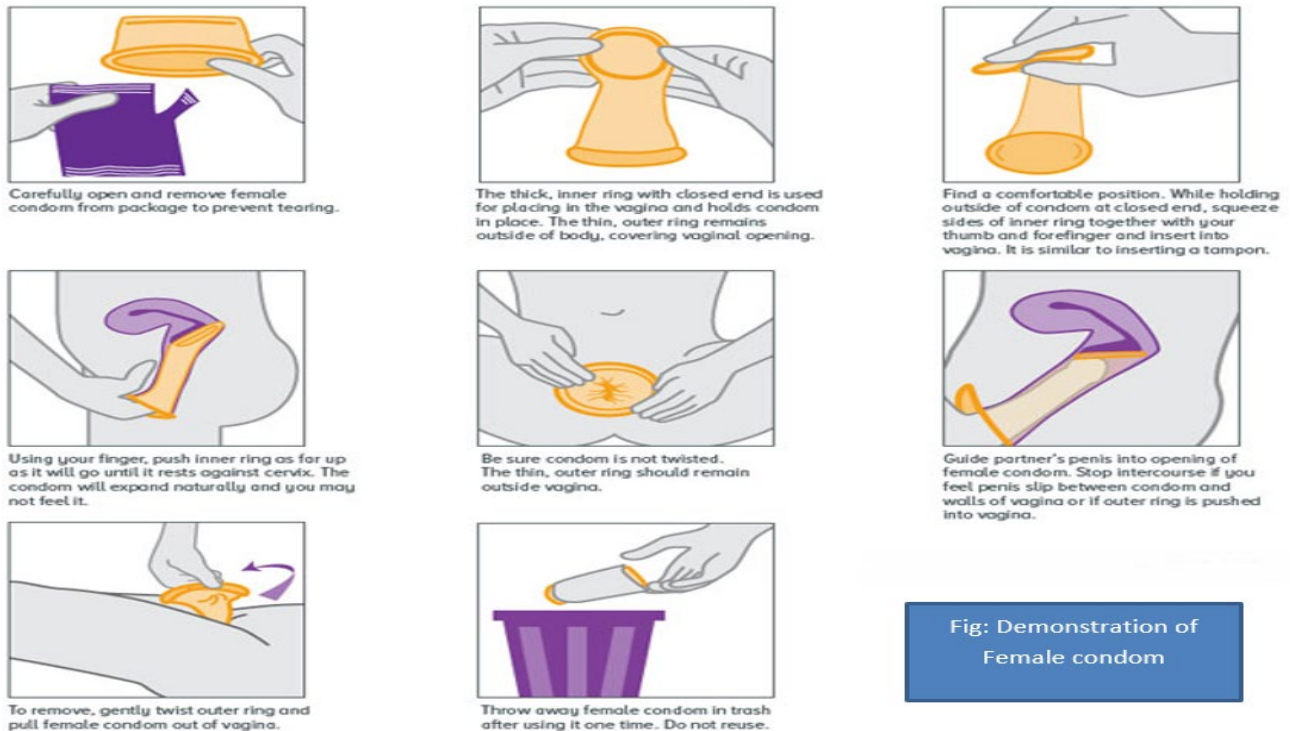


Figure 3.3. Demonstration steps on correct use of female condoms

g) Prevention of mother to child transmission of HIV

There are four prongs to eliminate the vertical mother-to-child transmission of HIV infection.

- Prong 1:** Primary prevention of HIV infection - focuses on keeping parents-to-be HIV negative,
- Prong 2:** Prevention of unintended pregnancies among women infected with HIV,
- Prong 3:** Prevention of HIV transmission from women infected with HIV to their infants addressing care for infants born to HIV positive women and their mothers during pregnancy, labor and childbirth, and the postpartum period.
- Prong 4:** Provision of treatment, care and support for women infected with HIV, their infants and their families.

Access to quality and comprehensive maternal and newborn health services (i.e., antenatal, labor and childbirth, postpartum and newborn care services) which integrate access to HIV testing and counseling is central to any effort to eliminate mother-to-child transmission of HIV.

h) Pre – Exposure Prophylaxis (PrEP)

Oral Pre – Exposure Prophylaxis (PrEP) is the use of ARV drugs by people who are not infected with HIV but on continuous risk to block the acquisition of HIV. PrEP is an evidence-based HIV risk-reduction intervention that is offered to all people at **substantial risk of acquiring HIV**. Substantial risk of HIV infection is provisionally defined as an incidence of HIV higher than 3 per 100 person-years in the absence of pre-exposure prophylaxis (PrEP). The Target population for PrEP service are FSWs and HIV negative partners of sero-discordant couples with substantial risk of acquiring HIV.

i) Prevention of transmission through non-sexual means

- Avoid sharing contaminated needles, razor blades or tattoo instruments,
- Carefully clean blood or body fluids and safe dispose of waste contaminated with body fluids.
- Clean and bandage any open sores or cuts.
- Protection of Health workers and Caregivers from exposure through Testing of blood, sterilization of materials, Standard Precautions and Post Exposure prophylaxis which will be discussed in chapter 16 of this manual.

EXERCISE 3-3: Linking ART with increased prevention



Tulema is on ART. He would like to get married in the future. The nurse counsels him to use a condom when he has a new partner, but he says he finds it very difficult to do so. The nurse listens carefully and sympathetically to Tulema's concerns.

Q. What would you do for Tulema?



Gemechis is HIV-positive and he is on ART. When Gemechis travels, he often has a partner for just one night.

Q. What would you tell Gemechis?

j) Linking ART with increased prevention

Although ART can reduce the level of HIV virus, it does not eliminate the virus, and transmission can still occur. Patients on ART need to understand that risks remain, and they need to work even harder to prevent transmission. This should be part of the agreement in deciding to take ART.

- **Risk of the patient being re-infected with another strain of the HIV virus.** Even though they are already infected, they can be infected a second time with a different strain that can cause further problems for their immunity.
- **Even-though the viral load decreases after getting treatment, there is still risk of infecting others.**

- There is also the **new risk that, if the patient has poor adherence to ART and has developed a drug-resistant HIV virus, transmission of the resistant virus can occur.** Drug-resistant HIV is more difficult to treat. Therefore, we need to continue to emphasize safer sex to prevent transmission of drug-resistant HIV.

ART access can:

- **Reduce the fear, stigma and discrimination that surround HIV/AIDS.** Being able to provide ART should make it easier for healthcare workers to discuss HIV infection with their patients, including how HIV is spread, and how transmission can be prevented. Administering ART also requires providing more prevention counselling during your clinical encounters, not less! Integrating prevention counselling with care and treatment scale-up is also important.
- **Increase the use of HIV testing and counselling.** It is very important that you routinely recommend HIV testing and work with HIV positive patients to have their partners and family tested, and to educate and counsel them before and after testing.
- **Help people change behaviours to avoid further transmission.** This requires health workers and community actors to educate the public and support preventive interventions.
- **Can lead to myths about a weaker virus, or that safer sex is no longer needed in patients taking ART.** This could lead to an increase in risky behaviour and more, not less, transmission. So we need to be active in our prevention efforts when we introduce ART.

What to do at your health facility:

Think of ways that your health facility can be involved in new projects that reach out to those most affected by HIV and help provide both prevention and care and treatment.

- **Provide better information and more effective education and counselling, to be sure that we get the prevention benefits from introducing ART.**

- **On every clinic visit, reinforce prevention** — by educating, checking up on key prevention interventions that apply for each patient, providing condoms, demonstrating condom use when needed, and supporting PLHIV in positive prevention.

EXERCISE 3-4: Practice on how to demonstrate correct condom use

Chapter summary

- Although ART can reduce the load of HIV virus, it doesn't eliminate the virus and transmission can occur.
- Even for HIV positives, there is a risk of acquiring new strain of HIV and/or the new virus could be a drug-resistant virus
- Anal sex has the highest rate of infection followed by vaginal sex and oral sex.
- The three common modes of HIV transmission are: unprotected sexual intercourse, use of contaminated sharps and needles and from mother to child.
- Access to ART can reduce stigma and discrimination, increase the utilization of HIV testing and counseling, help people change behavior to avoid transmission and can lead to assumption that safer sex is still needed while on ART
- The presence of STI increases the risk of acquiring and transmitting HIV.
- On every clinic visit, reinforce prevention with positives and the concept of PHDP
- Many HIV positive people assume that their partner is already infected but this is not always true
- Combination prevention modalities like condom, PrEP, VMMC, STI treatment can be administered in combination as appropriate,
- PrEP is useful prevention modality for FSWs with substantial risk and HIV negative discordant partners until the positive partner has sustained viral suppression,

Chapter 4: HIV Diagnosis in Adults and Children

Chapter Description:

This chapter describes how HIV infection is diagnosed and the national HIV testing service. The national strategies and approaches implemented in the country to identify HIV infected persons are also detailed in this section.

Primary objectives: At the end of this chapter participants will be able to provide HIV testing services in adult and children.

Enabling objectives:

By the end of this chapter participants will be able to:

- Explain the rationale of HIV Testing Services (HTS)
- Describe the values of HTS guiding Principles and List basic counseling skills
- List the steps in Provider Initiated Testing and Counseling (PITC) adults and children
- Explain the PITC protocols/ steps and tasks for adults and children,
- Describe the national three test HIV testing algorithm,
- Describe Partner and family index testing Services,
- Offer partner testing service to index,
- Offer HTS for partners and children of index,
- Offer index case testing (partner and children) for FSWs
- Describe innovative testing approaches: response driven HTS, Sexual network strategy, HIV self-test and risk assessment,
- Describe HIV diagnosis in infants, children and adolescents

Chapter Outline:

- Introduction to HIV testing service
- Provider Initiated HIV Testing and Counseling
- Partner and family based Index case testing
- Diagnosis of HIV in Infants, Children and Adolescents
- Laboratory diagnosis of HIV in infants younger than 18 months of age

Allocated time: 250 minutes

Learning activities

- Reading
- Discussion,
- Demonstration,
- Role paly,

4.1. Introduction to HIV Testing Services

HIV testing services (HTS) refer to the full range of services that should be provided with HIV testing, including counselling (pre-test information and post-test counselling); linkage to appropriate HIV prevention, treatment and care and other clinical services; and coordination with laboratory services to support quality assurance (QA) and the delivery of accurate results.

The HIV testing services (HTS) in Ethiopia are either facility based or community based depending on the delivery points. HTS initiated by Client (VCT) or health care provider (PITC) can be accessible at static public, private and NGO's health facilities. Community level HTS are provided through Home-based testing targeting specific sub-group or family members of Index case, Outreach HTS for high prevalence areas (hot-spots) and Work-place VCT service in big farms, factory or construction sites.

4.1.1. The Rationale

HIV testing is the critical first step in **identifying** and linking PLHIV to the treatment cascade; it also provides an important opportunity to reinforce HIV prevention among the negatives. To identify all HIV positives and attain universal access of comprehensive HIV/AIDS care and treatment services, it is mandatory to expand targeted HIV testing services to eligible clients coming to get different medical and prevention services. It is also important to target partners and families of HIV positive person who get treatment and care at ART/ PMTCT clinics. By performing HIV testing for family members of clients attending ART / PMTCT clinic, it is possible to identify those partners and children who are already infected but looks healthy and provide them the HIV care and treatment as early as possible. Furthermore, identifying those still HIV negative partner help to use prevention methods and support each other to cope with the situation.

Importance of HTS:

- **Identify** people with HIV through the provision of quality testing services for individuals, couples and families
- Support to **effectively link** individuals and their families to HIV treatment, care and support, as well as HIV prevention services, based upon their status,
- Scale up of targeted testing including innovative case detection approaches to **reduce** HIV transmission and HIV-related morbidity and mortality,
- Identify **sero-discordant** couples and help **prevention** of further transmission,
- Identify the HIV negatives and help them **to remain negative** by following ABCD principles of HIV prevention
- Reduce the **stigma** related to HIV–clients passing through the pretest counselling, education and posttest counseling will help them understand HIV and hence reduce stigma

4.1.2. The five guiding principles of HIV Testing and Counseling

Counseling is a two-way communication process that helps an individual to examine personal issues, make decisions and make plans for taking action. In the context of HIV/AIDS, it is a confidential two-way communication between a counselor/ provider and client(s) aimed to make personal decisions related to HIV/AIDS.

All forms of HTS, including HIV partner services, should adhere to the WHO Principles of HIV testing and counseling (The 5 Cs: **Consent, Confidentiality, Counselling, Correct test results** and **Connection**). Coerced or mandatory testing is never appropriate, whether that coercion comes from a health-care provider or from a partner, family member, or any other person. These principles apply to all HTS and in all circumstances.

- Consent:** People receiving HTS must give their informed consent to be tested and counselled, Verbal consent is sufficient. They should be informed of the process for HIV testing and counselling and of their right to decline. In addition, it is essential that all people with HIV are informed that assisted partner service is also voluntary, and that partners of HIV-positive clients are made aware that HIV testing is voluntary too.
- Confidentiality:** HTS must be confidential. Discussions held between the HTS provider and the client should not be disclosed to a third party without the expressed consent of the person being tested. Although confidentiality must be respected, it should never be used to

reinforce secrecy, stigma or shame. Counsellors should always ask clients, among other things, who they wish to inform and how they would like this to be done. Shared confidentiality with a partner or family members (trusted others) and health-care providers are often highly beneficial to HIV-positive clients.

- c. **Counselling:** Pre-test information and post-test counselling can be provided in a group setting if appropriate; however, all persons should have the opportunity to ask questions in a private setting if they request it. All HTS must be accompanied by appropriate and high-quality post-test counselling, based on HIV test results. Quality Assurance (QA) mechanisms, as well as supportive supervision and mentoring systems, should be in place to ensure the provision of high-quality counselling.
- d. **Correct test results:** Providers of HTS should strive to provide high-quality testing services and QA mechanisms that ensure people receive a correct diagnosis. QA comprises of both internal and external measures and should include support from the national reference laboratory. Additionally, all people who receive a positive HIV diagnosis should be retested to verify their diagnosis before initiation of ART or HIV care. Individuals at high ongoing-risks, or those who are using anti-retroviral drugs for pre-/ post exposure prophylaxis, should be encouraged to periodic retests.
- e. **Connection:** Linkage to prevention, treatment and care services should include the provision of effective and appropriate follow-up. Providing HTS in situations where there is no access or poor linkage to care, including ART, has limited benefit for those with HIV.

4.1.3. Elements of good counseling:

1. **Respect** for the client's beliefs, attitudes, values and culture should be maintained all the time even if they differ from that of the counselor/provider.
2. The counselor should be open minded, authentic, honest and congruent during the entire process of counseling. **Genuineness** promotes trust and positive relationship between the client and the counselor.
3. **Confidentiality** forbids any reference to, or discussion about a client, except within a professional relationship, and then only with the consent of the client. The counseling environment should also provide privacy for the client. **Shared confidentiality** is a form of

confidentiality in which the discussion matters will be limited to the counselor, the client and to those individuals who will be involved in the care and management of that specific client.

4. Privacy is important in the counseling interaction. This includes location (conducive, maintain confidentiality) and understanding the fact that the client request for counseling help in a personal capacity.

5. Unconditional positive regard is another attitude the counselor must adopt to express empathy. Counselors should view clients as individuals with problems and respect them without judging or condemning their past behavior.

6. Acceptance: counselors should not be judgmental of clients, but rather should try to accept clients, regardless of their socioeconomic, ethnic or religious background, occupation, sexual orientation or personal relationships.

7. Autonomy refers to the liberty to choose one's own course of action, to take full responsibility of the outcome of action and it is the cornerstone of the client's right of participation. Counselors promote the client's control over his/her own life, respect client's ability to choose, decide, and change in the light of his/her own beliefs, values and circumstances.

8. Empathy is showing warmth, concern and caring attitudes and responses. It is understanding the other person's point of view. It is being able to think and feel through the other person's (client's) perspective. It is an active process practiced by the counselor by clarifying communication, reflection of feeling, and imagining others' thoughts. So, empathy involves '**being with the client**'. Empathy is not synonymous with "sympathy".

4.1.4. Basic counseling skills for HIV testing

The following are the most important counseling skills that counselors use in HIV counseling:

1. Establishing rapport, or 'joining' with client, is crucial a key step in developing a trusting relationship. Developing rapport demonstrates the counselor's interest in and respect for a client's issues and concerns.

2. Showing empathy, the ability to empathize is one of the most essential counseling skills. Empathy involves understanding client's thoughts and feelings, and communicates from their point of view.

Example: A patient may say: *"I am coughing the whole night and feeling weak and cannot perform hard work but my employer wants me to work in the department which includes a lot of laborious works."*

Clinician's empathic response: *"So you feel very tired at the end of the day. That must be very difficult for you."*

3. Acknowledging difficult feelings, the presence of difficult feelings is a substantial and unavoidable component of counseling.

4. Affirming is congratulating or complimenting clients on the positive actions that they have been able to take. It is important to encourage success. Complimenting client helps them feel respected and valued, and it encourages them to make decisions about testing for HIV, and take a better care of themselves. They also may be more willing to share information about other actions they have taken.

5. Correcting false information/misconception Providing correct information to a client and correcting any myths and false information is important. There are many incorrect facts about HIV, AIDS and STIs, and they should be corrected.

6. Using third-person technique/impersonal statement: In making a general point, using a third person technique, also known as 'impersonal statement,' can be helpful in reflecting clients' feelings that are unspoken but nonetheless perceived. This technique is very useful in acknowledging, reflecting on, and normalizing the client's feelings and helps to avoid defensiveness in a client.

In this chapter, special emphasis will be given for targeted testing through PITC approach and index case testing.

4.2. Provider Initiated Testing and Counseling (PITC) Service

This is HIV testing service provided at clinical service setting points initiated by service providers in an opt-out approach for eligible patients coming to the facility for other medical

reasons. Generally, the direction is to test people who are more likely to be infected with HIV. By targeting high burden geographic areas and focusing on high risk population groups, and further targeting the service through innovative approaches like partner and family based index case testing, respondent driven HIV testing, HIV self-testing, and applying HIV-risk screening for identifying high risk individuals, it is possible to improve the overall HIV testing yield and linkage of positives to care.

4.2.1. Steps of Provider Initiated HIV Testing and Counseling Services

Service providers offering PITC to clients shall follow the following steps:

- Provide basic **pre-test information** on HIV and AIDS in groups or individually
- **Identify target** groups or individual client and offer testing for HIV by using opt-out approach
- Perform HIV test and interpret the result
- Provide HIV test result
- Provide post-test counseling
- Link those who are HIV positive to care and treatment and provide risk reduction counseling and link to appropriate services for those who are negative

4.2.2. Priority and target groups for PITC

Based on the prevalence of HIV among population groups and raised risk for HIV infection, the following groups are target for PITC.

Table 4.1. List of target groups for routine HTS through PITC

The eligible Target clients for routine HTS by using PITC approach are*:

1. All pregnant at first ANC Visit, laboring and postpartum women with unknown HIV status
2. Sexual Partners of HIV Positive pregnant/ postpartum women
3. Sexual partners of Adult Index case, biological children, and siblings (<19 years old) and biological parents of index children
4. FSW and PWID with unknown HIV status
5. All Tb patients and presumptive TB cases with unknown HIV status
6. All STI patients and their sexual partners with unknown HIV status
7. Children orphaned by AIDS
8. Patients with clinical S/S of HIV/AIDS visiting HFs

NB *The above mentioned target population are directly eligible for HTS. Other patient/client should be screened by national HIV risk screening tools. Its mandatory that all health care providers MUST use HIV risk screening tool at every POC/SDPs.

Note: The above mentioned target groups are not equally accessible at all service delivery points in health care facilities. Health care providers shall be well aware of the target groups frequently visiting respective service outlets.

- a) At PMTCT clinics at ANC clinic, Labor and delivery ward, postnatal clinic:
 - 1. Pregnant and lactating women with unknown HIV status at current pregnancy,
 - 2. Partners of HIV positive pregnant/ postpartum women and
 - 3.
- b) At ART and PMTCT clinics:
 - 1. Eligible Family members and sexual networks of Women/ men living with HIV in ART/ PMTCT clinic:
 - Biological children <19 years old
 - their parents (both spousal and non-spousal if available)
 - 2. Eligible sibling(s) and parent(s) of Children living with HIV in ART clinic:
 - Siblings (brothers and sisters) <19 years old,
 - Mother and father of index child,
 - 3. Regular repeat test for Discordant couples at ART/ PMTCT clinics:
 - HIV negative partner(s) of newly diagnosed person,
 - HIV negative partner(s) of PLHIV in care at ART and PMTCT clinics,
- c) At General Out- patient clinics, emergency and Inpatient units/ admission wards:
 - All TB patients with unknown HIV status and Presumptive TB Cases,
 - All STI patients with unknown HIV status visiting health facilities, and their partners and Sexual networks,
 - Children Orphaned by AIDS visiting health facilities,
 - Patients visiting health facilities with clinical signs and symptoms of HIV/AIDS,

a. , At community care and treatment services/ organizations:

- Children Orphaned by AIDS ,
- Commercial sex workers and their clients,
- Family members of PLHIV,

d) At DICs and KP friendly service points:

- Commercial sex workers and their clients

Exercise 4.1: Decide whether a given client is eligible for targeted PITC:

1. A 12 year old girl at OPD
2. A 34 year old married woman coming to FP unit,
3. A 3 month old infant brought to immunization clinic,
4. A pregnant woman coming for ANC,
5. Adolescent street boy brought to emergency by friends,
6. A daily laborer at emergency,
7. A young waitress at local restaurant,
8. Newly diagnosed HIV positive client coming to ART clinic,
9. A young girl coming for FP injection
10. Acutely sick truck driver

We use **substantial risk assessment** and target to the population group (KPP) for HIV testing. As there are different program units in a healthcare facility, the target individuals elicited for HTS also vary; the binding goal is that no HIV infected person shall leave the facility without knowing its status. The way we offer targeted PITC is defined with level of understanding and engagement of caretakers/ parents. Therefore, we use slightly different protocols for children and adults. The HIV test will also be provided using the national three- test algorithm.

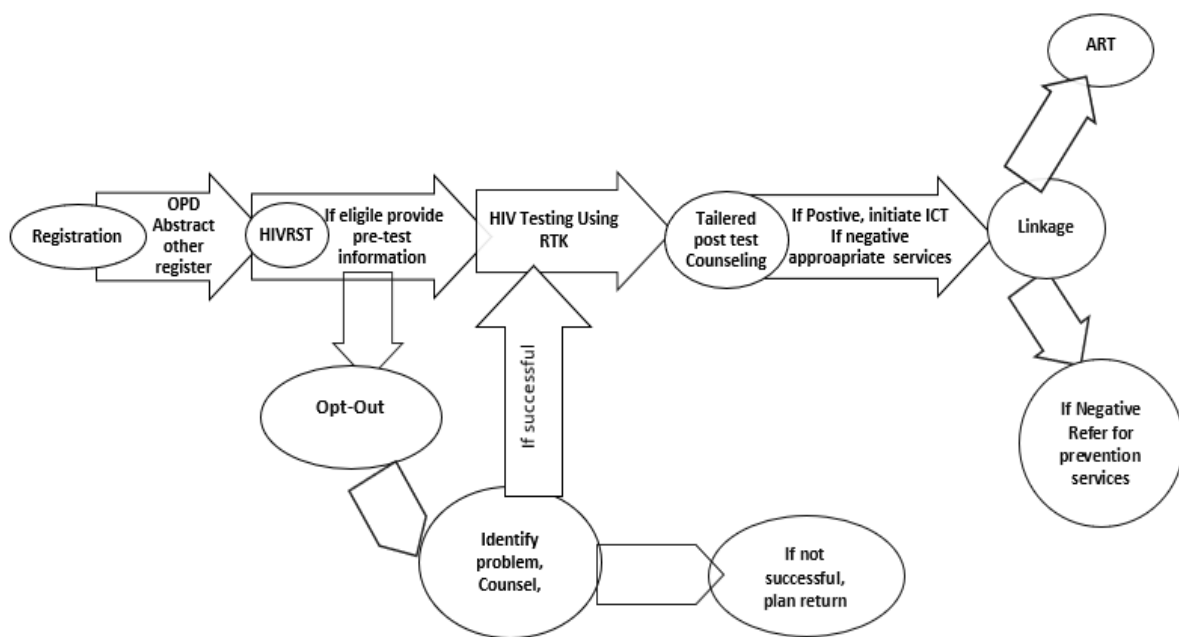


Figure 1: PITC Flow Chart

During provision of testing service, providers should follow specific procedures as outlined below. While following the procedures, providers need to utilize the national guideline, protocol, cue-card and job-aids.

A. Client registration

B. At service delivery points, clients will be registered using code numbers, names, UAN (Unique ART number)/ MRN (Medical record number). Conduct HIV risk screening: If eligible based on the HIVRST, provide pre-test information

C. Pre-test information

Pre-test information should be provided by health care providers or counselors. Couples should be encouraged to receive results together. Pre-test information for PITC can be provided in the form of individual or group information sessions. The relevant information that should be provided

a. Offer HIV Testing:

- *If a client accepts HIV testing when offered, proceed to HIV testing.*
- *If a client decline HIV testing, identify the problem and proceed to HIV testing if successful.*

- *If a client declines and the counseling is not successful, plan return back for testing*

b. Conduct HIV Testing

Based on the WHO recommendation(WHO, 2019a), Ethiopia is implementing the low prevalence (<5%) testing strategy. Currently we are using serial algorithm (three test algorithms instead of tiebreaker). In a serial testing algorithm, the samples are tested by a first test. The results of the first test determine whether additional testing is required. To improve the acceptability, uptake & quality of service delivery for HTS in all settings, rapid diagnostic tests (RDTs) should be used. These testing strategies have been developed assuming that all HIV assays used have a sensitivity of at least 99% and a specificity of at least 98%, resulting in an overall positive predictive value of 99%. The HIV testing must be done using nationally accepted RDTs following national HIV testing strategy as shown below. Same day results should always be respected irrespective of the type of delivery approach.

c. Providing HIV test results

Test results should be declared in person maintaining privacy (not by telephone, email or letter). Sites should not provide written HIV negative test results to clients to avoid misuse of results. Clients with HIV positive results requesting or requiring referral to other facility should be referred to the appropriate institution with pertinent information.

d. Provide Post-test counseling

All clients undergoing HIV testing should be provided with post-test counseling in person as individual or couple.

The messages and information included in post-test counselling and their intensity need to be tailored according to the test result. For those with negative results, this will further depend on whether they face ongoing HIV risk. Messages should always be clear and concise and client-centered, acceptable to the client and delivered in a safe, confidential, and non-judgmental manner.

Applying risk screening for high-risk groups or individuals, for more targeted PITC

A risk screening-based HIV testing which enables service providers to identify risky groups during counseling and gives opportunity for being targeted HIV testing. HIV risk screening tool is a tool, having a set of questions, used to identify the clients with specific risks for HIV transmission.

Risk screening tool could be used among patients in OPD setting for more consistent targeted testing services to identify at risk <15 and 15 plus and test them for HIV focusing on case detection. Risk screening tools have been utilized in other countries to identify who needs to be tested and maximize HIV case detection and increase efficient utilization of the limited RTKs

Figure 4.1. Provider-Initiated HIV Testing and Counseling Protocol for ADULTs

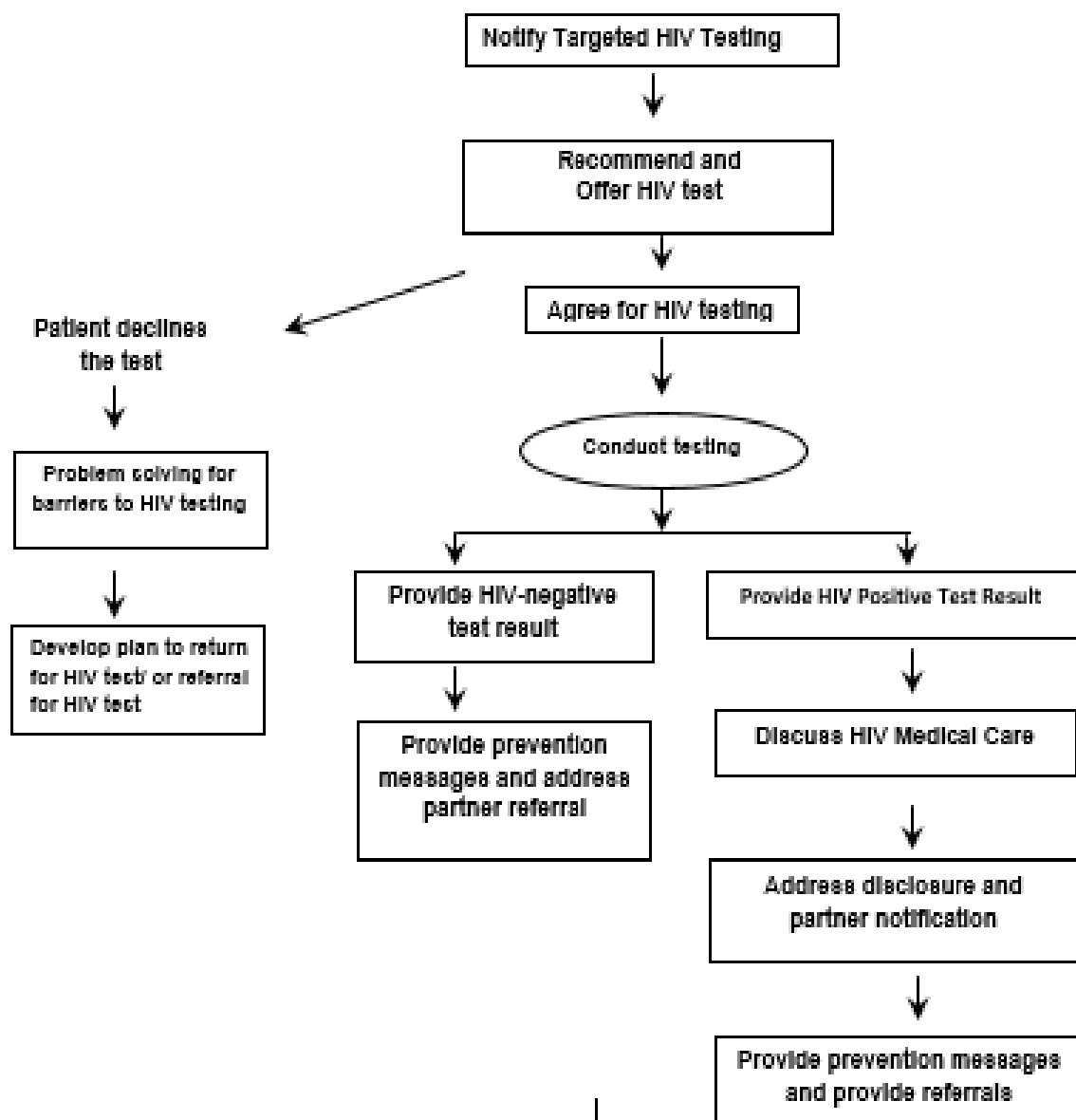
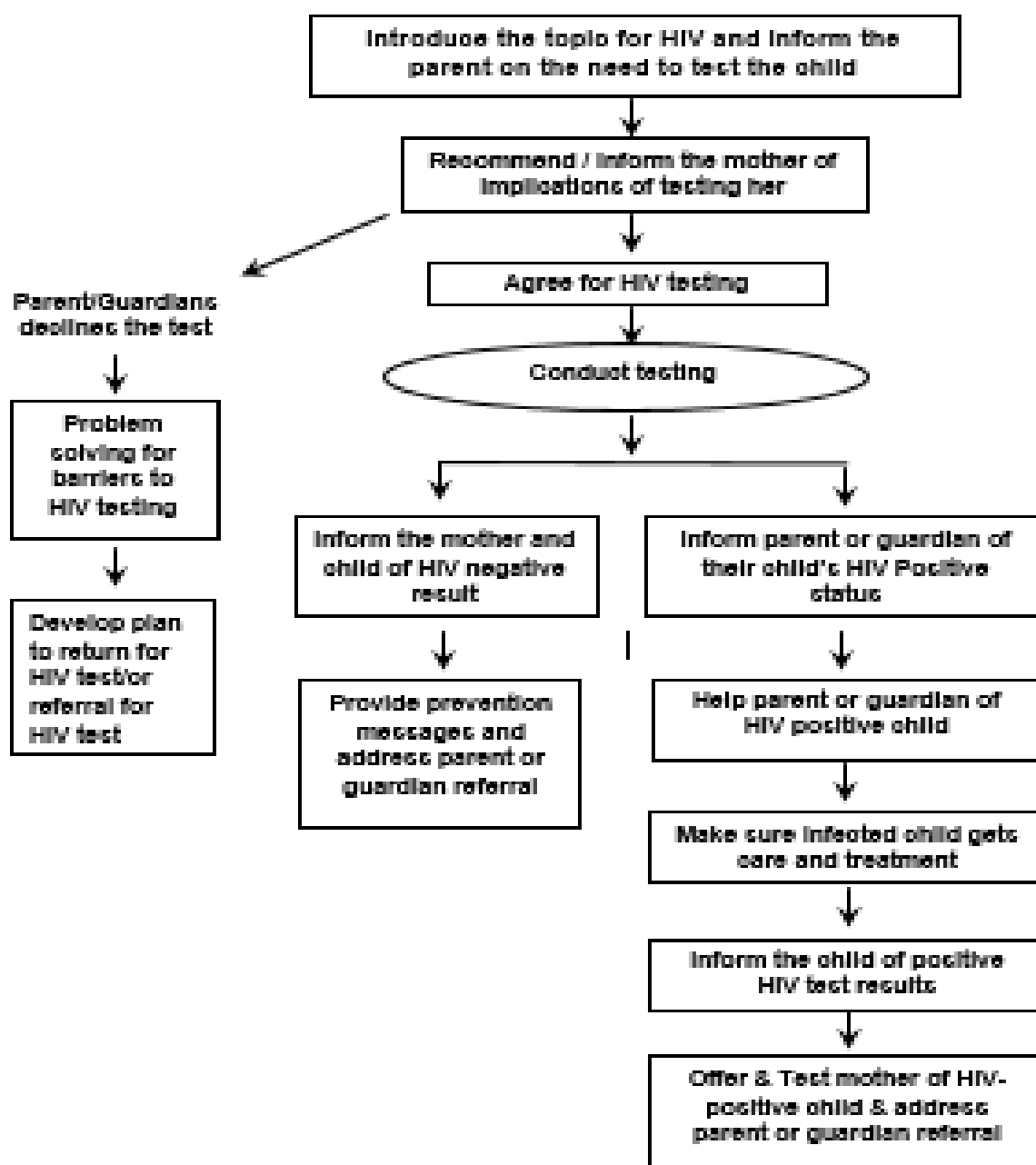
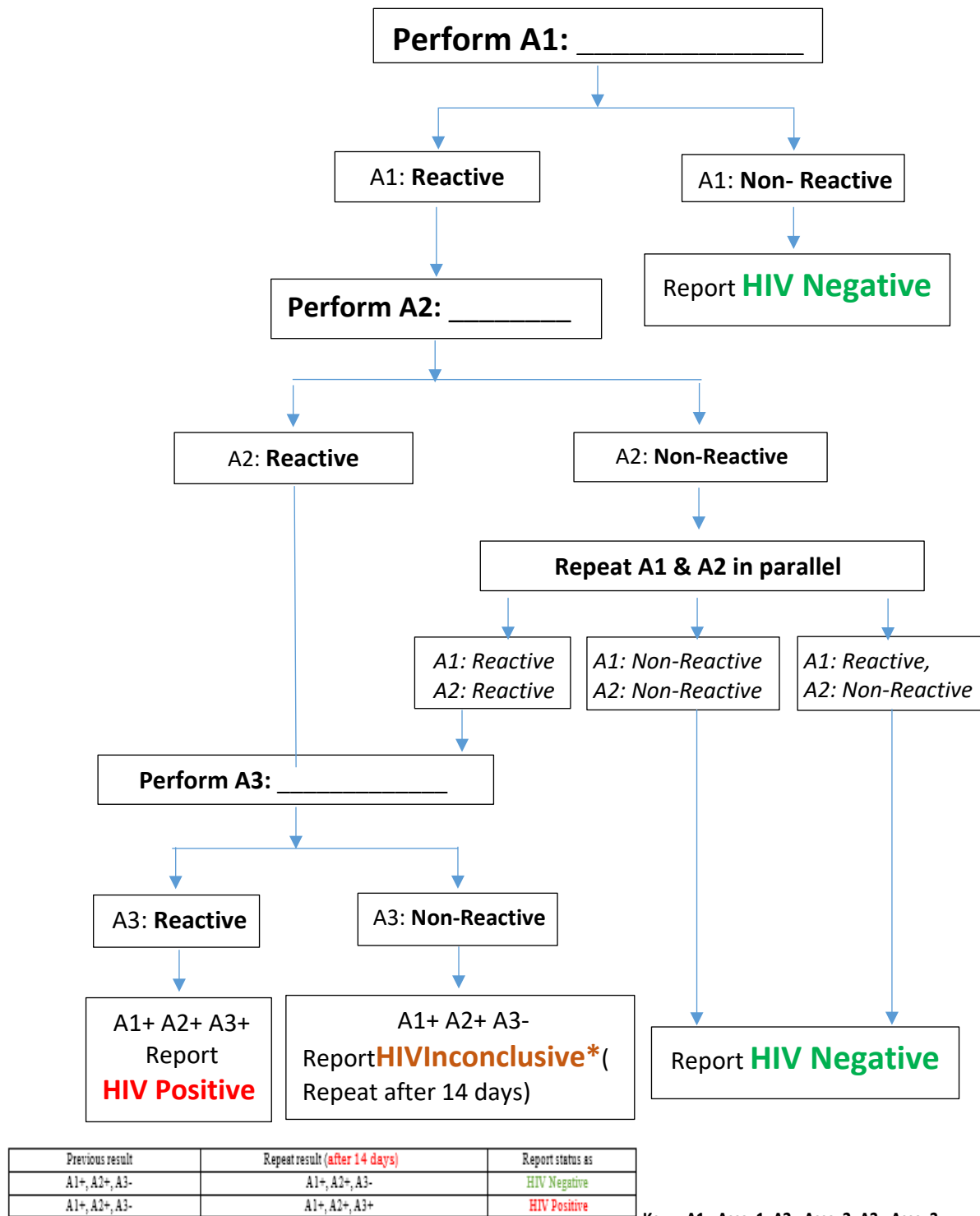


Figure 4.2. PITC protocol for pediatric



Provider initiated HIV testing and counseling (PITC), which is provided by opt-out approach at clinical service points for eligible patients (who come to the facility for other medical reasons). All health facilities should provide PITC service for eligible clients at outpatient and inpatient departments, using the national testing algorithm. The national test algorithm is displayed on the next figure.

Figure 4.3. HIV Rapid testing strategies and the three test algorithm



Inconclusive result interpretation after 14 days

NB:- Although it is highly unlikely to get two different results from the same assay and the same specimen if properly performed at a time, in extremely rare cases

A1: Non-reactive and A2:Reactive may happen during repeat parallel testing due only to unskillful practices. In such cases perform the test again (for the 3rd time) starting from A1 and strictly follow all the required remaining steps in the algorithm.

4.3. Partner and Family Based Index Testing Service

While zooming the HTS close to those with potential risks, the first and promising step is addressing those close to the HIV positives newly identified or on care, i.e., sexual partners (spousal and non-spousal) and their children.

Definitions:-

Index client is an individual newly diagnosed as HIV-positive and/or an HIV-positive individual who is enrolled in HIV treatment service.

Index testing is a voluntary process where health care providers ask index clients to list all of their sexual partners within the past year and biological children < years for HIV testing. All index testing must be consensual, confidential, and include counseling, correct test results and connection to treatment or prevention services (5C's). If the index client agrees, each listed partner is contacted, informed that he/she has been exposed to HIV and offered voluntary HIV testing services (HTS) while their biological children will be tested for HIV.

Sexual Partner is defined as Person(s) with whom index client has had sex contact at least once.

Non-spousal sexual partner is defined to include any sexual relationship with a partner outside the context of marriage.

Goal of index testing is to break the chain of HIV transmission by offering HTS to persons who have been exposed to HIV and linking them to prevention if they are HIV Negative and to care and treatment if they are HIV Positive. It is a proven effective case finding strategy to reduce future transmission.

Index testing enable us to identify partners and children with undiagnosed infection, hence early identification of healthy children living with HIV. While communicating with index parent, you can state this:

“Disease progression in children is faster than those of adults. It is advisable to test them early so that they can benefit from care and treatment services that are available in the facility if they are infected. I strongly recommend if you agree for the testing of your child/ren so that the status of the child can be known early.”

The prime targets of partner and family based index case testing service are biological children less than 19 years of age, and sexual partners (spouse and non-spousal if any). First, we will deal in detail with partners.

As disclosure is a concern, there are two approaches for elicitation of partner- the traditional partner testing using client's referral and the new innovative assisted partner testing whereby partners are notified their own risk & invited for HTS.

Client Referral= simply the index client takes responsibility for disclosing their HIV status to partner(s) and encouraging partner(s) to seek HTS. This is traditionally done with invitation letter or referral slip.

The Innovative Assisted Partner Testing Approach is implemented using contact/ index, provider or dual referrals.

Contract Referral= the index client enters into a "contract" with the provider whereby the index agrees to disclose his/her HIV status to his/her partner(s) and refer them to HTS within 14 days. If partner(s) do not access HTS within this period, in consensus with the index, the provider will contact the elicited partner(s) directly and offer voluntary HTS.

Provider Referral= with the consent of the index client, the provider directly contacts the client's partner(s) and offers them voluntary HTS while maintaining the confidentiality of the index client. Disclosure is reserved in this approach.

Dual Referral= a trained provider sits with the index client and his/her partner (s) to provide support as the client discloses his/her HIV status. Then the provider also offers voluntary HTS to the partner(s).

The assisted partner notification services, using provider and/or "contract referral", is a particularly effective strategy currently recommended for increasing uptake of partner testing and identifying and treating new HIV-cases.

Principles of Partner and family based index testing service

Partner and family based index testing should be focused on the needs and safety of the index client and his or her partner(s) and child (ren) (**Client-centered and client focused**). Both the **confidentiality** of the index client and all named partners and children should be maintained at all times. Client information including that of partner(s) shall be secured (registers and information sheets locked in filing cabinets, electronic document secured/ locked with

password). Service providers authorized to access case-specific information shall be responsible for protecting it (individual security). **Shared Confidentiality:** It is very important to discuss with index client that his/her information will be shared among other relevant professional to get appropriate care of his/her need. In this case, two organizations like a health facility and community- implementing partner may share a clients' personal information in order to provide the care.

Example 1: A provider in the facility may interview the index client to get the name of his/her partner(s) and shares these names with the community organization who goes out in the community to partner's home and provides HTS based on partner will. In this case, both the facility and community partner must "share the confidentiality" of the client's information. They should have a data sharing agreement which includes a description of how they will maintain the confidentiality of client information before beginning the services.

Example 2: A provider in the facility may interview the index client to get the name of his or her partner(s) and child (ren) and shares their name with other health professionals working in other health facility nearby the index & or partner residency for index client living out of the jurisdiction and provides HIV testing services to partner and child(ren).

Partner and family based index testing service is always **voluntary** with consent; mandatory or coercive approaches will never justified. The service shall be delivered in a **non-judgmental** manner, free from stigma or discrimination. Providers shall be well aware of the culture and norms so that the service they provide is **culturally and linguistically appropriate** preferably in the language spoken by our clients. The service should be **accessible and available to all** index clients regardless of where they are diagnosed (health facility or community settings) with strong referral and linkages to HIV treatment and prevention services. When a partner tests HIV-positive, he/she becomes a **new index patient**, and the process starts over from the beginning.

Exercise 4.2: Exercise on Confidentiality

Comment on the cases below whether there is a "Breach" or "No Breach" in confidentiality:

1. You have referred a partner, “Girma,” for follow up HIV services. During the session, Girma asks which of his partners gave his name. You respond by saying, “By law, I am not allowed to give the name of the woman who gave me your name.”
2. You are counseling Workneh, who was referred in through Partner and family based index case testing services and just learned that he is HIV-infected. Workneh has agreed to participate in Partner and family based index case testing service to name and notify Wudinesh as a partner. You recognize Wudinesh’s description and recall she has already been in for HIV prevention counseling and testing. You notified and tested Wudinesh 2 months ago; she is infected. You tell the Client not to worry; this partner “has already been taken care of.”
3. You are re-interviewing a Client that had previously shared information on three partners. You have already contacted two partners and referred them into the clinic for testing and treatment. However, since you haven’t been able to locate the third partner, you want to ask the Client for additional locating information. The Client gives additional locating information and then asks, “Hey, did you talk with partner #1 and partner #2?”
4. You are having dinner at home with your spouse. You are telling him/her about a Client you saw at work. You are concerned because your Client has so many problems. You explain that your Client is from a rural village, is in her late 20s and now lives in poor housing in the crowded part of the city. She just tested positive for HIV and shows some symptoms. She is a single mother with two very young children and is pregnant. She has a job as a janitor but has been out of work many days because of her sickness and she could lose her job. Your spouse says, “Oh my! That sounds just like the woman who works across the hall from my office. Her name is Gete, isn’t it? Does she have HIV? I won’t tell anyone.”
5. You have found Samira and confirmed that she is the partner you are looking for. She invites you inside and you feel confident that no one else is at home, so you say, “Semira, I’m here to let you know that someone you’ve had sex with has tested positive for HIV.”

When shall Partner and family based index testing be provided?

It is always important to consider **Partner and family based index testing** service whenever a HIV positive person is identified or coming to facility seeking care.

Information on importance of HTS for partner(s) and children shall be provided during post test counseling for new HIV positives at all testing points. Partner and family based index

testingservice shall be delivered at HTS point, ART & PMTCT clinics immediately after HIV diagnosis or at least annually as part of HIV treatment services and whenever there is a change in relationship status.

There are two types of interviews done with index client on soliciting partner(s) and accessing the solicited partner(s) to provide HTS. The initial interview is the first encounter you contact the index client and provide information about index case testingservice; discusses on how to elicit partner(s)'work together on how, when and where to notify for partner(s) and test child(ren). The Purpose of initial interview is to build rapport; provide the necessary information about the service; assess and determine motivation of the index; identify and resolve concerns and obstacles; encourage index client to elicit non-spousal partner(s) and prepare the index client to notify partner(s) about HIV exposure and to bring all his/her child(ren) (under 15 years) for testing.

Fourteen days following the initial interview, the second interview will be done with the index and status of the contractual referral selected by index client will be assessed. At this time, it is possible to identify any additional partner not revealed during the initial interviewing period and follow up of the newly identified client for HIV care and treatment services (among partner(s) & children tested positive for HIV).

Partner and children elicitation and testing is not a onetime event, it is important to review/update the patient information on patient intake form and further explore if there are partner(s)/child(ren) with HIV status not determined and help the index to get them tested for HIV.

As ART and PMTCT provider, you're expected to be well aware and well equipped on the techniques and potential challenges to provide the Partner and family based index case testing service effectively. Adherence supports, case managers and Mother Support Groups can support on offering partner and family based index case testing to the index client. HEPs can also help in this service at community level.

To improve service access for elicited partner(s) and child(ren) of index clients, offer options of coming to the health facility for HIV test or getting tested for HIV at other health facility or even getting community level service through home based testing where the index client proposed or the elicited partner(s) preferred.

Provide basic information to elicited partner on- when, where and who to contact and what to ask when they come to facility. All staff working in the ART/ PMTCT clinic shall be well aware on how to guide and facilitate the elicited partners coming to HF ART clinic to contact with the eliciting ART/ PMTCT provider. Once the elicited partner reached your facility, and welcome at ART/ pMTCT clinic, continue counseling and testing as per the protocol.

Providing partner and family based index case testing service as quickly as possible after detection of infection will help to prevent continued transmission. The timeframe to elicit the names of partner(s) of index client is sexual contact during the past 12 months, and for index client who doesn't have sexual contact during the specified period, explore further up to 24 months. Index have to be told to get all children with unknown HIV status during the first encounter and all children without HIV status documentation should be tested.

4.3.1. Offering partner testing service to index case

There are five major steps in offering partner HIV testing to the index case:

Step 1: Introduce partner and family based index case testing services to index client

Participation is voluntary and respecting confidentiality. As the index client benefits knowing own status and starting ART treatment, partner(s) identified with this mechanism will also benefit knowing their status for early treatment. While offering the partner testing service to index case, ensure the index that

- Partner(s) will not be told the index client's name or test results,
- The index client will not be told the HIV test results of their partner(s) or whether or not their partner(s) actually tested for HIV,
- Testing services and results for children will not be shared with others,
- Providers will not contact elicited partner(s) and child (ren) without their permission.
- They will continue to receive the same level of care at this health facility regardless of whether they choose to participate in index partner/family testing services or decline.

At ART and PMTCT clinics, clients newly enrolled and on follow up will be screened for partner and children testing at every visit not to miss any who is at risk (Annex 20). Address any questions raised by the index client, obtain verbal consent and fill contact information on case investigation form. (Annex 21)

It is advisable to enhance index client's motivation to participate in index testing services by:

- Affirming the client's autonomy and capacity for self-direction,
- Increasing client awareness about the importance of partner/family testing,
- Exploring and resolving any ambivalence or barriers to naming contacts,
- Identifying and encouraging behavior change.

During counseling in the ART/PMTCT clinic on partner and family based index case testing service, inform the client that there are 4 options for contacting their partners or Notifying their Partner about HIV Testing:

1. Index client can contact the partner themselves to let them know they should be tested for HIV,
2. Index client can bring the partner to the facility for an HIV test,
3. Index client can enter into a contract with a provider to contact the partner within 14 days. If the partner has not been tested by the end of 14 days, with consent the provider will contact the partner,
4. With consent of the index, the counselor/provider can contact the partners directly without telling the index name (this will be done anonymously).

The client can choose different options for different partners. The client is also free to change strategies. For example, maybe they initially choose client referral but later on can decide that they prefer dual referral.

"One of our services is to provide "partner and family based index testing services" to partners & children of index cases. We as provider, elicit the partners of index cases, create plan for notification and coach clients on referring their partners for partner services. We provide ongoing counseling and support to index cases on disclosure and partner notification options such as referral by clients themselves or with the support from us, providers (assisted partner referral). We strongly advise that all partners of index cases receive these services. Therefore, you and I need to plan how to communicate your partner(s)."

Step 2: Obtain List of Contacts (Elicitation)

Ask index client for permission to talk about index testing. Engage index client in discussion, and use open ended questions to elicit information from the client on perceived challenges on naming or informing partner(s) for they are married or no longer in contact or emotionally not ready to deal with diagnosis or disclose their status.

Ask the index client to tell the names and contact information of all the persons they have had sex with in the last 12 months. Begin, by asking about the most recent sex partner and working backwards. Then ask if there are any other partners that he/she can remember having sex with in the last 12 months like

“Who is the last person you had sex with?”

“Who was the person you had sex with before that?”

During this, encourage the client to list names and contact information for main partner(s) as well as casual partner(s), even if they only had sex once. If there is no in last 12 months, go for additional 12 months.

Use affirming statements to normalize the client’s challenges by saying,

“Many people aren’t quite sure how to tell their partner, especially partners you may not speak to anymore...”

For clients who have concerns about others learning their HIV status, provide reassurance that their confidentiality is protected. Their identity or status won’t be disclosed to anyone contacted. And provide advice such as:

“Are you aware that your partner can be notified about the need to seek an HIV test without you having to disclose your HIV status to him?”

“It takes courage to face your HIV diagnosis and help others to know theirs.”

Regarding children, elicit all children less than 15 years old and siblings despite their testing status. Use case investigation form to record all the contacts’ names, address and testing status.

Exercise 4.3.

Practice in group of three as index client and a counselor with the following scenario.

Case Study: You are interviewing a woman who has recently tested positive. She indicates that she has had two partners during the interview period. She is willing to talk to you about notifying one partner, but does not want to discuss the other at all because he is married and his wife is pregnant.

Q1. What challenges did you encounter during the interview?

Q2. What are some strategies you might use to ensure that both partners get tested?

Q3. What messages would you give this client?

Step 3: Conduct Intimate Partner Violence (IPV) Risk Assessment

Intimate Partner violence (IPV) is behavior by an intimate partner that causes physical, sexual, or psychological harm, including acts of physical aggression, psychological abuse and controlling behaviors and sexual coercion. Screening for IPV risk is a standard for partner and family based index testing services. Primary goal of the IPV Risk assessment is to ensure no harm comes to the index client as a result of partner and family based index testing services.

To protect the safety of the index client, partners who pose higher risk of IPV may need to be excluded from notification and testing services and the index client provided with and referred to appropriate care and support services for IPV.

Respond to clients experiencing IPV by listening to the client with empathy and a nonjudgmental attitude and provide available services and refer for appropriate GBV services including psychological, legal support services and GBV prevention. During counseling, determine if the index client is thinking of harming him/ herself or others as a result of the diagnosis. Use the “Snake in the house” analogy.

“We do not care how the snake got in the house, we just know we need to deal with it. HIV is like that snake. It does not matter how it entered our life or who infected us. What matters is getting on treatment and taking other positive steps to lead a healthy life.”

As a provider, screen all named partners for IPV by using three standard questions included on case investigation form.

1. Has [partner's name] ever hit, kicked, slapped or otherwise physically hurt you?
2. Has [partner's name] ever threatened to hurt you?
3. Has [partner's name] ever forced you to do something sexually that made you feel uncomfortable?

If the client answers “yes” to any of the screening questions, the provider should work with the client to see which notification strategy may be most appropriate. Remember that provider referral can be done completely anonymously without the client having to disclose his/ her HIV status to the partner. If the client's safety is at risk, then he/ she may not be able to participate in index partner services. Link or refer the client with IPV to Gender Based Violence (GBV) services.

Step 4: Determine the preferred method of Partner Notification for each Named partner

There are 4 options for notifying and testing their partner(s) for HIV, they can choose one or combination.

- Client Referral- Index client tells partner(s) and encourage him/ her to go for HTS & treatment,
- Contract Referral- 14 days are given for the index to tell his/her partner then provider contacts,
- Dual Referral- Index client and service provider will notify his/her partner together,
- Provider Referral (Optional)- provider calls or sends text message to solicited partner

Based on index client preference, document the chosen referral method for each listed partner on the case investigation form.

Step 5. Contact all named/ elicited partners based on client preferred method

If the index client chooses **client referral**: Help the client to make a plan (where, when, using what words) and provide conversation starter. Index can use partner invitation card to invite main partner. Invitation card explains how the facility wants to discuss on the disease prevention and control service it provides for the community and get feedback from him/her to further make the service improved.

Brainstorm some questions/reactions that partners might have and help the client determine some possible answers/responses. Allow the index client to practice with you until he/she feels

confident that they can say the words. Set an appointment with the index client to follow up and confirm that the partner has got tested.

If the client chooses **contract referral**: Following the above steps, an agreed date is identified 14 days after initial interview, that the client already notify and refer his/her partner(s) for HIV testing. If the index fails to do that or the partner(s) do not come for HTS by 14 days, the provider can directly call/contact the partner(s) with permission from the index client. If the client does not provide permission to contact his/her partner(s), record the outcome on the case investigation form.

If the client chooses **provider referral**: Contact and invite named partner for health services to the facility using telephone from ART or PMTCT clinics. Give the necessary information (when, where, and who to contact) when he/ she arrives at the facility. While contacting partner(s) via telephone, using the following “Script for Partner contact: Phone Call”.

Tips:

- Secure a private place and plan for what you will say.
- Gratitude and ask the partner the convenience to talk.
- Confirm the partner’s identity by asking him/her father and grandfather name and home address.
- Do not give any information to anyone other than the partner.
- Respond to questions from the partner about how you obtained his/her information.

Telephone, text message (SMS) and face to face may be used to contact a partner.

Telephone: Partner and family based index case testing services provider may use telephone to contact named partner and make a plan to meet in person. It is recommended to call early morning or evening to reach partner when not on work.

Telephone call Procedure:

- Greet mentioning his/her name,
- Introduce yourself by work address & responsibility, give your name
- Systematically confirm the person answering the phone is the partner you are looking for – like asking full name and residence, If unable to confirm, set up a face-to-face meeting
- Inform partner that you need to meet with him/her and plan meeting as soon as possible in your health facility,

Responding to incoming calls:

Partners may call back for missed call(s) you made. So, you need to be prepared to respond to incoming calls appropriately.

- Ask caller’s name.
- Introduce yourself by work address and responsibility, and
- Ask the caller to hold and check your records to confirm the caller is a partner.
- After confirm the caller’s identity, provide information and plan time to come to health facility for counseling if partner agrees.

Example Script for telephone notification:

Hello. My name is _____ and I am a health care provider at [Facility Name] _____.
 Am I speaking with [partner’s name] _____? [IF NO]: Is [partner’s name] _____ available?
 [If partner is not available], Thanks. I’ll try back later. [If YES], I have some important information for you that I need to share with you in person. What time today or any other day can we meet?
 [Plan time, confirm meeting location and end call by thanking the partner.]

Text messaging /SMS: It is a useful way of communication when a named partner is not responding to phone call.

Example Script for text message:

“I am health worker/counselor working at Health center/Hospital and I need to speak with you as soon as possible”. Please call me at #.

Hi [partner’s name], “I am XX with the XX Health Facility and I need to talk to you about an important health matter”. Please call me at #.

Responding to partners when they text you back:

“This is important matter. I can tell you more when you call. Please call me at #. Thank you”.

During the partner contact attempt, there has to be at least three attempts at varying times of the day and using different methods. If the contact attempts were successful, schedule for face to face notification and assist partner to get the service with the consent of index client.

The goal of sending a text message is to motivate the individual to call the health care provider so important and sensitive information can be exchanged. This should be considered when patients or their partners are not responding to phone call or client referral.

Text messaging partner notification should NOT BE conducted using personal phones, should be sent from work phone or computer.

- Work cellphone/ wireless phone should be secured and turned off when not in use,

- Once you no longer need to continue communicating with the individual, all messages should be deleted

If the client chooses **Dual referral**: Together with the index client, arrange an appointment to bring their sexual partners to health facility. The partner may come knowing HIV exposure (notified) or not knowing HIV exposure. The couple may get HIV testing together if the partner is not notified. If the partner is notified of the exposure, reaction management is critical, and voluntary HTS will be provided when the partner agrees. The testing will be done at ART and PMTCT room.

If an index client fails to bring elicited partner and child (ren), as you agreed, share information to community provider to contact them and bring back to health facility for notification/ HIV testing.

4.3.2. Offering PITC for the partner(s) of HIV positive person (NOTIFICATION):

Notification to partner under this Partner and family based ICT service means letting a partner to be aware that he/she has close risk and benefit from HTS and knowing his/ her own status. It does not have anything with disclosure of index status. Notifications should always be conducted in accordance with local procedures especially with respect to contacting people outside the provider's jurisdiction.

Notification of disease or exposure to disease supports epidemiological efforts and serves as both a surveillance and control tool.

a. Principles of Notification:

The principles of notification addresses confidentiality, Organization, promptness, thoroughness, communication and creativity/ flexibility.

Efforts to contact and communicate with infected patient, partners, and those at risk must be done in a manner that preserves the **privacy** of all involved. No information should be shared with unauthorized persons that could lead back to the identity of the original patient (**Confidentiality**). Being **prompt** with taking action that lead people found at risk of having or having HIV are protected or treated as well as followed-up timely. Explore all realistic possibilities to find and notify partners and others at risk for infection (**Thoroughness**). Organize room and medical record activities appropriate. Use **effective communication** and be clear on what they wish to communicate while speaking at the person's level of understanding (**Listening**). **Creativity** is essential so that individuals exposed to HIV are found and tested in a

timely manner. Be creative and follow all notification leads and determine additional ways to locate using specific investigative practices. Being flexible requires the ability and willingness to change the course of investigation at any time. It is up to you to utilize baseline information and initial clues to expand the amount and quality of locating information in an unwavering attempt to locate, identify, and inform someone who have been exposed to HIV.

Face to Face Notification is a discussion with partner to inform him/her of exposure and provide counseling service that will help him/her to agree accessing HIV testing. While conducting face to face notification, you should review the record and memorize all pertinent data, store confidential forms in a secure place, identify private convenient place for partners and confirm identity then proceed with the notification procedures.

Introduce yourself:

“My name is [name] _____. I work at this clinic [name of health clinic] _____ as disease prevention and control focal. As part of this I work on HIV program to help people who could benefit from HIV prevention service that includes HIV testing and STI treatment”

Notify of exposure, process reaction, answer questions:

Explain confidentiality and assure a private setting

“This is a place where we can speak privately without others hearing or interruption.”

“Our conversation is private, meaning I need to speak with you only and will never share with to others your personal information that you will share with me.”

Provide the notification and process reaction: “Recently our facility is planned to provide integrative service to the community and prevent and control communicable diseases one of it is HIV. So it’s important you get tested for HIV.”

Plan for immediate HTS “If you agree, we will provide you the HIV testing service today. This will take about 30 minutes.”

If the partner refuses the test or test result is negative, Counsel on behavior change & risk reduction prevention packages.

If test result is positive, re-emphasize on ART, Link HIV+ partners to ART service and continue another cycle with the client as a new index client.

b. Notification Problems/ Challenges:

The following are Typical Notification Problems and Suggested Responses. Partner may ask:

1) “How did you get my name?”

Response: “From someone who benefits HTS and treatment and who cares enough about you and wants you to get a better health service.”

“The health facility cares about the health of individuals in the catchment and want to notify their exposure.”

2) “Who gave you my name?”

Same reply as above, or if pressed further for an identity,

“Ethically the procedure does not allow to give the name. The good thing is, your information is confidential too; just like I can’t share this information—I also can’t share yours.”

3) “Do I have the disease?”

“We won’t know until you get tested. We will arrange this as quickly as possible. How about going in right now?”

4) “I haven’t had sex with anyone in over a month.”

“This exposure may not be recently may be more than months ago you could have sexual contact.”

5) “I haven’t had sex with anyone but my husband/wife.”

“Someone close to you had this disease and is concerned about your health. Your health and getting test is the most important thing right now.”

6) “But I feel fine. I haven’t had any sign and symptom.”

“That’s a good sign, and I really hope it means everything is OK. But, only testing can tell us for sure. Many people don’t have any signs or symptoms.” OR

“You’ve been exposed. If you get treated right away you may be able to prevent problems! We can get you medication to prevent them.”

7) “There isn’t anything wrong with me. I went to my doctor just last week.”

“Doctors often don’t have enough information to give you holistic service preventive treatment.”

Then ask the following questions, one question at a time,

“What’s your doctor’s name?” “What tests did you get?” “What treatment did you get?”

If the answers indicate the person might have gotten the appropriate diagnosis and treatment, consider contacting the physician immediately or consult the physician later on.

If it did not sound like the person was neither examined nor got treated for this infection, consider setting up an appointment as soon as possible to discuss afresh.

8. “Fine, I’ll take care of it with my doctor and you don’t need to get involved. You just trust me that I will take care of it.”

“Thanks for wanting to take care of this all on you own. In order for you to get the best possible care at your doctors, I will need to give him or her some medical information. What is his/ her phone number?”

In this instance, try to arrange or confirm the appointment personally. Try to get a signed release of information form so that the test results and treatment can be easily confirmed.

4.3.3. Individual offering of HIV testing for partner

a. Pre-test information

“HIV is a germ that destroys the part of your body needed to defend a person from illness. The HIV test will determine whether you have been infected with the HIV virus. It is a simple blood test that will allow us to make a clear diagnosis. Following the test, we will be providing counseling services to talk more in-depth about HIV AIDS. If your test result is positive, we will provide you with information and services to manage your disease. This will include antiretroviral drugs and other medicines to manage the disease. In addition, we will help you with support for prevention and for disclosure. If it is negative, we will focus on ensuring you have access to services and commodities to help you remain negative.”

b. Confidentiality

“The results of your HIV test will only be known to you and other team members who will be treating/helping you. This means the test results are confidential and it is against our facility’s policy to share the results with anyone else, without your permission. It is your decision to tell other people the results of this test.”

You are offered to be tested unless you decline?

c. Test Result delivery:

After running the test per the national algorithm, communicate the final result, in clear and understandable way to the client. The result of the test can be Positive, Negative or Inconclusive.

1. Negative Result:

"The test result is negative, which suggests you do not have HIV in your blood at this time." However, here is a slight possibility that a recent risk behavior (especially in the last month) may have resulted in the client becoming infected without the infection being indicated in this test result. However, be reminded that the current result represents all other, sometimes years of previous risk behavior. It is important to be very careful with the message "retest." If there is not any significant risk in the previous 6 weeks to 3 months, then no additional test is indicated unless the client has a subsequent risk. If there is a very recent and significant risk exposure, there is a small chance that the client could have been infected by that exposure and it was not yet detected on the current test. Avoid technical discussions of this information and recommend, when necessary, a specific time for possible retest linked to the specific previous but recent risk of exposure. You can also give additional information about staying uninfected.

"As you probably know, you can get HIV infection from having unsafe sex with your partner. To prevent transmission of infection from your partner, practice the following prevention strategies:

- Not having sexual intercourse with your positive partner.
- Support your positive partner to adhere to ART, to have regular STI and other OI screening and management, to have regular viral load monitoring
- Using condoms correctly and consistently every time you have sex with your positive partner having early treatment for sexually transmitted infections"

"We have condoms available in the clinic and you are welcome to take some. When the need arises, some other time, you can also get condoms from any other health facility or you could buy it for a very small amount from local shops or pharmacies. Would you mind if I demonstrate to you how to use condom?"

NB: Follow the national HTS guideline regarding specific retesting recommendations. E.g. Occupational exposure, and retesting of FSWs.

2. Inconclusive Result

"It occurs rarely, **Inconclusive** result means that the test was run but didn't provide a clear negative or positive result. People with inconclusive HIV test results can be HIV infected and in the process of seroconversion. All clients with an HIV inconclusive status should be encouraged to return in 14 days for additional testing to confirm their diagnosis."

3. Positive Result

The test result is positive, which suggests you do have HIV in your blood system. Ensure the client understands the result and provide the service readily available to keep him/ her healthy.

“This time, there is treatment for HIV that control the replication of virus and improve health condition. In addition to getting support from family and friends, you need medical care that can help you feel better and live longer. Now I will provide the ART service here or accompany and refer you to the ART clinic.”

Post-test Counseling to HIV positive clients and key talking points to facilitate linkage to treatment and long-term care:

- Provide the client with HIV Positive test results in a clear, compassionate and supportive manner.
- Provide posttest counseling, including on the following points:
 - Treatment is now recommended for all HIV positive individuals regardless of CD4 count and clinical stages,
 - ART is initiated on the same day of tested HIV positive unless there is a readiness problem,
 - ART is a life-long treatment, Sooner you start, healthier you will be, less likely to develop other serious opportunistic illnesses.
 - With treatment less likely to pass HIV to partner, breast feeding baby and unborn child
 - It is recommended that you tell your partner or someone close your status and bring them to clinic, they can help you remember the information and support you with treatment,
 - Also, if you/your partner are pregnant or planning to get pregnant, you should tell your healthcare provider at the HIV clinic so that he/she can talk to you about protecting your unborn child from getting HIV.
 - By the way, if you have children it is important to get them tested for HIV and know their status; they need to start treatment if they tested positive for HIV,
 - Remember to use condoms, encourage your partner to be tested and reduce number of partners,
 - Based on your faith, you can go to the holy water, churches, mosques etc. but never stop ART

Therefore, Linkage to care and treatment is important and all HIV positive clients shall be linked to ART through accompanied referral/ escorting by using referral format.

d. Providing test result and post-test counseling for partner coming with index (couples)

Be aware and recognize the complexity of couple posttest counseling in order to have an optimal counseling service. Ask the willingness of clients if you want to involve VCT counselor. As provider your role has the greatest influence and prevention success for couples with discordant result, your advice and education can support these discordant couples remain discordant (i.e., prevent the negative partner from HIV infection for life). Follow the following steps of posttest counseling for couples (discordant and concordant couples).

Step-1. Provide discordant and concordant test results:

<i>Task for discordant positive</i>	<i>Task for concordant positives</i>
1. Inform the couple that their results are available.	1. Inform the couple that their results are available.
2. State that the couple has received results that are different. Pause briefly for the couple to absorb the implications of the results.	2. Provide a simple summary of the partner test results: meaning both are positive, which indicates that both partners are infected with HIV.
3. Convey support and empathy.	3. Allow the couple time to absorb the meaning of the results.
4. Ask the couple if they understand their respective results.	4. Ask if the couple understands the results.
5. Review the explanation of how couples can have different results,	5. Encourage mutual support and diffuse blame.

Step-2. Discuss coping mechanisms and mutual support

<i>Task for discordant positive (Staying discordant for life!)</i>	<i>Task for concordant positive</i>
1. Invite both partners to express their feelings and concerns.	1. Invite both partners to express their feelings and concerns.
2. Validate and normalize the couple's feelings and acknowledge the challenges of dealing with different results.	2. Validate and normalize the couple's feelings and acknowledge the challenges of dealing with a positive result.
3. Ask the uninfected partner how he or she could best support his or her partner.	3. Ask how the partners can best support each other.

4. Recall the couple's strengths. Convey optimism that the couple will be able to cope and adjust to the situation.	4. Recall the couple's strengths. Convey optimism that the couple will be able to cope and adjust to living with HIV.
5. Address the couple's immediate concerns.	5. Address the couple's immediate concerns.

Step-3. Discuss positive living, HIV care and treatment

Task for discordant positive	Task for concordant positive
1. Discuss positive living and the importance of getting care for the HIV infected partner.	1. Discuss positive living.
2. Discuss positive living.	2. Address the need for preventive health care.
3. Address the need for preventive health care. a. Encourage immediate visit to the HIV clinic b. Avoid myths about treatment eligibility	3. Encourage immediate visit to the HIV clinic
4. Encourage the infected partner to access appropriate care and treatment services.	4. Avoid myths about treatment eligibility
5. Encourage the uninfected partner to serve as an advocate for the infected partner.	5. Encourage the couple to access appropriate care and treatment services.
6. Provide needed referrals to the HIV clinic and other services. Identify and problem-solve obstacles.	6. Provide needed referrals to the HIV clinic and other services. Identify and problem-solve obstacles.

Step 4. Discuss things to do at home to keep healthy

Task for discordant positive	Task for concordant positives
1. Discuss with the couple the need to live a healthy lifestyle. Discuss things that they can do right away to keep healthy.	1. Discuss with the couple the need to live a healthy lifestyle. Discuss things that they can do right away to keep healthy.
2. Discuss the importance of having safe drinking water to prevent diarrhea. Inform couple about where to get more information or obtain supplies.	2. Discuss the importance of having safe drinking water to prevent diarrhea. Inform the couple about where to get more information or obtain supplies.
3. Discuss the importance of using bed nets to prevent malaria (when applicable). Inform couple about where to get more information or obtain supplies.	3. Discuss the importance of using bed nets to prevent malaria (when applicable). Inform couple about where to get more information or obtain supplies.

4. Discuss the importance of good nutrition. Inform couple about where to get more information.	4. Discuss the importance of good nutrition. Inform couple about where to get more information.
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Step 5. Discuss the need for protecting the negative partner from HIV and the risk reduction (in case of concordance)

<i>Task for discordant positive</i>	<i>Task for concordant positive</i>
1. Address risk reduction within the couple. Explore long-term measures to reduce the risk of HIV transmission to the uninfected partner. Rapid viral suppression with ART, PrEP for negative partner until sustained viral suppression is attained in the positive partner. Proper and consistent use of condom.	1. Discuss the importance of being faithful and not having sex with outside partners.
2. Assess condom-related issues, including: a. History of condom use b. Condom skills c. Provide condom demonstration.	2. Inform couple of the need to safe sex using condom consistently and appropriately and then provide condom demonstration.
3. Address regular HIV testing for HIV-negative partner.	
4. Inform couple that condoms must be used always. Address the possibility that any other partners should be tested for HIV.	

Step 6. Discuss Children, Family Planning, and PMTCT Options

<i>Task for discordant positive</i>	<i>Task for concordant positive</i>
1. Revisit the couple's intentions concerning having children. Address the risk to the uninfected partner if the couple decide to have a child.	1. Discuss the issue of HIV testing of children.
2. Discuss the couple's reproductive options. If the woman is negative, Emphasize the increased risk of HIV transmission to fetus or infant if new infection happen to the HIV negative women during pregnancy or breastfeeding.	2. Revisit the couple's intentions concerning having children. Discuss the couple's reproductive options. Emphasize that having high/ unsuppressed viral load increase the risk of HIV transmission to fetus or infant during pregnancy or breastfeeding
3. Describe the country's PMTCT programs and services and identify where couples can access services.	3. Describe PMTCT programs and services and identify where the couple can access services.

4. Address issue of testing of young children if the woman is HIV-positive.	4. Address the couple's questions and concerns regarding PMTCT services.
5. Provide needed referrals. PMTCT/ FP • Family planning • ANC clinics (if woman is pregnant) • MCH clinic (if woman has young children and/or if she is breastfeeding)	5. Provide needed referrals. • Family planning • ANC clinics (if woman is pregnant) • MCH clinic (if woman has young children and/or if she is breastfeeding)

Step 7. Discuss disclosure and getting support

<i>Task for discordant positive</i>	<i>Task for concordant positives</i>
1. Explain the benefits for the couple to disclose their HIV status to others.	1. Explain the benefits for the couple to disclose their HIV status to others.
2. Explore couple's feelings about sharing their results with a trusted friend, relative or clergy. • Identify who could provide additional support. • Address confidentiality and disclosure concerns.	2. Explore the couple's feelings about sharing their results with a trusted friend, relative or clergy. • Identify who could provide additional support. • Address confidentiality and disclosure concerns.
3. Discuss disclosure basics.	3. Discuss disclosure basics.
4. Reinforce that the decision to disclose is mutual.	4. Reinforce that the decision to disclose is mutual.
5. Explore the possibility of participating in a support group and additional counseling sessions.	5. Explore the possibility of participating in a support group and additional counseling sessions.
6. Answer remaining questions and provide support. • Summarize.	6. Answer remaining questions and provide support. • Summarize.

Step 8: Record outcome of partner and family index case testing services

Document the process and outcome of each partner and child elicited on the case investigation form- record the method(s) of contact used and the outcomes-notified, tested, result; and for HIV-positive partner/ child-record whether he/she has been started on ART.Follow to confirm whether:

- Elicited partners received HIV testing and know their HIV status (result documented),
- Partners tested HIV+ are linked to HIV treatment (ART documented),
- Partners are examined and treated for STI, as applicable, and
- The above services are provided within “reasonable” amount of time (early testing- early prevention).

When HIV-positive partner is enrolled to ART, reinstitute the same process of Partner and family index case testing service to the client as index and elicit partner and children.

Step 9: Continually update outcome and check for new partners every time the index client visits ART clinic with appointment

At each ART refill visit, review client's chart and update for partner and family based ICT service. Check the intake form for documentation of HIV status of all the partner(s) and biological children and discuss on any gaps by asking the index client.

During the ART initiation encounter to new client, provide the partner and family based ICT service as soon as possible to have maximum impact on controlling the transmission.

4.3.4. Offering Children Testing Service to Index case

Initial interview steps:

For testing of biological children of index client, the preferred way is client referral in which service provider and the index client work together to test his/her child (ren) the sooner the better. The index Parent will be given 14 days to bring his/her child (ren) to the facility for HIV testing. If the index failed to bring his/her child(ren) for HTS, it is recommended to repeatedly advise the index to get his/her child(ren) tested by any means including with home testing service available in the community.

Step 1: Introduce Index Children Testing Services to the Index Client

Explain the importance of knowing the HIV status of all biologic children to the index client and encourage him/ her to ask any fear or concern he/ she might have.

“Unless tested for HIV, it is not possible to know if a child is infected with HIV or not.

HIV-positive children shall start HIV treatment to help them stay healthy and thrive.

Children of HIV positive parents are at higher risk of having or being HIV infected.”

Inform the care giver/ index client that there are two options for testing any biological children < 19 years with unknown status, either bringing them to facility for HTS or get tested at home or nearby when healthcare providers make community visit. Also inform that His/her HIV status will not be disclosed to the children and all HIV status will be kept confidential.

If Facility Based testing is chosen, the client shall agree to bring his/her children to the facility for testing within 14 days. The index can bring all children or a child at a time. The service provider will remind the index client to identify barriers for not bringing the children for test

and support him/her to develop plan until all children know their status. When the index brings his/her child (ren) to the facility for HIV testing, sit with the client (and the child if appropriate age) and explain the importance of knowing the child's status and the treatment available for positives and test the child for HIV. Enroll the positive child to ART and initiate treatment.

Based on index client's preference and consent, you can also communicate with community level testing provider (in our case urban health extension professionals or community level support organization) to visit the index client and perform HIV testing to his/ her child (ren) at home.

Step 2: Obtain List of Biologic Children

Ask the index client to list the names of all his/ her biologic children, and record it on the case investigation form. Talk with the index client about any fears or barriers.

“Children will benefit from getting HIV treatment earlier even if they seem healthy and they can take steps to remain HIV-free as they grow. Not only children living with you, but a child under 19 years of age living with grandparents/relatives elsewhere also shall be tested for HIV and status determined.

Steps 3: Determine Caregiver's/ parent's testing Plan

Once names of children are listed, arrange with the consented caregiver/ parent on options for HIV Testing for each under 19 child. Document options on the case investigation form.

If the client prefers to bring children to the facility, counsel and schedule a convenient appointment preferably within 14 days to bring the children and get tested for HIV. If the caregiver/ parent chooses HIV testing for his/ her children at home or near his/her village/ community, **counsel** and discuss on:

- Willingness of the caregiver/ parent if visited by a community provider/ HEP or outreach team at his/ her home or village,
- Scheduling a convenient date and time when all his/her children will be available at home for test,
- Confirming functional mobile phone number(s) and detail of home address (include landmarks) so that health care workers/ HEP will reach out.
- The list of his/her children will be given to UHP/ care providers who provide home or community level HIV testing service.

Step 4: Record Outcome of Family Index child Testing Services

It is important to document the outcome of each named child by caregiver/ parent on the case investigation form- test status, results and ensuring initiation of ART if the child tested positive.

4.3.5. Considerations for PLHIV Female Sex workers as index

Ways to prepare Female Sex worker for partners and children elicitation

Establishing trust and open communication has paramount importance when offering Partner and children index case testing service to female sex worker. Inform and make aware that partner elicitation is voluntary and all information will be confidential. Female Sex worker may usually be hesitant to share partner information if they do not trust the service providers. Many FSWs can have established relationship with common 'special boyfriends' who passively or coercively engaged with them as protectorate or fugitive besides to their paying clients. These partners are at higher risk as they have an established form of non-protected and non-paying sexual relationship with more than one sex workers at or over a period.

For fear of stigma –discrimination and rejection, FSWs are not usually comfortably open when they know they have HIV. When approaching FSW for partner index case testing, priority should be on eliciting non-paying partner(s) of female and male sex workers for it is easier to obtain contact information.

Biological children of FSWs are equally important when offering index case testing service to FSWs. All biological children (< 19 years with unknown HIV status) of HIV positive FSWs shall be tested for HIV. If the HIV positive FSW (index client) has child/ren, it is important to arrange with the index FSW to bring her child/ren and get tested in the facility. It is a common practice that child/ren of FSW are sometimes left in the countryside under the care of grandparents (FSW's mother). In such cases, it is important to inform the FSW that all her biological children shall be linked with OVC service providers in the town where the children live* and get tested for HIV.

*Note: A strong referral and treatment service linkage system should be in place before actively eliciting biological children of FSWs, it could be done either through services co-located within a trusted FSWs community space or within an established network of health facilities where providers have undergone KP sensitization trainings.

If a FSW mentions or screens positive for intimate partner violence (IPV) when being interviewed for partner and children index testing service; ensure linking this FSW to other services available like psychological/social support services, legal aid, GBV prevention, etc.

A mechanism should be in place for FSWs/beneficiaries to anonymously report any adverse event or other risk experienced as a result of participating in index testing.

As health care provider working with FSWs, consider developing a list of common slang terms used by FSWs so as to be familiar and build smooth rapport with them and avoid miscommunication/ misinterpretation.

Confidentiality and its implications shall be emphasized properly for FSWs. Be aware of the legal environment where they operate and how FSWs may be adversely impacted from disclosure of their “status”. Emphasize to FSWs that their personal identity and other information is kept protected and strictly confidential. Always inform FSWs that only limited information pertinent to contributing directly to the continuum of their care is shared among service providers (shared confidentiality).

Use unique IDs of index cases on a register that lists named contacts and children with the nature of their relationship to ensure confidentiality of index clients. Both identity of index & elicited partners shall not be displayed altogether.

4.4. Additional HTS Approaches and Tools to Strengthen Targeted HIV Testing

The following HTS approaches are recommended to reach the 2030 global target to end HIV/AIDS epidemic.:

a. Social Networking Strategy (SNS)

Social network strategy (SNS) is a case-finding strategy that uses social network connections to locate individuals engaging sexual partner/s and it can also be applied for drug injecting partners and social contacts of key population members with HIV and of those who are HIV-negative and at ongoing risk.

Social network is a set of individuals linked by one or more specific types of relationships (Wasserman, 1999). SNS has been used to find new HIV cases among key populations who have difficulty accessing HIV testing services. It has also been used to find previously diagnosed individuals who were out of care, and who may then be linked to ART for the first

time or re-engaged in ART services. SNS is a tool for peer-driven HIV testing services. By addressing people's confidentiality concerns and broadening the reach to social contacts, social network-based HIV testing approaches can improve the acceptability of partner services among key populations and so reach more people who may not otherwise test for HIV. Social network-based HIV testing approaches for key populations is recommended. (WHO Policy brief, Nov 2019).

SNS is a recruitment strategy for reaching and providing HIV counseling, testing, and referral services to persons who are unaware of their HIV infection by using social network connections to locate individuals at the highest risk for HIV

The underlying assumption of SNS is that people in the same social network share similar risk behaviors for HIV. This is not limited to current behaviors but includes past behaviors.

In Ethiopian context SNS can be offered as HTS approach for sexual and social networks of female sex workers and PWID.

The overall approach is to enlist HIV-positive and high-risk, HIV-negative persons to recruit individuals from their social, sexual, and drug-using networks for HIV testing services. Once recruited and tested, network members become recruiters and refer members of their networks for testing. This process continues, creating chains of recruitment that can penetrate hidden networks.

This strategy uses inclusion and exclusion criteria to identify the appropriate target/recruitment process and enhances success of the SNS.

Inclusion criteria for recruiters/seeds include:

- HIV positive and/or high-risk negative
- Willing to refer network members to HTS
- Comfortable talking about HIV
- Knowledgeable about HTS and testing locations

Exclusion criteria for recruits include:

- Persons who might pose a risk for violence,
- Persons with significant mental illness who might interfere with appropriate participation,
- Known HIV positive cases,

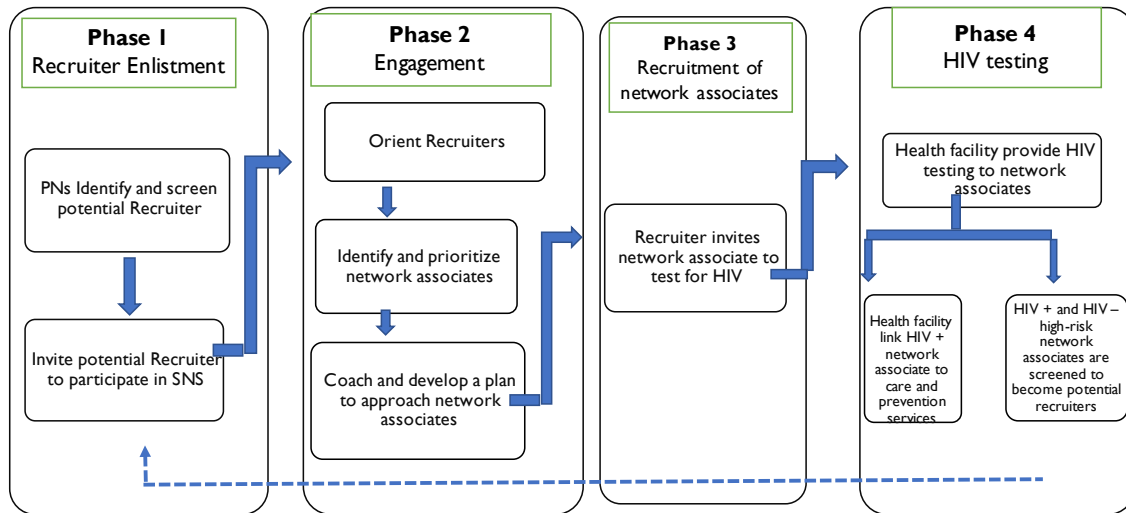
There are four major phases to a social networks program. These phases are

1. **Recruiter Enlistment:** In this phase, HIV-positive or HIV-negative high-risk persons who are willing to recruit individuals at risk for HIV infection from their social, sexual, or drug-using networks are enlisted into the program. To identify recruiters, approach HIV positive clients and identify additional people through the existing HTS or ART/PMTCT services.
2. **Engagement** (Orientation, Interview, and Coaching): After recruiters are enlisted into the program, they are provided with an orientation session that explains the nature of the program and the social network techniques that might be used to approach their associates and discuss HIV testing with them.
3. **Recruitment of Network Associates:**Next, recruiters will refer individuals for testing who they have identified as being at risk for HIV infection. All individuals should be approached by the recruiter alone, without the provider).

***HTS:** The next phase involves providing HIV testing and counseling to the network associates identified through the social network strategy.*

SNS workflow chart

Four Phase of SNS for HIV Testing



b. HIV Self Testing

HIV Self-Test (HIVST) is an innovative approach to deliver HIV testing services and contribute more for the national case finding efforts. HIV self-testing should be offered as an additional approach to HIV testing services.

HIV self-testing (HIVST) refers to a process in which a person collects his or her own specimen (oral fluid or blood) and then performs an HIV test and interprets the result, often in a private setting, either alone or with someone he or she trusts. As with all approaches to HIV testing, HIVST should always be voluntary, not coercive, or mandatory (WHO, 2016).

Approaches for HIV Self Testing

HIVST can be delivered through two distinct approaches. The approaches vary in terms of the level and type of support provided. Both approaches build public trust and mitigates issues related to stigma and discrimination.

Directly assisted HIVST refers to trained/oriented providers, or peers giving individuals an in-person demonstration before or during HIVST on how to perform the test and interpret the test result. Directly assisted HIVST for children through parent/ care giver or guardian Vs provider assisted from 2 to 15 years old will be piloted in few high burden health facilities and following review and evaluation of its results/findings decision will be made whether to broadly implement it or not.

Unassisted HIVST refers to when individuals self-test for HIV and only use an HIVST kit with manufacturer-provided instructions for use.

HIV Self -Testing, as a key strategy in the context of COVID-19,

- Expanding target population for HIVST, to include STI patients and their sexual partners,
- HIVST should be utilized optimally as additional testing strategy to reach clients undiagnosed otherwise and those who could not be reached timely because of MMD and COVID-19 situation.
- Program adaptations need to consider virtual (telephone, text, etc.) tracking based on client's preference and virtual implementation follow up at each level.

Both directly assisted and unassisted HIVST may include additional support tools:

- Manufacturer's instructions and brochures.
- Brief in-person demonstration (one-on one) before testing.
- In-person assistance during procedure.
- Hotline contacts for HIVST information.

c.Compulsory and Mandatory HIV Testing

Compulsory HIV testing can only be performed for specific reasons with individuals or groups when requested by the court. In all cases of compulsory HIV testing, individuals shall be informed of test results.

HIV is a blood-borne pathogen readily spread by blood transfusion or tissue/organ transplantation: therefore, it is mandatory to test blood or tissue for HIV before transfusion/transplantation/grafting. Mandatory screening of donated blood/organ/tissue is required prior to all procedures involving transfer of body fluids or body parts, such as artificial insemination, corneal grafts and organ transplant. Donors should be specifically informed about HIV testing of donated blood/organ/tissue.

d. Applying risk screening for high risk groups/ individuals for more targeted PITC

A risk screening based HIV testing which enables service providers to identify risky groups during counseling & gives opportunity for being targeted HIV testing. HIV risk screening tool is a tool, having a set of questions, used to identify the clients with specific risks for HIV transmission. Risk screening tools have been utilized in other countries to identify who needs to be tested and maximize HIV case detection and increase efficient utilization of the limited RTKs. In our setup risk screening can be applied for vulnerable adolescents/Youth and HIV negative pregnant/lactating women to provide risk based HIV testing for their partners. The following questions can be considered as examples for risk assessment:

- Have you ever tested for HIV in the past 12 month?
- Have you had unprotected sex?
- Did you drink a lot during night time & have sex with clients?
- Have you ever experienced symptoms of sexually transmitted infections?

4.5. Diagnosis of HIV in Infants, Children and Adolescents

More than 90% of HIV infection in children is due to transmission from mother during pregnancy, labor, and delivery and through breast-feeding. Early recognition of HIV infection in infants and children is crucial since there is fast progression of illness with 50% mortality by two years of age. Virologic tests are used to diagnose HIV infection in infants less than 18 months of age. Rapid HIV antibody test is used for infant ≥ 18 months of age.

Opportunities to identify children, who are in need of HIV testing should not be missed.

Potential entry points for testing children are:

- PMTCT program
- EPI units
- Nutritional therapy units
- TB clinics
- Pediatric outpatient and in-patient units
- Children of adults on ART/HIV care
- Siblings of children enrolled to HIV care
- Orphanage

It is currently recommended that HIV testing and counseling should be offered to vulnerable* adolescents, orphans, children and adolescents with malnutrition, children and adolescents with HIV indicator conditions and all under five children visiting health facilities. Diagnosis of HIV infection in a child implies potential HIV infected case among family members. Hence, HIV counseling and testing should be offered to family members of HIV infected child.

*Vulnerable adolescents includes those living in the street, orphans, adolescents in child headed household, girls engaged in sex with older men or in multiple & concurrent sexual partnership, out of school youth & adolescents who are sexually exploited.

HIV testing services for adolescents (10-19) and young people (15-24)

Vulnerable adolescents are often underserved and given insufficient priority in many HIV programs, with poor access to and uptake of HIV testing and counselling and linkage to prevention and care. Adolescents with HIV include those surviving perinatal infection and those newly acquiring infection as they become sexually active or are exposed through sexual assault, and blood transfusions. In generalized epidemic settings, many vertically infected infants are not diagnosed through program for PMTCT and would benefit from earlier HIV diagnosis and treatment. In many settings, adolescent girls and adolescents from key populations are also vulnerable to HIV infection and would benefit from access to acceptable and effective HIV services, including HIV testing and counselling.

Mature minors (Children aged 13-15, who are married, pregnant, commercial sex workers, street children, head of family, or sexually active) and adolescents above 15 years can access HTS service by giving self-consent.

HIV testing approaches to improve HIV case finding among children and adolescents

- HTS with other prevention services and linkage to treatment and care is recommended for vulnerable adolescents and youth age 15-24 years
- Provide HTS to all children and adolescents presenting with indicator conditions or signs and symptoms that suggest HIV, including oral candidiasis, failure to thrive, chronic cough and skin conditions.
- Provide HIV testing to all children and adolescents attending TB clinics and malnutrition services.
- Provide HTS to all children with parents or siblings receiving any HIV service (for example, PMTCT, ART) through home-based or facility-based HTS.
- Provide HTS to all children and adolescents with HIV indicator conditions attending pediatric inpatient health services.
- Provide HTS to all children and adolescents receiving orphan and vulnerable children (OVC) services.
- Provide HTS to adolescents from key populations in all settings

In addition to standard messages for all people diagnosed with HIV, post-test counselling for adolescents whose test result is HIV positive should include the following:

- Tailored advice with linkage to HIV treatment and long-term care
- Counselling, referral and linkage to specific psychosocial and mental health services tailored to both the situation in which infection happened and the developmental age of the individual
 - Information on adolescent rights and responsibilities, especially the right to confidentiality
 - Counselling, referral and linkage to specific sexual and reproductive health services, including contraception, and an opportunity to ask questions and discuss issues related to sexuality;
 - Individualized planning on how, when and to whom to disclose the HIV status (Adolescent HIV disclosure is described in chapter 13); and
 - Referral for group counselling and peer support groups.

HTS Counseling Practicum:

There will be Role play among participants for HTS information & counseling

In this role play, you will practice on how to counsel and elicit partner and family based partner index case testing the Sarah. While the counselor interview 'Sarah', the observer shall check whether the provider follows all the steps and skills of interviewing the index.

Case 1: Sarah is a 34 year old married woman who was recently diagnosed with HIV during her first antenatal care visit for her current pregnancy. Her youngest child is now 16 months old and she has older children aged 3 and 6years old. Her husband, Abraham, is a field worker with Communication Company.

Using the following Checklist, evaluate the counselor for practicing the steps mentioned above.

1. Explain the partner and family based index case testing service to Sarah on:
 - Importance of ICT,
 - Confidentiality for the index,
2. Identify and Resolving any barrier to name partner/ child(ren)
3. Explain Ways of contacting partner (notification) for HTS?
4. Whether the counselor set consented appointment time for testing of partner/ children?
5. What information/advice did you give the client?
6. What challenges did you encounter during the interview?
7. What are your thoughts for how you could address those challenges?

4.6. Laboratory Diagnosis of HIV Infection in infants younger than 18 months of age

Different laboratory tests are available to diagnose HIV infection. Before you can assess and classify an infant or child for HIV infection it may help you to understand the different tests which detect anti-bodies and the virus itself, and how to understand the results of these tests. This information is summarized in the following table.

Antibody versus Virologic tests

Antibodies tests, including rapid test	Virological assays (RNA or DNA PCR)
▪ Detect antibodies made by immune cells in response to the virus,	▪ Directly detect the presence of the HIV virus or products of the virus in the

- Antibodies from the mother passing on to child do not completely disappear until the child is 18 months of age - This means that a positive antibody test in children under the age of 18 months is not a reliable way to check for infection of the child, - Appropriate for adults and children older than 18 months of age, - Refer the annexed HIV rapid test SOPs	blood - Positive virological test can therefore reliably detect HIV infection at any age, even before the child is 18 months old - If the tests are negative and the child has been breast-feeding, this does not rule out infection as the baby may have just become infected. Tests done six weeks or more after completely stopping breast feeding are thought reliably rule out infection
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Interpretation of HIV test results for children less than 18 month of age

Rapid HIV test done before 18 months of age does not tell us infection status of infant as the maternal antibody existing in the infant body. The breast milk of an HIV positive mothers can transmit HIV infection. The main point that you need to remember is a positive virologic (DNA PCR) test at any age implies that the baby is HIVINFECTED, and a positive ANTIBODY test at 18 months or ae implies the child is HIVINFECTED.

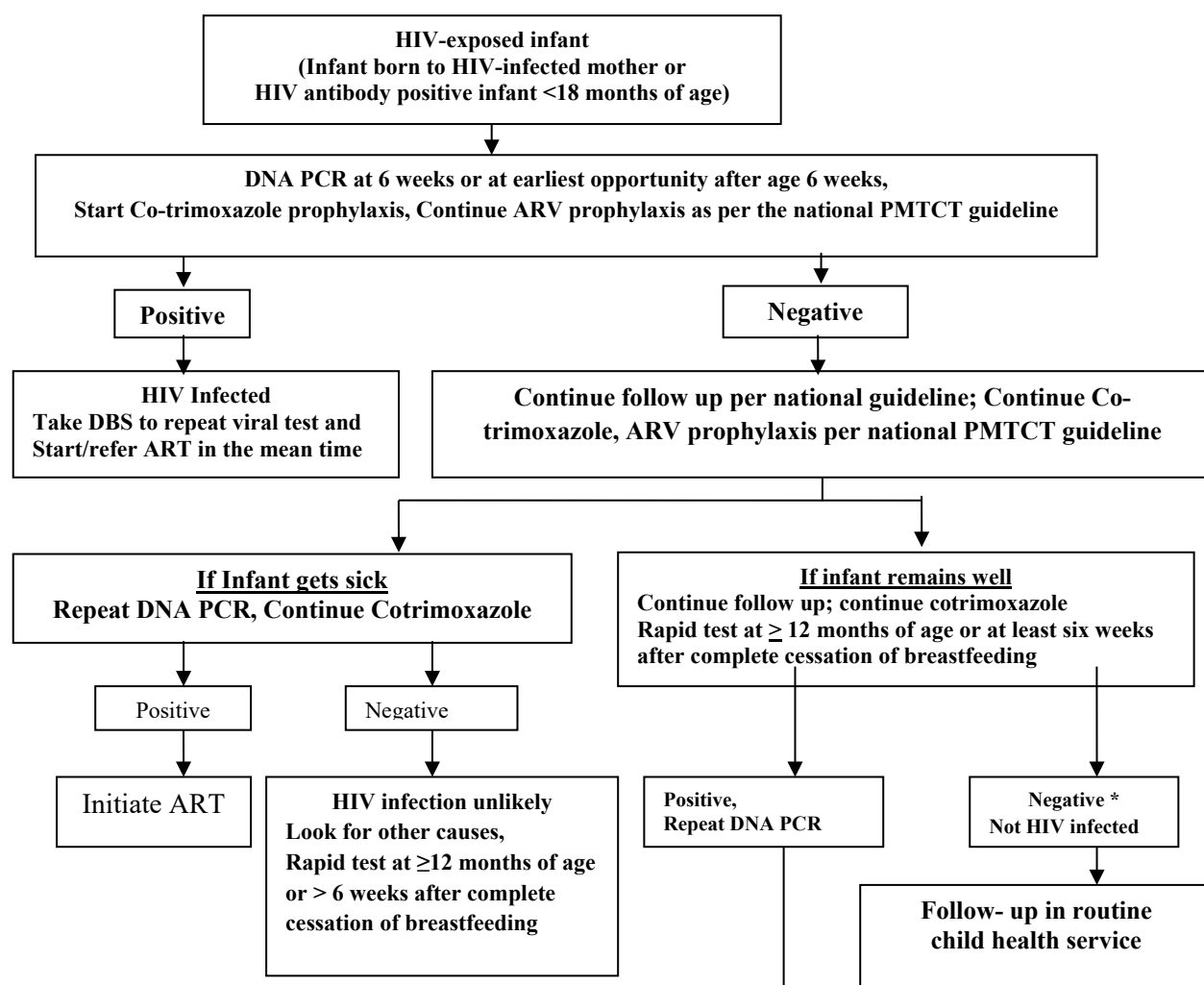
The table below provides details about how rapid antibody tests and virological tests should be interpreted in young infants & children. It takes into consideration whether the child is breastfed or not.

Table 4.2. HIV testing in children born to known HIV positive women

HIV TESTING IN CHILDREN BORN TO KNOWN HIV POSITIVE WOMEN			
Age	HIV testing	What results mean	Considerations
<18 months	Rapid HIV antibody test	If positive, result indicates presence of either mother's or child's HIV antibody. Can NOT confirm HIV infection!	Confirms child has been exposed to HIV, as passive transfer of maternal antibodies can cause positive test results.
		If negative and not breastfed: = Not infected . If negative but still breastfeeding: = repeat test 6 weeks or more after complete cessation of breastfeeding.	Negative test usually rules out infection acquired during pregnancy and delivery. But, the child can still be infected by breastfeeding (continued exposure).

	HIV virologic test (DNA PCR)	Positive virological test at any age implies HIV infection and start HAART and take DBS for Repeat DNA PCR	Best to perform when child is 6 weeks old or more
		Negative* virological test and never breastfed or not breast-fed in the last 6 weeks means the child is not infected. However, if a child with negative virologic test become symptomatic there is a need to repeat the test	Negative* results still on breast feeding need to be confirmed 6 weeks or more after breast feeding discontinued. If older than 9-12 months by this time antibody testing can be used before doing another Virological test, as only children who still have HIV antibody need another virological test
≥18 months	HIV antibody test (rapid test or ELISA test)	Valid results as for adults. Negative= child is not infected; Positive= child is infected.	If negative and still breastfed – repeat test once breastfeeding discontinued for 6 weeks or more.

*Continue follow up per national guideline; continue Co-trimoxazole, ARV prophylaxis per national PMTCT guideline



NB:

If the first virological test is Positive and the confirmatory virological test is Negative, a third test is needed to resolve the discordance between the two virological tests.

(WHO recommendations on the diagnosis of HIV infection in infants and children, 2010).

EXERCISE-4.3 *Answer the following questions:*

- 4.7. A 20-month-old baby has a positive virological test. Is the child HIV infected or not?
- 4.8. A 2-month-old baby has a positive virological test. Is the child HIV infected or not?
- 4.9. A 2-mo. breastfeeding baby has a positive virological test. Is the child HIV infected?
- 4.10. A 2-mo. breastfeeding baby has a negative virological test. Is the child HIV infected?
- 4.11. A 2-mo. old breastfeeding baby has a positive antibody test. Is the child HIV infected?
- 4.12. 18-mo. old breastfeeding child has a positive antibody test. Is the child HIV infected?
- 4.13. A 9-mo. breastfeeding baby has a negative virological test. Is the child HIV infected?
- 4.14. A 9-month-old non-breastfeeding baby has a negative virological test and the baby last breastfeed 3 months ago. Is the child HIV infected?
- 4.15. An 18 months old child has a negative antibody test and the baby last breastfed one week ago. Is the child confirmed HIV uninfected?
- 4.16. A 13 months old non breastfeeding baby has a positive HIV antibody test. Is the child HIV infected? How can you confirm the child HIV status?

Chapter Summary

- Children and partners of HIV positive clients are at higher risk of HIV infection or already being infected with HIV. Partner and family based Index case testing service will enable identify HIV infected partners and children and start ART.
- Goal of Partner and family based Index case testing service is to break the chain of HIV transmission by offering HTS to persons who have been exposed to HIV and linking them to prevention if they are HIV Negative and to care and treatment if they are HIV Positive. It is a proven effective case finding strategy to reduce future transmission.
- HIV testing is a critical initial step in identifying and linking PLHIV to treatment,
- All forms of HTS including HIV partner services should adhere to the WHO 5Cs,
- Innovative and targeted HTS are key for case detection, reducing HIV transmission and HIV related morbidity and mortality.

N.B. There will be practical exercise on rapid HIV testing using national algorithm after completing the content of chapter 5.

Chapter 5: Care of HIV Exposed Infants

Chapter Description:

This chapter describes the care package and early diagnosis of HIV for Infants born to HIV positive women. The national timeframes for follow up and immunization are also detailed in this section.

Primary objective: At the end of this chapter participants will be able describe how to follow an HIV exposed infant

Enabling objective:

By the end of this chapter, participants will be able to:

- List the components of care of HIV exposed infant
- Describe the timeframe for follow- up visits of HIV exposed infants
- Discuss the benefits of Co-trimoxazole Preventive Therapy (CPT) as part of the care package for exposed infants
- Describe the schedule and contraindications for immunization in HIV exposed children
- Describe DBS sample collection, drying, packaging, transportation and use on results for decision making.
- Explain how the HIV Exposed Infant follow-up form is used
- Fill out HIV Exposed Infant Follow up form

Chapter outline:

- Component of clinical care for HIV exposed infants
- Early Infant diagnosis using DBS for DNA PCR
- HIV exposed infant follow up form

Allocated time: 110 minutes

Learning activities

- Reading
- Question and answer,
- Demonstration,
- Group Discussion,
- Exercise,

Introduction

Infants born to HIV positive pregnant women by definition are HIV exposed and these infants can be infected with HIV during pregnancy, labor or after birth through breast feeding. All HIV exposed infants (infected and non-infected) will test antibody positive during the first few months of life. While the child with HIV infection can often be identified during the first months of life, HIV infection often cannot be excluded until after one year of age particularly in breast feeding babies.

Pediatric HIV disease can progress very rapidly and may require treatment before a positive diagnosis can be confirmed. HIV infected infants are susceptible to many opportunistic infections including PCP, TB and other bacterial infections that are associated with high rates of mortality. In the provision of care for these children, we use the national HIV exposed follow-up card.

5.1. Components of clinical care for the HIV exposed infant:

1. **History:** Take relevant history
2. **Physical examination:** Perform physical examination thoroughly
3. **Growth assessment:**(Will be discussed on chapter 9)
 - Growth is the most sensitive clinical indicator of HIV infection in infants and young children.
 - Children with HIV infection are at high risk for poor growth
 - Growth should be monitored closely for all HIV exposed and infected infants
4. **Developmental assessment:** use developmental check list to assess growth and development
5. **Infant feeding:** nutrition and feeding history should be assessed regularly.
6. **Immunization:** all HIV exposed infants should be immunized according to EPI recommendations.

Table 5.1. Immunization schedule for HIV exposed infants

Immunization schedule for infants and children recommended by the WHO EPI
--

Vaccine at age of	Birth	6 wks	10 wks	14wks	9 months	15 months
BCG*	x					
Oral polio	x	x	x	x		
DPT-HepB-Hib		x	x	x		
Pneumococcal vaccine (PCV)		x	x	x		
Rota		x	x			
Measles					x	x
*Infants with symptomatic HIV should not receive BCG vaccine. DPT-HepB-Hib = Diphtheria/ Pertussis/ Tetanus, Hepatitis B, Homophiles influenza type B, PCV-10= Pneumococcal conjugate vaccine,						

7. ARV prophylaxis for HEI:

Considering all HEIs as High risk infants and recommending enhanced postnatal prophylaxis (ePNP) for all HEIs. The ePNP is a combination of NVP and AZT for 6 weeks with continued NVP prophylaxis for additional 6 weeks (i.e., AZT for 6 weeks and NVP for 12 weeks) for all HEIs irrespective of the risk. **NVP & AZT Prophylaxis syrups shall be started within 1 hour of birth to the HEI.** The duration of ARV prophylaxis administration for HIV exposed infants identified after birth (through infant or maternal HIV antibody testing) depends on the status of Breast feeding:

- For Infants on breastfeeding:
 - Initiate ART for the mother,
 - Provide NVP+AZT daily for 6 weeks and continue NVP for additional 6 weeks,
 - Collect specimen for DNA PCR testing at 6 weeks of age,
- Infant not breast feeding:
 - Initiate ART for the mother,
 - If infants are receiving replacement feeding, provide NVP+AZT daily for 6 weeks,
 - Collect specimen for DNA PCR testing at 6 weeks of age,

Table 5.2. Summary of maternal & infant ARV prophylaxis for different clinical scenarios

Scenario	Maternal ARV Prophylaxis ^a	Infant ARV Prophylaxis	Duration of infant ARV prophylaxis
Mother diagnosed with HIV during pregnancy	Initiate maternal ART	NVP+ AZT	NVP+AZT for the first 6 weeks and NVP for additional 6 weeks (total 12weeks of NVP):
Mother diagnosed with	Initiate maternal		NVP+AZT for the first 6 weeks

HIV during labor, within one month before delivery or postpartum within 72 hours regardless of feeding plan	ART	NVP + AZT	and NVP for additional 6 weeks (total 12 weeks of NVP):
Infant identified as HIV exposed any time after birth (through infant or maternal HIV antibody testing) and is breastfeeding	Initiate maternal ART	NVP+AZT	-NVP+AZT for the first 6 weeks and continue NVP for additional 6 weeks, PLUS -Take DBS specimen for DNA PCR for EID same day if infant older than 4weeks.
Infant identified as HIV exposed after 72 hours after birth (through infant or maternal HIV antibody testing) and is NOT Breastfeeding	Initiate maternal ART	No prophylaxis needed	-Do DNA PCR test in accordance with national recommendations on early infant diagnosis; -no infant ARV prophylaxis; -Initiate treatment if the infant is infected
Mother receiving ART but interrupts ART regimen while breastfeeding (such as toxicity, stock-outs or refusal to continue)	Determine an alternative ART regimen or solution; counsel regarding continuing ART without interruption	NVP + AZT	-Until 6 weeks after maternal ART is restarted or until 1 week after breastfeeding has ended, -Do DNA PCR test in accordance with national recommendations on early infant diagnosis

Table 5.3. Dosage of NVP and AZT syrup for different age groups

Infant age	NVP daily dosing	Dose in ml	AZT daily dose	Daily dose in ml
Birth to 6 weeks				
Birth weight <2000g	2mg/kg daily once	0.2ml/kg once daily	2mg/kg twice daily	0.2ml/kg twice daily
Birth weight: 2000-2499 g	10 mg once daily	1ml once daily	10mg twice daily	1ml twice daily
Birth weight ≥2500 g	15mg once daily	1.5ml once daily	15mg twice daily	1.5ml twice daily

Age 6 weeks to 12 weeks				
Age 6 weeks to 12 weeks	20 mg once daily	2ml once daily or half a 50mg tablet once daily		

- Concentration is 10mg/ml for both NVP and AZT
- Dose listed in the table is given once daily for NVP and twice daily for AZT
- Increase the dose administered as weight increases.
- Follow the manufacturer's instruction for the duration of use following opening. The bottle should be labeled with the date on which it was 1st opened.
- Low birth weight infants (<2000mg) should receive mg/kg dosing, suggested dose is 2 mg/kg once daily for NVP and 0.2mg/kg twice daily for AZT
- NVP and AZT infant doses: The oral syringe should not be placed directly in to the bottle. Infant dose should be measured by pouring a small amount of syrup into a cup, and then draw the actual dose with oral syringe. Discard the leftover suspension in the cup.
- Dosing beyond 6 weeks of age in special situations in which prolonged NVP dosing of up to 12 weeks should be considered (such as the mother having had limited ART and not being likely to be virally suppressed; the infant is identified as HIV exposed after birth and is breastfeeding).

8. Co-trimoxazole Preventive Therapy (CPT) for HEI

Using pediatric co-trimoxazole in all HIV exposed infants significantly reduces the rate of PCP and other bacterial infections and in turn reduces infant morbidity and mortality rates. Start co-trimoxazole to all HIV exposed infants from 6weeks of age. CPT for HIV exposed infants should be continued until the child is confirmed not to have HIV infection.

Table 5.4. Dose of CTX for prevention of PCP in infants and children

Age	Suspension Per 5ml (200/40mg)	Pediatric tablet (100/20mg)	Single Strength adult tablet (400/80mg)	Double Strength adult tablet (800/160mg)
< 6mo	2.5ml	1 tablet	1/4 tablet	--
6 mo- 5yrs	5 ml	2 tablets	1/2 tablet	--
6-14 yrs	10 ml	4 tablets	1 tablet	1/2tablet
>14 yrs	--	--	2 tablets	1 tablet

9. TB risk Assessment for HEI

At each visit, the infant should be evaluated for Tuberculosis. We need to ask for household exposure with an adult who has tuberculosis and symptoms suggestive of the disease and chest radiograph (CXR) if clinically indicated. Young children living in close contact with a source

case of smear-positive pulmonary TB are at particular risk of TB infection and disease. The risk of infection is greatest if the contact is close and prolonged (such as contact of an infant or toddler with the mother or another caregiver in the household). If the mother is diagnosed PTB and started treatment less than 2 months before delivery or diagnosed after birth:

- Give the baby 5mg/kg isoniazid (INH) orally once a day for 6 months to the baby.
- Do not give BCG vaccine at birth. Delay BCG vaccination until INH prophylaxis is completed, or **repeat** BCG after complete course of IPT.
- Reassure the mother that it is safe to breastfeed the baby.
- Follow up the baby every 2 weeks, or according to the national guidelines, to monitor growth.

Contact: *Any person who has been exposed to an index case.*

Close contact: *A person who is not in the household but who shared an enclosed space, such as a social gathering place, workplace, or facility, with the index case for extended daytime periods during the 3 months before the start of the current treatment episode.*

Household contact: *A person who shared the same enclosed living space as the index case for one or more nights or for frequent or extended daytime periods during the 3 months before the start of the current treatment episode.*

10. Early Determination and evaluation of infection status for HEI

One of the goals of follow up of HIV exposed infants is to identify and treat HIV infected ones early. All HIV exposed infants should have virologic testing at 6 weeks of age or at earliest opportunity thereafter. For negative virologic test at 6 week, follow up will continue until antibody test is done at the age of 18 month to declare and discharge as HIV free.

11. Current assessment and plan

At each visit based on the findings on history, physical examination (that includes growth and development) and/or laboratory investigations, we need to have the assessment of the infant and we should plan our next steps in their management and follow-up.

12. Follow Up Visits and Schedule

Follow up of HIV exposed infant is recommended to be done monthly for the first six months of life and then every 3 months until final status is determined at age of 18 months. *See table below for details of follow up schedule and the care components that should be carried out at each visit.*

Table 5.5. Follow up visit schedule for HIV exposed infants*

Age in weeks/months	At birth	6 wk	10wk	14wk	5 m	6 m	9 m	12 m	15 m	18 m
History	x	X	x	x	x	x	x	x	x	x
Physical exam	x	X	x	x	x	x	x	x	x	x
Growth assessment	x	X	x	x	x	x	x	x	x	x
Developmental	x	X	x	x	x	x	x	x	x	x
CounselInfant Feeding	x	X	x	x	x	x	x	x	x	x
Determination of HIV status		DNA PCR	Do DNA PCR if the test is not done at 6 weeks** Repeat DNA PCR if infant is sick or the first DNA PCR test is positive					Perform rapid antibody test at least 6 weeks after cessation of breastfeeding		
Co-trimoxazole Preventive Therapy		X	Continue until HIV is excluded, and infant is no longer at risk from breastfeeding.							
TB Risk Assessment	At each visit									
Immunizations	x	X	x	x			x			
Adherence counseling	x	X	x	x	x	x	x	x	x	x
* This is the minimum; children should be seen more frequently if clinically indicated.										
** If the infant is between 9-12 months, first do Antibody test and if positive collect DBS for DNA PCR										

5.2. Early infant diagnosis using DBS for HIV DNA PCR

HIV disease progresses fast in infants and children and nearly 50% of children will die in their first two years of life if not diagnosed and enrolled into care. This shows the need for early diagnosis and treatment of HIV in infants and children. The primary goal of early infant diagnosis is to identify the HIVinfected child early, prior to the development of clinical disease, during the first months of life so that interventions and treatment can be started early.

DBS for HIV DNA PCR

DBS is one of the sample taking technique for virologic HIV testing in infants and children < 18 months of age. It has been used for genetic screening since 1960s. It is whole blood sample dried on a filter paper and sent to regional laboratories for DNA PCR test. Its advantage is that it requires very small amount of blood sample, less infectious (less biohazard), easy to store and transport the sample and has good sensitivity and specificity.

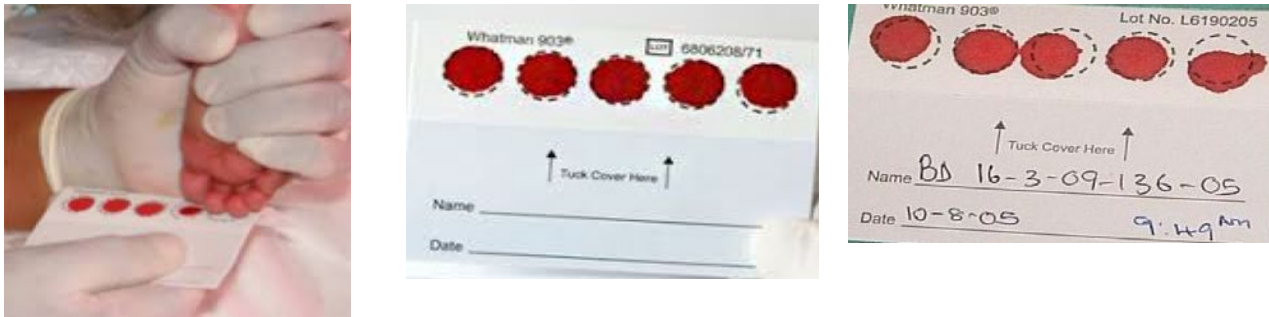


Figure 5.1. DBS samples taken on Whatman paper

There are three major activities in performing DBS: Sample collection, Sample drying and Sample packaging- storage and transportation.

1. DBS sample collection

In order to collect sample using DBS, the following supplies are needed: Gloves, Pen, Lab request forms & HEI register, DBS card (filter paper), Lancet, Disinfectant for skin, Gauze or cotton wool and waste container

NB Before stating the sample collection make sure that all supplies are complete, fill the required information correctly on laboratory requisition form and the DBS card (infant's name, unique identifier, date of sample collection, facility name).

The best site to take sample for DBS is either the heel or toes. If the child is < 2 months (below 4 kilo grams), it is preferred to prick at the heel to avoid bone damage; If the infant is >2 months but <2 years the medial face of the big toe is the best site for pricking and blood collection. Draw an imaginary line from the midpoint of the big toe to the heel, and one from between the 4th and 5th toe to the heel. The arrows indicate safe areas for puncture site - namely X, V and Z. Do not puncture the back of the heel, Achilles tendon, or lateral aspect of the big toe.

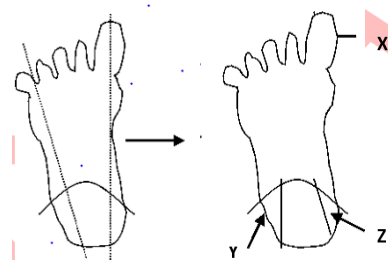


Figure 5.2. DBS sampling anatomic areas on the feet

Steps of Heel/ Toe prick DBS Specimen Collection (see pictures below)

1. Mother warms the foot by rubbing it gently with her hand.
2. Wash hands, put on powder-free gloves (do not allow water, powder from gloves, antiseptic solution to come in to contact with DBS card before or after collection of sample since this will affect the result).
3. Position baby with foot down
4. Clean area, and let it dry for 30 seconds
5. Press lancet into foot and prick the skin
6. Wipe away the first drop of blood with sterile gauze or clean cotton ball
7. Allow another large droplet of blood to form at the puncture site
8. Gently touch the printed side of the filter paper against the large drop and allow it to completely fill the circle on the paper. The first drop should fill the circle. Do not press the paper against the hill.
9. Fill at least 4 circles of the filter paper in the same manner with successive drops of blood. If there is not enough blood, you may gently press the area around the puncture site. Do not milk the area, or you will get tissue fluid instead of blood.
10. Clean the puncture site and press with a cotton swab to stop bleeding. Do not use a bandage.

a. Mother warms the foot



b. Press lancet into foot and prick the skin



c. Let the blood drop to the card



d. Fill at least three circles of the filter



Figure 5.3. Technique of taking DBS blood sample(a-d)

2. DBS sample drying

Immediately after taking the blood sample on the filter paper allow the specimen to air dry horizontally on a rack for at least 3 hours. In case there is humidity in the room keep the sample over night before storing. Don't touch or smear the blood spots as this will may cause contamination and result in invalid sample or false results. Keep the sample from direct sunlight, dust and bugs. Do not stack, heat or allow the filter paper to touch other surfaces during the drying process. (NB. Drying is the most important process to assure the stability of the HIV viral DNA in the paper. If the DBS is not well dried after staying overnight at room temperature, the silica contained in the desiccant packets packed together with the DBS can remove all the remaining moisture.)

3. Package, storage and transportation of DBS sample

Supplies needed for sample packaging and storage are: Proper Dried DBS, Weighting paper Sealable impermeable plastic bags, Humidity cards, and Desiccant packs.

Steps in package and storing of DBS sample (see pictures below):

1. At the end of each day, pack the dry DBS (leave it to dry overnight if not dry after 3 hrs)
2. Wrap DBS card in a weighing/glycine paper so that DBS cards from different infants are not touching one another.
3. Insert up to 15 DBS cards into sealable plastic bag
4. Add 1 desiccant per DBS card in the bags (to remove any residual moisture from cards)
5. Add 1 humidity card per bag (as per the manufacturing instruction)
6. Press as much air out of bag as possible and seal it shut.
7. The humidity indicator card and the desiccant packs have a color indicator. All cards and packs should be replaced with fresh material before they develop color change.
8. Put in an envelope at room temperature in darker place (avoid exposing DBS to light) if it is not going to be shipped to the lab the same day
9. DBS should be transported to the Central Laboratory or testing site on the scheduled day.
10. To prepare DBS for transport, remove bagged samples from the drawer and remove old desiccants. Add fresh desiccants and reseal bag.
11. Use the specimen delivery checklist to verify you have a request form for each DBS.
12. Place the bag of DBS, request forms, and the specimen delivery checklist into a large cardboard or padded envelope, and seal.
13. Label the envelope with your clinic name, address and contact person. Infant DBS specimens Date you are sending to lab
14. Transport the bag by the fastest means possible to the laboratory. Avoid exposure to extremes of heat during transportation.
15. If a cooler is available for transport this will protect the samples from short periods of high temperature.



Figure 5.4. Packaging DBS samples

Add 1 desiccant per DBS card into the bags & Insert cards in to sealable plastic bag wrapped with weighting paper with the assistance of lab personnel and your facilitator, you will see a video clip on DBS sample collection and practice how to take DBS sample collection, drying, packaging and store for transportation based on the steps mentioned above.

4. Using DBS results for decision making:

- a. Negative results:
 - Continue cotrimoxazole prophylaxis and other components of care
- b. Positive Results
 - Initiate ART and collect sample for second DBS test
 - Continue CPT and other components of care

5.3. HIV Exposed Infant Follow up form:

HIV Exposed Infant follow up form is used to follow HIV exposed infants. The first (top) section of the follow up form is filled on first visit. This top section includes patient identification, type of prophylaxis provided for the mother and infant and immunization.

Following this there is a section to document evaluation of the infant. This section has rows to document weight, length, head circumference, infant development, infant feeding option, clinical diagnosis, CPT, infant laboratory diagnosis, conclusion, decision and next follow up date.

Case study 5.1

Fasil Hailu is a 6 weeks old infant brought to Bole Health Center. He was born at the health center on Tir 5, 2005. His mother Gebeyanesh Bekele lives in Addis Ababa Bole Kifle ketema Woreda 5. Her house number is xxx and her mobile number is 0911.99.99.98. At birth Fasil was provided with OPV0 and BCG vaccine, the nurse also provided him nevirapine syrup for six weeks.

Gebeyanesh started HAART on the 3rd month of her pregnancy and her husband died from unknown illness a year before.

The nurse in the health center evaluated Fasil. He is on exclusive breastfeeding since birth; his mother said that he is growing well and there is no abnormal physical finding. His weight is 4.8kg, length 58 cm and head circumference of 38 cm. The nurse collected specimen for DNA PCR and provided cotrimoxazole syrup for a month.

Q1 – How would you fill this visit?

Q2 – How do you interpret the growth of Fasil?

Q2- What will be the next appointment date?

Discuss and explain the HIV exposed infant follow up form with the case given

HIV-Exposed Infant Followup Card Federal Ministry of Health

Facility Name: _____		Infant's SMRN: _____		Enter Enrollment date: (/) (DD/MM/YYYY)(EC)													
HEI Code: _____		Infant referred from if referral _____															
Infant's Name: _____		Date of birth: (/ /)		Place of birth: _____ Age at enrollment: () Birth Weight: (gm.) Sex: M () F ()													
Mother's/Caretaker's Name: _____		Mother's SMRN: _____		If "Caretaker" describe relationship _____													
Addresses: Region: _____ Sub-city _____		Woreda/Kebelle: _____		House No. Telephone No of the family/guardian _____													
Parents Status: Mother? () A live/() Dead If alive mother is: () Enrolled in HIV/ART care/ () Not enrolled in HIV ART care, If "enrolled" () Within the facility/() Out of the facility (specify) _____																	
Mother's PMTCT Interventions: () HAART () None If on ART, duration on ART (Specify Regimen and month on ART): _____																	
HAART (Specify regimen) Start date Unique ART No. of the mother: _____																	
Infant ARV Prophylaxis: () NVP for 12 weeks + AZT for 6 weeks () NVP for 6 weeks + AZT for 6 weeks () NVP daily for 6 weeks () NVP daily for 12 weeks () AZT for 6 weeks () None																	
Father HIV Status: () Positive / () Negative / () Unknown, () Alive/() Dead If alive, enrolled in: () HIV/ART care/ () Not Enrolled in HIV/ART care If on ART Unique ART No. _____																	
Immunizations: (circle) BCG () OPV(0) (1) (2) (3) Pentavalent(1)(2)(3) PCV(0) (1) (2) (3) Rota(0) (1) (2) Measles ()																	
Date of visit	Age	Anthropometric measurements			Growth pattern (0/1)	Developmental mile-stone Red Flags (0/1)	Infant Feeding practice (1/2/3/4/5/6)	Mother's Breast condition; if breast feeding (0/1/2/3/4/)	Abnormal findings or diagnosis that may suggest HIV infection (0/1/2/3/4/5/6/7/8/9)	Treatment/ medication given for infant	Cotrimoxazole prophylaxis		HIV test/s Done (0/1/2/3/4/5/6)		Conclusion (0/1/2/3/4/5)	Decision (1,2,3)	Next visit date
		Weight	Length	Head circumference (HC)							Dose	Adherence (G/F/P)	DNA PCR test	Rapid Antibody test			

Health Care provider referred the infant for HIV care/ART or discharged from follow up: Name _____	Signature _____	Tel.# _____	Date / / (DD/ MM/ YYYY)(EC)
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Figure 5.5. HIV Exposed Infant follows up form front page,

HIV Exposed In															
Date of enrollment: It is the date (dd/mm/yy) when the HE is linked to & registered on the integrated MNCH/PMTCT register (MBPC register). and enrollment shall be done as soon as after birth before discharge. HEI Code: Infant's Full Name Initial / Mother Full Name Initial / Regional Code / Facility Type / Specific Facility Code / Year / Month / Serial Number on integrated MNCH/PMTCT (MBPC Cohort) Register and unique ART number of the mother NVP: Extended NVP should be provided for 12 weeks for those mothers who received ART for less than 4 weeks before delivery or start taking ART after delivery.															
Date of Enrollment	A	Anthropometric measurement	Developmental (Dev't) Milestone = Red Flags												
Enter in Ethiopia Calendar (dd/mm/yy)	Enter current age of the infant in months on the day of visit	Wt: enter current weight of the infant in Kg Lt: enter current length of the infant in cm HC: enter current head circumference in cm	Code as: 0 = No Red Flags 1 = Red Flags At: 2 month: Unable to smile in response to your smile 3 month: Does not respond to sound by quieting 4 month: Does not grasp rattle put in top palm or hold head steady when you sit him up 6 month: Does not reach for object held at eye level in front of him 9 month: Does not bear weight and stand with hand held 12 month: Does not stand alone 14 month: Inability to walk alone 15 month: Does not use one word meaningfully 18 months: Does not use at least 3 words, does not point to what he wants												
Date of visit															
Enter date of visit in Ethiopia Calendar in dd/mm/yy formation															
Infant feeding practices: For infant less 6 months: "1" Infant is on exclusive Breastfeeding (EBF), "2" Infant is on exclusive replacement Feeding (ERF), "3" Infant is on mixed feeding (MF): Should be corrected immediately For infant older than 6 months "4" Infant is on breastfeeding and complementary feeding, "5" Replacement feeds with complementary feeds <u>Breast condition – If mother breast feeding</u> "0" Normal, "1" Cracked nipple(s), "2" Mastitis, "3" Breast Abscess "4" Other (Specify) "5" Not Applicable (for ERF mothers)	Growth Pattern "0" if normal "1" if there is growth failure evidenced by: - Crossing two percentile - Weight for age < 3 percentile - Length for age < 3 percentile - Head Circumference < -3SD	HIV Test Done "0" Not tested "1" Sample collected for DNA PCR "2" DNA/PCR result "Negative" or "Not detected" "3" DNA/PCR result "Positive" or "Detected" "4" Rapid antibody test result negative "5" Rapid antibody test result positive													
	Abnormal findings of diagnosis that may suggest HIV infection (enter all appropriate findings) "0" none, "1" Generalized lymphadenopathy "2" Oral Candidiasis "3" Purulent ear discharge "4" Findings suggestive of pneumonia or lower respiratory tract infections "5" Persistent diarrhea "6" Hepatosplenomegaly "7" Skin lesions "8" Persistent Fever "9" Other (specify)		Co-trimoxazole Prophylaxis Mark * before the dose at start date Enter DC instead of dose when treatment discontinued												
			Cotrimoxazole Prophylaxis Adherence grading <table border="1"> <thead> <tr> <th>Adherence</th> <th># of missed doses per month</th> <th>% of missed / month</th> </tr> </thead> <tbody> <tr> <td>G (Good)</td> <td>< 3 doses</td> <td>> 95%</td> </tr> <tr> <td>F (Fair)</td> <td>5-8 doses</td> <td>85-94%</td> </tr> <tr> <td>P (Poor)</td> <td>≥ 9 doses</td> <td>< 85%</td> </tr> </tbody> </table>	Adherence	# of missed doses per month	% of missed / month	G (Good)	< 3 doses	> 95%	F (Fair)	5-8 doses	85-94%	P (Poor)	≥ 9 doses	< 85%
	Adherence	# of missed doses per month	% of missed / month												
G (Good)	< 3 doses	> 95%													
F (Fair)	5-8 doses	85-94%													
P (Poor)	≥ 9 doses	< 85%													
Decision 0 = Continue Follow Up, 1 = Referred for Pediatric HIV care within facility 2 = Referred Pediatric HIV care outside facility, 3 = Discharged Negative for mcare (HIV Free)		Treatment given: List drugs/ medication given on each visit													
Conclusion 0 = No Clinical or laboratory evidence of HIV 1 = Clinical evidence of HIV, 2 = Laboratory evidence of HIV (DNA PCR / Antibody tests) 3 = Lost to follow up, 4 = Died, 5 = Other (Specify)															

Figure 5.6. HIV Exposed Infant follow up form back page,

Chapter summary:

- An HIV exposed infant needs regular follow-up until HIV infection is excluded
- The major aim of the follow-up is to identify those who are HIV infected early and put them on life saving ART.
- During the follow-up, among others they should be provided with CPT, their growth and development monitored at regular intervals and assessed for any sign of HIV infection.
- DBS for DNA PCR should be done at 6 weeks of age or at the earliest opportunity thereafter.
- High risk pregnancies and high risk infants have higher risk of acquiring HIV infection,
 - Risks:
 - Combination prophylaxis,
 - Longer duration

Chapter 6: Introduction to Chronic HIV Care

Chapter Description:

This chapter defines and describes the chronic care for HIV. The communication skills and principles are also detailed in this section.

Purpose: To learn how to provide good chronic care for PLHIV

Enabling objectives

By the end of this chapter, participants are able to :

- Define Continuum of care/ chronic HIV care
- Describe the difference between acute and chronic care
- Describe effective Communication skills
- Explain the general principles of good chronic care
- Demonstrate 5A's of chronic HIV care
- List out Key elements of chronic HIV care

Chapter outline:

- Introduction to continuum of care
- Acute and chronic care and prevention
- General care package for PLHIV

Allocated time: 80 minutes

Learning activities

- Reading
- Discussion,
- Demonstration and Role paly,

6.1. Introduction to Continuum of Care

People living with HIV and their families have medical, emotional, social, physical and spiritual needs that change over time. They often must cope with the effects of stigma and discrimination, poverty, loss, neglect and abandonment. The purpose of the Continuum of Care is to address HIV as a chronic disease and develop systems that provide human, effective, high-quality comprehensive and continuous care to PLHIV and their families.

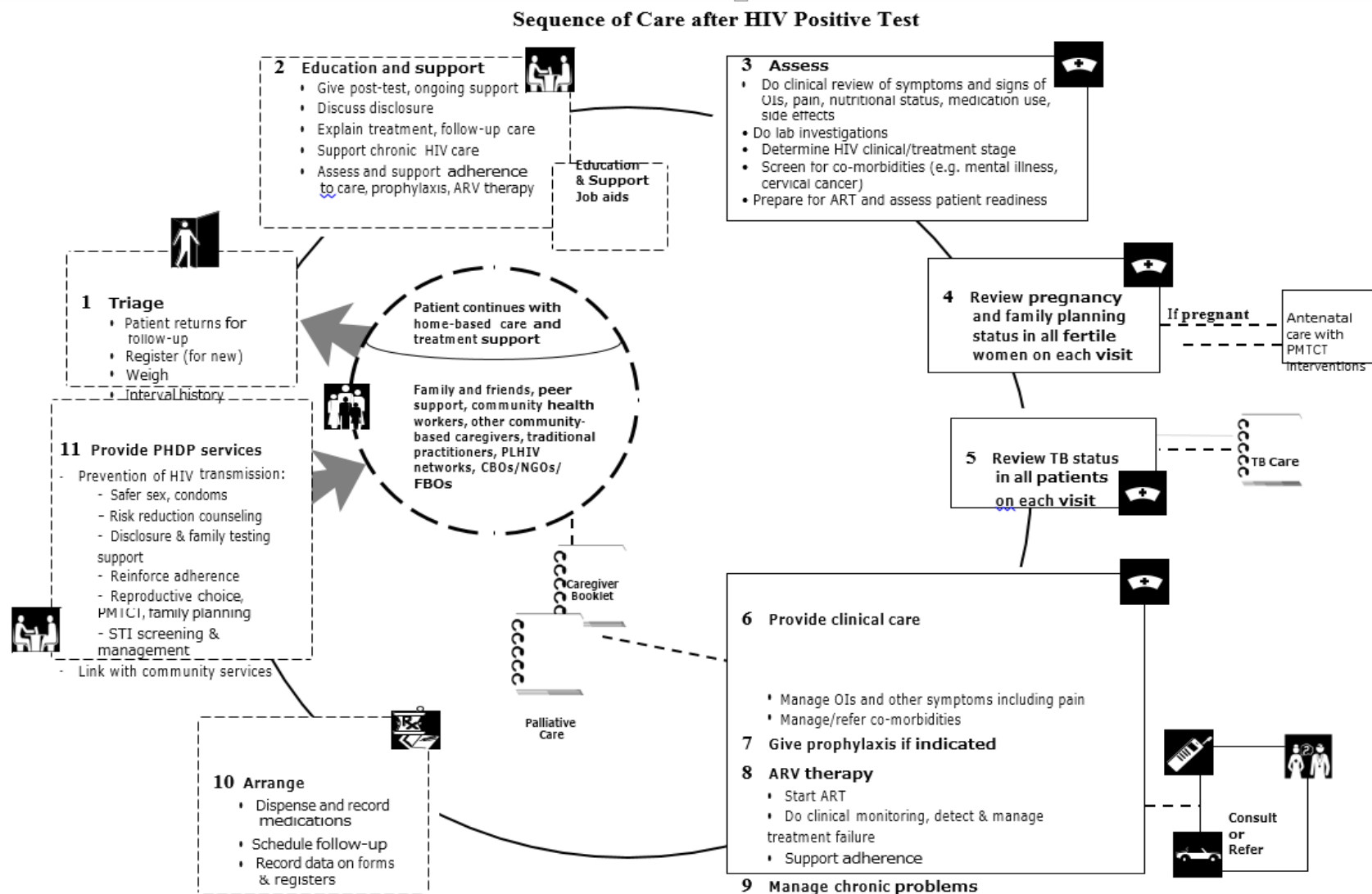


Figure 6.1. Sequence of Care after HIV positive test

Continuum of HIV services refers to a comprehensive package of HIV **prevention, diagnostic, treatment, care and support** services provided for people at risk of HIV infection or living with HIV and their families.

The Continuum of Care has two defining characteristics:

*The Continuum of Care is **a network** that links, coordinates and consolidates care, treatment, and support services for PLHIV. These services are provided in their homes, in the communities where they live, and in the health facilities that serve them.*

*The Continuum of Care is also **the group of services themselves** that together provide comprehensive support to PLHIV and their families. While these services are generally provided by a number of different organizations, the system that links, and coordinates them is planned and managed by the committee whose members include government officials, service providers, and representatives from non-governmental organization (NGO) representatives, PLHIV, and other stakeholders.*

Comprehensive care for people living with HIV requires both clinical care based at the health facility and home-based care involving the patient, family and friends, health extension workers, other community-based caregivers, traditional healers, and community-based organizations such as NGOs and FBOs.

6.2. Acute and Chronic HIV care and prevention

Acute care includes the management of the common illnesses that are problems for children, adolescents and adults. These include common bacterial infections, skin, neurological, mental health problems, drug toxicities, nutritional problems, etc. HIV infection causes a chronic disease, but if we only care for the patient during episodes of acute illness, then we are only providing acute care. We are not yet providing good chronic care. For chronic conditions including HIV, health care needs to be organized and delivered in a way such that the patient and family participate actively in the care provided to the patient. Acute care will always be necessary (even chronic conditions have acute episodes), but at the same time health care must be able to address long-term health problems through chronic care.

To improve the comprehensive care of PLHIV at health centers and hospital / ART clinic, it is important to be able to provide both good acute and chronic care and prevention at the health facilities, and to link with home and community-based care.

Communication Skills are very important for effective acute and chronic care delivery.

Effective communication skills are:

1. **Active listening:** listening involves not only hearing sounds but, as much as possible, also accurately understanding the meaning. As such, it entails hearing words, being sensitive to vocal cues, absorbing movements and taking into account the context of communication.
2. **Attending:** refers to the behavioral skills of paying attention to the client by limiting distractions and equalizing the power between the counselor and the client.
3. **Paraphrasing:** is a verbal statement that is interchangeable with the client's statement and is concerned with the cognitive (thought) content of the client.
4. **Reflection of feelings:** is similar to paraphrasing, but the focus is more on feelings (emotions) of the client.
5. **Summarizing:** is the gathering together of a client's verbalizations, behavior, and feelings and presenting them to the client in an outlined form.
6. **Questioning:** is one of the most important tools the counselor uses to guide the client through the counseling process.
7. **Reframing:** refers to the client's individual experience pictured from the counselor's point of view. Clients interpret events as they see them, but often from a position of depression or low self-esteem.
8. **Confrontation:** is a communication technique used to reflect a contradiction expressed by a client. Contradictions include differences between self-perception and behavior; between verbal and nonverbal messages; or between two different verbal messages.
9. **Self-disclosure:** this is a situation where the counselor communicates to the client his/her feelings or perceptions about her/himself also revealing something about his/ her status to someone trusted.

6.2.1. General Principles of Good Chronic HIV Care

Good chronic HIV care recognizes the fact that the patient and family/care taker should understand and learn to manage the patient's chronic condition. HIV infection and its progression to AIDS require much education and support to give patient and family/care taker

the skills to self-management. Although the clinical team and others at home and in the community can help, it is the patient that needs to learn to cope with their infection, to disclose to those they trusts in order to get further help, to learn to practice prevention and positive living, and to understand and use prophylaxis and ART and other treatments.

We use the following general principles in managing many chronic diseases or risk conditions that include HIV/AIDS:

1. Develop a **treatment partnership** with your patient,
2. Focus on your patient's **concerns** and priorities
3. Use the **5 A's**—Assess, Advice, Agree, Assist, Arrange
4. Support patient **self-management**
5. Organize proactive **follow-up**
6. Involve **non-clinical teams**(case manager/adherence support /Data Clerk/ support staff) in your health facility(*Refer MDT chapter 17*)
7. Link the patient to **community-based** resources and support (*Refer chapter 17*)
8. Use **written information**—National Pre-ART and ART registers, Intake and Follow-up forms, appointment register, inter facility referral forms, transfer out forms and laboratory request forms (*Refer chapter 9*)
9. Work as a **multidisciplinary** clinical team (*Refer chapter 17*)
10. Assure **continuity** of care.

a) Develop a treatment partnership with your patient

What is a partnership? A **partnership** is an agreement between two or more people to work together in an agreed way towards an agreed goal. For good chronic HIV care, the partnership is between the health worker or clinical team and the patient and family/care taker. In a partnership, both parties share responsibility for the agreement. Each partner knows what role he or she plays in the partnership. Partners treat each other with respect. One partner does not have all the power.

Examples of a partnership:	What is not a partnership?
• Doctor and nurse working in the surgical theatre • Football team • Cook and waiter working in a restaurant • Couple • Actors working in the same play	• Mother and young child • Teacher and young students • Police and criminals • Officer and recruits in the army

b) Focus on your patient's concerns and priorities

Often we focus only on the obvious signs or symptoms of illness and may miss the real reason that the patient came to the clinic. It is important to find out why the patient/care taker has come and make sure it is addressed.

c) Use the 5 A's—Assess, Advise, Agree, Assist, Arrange

The 5 A's are a key part of good chronic HIV care. They are a series of steps to use in caring for patients.

'**Assess**' includes asking what the problem is, listening to the answer, asking further questions, then examining the patient by looking, listening (for example, to hear a wheeze), and feeling as appropriate for the patient's symptoms.

In acute care, we usually assume that the patient's/care taker's goal for the visit is treatment for their main symptoms. It is common to skip over assessing behavioural risk factors or to ask about how the patient/care taker is managing the patient's chronic condition. Sometimes we even miss the real reason the patient came to the clinic.

For good chronic HIV care, this step needs to be expanded in chapter 7.

Advice includes recommending the treatments to the patient, educating the patient/care taker, and preparing the patient/care taker for self-management. It is important to discuss the options, not to just tell the patient/care taker what to do. It is also important to evaluate how ready the patient/care taker is to adopt the treatment and to ask checking questions to make sure that they have understood.

'**Agree**' means that the patient/care taker understands, wants, and agrees to the treatment plan. **This is a step we often skip!** It may be logical to skip this step during emergency care for trauma or a very sick patient. They have come (or been brought) for care and are too sick for a discussion or to make a choice.

However, most care is not an emergency and we are learning about how to best provide care for patients. For chronic care, AGREE is the key step in the process since it is the basis for forming a partnership with the patient/care taker and supporting good patient self-management.

‘Assist’ includes treatments (medication and other treatments), and advice and counselling, but also includes help that you can provide the patient/care takers in terms of skills to carry out the treatments or to overcome barriers. An important way to assist the patient is to get other help, by linking to available support in the community or to peer support groups, or by involving someone to help support them in their treatment. Usually we (and the patients) focus only on the pills or the injection. There is much more that is needed than this for good chronic care, especially for life-long treatment like ARV therapy.

Even the best plans for treatment often run into problems. When the patient returns, they may need more assistance to solve problems and overcome obstacles. We often assist the patient/caretaker only with treatments and skip other ways of assisting that may be as important.

ARRANGE a definite follow-up, scheduling a group appointment, arranging how the medication can be picked up on the next visit, and recording what happened on the visit are all parts of arrange.

d) Support patient self-management

Whenever talking about how patients can have their needs met and who can support them, health workers should always try to leave patients as much in charge of their own care as possible. This is very important for adolescents and adults. Patients with a chronic condition such as HIV have pivotal role in the management of their own care, i.e., **self-management**. Self-management means the patient taking responsibility for their own health care. Self-management recognizes that the patient/caretaker takes responsibility for the daily treatment of their condition- taking medication (such as co-trimoxazole or ARV drugs). Every day he or she makes choices about their diet, exercise, and other lifestyle issues that protect or damage their health (when the patient living with HIV make choices to protect their health, this is often called **positive living**). He or she makes choices on how to practice safer sex, use condoms, and prevent sexually-transmitted infections and unintended pregnancies. (This is called '**prevention**.')

The care team helps patients understand their options and the consequences of their decisions. The *patient* is the one who, on a daily basis, makes lifestyle choices and chooses to follow the care plan.

e) Work as a multidisciplinary team (MDT)

Providing good chronic care requires teamwork. To be able to deliver ART, we need to form a clinical team that includes a doctor/health officer/ nurse, pharmacy personnel, laboratory personnel, non-medical caregivers and data clerks. This team may work together differently depending on where they are located.

In the health facility ART Clinic, the whole clinical team can work together in the same clinic. They can consult with each other easily every day. We will learn more about how MDT can operate at the end of the second week of this course.

6.3. General Care Packages for PLHIV

As all people living with HIV are eligible for ART, enrolment in care provides an opportunity for close clinical and laboratory monitoring, early assessment for opportunistic infections (OIs), comorbidities and timely screening and management of OIs. Many care interventions are relevant across the full continuum of care, including HIV-exposed individuals and people living with HIV before initiation and during ART. The following critical interventions need to be addressed at initial encounter:

- Confirm HIV status by retesting and enrolling into HIV care (including recording into HIV positive tracking and pre-ART registers).
- Ensure any OI and other clinical problems that may delay ART initiation are ruled out or addressed.
- Ensure initiation of ART within two weeks after TB treatment is started except when signs and symptoms of meningitis (both TB and/or cryptococcal meningitis) are present.
- Ensure that barriers to adherence and treatment continuity are assessed and addressed accordingly.
- Ensure client is fully aware and makes informed decision for early initiation and continuation of treatment

NB. There will not be a regular pre-ART care service, however, all confirmed HIV patients should be enrolled on the pre-ART register initially and transferred to ART register as they start the HAART.

Key elements of chronic HIV care are:

1. Retesting for verification
2. Complete clinical assessment (history taking, complete physical examination and relevant lab tests)
3. WHO clinical staging
4. Prevention, screening and management of opportunistic infections and co-morbidities (see chapter 11)
5. Rapid ART initiation
6. Patient monitoring and follow up
7. Support for disclosure and assisted partner notification
8. Risk reduction counseling and combination HIV prevention approaches
9. Screening for and managing mental health problems and substance use
10. Adherence and psychosocial counseling and support
11. Nutritional assessment and counseling
12. Screening for other STIs
13. Prevention screening and treatment of cervical cancer.
14. Management of pain and symptoms.
15. Pregnancy status, family planning and contraception.
16. Document all relevant client information

Exercise 6.1:

Role play on good and bad use of 5A's with Expert Patient Trainer (EPT)

Under this roleplay, you will start practicing client assessment exercises with expert patients using checklist. Expert patients are trained PLHIV already trained on the cases and they act accordingly. These expert patients will assess and give you feedback on your approach. You are expected to exercise these skills during subsequent encounters to EPTs.

Chapter summary

- Chronic care is essential for the management of chronic diseases including HIV.
- Acute care is important in the management of chronic diseases because they may have acute illnesses related with their chronic problem.
- Applying the ten principles of good chronic care is fundamental in providing successful chronic care.
- Self-management means the patient taking responsibility for their own health care.

Chapter 7: Clinical Evaluation and WHO Clinical Staging

Chapter Description:

This chapter describes the clinical and history evaluation of HIV positive individuals to determine progress of the disease. The WHO clinical stages and clinical assessment while on treatment are also detailed in this section.

Purpose:

Understand evaluation of HIV positive individuals and do WHO Clinical Staging.

Enabling objectives

By the end of this chapter, participants will be able to:

- Explain how to take history and do physical examination
- Describe relevant laboratory tests required for baseline assessment
- Describe the WHO clinical staging

Chapter outline:

- Clinical evaluation
- WHO clinical staging

Allocated time: 120 minutes

Learning activities

- Reading
- Discussion,
- Demonstration,
- Practical Exercise at Skill station,

Background

HIV infection is a disease with variable manifestations. Clinical assessment should be conducted in a stepwise and standard approach. Perform rapid assessment to ensure that there are no serious life threatening conditions. Be aware that people with HIV may present with serious illness as a result of advanced HIV disease, opportunistic infections or drug toxicity. HIV infected infants and young children have rapid progressive course of illness reflecting

immunological immaturity to contain infection. Therefore, taking proper history and performing complete physical examination, conducting basic laboratory investigations and WHO clinical staging are integral parts of comprehensive HIV care. A health care provider has to be equipped with basic clinical skills in assessing and evaluating clients coming for HIV care as per the guideline. It includes the clinical evaluation by history taking and physical evaluation and based on the finding categorizing according to the WHO clinical stage.

7.1. CLINICAL EVALUATION

a. History taking: There are many problems related to HIV infection and treatment. Take good medical history to screen for problems and to identify those patients requiring rapid treatment planning or isolation for infection control purposes, there shall be systematically structured way of interviewing the client.

1. **Identification:** Important information to be gathered in this section are: Client name, age, sex, occupation and contact address. In case of children Parents'/caregiver's name, age and educational status should be gathered.
2. **Chief complaints:** This component of the history contains the main reason for the patient or parents/caregivers to seek medical attention Chief complaint may be more than one.
3. **History of present illness:** The chief complaints are elaborated on this part of the history. Deep interviewing is made to solicit organ systems involved. The person in charge of history taking has to be equipped with skill on how to interpret information gathered from client/parents/care givers.
4. **Past medical history:** Ask for history of past major illnesses like Tuberculosis, persistent fever, diarrheal illness etc., if there was any previous hospitalization that has documented. For pediatric age groups ask for maternal health during pregnancy, whether Mother had been enrolled to PMTCT care and PMTCT care provided to the baby.
5. **Nutritional History:** Ask about loss of appetite, weight loss or gain. For children the history should include nutritional history from time of birth till current age. Detailed history about breast feeding; history of mixed feeding, time and type of formula feeding initiated; type of complementary diet and age of start and current diet
6. **Developmental history (In children):** Developmental delay is one of the commonest manifestations in HIV infected infants and children. Ask parents for developmental milestones using developmental check list. For example, ask whether

the child started to show social smile, roll over, sit without support, is able to creep and crawl, stand alone, walk alone, utter words etc. For school age children, ask for school performance and learning difficulties if any.*Use simplified WHO motor skill development assessment checklist for growth and development assessment(chapter 9).

7. **Immunization history:** Document type of vaccine administered. If the child's vaccination is not up to date, ask the reason why and manage the infant/child accordingly.
8. **Family history:** Ask for family size, spouse, and number of children and presence of transmissible disease like tuberculosis. Ask whether HIV test has been done for the family members and whether disclosure has been performed to family member.

b. Physical Examination: After a thorough history undertake a complete physical examination at the first visit and targeted examination can be used during follow up visits. The examiner should perform the examination in a gentle manner. Performing physical examination in children is not an easy task and usually it requires a great deal of patience and time. Just by attentively looking at the child much relevant information pertaining to the child health status can be gathered. For example, the way how the child reacts to the environment, skin color, body status and breathing pattern etc.

1. **General appearance:** check whether client is healthy looking, acutely sick looking, and chronically sick looking.

Check also the functional status for adult client and development for children

Adults functional status	Developmental assessment -Children
W = working, capable of going out of home and do routine activities including the daily work	A=Development appropriate to age
A = Ambulatory, capable of self-care and going to the toilet unsupported	D=Development delay for age
B = Bed-ridden, cannot go even to the toilet unsupported.	R=Normal development milestones lost after the illness (Developmental regression)

Exercise 7.1. Now your facilitator will lead drill on patient functional status.

2. **Vital signs:** Count Pulse rate and respiratory rate. Take measurement of temperature and blood pressure.

Note:

- Blood pressure, pulse and respiratory rate vary with age
- Use appropriate size cuff to measure blood pressure.
- Use the axillary route to measure temperature

Table 7.1. Normal pulse and respiratory rate for different age groups of children

Age	0-3mo	3-6mo	6-12mo	1-3yrs	3-6yrs	6-12yrs	>12yrs
Pulse rate	100-150	90-120	80-120	70-110	65-110	60-95	55-85

Age	0-2mo	2-12mo	12-59mo	>5yrs
Respiratory rate	60	50	40	30

3. **Anthropometric measurements:** Measure height, weight and calculate BMI. For pediatric age group measurement should also include length/height, mid upper arm circumference (MUAC) and head circumference. Findings should be plotted on appropriate growth chart for age and interpreted. For children above two years measure height in standing position. Weight is taken using different scales for age.
4. **HEENT (Head, Ear, Eyes, Nose and Throat):** **Head:** look for whitish scaly or boggy lesions over the scalp **Eyes:** Look for color of sclera, pallor and inflammation of conjunctiva. **Ears:** Look for ear discharge. **Nose:** look for nasal discharge. **Mouth and throat:** look for oral thrush, cracks at angle of the mouth (angular cheilitis). Halitosis (bad odor) in cases of chronic HIV associated lung diseases.
5. **Lymph nodes/glands:** Palpate lymph nodes at cervical, axillary and inguinal areas. Describe size, consistency, tenderness and discharge from lymph nodes. Persistent generalized lymphadenopathy is defined as lymph nodes greater than 1cm in diameter at two or more noncontiguous sites. Look for parotid gland enlargement, as this is one of common features of HIV infection in children.
6. **Respiratory system:** Count respiratory rate. Assess whether the patient is in respiratory distress or not. Look for chest in-drawing in children. Listen for abnormal breathing sounds like wheezes, stridor etc. **Percussion:** check for any sign of consolidation or pleural fluid. **Auscultation:** check for abnormal breathing sounds like bronchial breathing sounds,

wheezes, decreased air entry, or crepitation.

7. **Cardiovascular system:** Look for cyanosis and clubbing of fingers. Cyanosis can be checked on fingertips, lips and tongue. Listen for heart sounds and abnormal heart sounds like murmur
8. **Abdomen:** Abdomen is divided into four quadrants: right upper and lower, left upper and left lower. Look for abdominal distension or ascites. Check for palpable liver, spleen or any other abdominal mass. Infants are best examined while sitting on lap of their mother/care giver.
9. **Limbs:** check for fungal nail infection (onychomycosis), joint movement, swellings, tenderness and hotness at the extremities.
10. **Skin:** many skin disorders are easily recognized by simple observation. The entire body surface, mucus membrane, hair and nail beds should be examined under adequate light. Carefully examine skin lesions and describe their nature: papular, macular, vesicular, pustular, bullous, nodular, crusty, scaly, etc. Look for warty skin lesions.

Primary skin lesions are:

- **Papules:** palpable solid lesions less than 0.5 cm
 - **Nodules:** palpable solid lesions greater than 0.5 cm in Macule: an alteration in skin color but cannot be felt
 - **Vesicles:** raised clear fluid filled lesions less than 0.5 cm in diameter
 - **Pustules:** skin lesions filled with purulent material
 - **Bullae:** fluid filled elevation of the skin more than 0.5 cm in diameter.
11. **Central nervous system (CNS)/developmental assessment:** check Consciousness, Orientation to Time, Place and Person, Meningeal signs, Weakness/paralysis, Signs of anxiety and /or depression.

In children, examination of CNS is crucial part of evaluation of HIV exposed and infected children. In relation to maturation of CNS, normal findings may differ with age of the child. To complete CNS examination cooperation of the child is vitally important. HIV Encephalopathy is one of the common features of HIV infection during infancy and early childhood. Manifestations of HIV encephalopathy include failure to attain or loss of developmental milestones impaired brain growth (microcephaly), motor deficit evidenced

by gait disturbance, muscle power weakness and depressed reflexes. During CNS examination look for the following signs:

- **Motor function test:** Check for muscle tone, muscle bulk, power and gait disturbance.
- **Sensory function test:** check sensation for touch, pain and temperature
- **Signs of meningeal irritation:** check for neck stiffness and kerning signs. In infants look for bulging fontanel.

NB: Level of consciousness in children will be either

- 1) **Alert:** patient is fully awake or
- 2) **Lethargic** or **unconscious:** spontaneous movements are decreased or No response to any stimuli.

c. Performing basic Lab tests:

Basic Lab assessment¹ performed before or after initiation of ART:

- baseline CD4,
- CBC, ALT, creatinine (if available).
- If presumptive TB diagnosis, do Gene Xpert. If no Gene Xpert, do sputum smear.
- Pregnancy and other tests as necessary.
- Cryptococcus antigen if CD4 cell count ≤ 100 cells/mm³ (Refer to Chapter 10.4.2)

Six month after initiation of ART, Perform initial viral load testing to monitor treatment response and continue the routine viral load testing yearly from time of initiation to detect treatment failure.

NB: However, absence of these testes or long Turnaround Time (TAT) of results should not delay ART initiation.

7.2. WHO CLINICAL STAGING

Using the history and physical examination findings, it is important to determine which WHO clinical stage fits to the client. It should be part of the baseline assessment (first visit) on entry into a care and treatment program and is used to guide decisions on when to start co-trimoxazole prophylaxis and monitoring patient response. In addition, there is T-staging which uses the same clinical parameters which are used for monitoring of ARV treatment success or failure after six months of therapy. The detail WHO clinical stages are displayed on table 7.2.

¹ ART initiation should not be delayed for the CD4 test, however do the test to assess the severity of immunosuppression, as baseline for ongoing monitoring and OI prophylaxis.

Definition of Common opportunistic illnesses

- A. **Persistent Generalized Lymphadenopathy (PGL):** a clinical diagnosis after exclusion of OIs and malignancies. It is defined as non-tender, enlarged lymph nodes of greater than 1cm in 2 or more non-contiguous sites (excluding inguinal) and persisting for ≥ 3 months.

Look at photo A.

- B. **Skin problems:** You are probably already familiar with many of these from your clinical work. Look at the photos referred to below. This describes some of the skin problems associated with HIV infections. **Itching rash**— that includes:

Seborrhoea: scaly skin eruption on the border between face and hair and side of the nose or chest. These are often greasy scales and redness.

Look at photo C.

Prurigo: itchy skin eruption on the arms and legs. Often there are small papules and scratch marks. They may leave dark spots with light centers.

Look at photo D.

- C. **Herpes Zoster:** common on the chest (thorax), but also occurs on a leg, arm or the face. The vesicles are in only one area on one side of the body never cross the mid line. There is intense pain. Sometimes the pain continues after the lesions heal. The pain is often a shooting pain post herpetic neuralgia.

The blisters can turn into lesions and later into crusts. They may be haemorrhagic (bloody). Generally, it takes 2-3 weeks before healing. There is often scarring.

Herpes zoster is a sign of clinical stage 2. It can be treated or the active lesions aborted only if treatment is provided within 72 hours after the onset of symptoms and /or signs.

Look at photo E.

- D. **Mouth/throat and oesophagus problems:** The fungal organism called Candida can cause several conditions: sores at the corners of the mouth, oral thrush, or oesophageal infection. The same organism also causes vaginal candidiasis.

Angular cheilitis: Candida can cause small chronic sores or cracks around the lips, often at the corners of the mouth. These are called angular cheilitis. This occurs early in HIV infection, in stage 2.

Look at photo B.

Recurrent mouth ulcers: These ulcers result from variety of causes but most commonly due to recurrent herpetic viral infection and idiopathic (none identified causes)

Look at photo F.

Oral thrush: White patches from candida can be removed. Sometimes they can appear mostly as red patches. Oral thrush is a sign of clinical stage 3.

Look at photo G.

Oesophageal thrush: Patients who have the white patches of oral thrush and who have severe pain on swallowing may have oesophageal thrush.

Oesophageal thrush is a serious infection since it can prevent the patient from eating. Oesophageal thrush also indicates more immune suppression than oral thrush only and is a sign of clinical stage 4.

Oral hairy leukoplakia: Finer white patches on the side of the tongue which cannot be removed are called oral hairy leukoplakia. These can appear as white vertical lines on the lateral surface of the tongue. This is different from candida and does not need to be treated with antifungals. However, oral hairy leukoplakia is a sign of clinical stage 3.

Look at photo H.

E. Pulmonary tuberculosis: Pulmonary tuberculosis considered as stage three when it is diagnosed after confirmed HIV infection

F. HIV wasting syndrome: Patient is extremely thin (lost > 10 % of body weight) with chronic fever and/or chronic diarrhoea.

NB:

1. Always consider significant weight loss when the weight loss is greater than 5%.
2. If the patient has clinical conditions that fall in more than one WHO clinical stage, stage the patient in the highest WHO clinical stage where there is a sign.

G. Mental health problems and substance use: Mental health problem, substance use or other problems could be barriers to adherence/initiation, appropriate support. Providers should screen for and manage mental health problems and substance use.

Table 7.2. WHO clinical staging for children and adults

Adults and adolescents	Children
Clinical Stage 1	
○ Asymptomatic ○ Persistent generalized lymphadenopathy	○ Asymptomatic ○ Persistent generalized lymphadenopathy
Clinical stage 2	
○ Moderate unexplained weight loss (5-10% of presumed or measured body weight) ○ Recurrent upper respiratory tract infections (sinusitis, tonsillitis, otitis media, pharyngitis) ○ Herpes zoster ○ Angular cheilitis ○ Recurrent oral ulceration ○ Papular pruritic eruption ○ Fungal nail infections ○ Seborrhoeic dermatitis	○ Unexplained persistent hepato-splenomegaly ○ Recurrent or chronic upper respiratory tract infections (otitis media, otorrhoea, sinusitis, tonsillitis) ○ Herpes zoster ○ Lineal gingival erythema ○ Recurrent oral ulceration ○ Papular pruritic eruption ○ Fungal nail infections ○ Extensive wart virus infection ○ Extensive molluscum contagiosum ○ Unexplained persistent parotid enlargement
Clinical stage 3	
○ Unexplained severe weight loss (>10% of presumed or measured body weight) ○ Unexplained chronic diarrhoea for longer than 1 month ○ Unexplained persistent fever (intermittent or constant for longer than 1 month) ○ Persistent oral candidiasis ○ Oral hairy leukoplakia ○ Pulmonary tuberculosis ○ Severe bacterial infections (such as pneumonia, empyema, pyomyositis, bone or joint infection, meningitis, bacteraemia) ○ Acute necrotizing ulcerative stomatitis, gingivitis or periodontitis ○ Unexplained anaemia (<8 g/dl), neutropaenia (<0.5 x 10⁹/l) and/or chronic thrombocytopaenia (<50 x 10⁹/L)	○ Unexplained moderate malnutritionnot adequately responding to standard therapy^a ○ Unexplained persistent diarrhoea (≥14 days) ○ Unexplained persistent fever (above 37.5°C, intermittent or constant, for longer than one 1 month) ○ Persistent oral candidiasis (after first 6 weeks of life) ○ Oral hairy leukoplakia ○ Lymph node tuberculosis ○ Pulmonary tuberculosis ○ Severe recurrent bacterial pneumonia ○ Acute necrotizing ulcerative gingivitis or periodontitis ○ Unexplained anemia (<8g/dl), neutropaenia (<0.5 x 10⁹/L) or chronic thrombocytopenia(<50 x 10⁹/L) ○ Symptomatic lymphoid interstitial pneumonitis ○ Chronic HIV-associated lung disease, including bronchiectasis

<i>Stage 4</i>	
○ HIV wasting syndrome ○ Pneumocystis (jirovecii) pneumonia ○ Recurrent severe bacterial pneumonia ○ Chronic herpes simplex infection (orolabial, genital or anorectal of more than 1 month's duration or visceral at any site) ○ Oesophageal candidiasis (or candidiasis of trachea, bronchi or lungs) ○ Extrapulmonary tuberculosis ○ Kaposi sarcoma ○ Cytomegalovirus infection (retinitis or infection of other organs) ○ Central nervous system toxoplasmosis ○ HIV encephalopathy ○ Extrapulmonary cryptococcosis, including meningitis ○ Disseminated non-tuberculous mycobacterial infection ○ Progressive multifocal leukoencephalopathy ○ Chronic cryptosporidiosis ○ Chronic isosporiasis ○ Disseminated mycosis (extra-pulmonary histoplasmosis, coccidioidomycosis) ○ Lymphoma (cerebral or B-cell non-Hodgkin) ○ Symptomatic HIV-associated nephropathy or cardiomyopathy ○ Recurrent septicaemia (including Non-typhoidal Salmonella) ○ Invasive cervical carcinoma ○ Atypical disseminated leishmaniasis	○ Unexplained severe wasting, stunting or severe malnutrition not responding to standard therapy^b ○ Pneumocystis (jirovecii) pneumonia ○ Recurrent severe bacterial infections (such as empyema, pyomyositis, bone or joint infection, meningitis, but excluding pneumonia) ○ Chronic herpes simplex infection (orolabial or cutaneous of more than 1 month's duration or visceral at any site) ○ Oesophageal candidiasis (or candidiasis of trachea, bronchi or lungs) ○ Extrapulmonary tuberculosis ○ Kaposi sarcoma ○ Cytomegalovirus infection (retinitis or infection of other organs with onset at age more than 1 month) ○ Central nervous system toxoplasmosis (after the neonatal period) ○ HIV encephalopathy ○ Extrapulmonary cryptococcosis, including meningitis ○ Disseminated non-tuberculosis myco-bacterial infection ○ Progressive multifocal leuko-encephalopathy ○ Chronic cryptosporidiosis (with diarrhoea) ○ Chronic isosporiasis ○ Disseminated endemic mycosis (extrapulmonary histoplasmosis, coccidioidomycosis, penicilliosis) ○ Cerebral or B-cell non-Hodgkin lymphoma ○ HIV-associated nephropathy or cardiomyopathy

^aDocumented loss of body weight of -2 SD, failure to gain weight on standard management and no other cause identified during investigation

^bConfirmed by documented weight loss of >-3 SD +/- oedema

Table 7.3.CLINICAL REVIEW FORM

1. History	2. Physical Examination (P/E)	6. Assess Family Status
If this is first visit: Review history. Check record for TB, other opportunistic infections, chronic problems. For all visits: • How have you been? • Have you developed any new symptoms or problems? • Have you had any of the following? <i>If yes</i>, evaluate the patient : ○ Cough? ○ Night sweats? ○ Fever? ○ STI signs (For all: genital discharge, sore, dysuria. For men: scrotal swelling, inguinal swelling. For women: Lower abdominal pain) ○ Diarrhoea? ○ Mouth sores? ○ New skin rash? ○ Headache? ○ Fatigue? ○ Nausea or vomiting? ○ Poor appetite? ○ Tingling, numb or painful feet/legs? ○ Any other pain? <i>If yes, assess using WHO stepladder</i> ○ Problems sleeping at night? ○ Sexual problems? • Additionally, in children ask ○ Unable to feed ○ Difficulty of breathing ○ Grunting ○ Leg swelling ○ Convulsion ○ Lethargy/coma ○ Ear pain • Are you sexually active? How many partners do you have? Do you use condoms? • Have you been feeling sad or unhappy or have you lost interest in your normal activities recently? If so assess for depression • Have you been feeling scared or frightened recently? • Do you drink ‘tella,’ ‘tej,’ ‘areke’, beer or other alcohol beverage? How many days of a week do you drink? How much do you drink? • Have you needed urgent medical care? If yes, ask for record/diagnosis. • Which drugs (anti TB, traditional, ARVs, illicit drugs, etc) are you taking and how often? • Assess adherence • How are things at home? • What usual physical activities are you doing? • Is there anything else you want to talk about?	In all patients: · Look for pallor. <i>If pallor</i>, check for hemoglobin. · Look for jaundice · Look for thrush. · Weigh. Calculate weight gain, or loss. If weight loss, ask about food intake. If any new symptoms: · Measure vital signs. · Measure weight, height, head circumference and MUAC · Check for nodes. If >2 cm, · Look for skin rash. · Look for evidence of violence. · Do further assessment of symptoms If first visit (also check every 6 months; skip if known problem): ○ Tell patient you want to check his memory. ○ Name 3 unrelated objects, clearly and slowly. Ask patient to repeat them: ○ Can he or she repeat them? (Registration problem?) <i>If yes</i>, wait 5 minutes and ask again, "Can you recall the 3 objects?" (Recall problem?)	○ For women: ○ Ask if she is pregnant, ask LMP ○ If she is not pregnant, ask for pregnancy intention ○ Assess for FP need ○ For men: ○ Ask about pregnancy status of their partner ○ Ask for intention to have a baby ○ Have you disclosed your HIV status to your partner or someone significant? ○ Is your partner tested for HIV? ○ Are your children tested for HIV?
		7. Review TB status
		○ Active TB ○ Newly diagnosed ○ On anti TB ○ Positive TB screen ○ TB suspected on previous visit, check for sputum result and respond as per the TB guideline ○ Order sputum examination ○ Negative TB screen ○ Give IPT
		8. Initiate ART
		○ Do adherence preparation, Determine readiness, ○ Initiate ART as soon as possible including same day ART
		9. Adherence Counseling
		○ Conduct ongoing adherence counseling
		10. PHDP
		○ Give client condoms at every visit ○ Agree on a prevention goal for the client
	3. Laboratory	
	Request relevant laboratory tests as baseline, But this should not delay ART initiation	
	4.WHO Clinical Staging	
	Determine the WHO Clinical Stage of your client	
	5.Functional Status /Developmental milestone	
	Determine the functional status for adult and Developmental appropriateness for children	

Exercise 7.2. Do photo booklet exercise.

EXERCISE 7.3. Write down the clinical stage of the cases described in the questions.

1. An extremely thin HIV+ patient has chronic fever for 3 months. _____
2. An HIV+ patient with pulmonary tuberculosis and purple lesions on the skin of the left leg, with oedema of the leg. _____
3. An HIV+ patient with oral thrush and diarrhoea for 1 month. _____
4. An HIV+ patient with tuberculosis of the cervical lymph nodes. _____
5. An HIV+ patient with recurrent big abscesses of the skin which extend to the muscle, with some yellow pus coming out of some of them. _____

Chapter summary

- Comprehensive clinical assessment at the beginning and each follow up visit is critical for better treatment outcome.
- Lab testing should not be a reason to delay initiation of therapy.
- Using the clinical review form will help for proper assessment at any visit.

Chapter 8: Assessment of Growth and Development

Chapter Description:

This chapter describes the approaches of assessing the growth and development of HIV exposed infants and children infected by HIV. The standards of measurements and interpretation of results are also detailed in this section.

Purpose: To learn about assessment of growth and development and use this to monitor disease progression in HIV exposed and infected infants and children

Enabling objectives:

At the end of the chapter, participants will be able to:

- Interpret measured weight, height and head circumference,
- Sketch on age and gender appropriate standard growth chart over time and interpret growth patterns
- Assess development using history, observation and developmental check lists and recognize developmental abnormalities
- Describe how HIV infection affects growth and development

Chapter outline:

- Assessment of growth
- Assessment of development

Allocated time: 60 minutes

Learning activities

- Reading
- Discussion,
- Demonstration,
- Exercise,
- Case study

Introduction

Malnutrition and delay in acquisition of developmental milestones are more common in HIV infected children. HIV exposed and HIV infected children need frequent monitoring of growth and development, in order to detect growth faltering and developmental abnormalities early enough before severe manifestations occur. The occurrence of growth and developmental abnormality may be an indication of HIV infection in HIV exposed infants or disease progression in those who are HIV infected and possibly treatment failure in those who are on HAART.

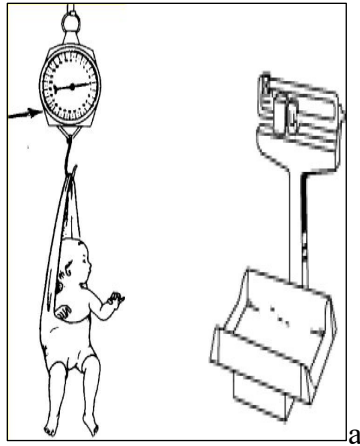
8.1. Assessment of Growth

8.1.1. Growth

Growth is an increase in size example increase in weight and height. HIV has severe effect on growth of infected child. **Growth abnormalities** are common in HIV infected infants and children and occur throughout the course of the disease. It can be a primary manifestation of HIV disease in exposed infants, or can be secondary to opportunistic infections or complication of HIV. May indicate treatment failure in children on ART or disease progression in HIV infected children.

Components of growth monitoring are measuring the Weight, Height/length and Head circumference of HIV infected child.

8.1.2. Measuring weight



Steps for taking the weight of a child/adolescent using electronic scales:

- Place the electronic scale on a flat, level surface.
- Check and readjust the weight reading to zero.
- Undress the child.
- Make him/her stand on the middle of the scale's surface.
- When the child is settled and the weight reading is stable record the weight to the nearest 100g (0.1). Make sure that nobody holds the child during weighing and that the child stands freely without holding anything.

Read and announce the value from the scale. The assistant should repeat the value for verification and record it immediately.



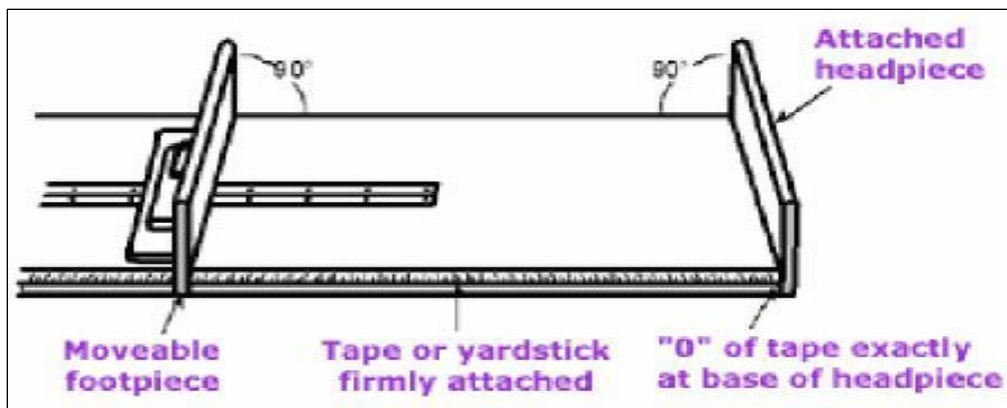
Figure 8.1. Pictures of commonly used weighing scales for children(a-d)

To enhance accuracy of weight measurement:

- Use same scale at each visit
- Scales should be zeroed daily and calibrated weekly
- Infant scales should be used for children <20kg
- Remove all clothing including diaper
- Weigh infants supine and older infants sitting
- Record weight to the nearest 0.1kg and Record measurement immediately

8.1.3. Measuring length/height:

- For children 0-2 years of age measure length
- Remove shoes when you measure length/height and use two people
- Keep the knees straight and ankles in neutral position when you measure length
- For children ≥ 2 years of age measure height
- When you measure length, make sure the legs are close to each other and the heels, the buttock and the back of the head touches the wall or are in straight line. The child should look straight parallel to the floor.
- Place a ruler or a hard paper on top of the head to perpendicular to the wall to take the measurement. If there is large hair press gently
- Record measurement of length/height to the nearest 0.5 cm

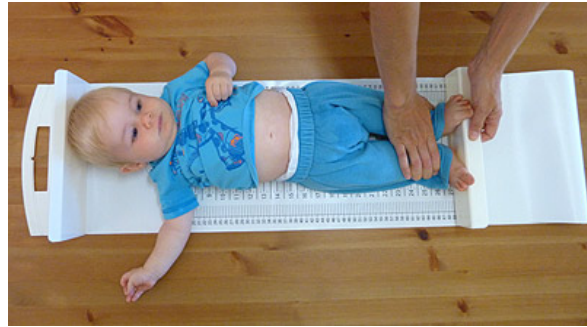


surement immediately

a.



b.



c.

Figure 8.2. Pictures showing length board and length measurement

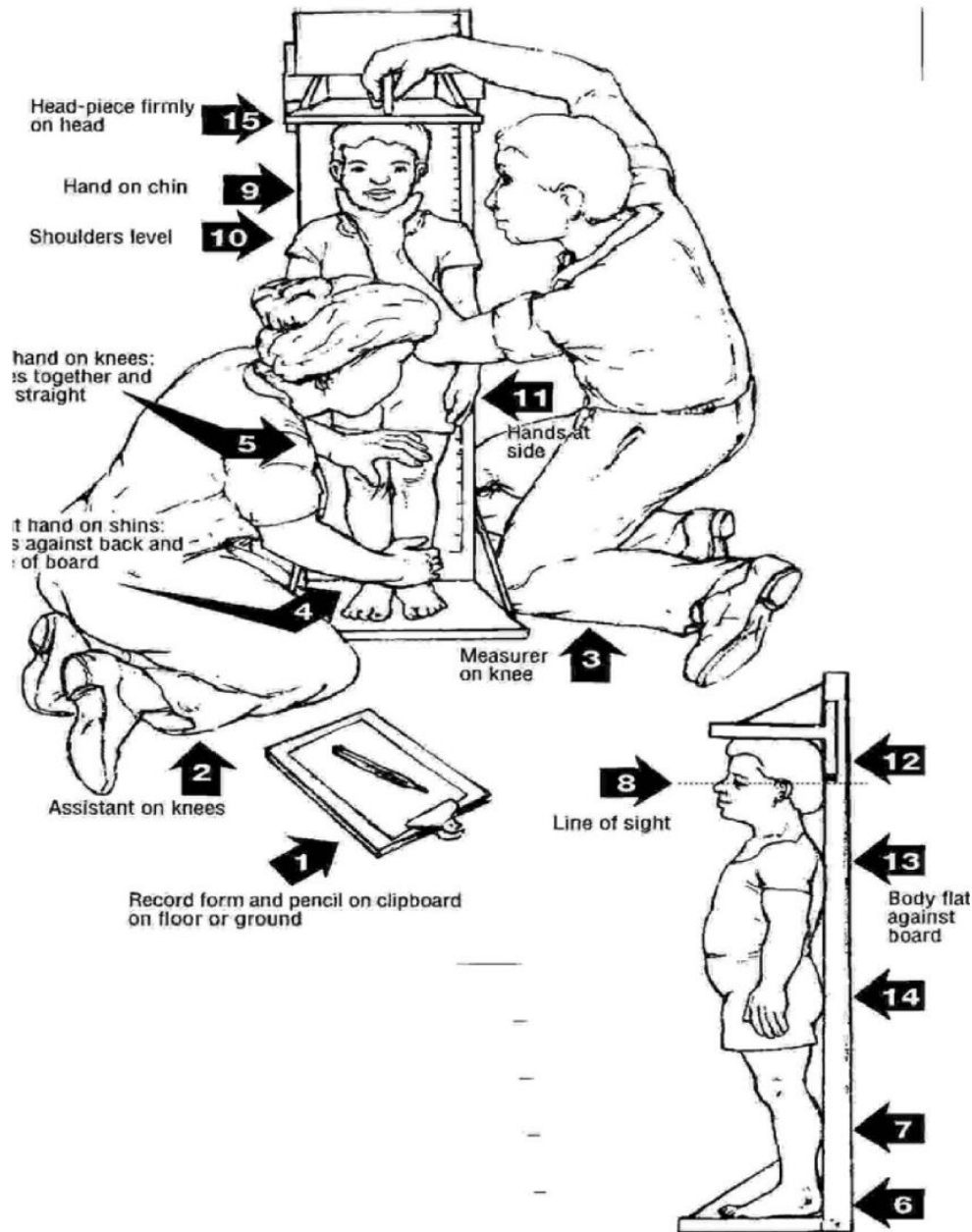


Figure 8.3. Picture demonstrating measurement of height in a child ≥ 2 years of age

8.1.4. Measuring head circumference: Measuring head circumference is an indirect way of measuring brain growth. HIV affects brain growth directly and is an important cause of acquired microcephaly in children.

How to measure head circumference?

- Use a non-stretchable plasticized tape

- Place tape at the mid forehead and extend circumferentially to include most prominent portion of occiput to obtain the greatest volume
- Record measurement to the nearest 0.5 cm
- Record measurement immediately

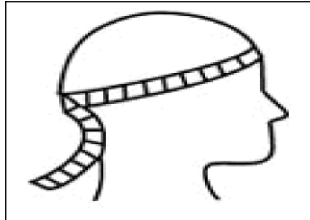


Figure 8.4. Picture showing measurement of head circumference

Measure the head circumference of children up to 36 months of age (3 years).

Exercise:

Practice measuring head circumference among yourselves after demonstration by the facilitator.

8.1.5. Growth chart/curve

A growth chart or curve is a graph that records changes in the child's growth over time compared to normative growth rates. Growth Chart data is collected from large numbers of children in a particular population. It is a normative data on weight, height and head circumference disaggregated by age and sex. Growth curves are most applicable in similar populations but there could be minor differences by race, geography and mode of infant feeding. Growth chart is an easy and systematic way to follow **CHANGES IN GROWTH OVER TIME** for an individual child.

a. Types of Growth charts:

There are a number of growth charts that are used to follow growth of children. We use the WHO growth charts with z-score in ART care and treatment service. The **WHO children Growth Charts** are prepared based on data collected from children in Brazil, Norway, Ghana, India, Oman, and the USA. The data collected from these countries demonstrated that children worldwide have the potential to develop to within the same range of height and weight independent of race or ethnicity if given an optimum start.

The growth indicators include weight for age, height/length for age, weight for height/length, head circumference for age and body mass index (BMI) categorized for boys and girls and with different age bands.

The commonly used WHO growth charts are:

- Weight for age z-score: boys and girls, birth to 5 years,
- Height/length for age z- score: boys and girls, birth to 5 years,
- Weight for length z-score: boys and girls, birth to 5 years,
- Head circumference for age z-score: boys and girls, birth to 5 years,
- BMI for age: boys and girls

The WHO z-score growth charts for Boys and Girls are annexed (See annex I)

The other commonly use growth chart for children and adolescents are the CDC Growth Curves. The CDC growth curves extend from birth to 20 years being disaggregated by age and sex. The following CDC growth curves are available:

- Weight for age: boys and girls,
- Height/length for age: boys and girls
- Head circumference for age: boys and girls
- Weight for height/length: boys and girls
- BMI for age: boys and girls

The choice among growth curves in monitoring the growth pattern of children does not have significant effect but as the WHO curve is prepared based on data from different developed and developing countries; therefore, it is the WHO z- score growth chart that we use in HIV chronic care.

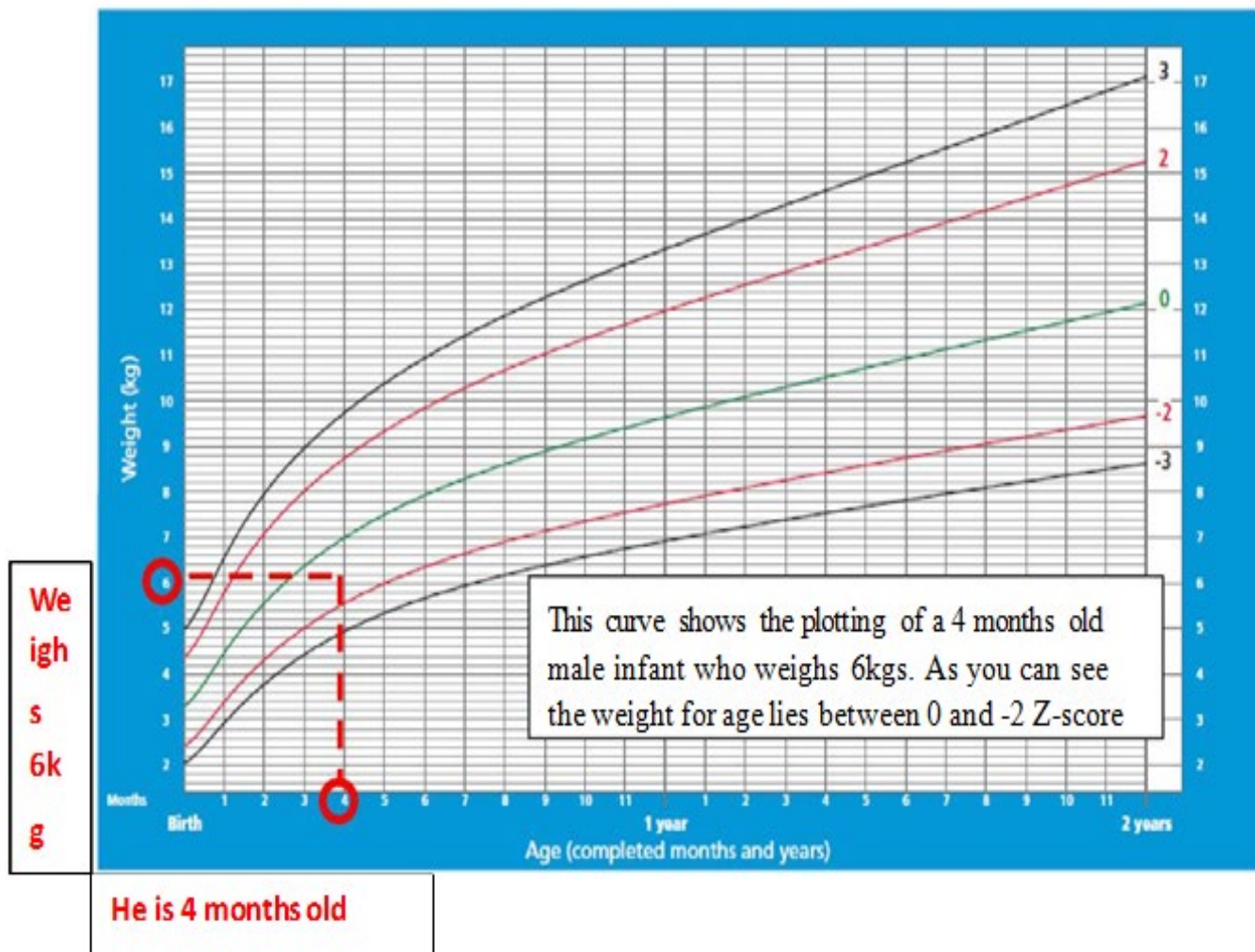
b. Plotting Anthropometric measurements on growth charts

- Measure and weigh child using same methodology at each visit
- Using age and sex appropriate charts, plot measurement on the vertical axis against age on the horizontal axis.
- Compare growth point with previous points

See examples below on how to plot anthropometric measurements on growth charts.

Weight-for-age BOYS

Birth to 2 years (z-scores)



Head circumference-for-age GIRLS

Birth to 5 years (z-scores)

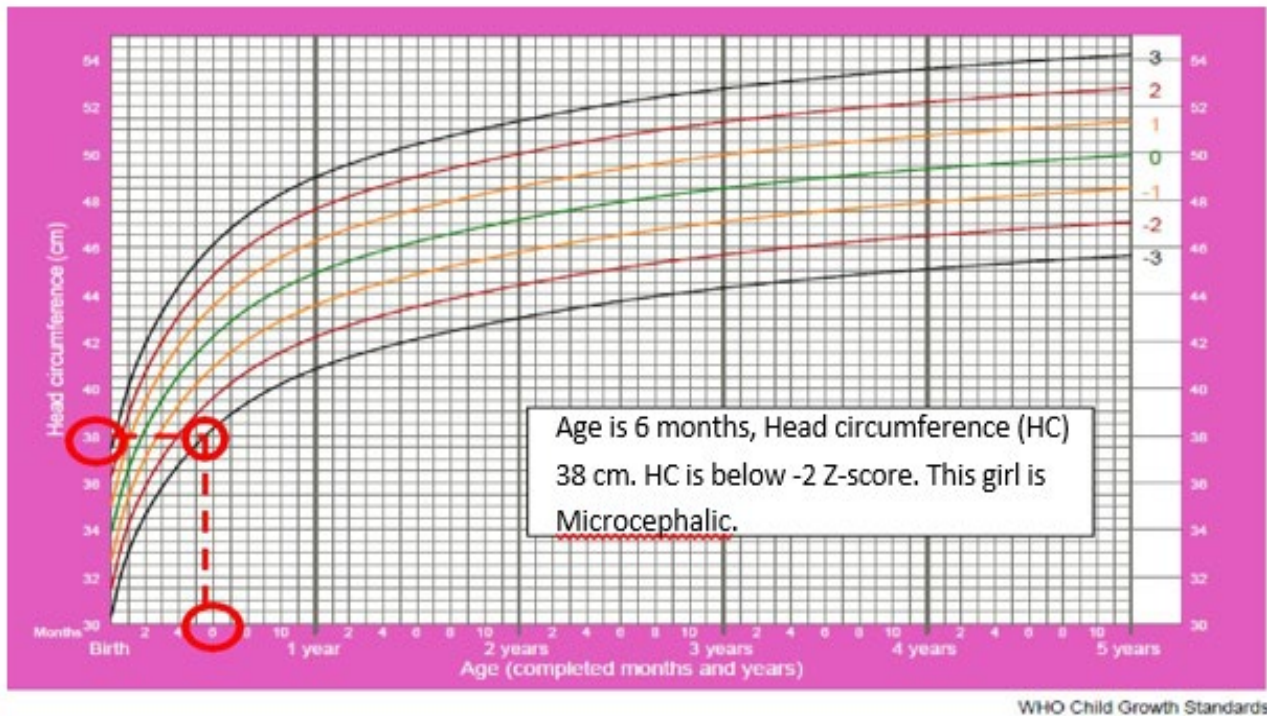


Figure 8.5. Interpretation of plotting on Growth curves

c. What are Z-scores?

Z-score speaks to the general population and addresses the question - what percentages of the population are of the same weight and height?

- If a child is on the 0 Z-score for weight for age, it means that 50% of children of the same age are heavier than this child and 50% weigh less than the child.
- If a child is on the -2 Z-score, the growth chart tells you that the child is small relative to other children of the same age - 95% of children of the same age are heavier,
- If someone is on the +2Z-score, they are generally a little bit fat. They weigh more than 95% of children their age.

Exercise:

Plot the following weight for age WHO growth chart: Birth Weight- 2.2 kg, 1 mo- 3kg, 2 mo-3.8kg, 3mo-4.6kg, 4mo-5.2kg and interpret the growth pattern for both sexes.

8.1.6. Growth velocity:

Looks at the rate (speed) of growth over time. Children are expected to gain a certain minimum amount of weight at certain time of their age:

0-3 months:	25-30gm/day
3-6 months:	20gm/day
6-12 months:	12gm/day
12-18 months:	8gm/day

Growth velocity is higher in the first year of life and peaks again during puberty. Children who are small may have normal growth velocity and follow along their own curve.

8.1.7. Classifying the child's growth

Children who are well and healthy should gain weight and length/height. Growth curves record the weight of children over time. Children who are growing normally follow a growth curve parallel to one of the standard growth curves. Weight loss or failure to gain weight can be identified by observing the child's weight over time. When weight —falters or the growth curve —flattens and is no longer parallel to the chart line this indicates the need for clinical assessment, management and nutritional intervention and possibly ART.

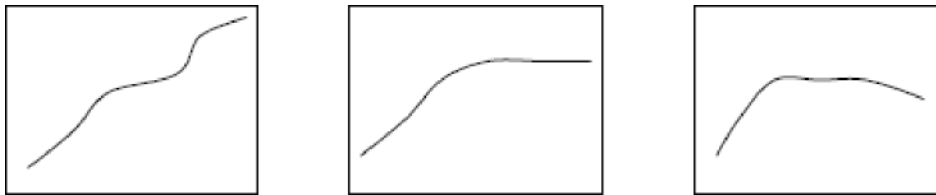
In children 6-60 months, **severe malnutrition** is marked by the presence of any of the following:

- weight-for-height is below **-3** z-score of the WHO growth standards, **or**
- there are signs of severe visible **wasting**, **or**
- **edema** of both feet is present, **or**
- MUAC is less than **110 mm** in children up to 60 months (5yrs) of age.

A child will be classified as having **poor weight gain** if any of the following are present:

- Reported weight loss, or
- Very low weight (weight-for-age less than -3 z-score), or
- Underweight (weight-for-age less than -2 z-score), or
- Confirmed weight loss (>5%) since the last visit, or
- Growth curve flattening

Examples of growing well, curve flattening (crossing growth lines) and losing weight (using weight-for-age charts)



If a child has poor growth (severe malnutrition or poor weight gain), the nutritional history should be more extensive than the one taken at a routine clinical visit including:

- Symptoms that might **impair** nutritional status (pain, oral lesions, GI losses, acute illness)
- Has **access to**/availability of food changed recently? Change in family income?
- Are there changes in **caregiver** or home circumstances?
- Has **mother noticed** that something is wrong? What does she think is the cause?
- The previous **24-hour diet recall**

Detail Physical examination like:

- Look for **oral lesions**
- Look for signs of **vitamin** or **micronutrient deficiency**
- **Skin changes** suggestive of severe acute malnutrition
- Localizing signs of **infection**
- Evidence of **HIV infection**- hepato-splenomegaly, parotid enlargement, and thrush etc.

Using the assessment and Determine etiology of failure-to-thrive:

- Lack of availability of food in the household or lack of knowledge
- HIV disease and opportunistic infection (these conditions may lead to inadequate intake, excessive loss or increase in metabolic rate).

Finally, develop a treatment plan as determined by available resources and Ethiopian guide to clinical nutrition care for children and adults with HIV:

- Treatment of acute illness

- Provision of food, referral to local resources
- Nutrition counseling
- Recommend HIV testing if infant/child is not tested previously.

8.2. Assessment of Development

Development: pertains to acquisition of various functions and skills with a gradual change from simple to complex. These spheres of development include language, psychosocial, Fine and gross motor skills and cognition.

Delayed development or loss of milestones may be the first sign of HIV infection in an infant or young child:

- While other causes are possible, abnormal development should raise concerns of HIV infection
- Infants are at high risk for HIV encephalopathy and severe neurologic disease
- The assessment should include Cognitive, Motor, Language, and Social skills

You can assess **Development** by:

- **Asking** Parental reports on milestones achieved,
- **Observing** child's activity/ movement- a 12-month old walks into the examining room, 6-month old sitting upright in mother's arms and
- **Perform Examination** like Neurologic exam, hearing and vision screen and Looking for Warning signs/red Flags- early warning signs

Table 8.1. Normal Development Milestones

Age	Milestones
1 month	Turns head when prone, responds to sound, follows moving object
2 months	Raises head slightly further when prone and head sustained in plane of body on ventral suspension, follows moving object 180 degrees, smiles socially, listen to voice and coos
3 months	Lifts head and chest when prone; head above plane of body on ventral suspension; sustained social contact; listen to music; says “aah,ngah”
4 months	No head lag on pull to sitting position; head steady; reaches and grasps objects and bring them to mouth, laughs out loud,
5 months	Rolls over from stomach to back; imitates speech sounds e.g. makes razzing sounds, reacts differently to strangers
6 months	Sits unsupported briefly, transfers objects from one hand to the other, babbles, reaches for familiar persons
9 months	Pulls to stand, uses finger and thumb to pick up small objects, waves bye-bye, says mama
12 months	Walks with one hand held; imitates actions; releases object to other person on request, a few words besides “mama,” “dada”
15 months	Walks alone; crawls upstairs; makes a tower of 3 cubes; follow simple commands; may name a familiar object
18 months	Runs stiffly; walks up stair with one hand held; imitates scribbling; feeds self; uses 10 words; identifies one or more parts of the body
2 years	Runs well; walks up and down stairs; circular scribbling; put three words together; often tells immediate experience
3 years	Rides tricycle; stands momentarily on one foot; copies a circle; imitate a cross; knows age and sex; helps in dressing; washes hands; speech is understandable
4 years	Hops on one foot; climbs well; draws a person; tells a story; plays with several children
5 years	Skips alternating feet; dresses and undress; names 4 colors; jumps over low obstacles; ask questions about meaning of words
Source: Nelson Textbook of Pediatrics 17th edition 2004	

Table 8.2. Abnormal Development that show delayed acquisition of milestones

AGE	WARNING SIGNS/Red flags
1 month	Does not regard face, no eye contact, , poor suck, floppy
2 months	Does not look at you with both eyes at least for a few moments, and does not follow with eyes if you move your face slowly from side to side
3 months	Does not respond to sound by quieting, no smile
4 months	Does not hold head steady for a few moments when you sit him up,
5 months	Does not raise head and support weight on arms when in prone position
6 months	Cannot reach for objects with both hands, Floppy, no response to sound, Poor social response to people
9 months	Unable to sit unsupported ,
12 months	Unable to bear weight on legs
15 months	Does not walk alone, is not using at least one word meaningfully
18 months	Does not use at least 3 words, and does not point to what he wants

Case Study1:

Discuss the following case in plenary by actively involving participants through questions:

Betty is a 3 months old HIV-exposed infant who is being exclusively breastfed. She is currently on co-trimoxazole prophylactic therapy and has been followed in the MCH clinic with her mother. Her mother is concerned she is not getting enough breast milk and wants to supplement with infant formula and porridge, she was 2.5kg at birth and currently weighs 4.4kg.

Plot the following on appropriate growth chart: Birth Wt: 2.5 kg, 6 weeks: 3.4 Kg, 3 mo: 4.4 kg.

Q1. What do you want to know?

Q2. What will you tell the mother?

Chapter summary:

- Growth and developmental monitoring is a cheap and effective way to identify children who require close follow-up.
- At each visit weigh, measure length/height, head circumference and plot the measurements on growth curve to monitor growth pattern and identify any growth abnormality.
- Assess development by asking parents, observing and examining infant
- Growth faltering or failure-to-thrive requires prompt work-up and evaluation of HIV status
- Delayed or loss of developmental milestones needs prompt evaluation of HIV status

Chapter 9: Introduction to HIV Care/ART Intake and Follow Up forms

Chapter Description:

This chapter describes the national HIV care recording and reporting formats. The national HIV care/ ART intake and follow up forms are also detailed in this section.

Purpose: Learn how to fill the National HIV Care/ART Patient Intake and Follow Up Forms.

Enabling objectives:

By the end of this chapter, participants will be able to:

- Explain how the intake form is used
- Explain how the HIV Care/ART follow-up form is used
- Fill out the HIV care/ART follow-up form

Chapter outline:

- Intake form
- Follow up form

Allocated time: 160 minutes

Learning activities

- Reading
- Discussion,
- Demonstration,
- Exercise,

Introduction:

Effective chronic HIV care including ART requires keeping track of what happened on previous visits. Any member of the clinical team who sees the patient needs to be able to know key clinical details, and what education and support the patient has been given on the previous visits, in order to know what to do on this visit. In this chapter, you will learn to use a treatment follow-up form that should stay in the clinic, for use on each patient visit. This form is called **the HIV Care/ART Follow-Up form**. It is designed to be used from the time that the HIV-positive patient registers for chronic HIV care. It is not only for ART client.

The advantage of the HIV Care/ART form is that a summary of the patient's chronic HIV care and ART visits is kept in one place. It helps the clinical team keep track of the individual patient and all patients receiving chronic HIV care. This will be important as the number of these patients increase.

You will also learn to fill out a register and report forms in detail covered in another section. Your facilitator will give you copies of the HIV care/ ART follow-up form to use during exercises.

A general principle of good chronic care:

Use written information—registers, treatment plan and treatment cards—to document, monitor and remind.

When an HIV-positive patient decides to have ongoing care in your clinic, fill out the card for the new patient. This is called enrolling in chronic HIV care. This does not happen automatically when the patient receives a positive HIV test result. Patients need to understand what is involved in HIV care and want to be cared for on an ongoing basis (with follow-up appointments). This is the first step in forming a partnership with the patient.

The top part of the follow up card contains the information that must be filled out on the first visit. On that visit, you will also fill out the first row.

Other information on the HIV card will be added on subsequent visits. Note that you need to check the written documentation of the HIV test before filling in the results. If only verbal results are given, the patient will need to have another rapid test performed at your facility.

9.1. ART Clinic Patient Record (Intake Form)

It is a checklist which captures the basic information necessary to provide chronic HIV care/ART. It captures the basic demographic data, contact person, previous treatment, current status & WHO staging at enrollment (history and physical examination), treatment plan, readiness assessment and adherence counseling.

The intake form has two sections:

- Form A- Socio demographics & family care and

- Form B- Past presenting illness & PHDP section.

On top of each form the basic demographic data should be documented.

The form is completed by the Multidisciplinary HIV/AIDS Team (MDT) working at the ART clinic i.e. the Medical Doctor /Health Officer/Nurse, the adherence/triage nurse and the data clerk.

You will be more familiarized with the intake form when you do exercise at the Monitoring and Evaluation training day. Your facilitator will show you the latest national HIV care/ART patient intake form.

How to fill the HIV care/ART patient intake form?

1. Write client's name in full, age and sex. If the client is a child, write mother's name, place, mode and date of delivery,
2. Write date HIV confirmed (write the re-testing date here) and type of test,
3. Write referral information,
4. Write CARE GIVER or EMERGENCY CONTACT INFORMATION (make sure client understand the purpose and give true address)
5. Write the DISCLOSURE status at Enrollment
6. Write list of Family members (Spouse and Children (<15 years old) and document the HIV Status at enrollment and update on follow up visits until status is determined and positives start ART. If the client is child, write list of parents and siblings (brothers and sisters under 15 years of age)
7. On the second section of the intake form, mark and past Opportunistic illnesses (OIs), Prophylaxis and treatments used and write the dates that apply for treatment or prophylaxis.
8. Write Client STATUS AT ENROLLMENT by marking boxes appropriate to client and dates that apply.
9. Write Presenting symptoms, physical examination findings by marking or/ and filling the symptoms/ spaced provided.
10. Based on the clinical symptoms and signs, write the WHO stage of the client on the space provided.
11. Complete the columns for issues under positive health dignity and prevention table.

9.2. HIV Care/ ART Follow-Up form

How to fill the HIV care/ART Follow Up form?

1. On the top box, write the patient's card number, Unique ART number, Name, Age and sex of the patient, and the complete address. The address includes region, Woreda/sub-city, Kebele, house number and telephone number (ensure correct number)
2. Write the dates of HIV test (date confirmed by retest) and client ready and mark type of HIV test, and height in cm.
3. On the clinical follow up table, Write the date of clinical follow-up visit and indicate if the specific visit is scheduled or not,
4. Write the duration of current treatment counting since start date of ART. Put the number of months. When ART is first started/ dispensed, write "0" in this column.
5. Record the patient's weight in kilograms (kg). Also check for bilateral edema and mark +/_ , Before ART is started: If weight loss compared to prior known weight, calculate % of weight loss as follows

$$\% \text{ weight loss} = \frac{\text{old weight} - \text{new weight}}{\text{old weight}}$$

When patients are on ART: Put weight at start of ART on the summary sheet. In patients who have lost weight before starting ART, it is useful to follow weight gain in the first year to see their response to ART. After starting ART, use the weight on the day ART was started to compare the current weight.

$$\% \text{ weight gain on ART} = \frac{\text{new weight} - \text{weight when ART started}}{\text{weight when ART started}}$$

Most patients should gain weight gradually over the first 6-12 months that they are on ART.

Write BMI result for adults and MUAC for pregnant or lactating women.

$$BMI = \frac{\text{weight in kilogram}}{(\text{height in meter})^2}$$

6. For women of childbearing age, write code for family planning status. **Ask at each visit:** *Are you pregnant now, or do you think you might be?* If the patient is pregnant, write **P** for

pregnancy. Childbearing age is 15 years to 49 years; it is important to ask these women about pregnancy in each visit. If pregnant, also write the estimated due date (EDD).

For all adolescent or adult patients, ask about family planning at each visit. Record FP or no FP. If on family planning, record the methods. You may use the codes provided below.

Ask both men and women about current family planning at each visit.

1= Condoms

2= Oral contraceptive pills

3= Inject able (e.g. Depo-Provera)

4=Implantable hormones

5= Intrauterine device

6=Vasectomy/tubal ligation/hysterectomy

7 =Abstinence (No sex)

For children, use this column to record Height and head circumference (if < 3years).

7. Record the code for **functional status** best applies to adult patient (W=Working, A=Ambulatory, or B=Bedridden) and **Developmental** status for children (**A=appropriate, D= Delay, R= Regression**).
8. Write the **WHO clinical Stage 1, 2, 3 or 4** (Use **T staging** for clients already on treatment for more than 6 months)
9. **Record the result of SCREEN FOR TB AT EVERY VISIT.**

A) TB SCREENING:

1. *current cough*
2. *fever*
3. *night sweats (for adults)*
4. *weight loss or (poor weight gain for children)*
5. *contact history with a TB case (for children)*

*Under screening column, Write **P (Positive Screen)** for YES to any one of the above questions OR **N** for Negative screen. A client with P result shall be Evaluate for TB*

- B)** Under *Xpert MTB/RIF (Gene Xpert)* and *LF_LAM* column, Write one (1) if specimen for GeneXpert is sent and write 2 if specimen for LF_LAM is sent and write 1 and 2 if specimen for both genexpert and LF_LAM are sent. Then write in the result column the codes for the result (**P=positive OR N= negative**) (e.g. **If the result for LF_LAM is positive and result for Gene Xpert is negative write as 2P,1N**)

- C) TB TREATMENT/PROPHYLAXIS,** under this column write the appropriate code:

6H-1-6=Currently on 6H prophylaxis

3HP-1-3=Currently on 3HP prophylaxis

(Specify month on treatment e.g. **6H4= 4th month on 6H prophylaxis and 3HP2=2nd month of 3HP prophylaxis**)

On completion of prophylaxis=6H C if on 6H or 3HPC if on 3HP prophylaxis

Discontinued for any reason=6H DC if discontinued 6H or 3HP DC if discontinued 3HP

Under the TPT Dispense column, write the number of doses dispensed **TB Rx=**
Currently on DOTS (Specify month on treatment e.g. TB Rx4 as 4th month on DOTS)

....On completion of DOTS=TB Rx C

....discontinued for any reason=TB RX DC

10. Cervical Cancer Screening: Use codes 0-3 for Type of screening visit (0-3)- 1-First time screening 2-Re-screening from previous negative 3-Post treatment follow up

Use codes (0-6) for Type of Cx Ca Screening **service status:** -

0. Cx Ca screening not done

1. HPV Negative

2. HPV Positive

3. VIA Negative

4. VIA Positive: eligible for Cryo

5. VIA Positive: Non-Eligible for Cryo

6. Suspicious for Cx Ca

Use codes 0-4 for management of cervical lesions:

0. No treatment / further evaluation & management service given

1. Cryotherapy

2. Thermal ablation treatment

3. LEEP service

4. Further evaluation & management service given to Suspicious for Cx Ca

Use codes 0-4 for **Referral for Cx Ca screening & management**

0. Not referred

1. Referred for Cx Ca screening

2. Referred for Cryotherapy /Thermal ablation Tx

3. Referred for LEEP service

4. Further evaluation & management suspicious cases

11. New OIs: Use the **Codes for new OIs** or write the whole word to record an OI that the patient currently has.

NOI=No OI or Opportunistic cancer

Z = Zoster

BP= Bacterial Pneumonia

PTB= Pulmonary TB

ETB= Extra-Pulmonary TB

TO=Thrushoral

EC=Oesophageal candidiasis

UM=ulcersmouth

DC or **DA** = Diarrhea Chronic/Acute

PCP= Pneumocystis Carini Pneumonia

CT= CNS Toxoplasmosis

CM= Cryptococcal Meningitis

NHL=NonHodgkins Lymphoma

KS=Kaposi' Sarcoma

CCa=Cervical Cancer

O=Other

12. Pain assessment: write the code appropriate for pain management if provided,

NP= No pain

S1= WHO Step 1

S2= WHO step 2

S3= WHO step 3

13. **CD4 result:** under this column enter the CD4 value in number for adults and in percentage for children under 5 years of age. On the date CD4 is requested, put (*) mark.

$$CD4 \text{ percentage} = \frac{\text{Absolute CD4 count}}{(\text{total WBC count} \times \text{Percentage lymphocyte})}$$

14. **Viral Load result:** Write viral load when investigation ordered and enter the amount of copies per ML when the result is available.
15. **Document any other labs.** Enter the value of any lab results here,
16. **Assess adherence and record dispensed dose of co-trimoxazole:** Enter G (Good), F (fair) or P (poor) for adherence of the client for daily cotrimoxazole. Later in the course, you will learn how to assess adherence and record it using percentages. Record in the *Dose* column the number of doses dispensed that visit.
17. **Fluconazole:** if the patient is on treatment or prophylaxis for cryptococcal meningitis, write the dose dispensed.
18. **Other medications dispensed, nutritional supplements.** If the patient is taking medicine other than ART, co-trimoxazole, Fluconazole or INH, list the names, doses, and frequency under this column.
19. **Assessment for ARV adherence:** Either record the numeric percentage or describe adherence as Good (equal to or greater than 95% or ≤ 3 doses missed per month), Fair (85-94% or 4-8 doses missed per month), or Poor (less than 85% or ≥ 9 doses missed per month). Write in 'Why' column, the code of reasons for fair or poor adherence as listed below.
- 1 = Toxicity/side effects
 - 2 = Share with others
 - 3 = Forgot
 - 4 = Felt better
 - 5 = Too ill
 - 6 = Stigma, disclosure or privacy issues
 - 7 = Drug stock out–dispensary
 - 8 = Patient lost/ran out of pills

9 = Delivery/ travel problems

10 = Inability to pay

11 = Alcohol

12 = Depression

13 = Other

- 20. ARV drugs: record the dose of ARV drugs dispensed, side effect and reason for change if any.** If there is a treatment interruption or change (ART is stopped, changed or the patient is lost) record this too. Under the *Dispense* column, write the number of doses dispensed and the regimen code. For patients who have stopped ART or are lost to follow-up, write whether Stopped or Lost, record the date and the reason code for why stopped. If ART is restarted, record the date of Restart.

If side effects occur, fill the codes accordingly. If decisions are made to Substitute first line ARVs or Switch to second-line, or third line; record the code of reasons "why" the regimen is changed under the respective column. Codes for why substitute or switch are:

1= Toxicity/side effects

2= Due to new TB

3= New drug available

4= Drug out of stock

5= Clinical failure

6= Immunological failure

7= Virological failure

8= Other reason (specify)

If the patient has transferred in from somewhere else but was on ART and you are continuing the ART, it is important to write the date of transfer, from where the patient transferred in and when ART was started.

- 21. Document Client set HIV prevention plan** for adult: at every visit document the code discussed or attained. STI screening is routine. Partner and children testing is routinely checked until all know their status.

D= Agreed to disclose to partner/family/friend

PT= Agreed to bring partner for testing

ChT= Agreed to bring children for testing

SSex= discussed and agreed to practice safer sex.

SubU= Decides to avoid or decrease substance use

ASS= Assesed for STI

SRX= Client managed for STI

For children, document codes for **Paediatric HIV disclosure status** under this column:

DS0= No disclosure started,

DS1=Stage1, partial, about the illness, medicine, keeping healthy,

DS2=Stage2, about germs, body soldiers

DS3=Stage3, use of terms like CD4, Viral Load, HIV,

22. Record the date for the next follow-up appointment.

Record the date of next appointment to return for clinical monitoring, ARV re-supply or any other reason. In addition, this date should be written down on appointment card for the patient to take home, as well as in a facility appointment logbook to facilitate retention with follow-up.

NB. It is good practice to mark the DSD model the client is currently enrolled in front of each appointment date, like- ASM, FTAR, CAG, PCAD, AHD, DSD for Adolescent, Specify other DSD models if any etc.

Now your facilitator will give you the latest copy of the National HIV Care/ART Follow up Form which you will be practicing about in almost all practical sessions of the training.

EXERCISE 9.1

For this exercise, use a blank HIV/ART follow up form and fill it in accordance with the case.

Part 1: Today's date is 22/6/09. Patient's name is Aster Abebe Bekele. Who was registered at the facility Millennium Health center with MRN of 014195. Aster is a 35 years-old married woman. She comes to the clinic for the first time. She shows you written confirmation of her rapid HIV test that indicates she has HIV and was tested on 12/04/09 at a stand-alone VCT center. The date she had her result, the counselor at VCT center recommended her to go to a health centre or hospital but since she felt "fine" she did not go anywhere. She came today because she has painful rash on one side of her chest extending to the middle of her back for the last 2 days that has not been helped by the cream she has at home.

You counseled her about the retesting to initiate ART and her test result found to be HIV positive. Aster tells you that she has never been on ART before but thinks she needs it now. Upon further questions, you determine that she lives at house number 910 in Woreda 10 of Bole sub-city of Addis Abeba. She has got mobile phone with the number 09.99.99.99.99, and she would like her husband Mulugeta 35 years to be her treatment supporter since he is well informed about her sero-status. His mobile number is 09.99.99.99.98.

She tells you that she has been married for 5 years and has one 4-year old son. Her husband is HIV+, has been in HIV care and is on ARV drugs for the past year. As per the recommendation of the VCT counselor, she took her son to get tested last month and found that he was positive as well. She tells you that she was shocked as he has not been sick more than other children. She made an appointment to bring him into HIV care after she felt better from her current condition.

She and her husband have decided not to have more children and they use condoms most of the time. She is able to do housework currently but is not employed. Aster has had no other medical problems in the past. In fact, she tells you that this is the first time she has such symptom which is why she came. Except for the rash, her clinical review is negative. Her weight is 54 kg. Her height is 170cm.

On physical exam, you see a dark red vesicular (blistering) rash over the right side of her chest extending to her back which does not cross the midline. She has no rash in her eyes or the rest of her body. CD4 and other basic Lab tests were done. Her CD4 count is 575 and other lab results were within the normal range.

You decided to give the patient acyclovir 400mg 3 times a day for 7 days, make an appointment for her in a week time to check the rash and her readiness to start ART.

NB.No assumptions should be considered while performing this exercise.

Questions:

1. Fill out **intake forms** based on the above hypothetical history.
2. Fill out the **HIV Care/ART follow-up form** for the first visit by summarizing intake form using a blank follow up form.
3. What clinical stage is Aster at this visit? _____
4. What is her functional status? _____

5. What other interventions need to be done for Aster? ____

Part 2: Aster returns one week later on 29/06/09 as scheduled and tells you that her rash has improved. You do a clinical review which is negative. Her weight remains the same, and her rash has visibly improved. She is still working at home. You asked her about her readiness and she said that let me think and come with the next appointment on 13/07/09 for ART initiation.

Question:

1. Fill out the HIV Care/ART follow-up form for the **second** visit.
2. What clinical stage is Aster at this visit? ____
3. What is her functional Status? ____

Part 3: Aster returns earlier than her scheduled follow-up on 06/07/09 because she has developed cough and fever. On further inquiry you find out that the cough was productive she also has loss of appetite and night sweats for the last 1 week. She tells you that she feels very weak; right now she is unable to do housework but is able to move around. Her weight is 52 kg.

At physical examination she has dullness and crepitation on the posterior left middle chest. You order her LF_LAM and Xpert MTB/RIF (GeneXpert), CD4 count and tell her to come back as soon as her result is ready.

Questions:

1. Fill out the HIV Care/ART follow-up form for the **third** visit.
2. What is the TB screening result of Aster at this visit? ____
3. What is her functional status? ____

Part 4: On 07/07/09 Aster returns for her lab results. Her LF_LAM and Xpert MTB/RIF (GeneXpert) result are positive and her CD4 count was 355. She still has cough and fever. Her weight is 52 kg and she still is unable to do work at home but is able to move around. You referred her to TB clinic and order her Cotrimoxazole refill for 2 month and sent for adherence counseling.

Question: Fill out the HIV Care/ART follow-up form for **fourth** visit.

Part 5: On 20/07/09 Aster returns after she took two weeks of TB treatment, counseled and reported as ready for ART. Her weight is 52 kg and she still is unable to do work at home.

You put her on **TDF-3TC-DTG** and appoint her to come after 2 weeks ordering her Cotrimoxazole refill for 2 weeks.

Question: *Fill out the HIV Care/ART follow-up form for **fifth** visit.*

Below are the National HIV Care/ART Intake and follow up forms. Your facilitator will give you copies of the latest national M & E tools for the purpose of practicing how to fill them based on exercise 8.1 and 8.2.



FEDERAL MINISTRY OF HEALTH OF ETHIOPIA

HIV Care/ ART clinic Intake form- ASocio demographics & family care

Facility Name _____ MRN _____ Enrollment date: ____/____/____(DD/MM/YY) Unique ART No. _____ Unit TB No. _____

Client Name _____ Fathers' Name _____ Garand Fathers' _____
 For child only:-Mother's Name: _____ Place of delivery: ☐ Health facility ☐ Home ☐ Other specify _____
 Mode of delivery ☐ Spontaneous vaginal ☐ Cesarean section ☐ Other specify _____
 Date of birth ____/____/____ age: ____years(enter months for a child<5 years of age) Sex: ☐ M ☐ F
 Client/ child's care giver) Marital Status: ☐ Never Married ☐ Married ☐ Divorced ☐ Widowed
 Religion: ☐ Orthodox ☐ Muslim ☐ Protestant ☐ Catholic ☐ Other _____
 Level of education: ☐ No education ☐ Primary ☐ Secondary ☐ Tertiary ☐ Other/specify _____ Occupation: _____
 Address: Region _____ Zone/Sub city _____ Woreda _____ Kebele _____ House No. _____ Telephone _____
 Client reside within the catchments area ☐ Yes ☐ No, if no; Is challenge anticipated to regularly follow in this facility ☐ Yes ☐ No
 Date confirmed HIV+/: ____/____/____(DD/MM/YY) Type of HIV Test: ☐ Rapid HIV tests ☐ DBS/PCR (for children)

CLIENT REFERRAL INFORMATION

From with-in the Facility

☐ Medical OPD ☐ Pediatric OPD ☐ TB Clinic ☐ MCH (PMTCT) ☐ VCT ☐
 Adult IPD
☐ Pediatric IPD ☐ Other Outpatient specify _____

Outside the Facility

☐ Health Centers ☐ Public Hospital ☐ Private Hospital ☐ NGO/FBO Hospital ☐ Private Clinic ☐ Self-referred ☐ Others _____

CARE GIVER/EMERGENCY CONTACT INFORMATION

Full Name: _____ Age: ____ Gender: ☐ Male/ ☐ Female Relation: _____
 Address: Region _____ Zone/Sub city _____ Woreda _____ Kebele _____ House No. _____ Tel: _____

DISCLOSURE AT ENROLLMENT:

Does anyone else know about your/your child's HIV Status? ☐ Yes ☐ No

If yes who knows your/your child's HIV positive status? ☐ Spouse (Wife/Husband) ☐ Own Child (ren) ☐ Parent(s)
☐ Sibling(s) (Brother(s)/Sister(s) ☐ Relatives ☐ Friends ☐ Others _____

FAMILY MEMBERS(Spouse/Parent and Children/Sibling) HIV STATUS AT ENROLLMENT AND FOLLOW UP Visit

(Enter date for appropriate HIV status in DD/MM/YY format*)

Figure 9.1. Intake form, section A, from page

At enrollment				At enrollment or follow up visit: update all black fields as status changed					
Family member	Age	Sex	Health status (healthy, chronically ill, dead)	Counseled for HIV*	Tested for HIV*	HIV test Result; P, N, IND	Enrolled in Care*	Started ART *	Unique ART #
Spouse/Parent									
Child1/Sibling									
Child2/Sibling									
Child3/Sibling									
Child4/Sibling									
Child5/Sibling									

INSTRUCTIONS: Intake form-A Socio demographics & family care

Note: All but family HIV status fields must be completed at enrollment & all date fields must use Ethiopian calendar in DD/MM/YY format

Health Facility Name – Health Facility name as registered at the Ministry of Health

Medical Record Number (MRN): Unique individual identifier on medical information provided by facility on folder

Enrollment date: Enter the date client was registered in HIV care using Ethiopian calendar in DD/MM/YY format

Unique ART number: Unique identifier assigned to a client when starting ART. It is composed of region number/facility type code/ specific facility code /client assigned 5 digit serial numbers starting as 00001

Region Code: 1.Tigray (TG) 2.Afar (AF) 3.Amhara (AM) 4.Oromia (OR) 5. Somali (SO)
6. Benshangul Gumuz (BG) 7. SNNPR (SN) 12. Gambella (GA) 13. Harar (HA) 14. Addis Ababa (AA) 15. Dire Dawa (DD)

Facility Type code: Use 08 for Hospital and 09 for Health center **Facility Specific Type code:** Assigned by RHB

Client Name: Enter client's given name, **Father's Name:** - Enter client's father's name. If not known enter NA **Grandfather's Name:** - Enter client's grandfather's name. If not known enter NA

For a Child ask and fill Mother's name, place and mode of delivery as:

Child's; motherName: Enter client's mother if the client is a child <15 years. If client's mother name is not known enter NA

Place and mode of delivery: Mark in appropriate circle that indicate the place and mode of child's delivery

Date of Birth: - Use Ethiopian calendar in DD/MM/YYYY format. If only month & year are known, enter 00 for day, if only year is known, enter 00 for day & 00 for month

Age: - Enter Client's current age in years. If Client is less than 5 years old, enter age in months.

Sex: - Mark in the appropriate circle that indicate sex of the client

Marital Status: - Mark in appropriate circle that indicate clients (child's care giver) current marital status

Religion: - Mark in appropriate circle that indicate religion of the client

Level of Education: - Mark in appropriate circle that indicate client's completed year of schooling

Occupation: Enter what the client's current works for living

Client Address: - Enter the current permanent address of the client including; Region, Zone/Sub-city, Wereda, Kebele, House number as well as telephone number

For clients out of the catchment area of this facility: confirm absence of barriers for regular follow up at this facility including, time, transport access, transport money

Date confirmed HIV+: Enter test date using Ethiopian calendar in DD/MM/YYYY format; if tested more than once, enter the most recent one

Type of HIV Test Mark in appropriate circle that indicate the type of test the client took

Client referral information; Mark in appropriate testing point if client linked from with the facility or indicate facility type if from outside facility

Care giver information: Enter the name, the address & relationship of care giver; to avoid unintended disclosure it is advisable if care giver is confided client sero-status

Disclosure at enrollment: Does anyone know client HIV Status? Mark appropriate response that indicate clients disclosure status

To who knows clients HIV status: Mark appropriate response that indicate for whom the clients has disclosed

Family members (spouse/parent /child/ren/sibling's) HIV status: at enrolment.

At Enrollment: Mark the relationship, enter the age, sex & health status of client's each family members and for those whose HIV Status is determined indicate the date

At follow up visits: if the status of the clients family member changed enter the date in DD/MM/YY format in the cell corresponding to the updated status

Figure 9.2. Intake form Section A, back page

HIV Care/ ART clinic Intake form-B
Past, presenting illness&PHDP FEDERAL MINISTRY OF ALTH OF ETHIOPIA
Facility Name _____ **MRN** _____

PAST OPPORTUNISTIC ILLNESS (OI), PROPHYLAXIS AND TREATMENT (MARK ALL THAT APPLY)
Past OI

☐ Pulmonary TB ☐ TB-Extrapulmonary ☐ Fever (>1 month; unexplained) ☐ Diarrhea (>1 month)
☐ Oral Candidiasis ☐ O pharyngeal candidiasis ☐ Wasting Syndrome ☐ OPneumocystis Carinii
☐ Pneumonia ☐ Pneumonia (recurrent) ☐ Cryptococcal Meningitis ☐ Minor Mucocutaneous manifestations
☐ Herpes-simplex(>1month) ☐ Toxoplasmosis (brain) ☐ Kaposi sarcoma ☐ Other specify _____

Past Prophylaxis/Treatment
Cotrimoxazole: Date started _____ / _____ / _____ If stopped; date stopped _____ / _____ / _____

INH: Date started _____ / _____ / _____ If completed; date completed _____ / _____ / _____

Fulconazole Preventive Therapy (FPT): Date started _____ / _____ / _____ If completed; date completed _____ / _____ / _____

ARV for PMTCT: - ☐ Yes ☐ No If Yes Health facility: _____ Regimen for mother: _____ Regimen for baby: _____

Lifelong ART/TI: ☐ Yes ☐ No if yes Regimen: _____ Start date: _____ / _____ / _____ Months on ART _____ Still _____ on Treatment: ☐ Yes ☐ No

Client STATUS AT ENROLLMENT
VITAL SIGNS: Temp (°C) _____ PULSE/HR (/m) _____ BP (mmHg) _____ RR (R/m) _____

ANRTOPEMETRY: -Height (cm) _____ Weight (kg) _____ BMI (kg/m²) _____ **for child only;** Head circumference (cm): _____

For child; anthropometry interpretation: ☐ Normal weight for age ☐ underweight

FUNCTIONAL Status: ☐ Working ☐ Ambulatory ☐ Bed ridden

Developmental status (for a child<5) ☐ Appropriate for age ☐ Delay ☐ Regression

PRESENTING SYMPTOM: ☐ Cough ☐ Weight Loss _____ % body wt ☐ Dyspnea ☐ Fever > 1 month ☐ Night sweat ☐ Diarrhea ☐ Mental Confusion ☐ STI Symptoms ☐ Dysphagia/ Odynophagia ☐ Nausea and/or vomiting ☐ Persistent Headaches
☐ Others specify: _____

CLIENT'S PREGNANCY STATUS at Enrollment: ☐ Pregnant: LMP _____ / _____ / _____ EDD _____ / _____ / _____ ☐ Not Pregnant ☐ Not Applicable

CLIENT GENERAL APPEARANCE OF AT Enrollment: _____

Physical Examination by system

System	Normal	Abnormal	Specify Abnormal Finding
HEENT			
Lymph nodes			
Chest			
Heart			
Abdomen			
Genitourinary System			
Musculo-skeletal system			
Skin			
Nervous System			

WHO HIV Clinical Stage at enrollment: _____ **why:** _____

POSITIVE HEALTH DIGNITY AND PREVENTION (PHDP) ISSUE ADDRESSED

Date	Basic HIV & TB transmission education,	Prevention: abstinence, safer sex, condoms	Positive living	Progression of disease	Available treatment/ prophylaxis CPT, IPT, FPT	Education on essentials of ART	Why complete adherence needed	Explain dose, when to take	What side effects & how to manage	What to do if one forgets dose	Conclusion

Figure 9.3. Intake form, Section B, front page

INSTRUCTIONS: Intake form- B Past and presenting illness&PHDP

Note: All fields must be completely filled in at enrollment then after should be update accordingly & all date fields must use Ethiopian calendar in DD/MM/YY format

Health Facility Name – Health Facility name as registered at the Ministry of Health

Medical Record Number (MRN): Unique individual identifier on medical information provided by facility on folder

Past Opportunistic Illness (OIs):– Mark all IOs that the patient has experienced before enrollment. **Note:** that this information can be obtained from the client, medical/lab records or referral form

Past Prophylaxis / Treatment:-

Cotrimoxazole (CTX): If client is on CTX at enrollment enter the start date; using Ethiopian calendar in DD/MM/YYYY

Isoniazid (INH): If client on INH at enrollment enter the start date and if client has completed the INH enter stop date using Ethiopian calendar in DD/MM/YYYY format

Fluconazole Preventive Therapy (FPT): If client is on FPT enter the start date and if client has completed the FPT enter stop date using Ethiopian calendar in DD/MM/YYYY format

ARV for PMTCT: Mark appropriate circle to indicate client exposure to ARV for PMTCTE, if yes enter facility name and the regimen given to the mother and the baby

Lifelong ART/TI: For all transfer in, mark ART status at enrolment, enter the regimen, how long the client was on ART, start date in DD/MM/YY format, & if the client is still on ART

VITAL SIGNS:–Check and enter the measurement: **temperature** in °C, **Pulse(PR)** in rate per minute **Blood pressure (BP)** in mmHg **Respiratory rate (RR)** in rate per minute

Anthropometric Measurement:–measure and enter: **Height** in cm, **Weight** in kg, **body mass index (BMI)** in KG/ M²

Functional Status:–Assess & mark appropriate circle: **Working** if able to perform usual work in & out of house, **Ambulatory** if Able to perform activities of daily living but not able to work **Bedridden(B)**=not able to perform activities of daily living

Developmental status for children <5 years:- **A= Appropriate** if a child has attained milestones for age: **Delay:** if the child failure to attain milestones for age **Regression:** loss of what has been attained for age

Presenting Symptom: Ask and mark all the symptoms that the client complain to have at enrolment

Client's pregnancy status at enrolment: If pregnant enter Date of last Menstrual Period (**LMP**) & Expected Date of Delivery (**EDD**) otherwise mark appropriate circle

Client general appearance at enrollment: Assess and describe the general prance of the clients in the space provided

Physical Examination: Assess and mark appropriate physical finding category for each system; describe all abnormal findings

WHO HIV Clinical Stage: Assess and stage the clients; enter the criteria (from past /presenting finding) used to put the client on a particular WHO clinical stage; **listed below**

WHO HIV Clinical Stage 1: -Clinically Asymptomatic Client -Persistent Generalized Lymphadenopathy (PGL)

WHO HIV Clinical Stage 2: -Minor Mucocutaneous Manifestations - Herpes Zoster - 5-10% body weight loss - Recurrent upper respiratory tract infection

WHO HIV Clinical Stage 3:

- Oral Candidiasis -Unexplained Chronic Diarrhea (>1 month) - >10% of Body Weight loss - Oral Hairy Leukoplakia
- Unexplained Prolonged Fever (>1 month) - Pulmonary Tuberculosis - Bacterial Pneumoni Unexplained anemia (<8g/dl), neutropaenia (<0.5x10⁹/l) and/or chronic thrombocytopaenia (<50 x 10⁹/L)-Other Severe Bacterial Infections (eg. pyomyositis)

WHO HIV Clinical Stage4:

- Extrapulmonary Tuberculosis -HIV Wasting Syndrome - Atypical Mycobacteriosis -Candidiasis (Esophagus, Trachea, Bronchi or Lungs) - Cryptococcosis Extrapulmonary - Cryptosporidiosis with Diarrhea (>1 month duration) -Herpes Simplex (mucocutaneous >1 month, or visceral - CMV Disease (other than liver, spleen, lymph nodes) -HIV Encephalopathy - Karposi's Sarcoma- Lymphoma - PML - Mycosis, Disseminated (i.e. Histoplasma, Coccidioides) - Pneumocystis Carinii Pneumonia (PCP) - Salmonella Septicemia, Non-typhoid - Toxoplasmosis of the CNS - Symptomatic HIV-associated - Invasive cervical carcinoma- nephropathy or cardiomyopathy - Atypical disseminated leishmaniasis

POSITIVE HEALTH DIGNITY AND PREVENTION PHDP) ISSUE ADDRESSED

Enter the date of adherence counseling in DD/MM/YY format and mark all issues raised and dressed during PHDP counseling

Figure 9.4. Intake form, Section B, back page

S/US	Monthson ART	Pregnancy Status/FamilyPlanningMethod	Functionalstatus	Client readiness											
S=Scheduled US=Unscheduled TBSCREEN SCREENFOR TB AT EVERYVISIT Adult&Adolescent 1. CurrentCough? 2. Fever? 3. Nightsweats. 4. Weightloss? P=(Positivescreen)-Yesto anyofthe above— EvaluateforTB. N=(Negativescreen)- Noto all the questions above—assessforIPTeligibility Children0-14years old 1. CurrentCough? 2. Fever? 3. Weightlossor poorweightgain? 4.ContacthistorywithTB patient? P=(Positivescreen)-Yesto anyoneofthe four—evaluatefor TB N=(Negativescreen)-Noto allfour—assess foreligibility to IPT XPert MTB/ RIF (1) P= Positive N= Negative LF LAM (2) P= Positive N= Negative	Durationin monthsintheinitiationofART: 0=ARTInitiationdate 1week=1 week 2weeks=2 weeks 3weeks=3 weeks 1= 1 month 2=2 months If not started on ART (Pre-ART)leave this column blank Pain Assessment & Management AssessforPain& Manageas NP= Nopain S1= WHOStep1 S2= WHOStep2 S3= WHOStep3 Nutritional Status (adults) BMI (wt/(h²)) (for non- pregnant/nonpostpartum) 1= Normal (18.5-24.9kg/m ²) 2= Mild (17-18.49kg/m ²) 3=Moderate malnutrition (16-16.99kg/m ²) 4=Severe malnutrition (<16kg/m ²) 5=Over weight (25-29.99 kg/m ²) NB: write the codes (1,2,3,4 or 5) MUAC (for pregnant/postpartum /bedridden) 1= Normal (>23cm) 2=Moderate malnutrition (19-23cm)3=Severe malnutrition (<19cm forpregnant and postpartum /<18cm for bedridden)	P=Pregnant(Ifpregnant,giveestimateddue date(EDD)) PMTCT=Referredto PMTCT&indicatelinkeage WP=wanttobecomepregnant NoFP=not pregnant& isnot usinganyFP methods FP=OnFamilyPlanning(entercode): 1=Condoms 2=Oralcontraceptivepills 3=Injectable 4= Implant 5=Intrauterinedevice 6=Vasectomy/ tuballegation 7=Abstinence(nosex) Length / height/ HC Measurelength/heightincm forchildren youngerthan14yearsatEVERYvisit. Measurehead circumferenceincm forchildren younger than3yearsatageatEVERYvisit NutritionalStatus(Children) W/H 1=Normal/ Appropriate (> -1Z-score) 2=Mild(<-1and> -2Z-score) 3=MAM- ModerateAcuteMalnutrition (<-2and>-3 Z-score) 4=SAM-SevereAcuteMalnutrition(<-3Z-score)	W=Workingable to performusualwork inor out of the house,harvest, go to school,for children, normal activitiesor playing) A=Ambulatory (ableto performactivities ofdaily Living) B=Bedridden (not ableto performactivities of daily living) DEVELOPMENTALMILESTONESFORCHILD A=Appropriate: Sittingwithout support3 to9months Standingwithassistance.....5to 11months Hands and knees crawling6 to1 3months Walkingwithassistance7 to14 months Standingalone.....8 to17 months Walkingalone..... 9 to18 months Delay: Failure toattainmilestonesforage Regression: Loss ofwhat has beenattained for age NutritionalStatus(Older children & adolescents) BMIforage(5-18yrs) 1=Normal/appropriate (>-1Zscore) 2=Mild(<-1and> -2Zscore) 3=Moderatemalnutrition(<-2and>-3Zscore) 4=Severemalnutrition(<-3Zscore) *BMIforage forolder childrenandadolescents.	Enter the date (dd/mm/yy) client is ready for ART initiation when client is counseled, adherence barriers addressed and client is willing to start ART Client Set HIVPreventionPlan D=Agreedto Disclosetopartner/family/friend, PT= Agreedto bring partner for testing ChT=agreedtobringchildrenfor testing. SSex=discussed&agreedto practicesafersex SubU=Decidestoavoidto decreaseSubstanceuse ASS=AssessedforSTI SRX=clientmanagedforSTI ForchildrenFill instageofHIVdisclosure DS0 = No disclosure DS1=Satge1, about the illness, taking medicine, keeping healthy, DS2=Stage2, about germs, body soldiers, DS3=Stage3, use of terms like CD4, Viral Load, HIV DISPENSE DOSE/ REGIMEN CODE ADULT FIRST LINE 1d = AZT - 3TC - EFV 1e = TDF - 3TC - EFV 1g = ABC + 3TC + EFV 1j = TDF + 3TC + DTG 1k = AZT + 3TC + DTG 1i = Other specify CHILD FIRST LINE 4d = AZT+3TC+EFV 4e = TDF+3TC+EFV 4f = AZT+3TC+LPV/r 4g = ABC + 3TC + LPV/r 4i = TDF + 3TC + DTG 4j = ABC + 3TC + DTG 4K = AZT + 3TC + DTG 4L = ABC + 3TC + EFV 4h = Other specify ADULT SECOND LINE 2e = AZT + 3TC + LPV/r 2f = AZT+3TC +ATV/r 2g=TDF + 3TC + LPV/r 2h= TDF + 3TC + ATV/r 2i= ABC + 3TC + LPV/r 2j = TDF + 3TC + DTG 2k = AZT + 3TC + DTG 2l= Other specify CHILD SECOND LINE 5e=ABC+3TC+LPV/r 5f=AZT + 3TC + LPV/r 5g=TDF + 3TC + EFV 5h=ABC + 3TC + EFV 5i= TDF + 3TC+LPV/r 5m= ABC+3TC+DTG 5n= AZT+3TC+DTG 5o= TDF + 3TC + DTG 5j= Other specify ADULT THIRD LINE 3a = DRV/r+DTG+AZT+3TC 3b = DRV/r+DTG+TDF+3TC 3c = DRV/r+ABC+3TC+DTG 3d = DRV/r+TDF+3TC+EFV 3e = DRV/r+AZT+3TC+EFV 3d = Other specify CHILD THIRD LINE 6c = DRV/r + DTG + AZT + 3TC 6d = DRV/r + DTG + TDF + 3TC 6f = DRV/r+DTG+ABC+3TC 6g = DRV/r + ABC + 3TC + EFV 6h = DRV/r + AZT + 3TC + EFV 6e = Other specify Referral for Cx Ca screening & management (0-4) 0, Not referred 1. Referred for Cx Ca screening 2.Referred for Cryotherapy /Thermal ablation Tx 3.Referred for LEEP service 4. Further evaluation & management suspicious cases											
TB PROPHYLAXIS TREATMENT 6HI-6=CurrentlyonNI prophylaxis (Numberrefers to monthson 6HI) 6HI-C=Completedtreatment 6HI-DC=discontinuedforany reason 3HP1-3=Currently on 3HP prophylaxis (Number refers to months on 3HP) 3HP-C=Completed treatment 3HP-DC=discontinued for any reason TB RX1 6=CurrentlyonAntiTB(numbers of LAM) OI/ Opportunistic Cancers NOI=No OI or Opportunistic cancer Z=Zoster BP=BacterialPneumonia PTB=PulmonaryTuberculosis EPTB=Extra pulmonarytuberculosis TO= Thrush oral EC=esophagealcandidiasis UM=ulcers-mouth DCor DA=DiarrheaChronic/Acute PCP=Pneumocystispneumonia CT=CNS Toxoplasmosis CM=CryptococcalMeningitis NHL=NonHodgkinsLymphoma KS=Kaposi'sSarcoma CCa=Cervicalcancer O=Other	ADHERENCE Estimateadherencesusingthe table below: <table border="1"> <thead> <tr> <th>Adherence</th><th>%</th><th># misseddoses</th></tr> </thead> <tbody> <tr> <td>G (Good)</td><td>≥95 %</td><td>(of 30 doses) <2 doses</td></tr> <tr> <td>F (Fair)</td><td>85-94 %</td><td>2-4doses 4-9 doses</td></tr> <tr> <td>P (Poor)</td><td><85 %</td><td>≥5 doses ≥10 doses</td></tr> </tbody> </table> IfFairorPooradherence,inwhy columnnotereason: 1. Toxicity/ Sideeffects 2. Share withothers 3. Forgot 4. Felt better 5. Too ill 6. Stigma,disclouser 7. Drugstockout 8. Lost/ ranoutofpills 9. Delivery/ travelproblems 10. Inabilitytotapy 11. Alcohol 12. Depression 13. Other	Adherence	%	# misseddoses	G (Good)	≥95 %	(of 30 doses) <2 doses	F (Fair)	85-94 %	2-4doses 4-9 doses	P (Poor)	<85 %	≥5 doses ≥10 doses	SIDE EFFECTS 1. Noside effects 2. Nausea 3. Diarrhea 4. Fatigue 5. Headache 6. Numbness/tingling/pain 7. Rash 8. Anemia 9. Abdominalpain 10. Jaundice 11. Fatchanges 12. Dizzy,anxiety,nightmare,depression 13. Other REASONSFORREGIMENCHANGE 1. Toxicity/ Sideeffects 2. DuetonewTB 3. Newdrugavailable 4. Drugstockout 5. Clinicalfailure 6. Immunologicfailure 7. Virologicfailure 8. Other Cx Ca screening, Tx & referral services Type of Cx Ca Screeningresult (0- 6): 0. Cx Ca screening not done 1. HPV Negative 2. HPV Positive 3. VIA Negative 4. VIA Positive: eligible for Cryo 5. VIA Positive: Non-Eligible for Cryo 6. Suspicious for Cx Ca	REASONSFOR STOPPINGREGIMEN STOP=StoppedART IfSTOP,Inwhy column,note reason: 1. Toxicity/sideeffects 2. Treatment failure 3. Poor adherence 4. Illness,hospitalization 5. Drugs out of stock 6. Patient lackfinances 7. Otherpatient decision 8. Other VIRAL LOAD Mark '*' under VL column when requested/ specimen collected, Write the amount and interpret as undetectable and detectable for clinical intervention Management of cervical lesions (0-4) 0. No treatment / further evaluation & management service given 1. Cryotherapy 2. Thermal ablation treatment 3. LEEP service 4. Further evaluation & management service given to Suspicious for Cx Ca
Adherence	%	# misseddoses													
G (Good)	≥95 %	(of 30 doses) <2 doses													
F (Fair)	85-94 %	2-4doses 4-9 doses													
P (Poor)	<85 %	≥5 doses ≥10 doses													
Inthefollow-up date, in2 nd columnif oneof theoptionsbelowapplies, userawnext to the last visit to enter the appropriate information: TO=Transferout LOST=not seensince ≥1 month,3 months DROP= lost tofollow-upfor >3 months STOP= When the clinician stop ART for different reason and patient is on follow up DEAD															

Figure 9.6. ART Follow up form, back page

Chapter summary:

- One of the general principles of good chronic HIV care is to use HIV Care/ART intake and follow up form properly, correctly and timely.
 - The intake form captures important information about patient previously and current at enrolment
 - The follow up form captures the necessary information a client receives at every clinical visit
 - New regimen codes added.
 - LF_LAM, Cervical cancer and 3HP additional parameters were added
-

Chapter 10: Adherence and Adherence Preparation

Chapter Description

This chapter describes what adherence is and the chronic care 5A's to prepare patients for ART adherence.

Purpose: Learn how to prepare and support PLHIV on adherence to treatment and care.

Enabling objectives:

By the end of this chapter, participants will be able to:

- Define Adherence;
- Use the chronic care 5 A's to prepare patients for ART adherence;

Chapter Outline:

- Definition of Adherence
- Adherence preparation

Allocated time: 100 minutes

Learning activities

- Brainstorming exercise,
- Case scenario,
- Discussion,
- Exercise,
- Practical Exercise at skill station,

Brainstorming Exercise 10:1: Case scenarios

Case scenario 1: Abebe is referred from OPD for ART initiation. The referral letter states that Abebe has TB. He has started anti-TB treatment and co-trimoxazole prophylaxis.



- How do you approach this patient?
- What information do you provide?
- What are the concerns to start ART?

Case scenario 2: Today is preparation session for GM. (CD4 count is 150; clinical stage 3).



The service provider assessed what GM already knew about ART. He mentioned knowing some friends on ART; and he knew it is important to get food support to start ART. So he knows some things.

- He agrees to start ART and he is optimistic on taking ART. What do you learn from this case?
- How do you get the knowledge & information GM has?
- As a provider what do you want to do?
- What do you tell him to implement adherence in his life?
- Do you give him all of the information at once or arrange another session?

10.1. Defining Adherence to ART

Definition: **Adherence** means accepting, agreeing and correctly following a prescribed treatment (participation of the patient).

Adherence implies understanding, consent and partnership between the care giver and patient. We have **clinical adherence** meaning regular attendance of the patient to clinical visit according to given appointment, and **drug adherence** which means taking the drug according to instructions given by the providers.

Being adherent to the HIV medications directly means that:

- The ART suppresses the **viral replication** to the maximum (=sustained viral suppression),
- HIV infection is prevented from advancing to **AIDS** (=healthy and improve quality of life),
- The risk of developing **OIs** is minimized, and
- The risk of **drug resistance** is reduced.

What does good adherence to care and treatment means?

HIV medication adherence means taking the right ARV medication, at the right dose, at the right time every day and exactly as prescribed for life long,

Clinical adherence is regularly attending clinical care follow ups and periodic laboratory monitoring with avoiding practices that interferes with the effect of medication.

Adherence can be difficult for many reasons. For example, side effects from some ARVs can make it hard to stick to an HIV regimen or dosing schedule might not fit well into a person's routine.

10.2. Adherence preparation

It is important to apply the 5 A's in the process of preparing adherence. First see this exercise.

Empathy Exercise 10.2:

Instruction: For moments imagine that you are diagnosed with HIV infection and your care provider briefly states about ART and prescribes the ARV drugs for you. While you are collecting your medication, the pharmacist tells you about the drug- some of the side effects are strange- bad dream.

Close your eyes and try to imagine your life, your medication, how you can cope.

How will you take the medication?

Whom will you tell about your problem?

What barriers do you anticipate you will face while taking your medication?

One of the principles of good chronic care is using the 5A's to prepare patients for ART adherence-**Assess, Advise, Agree, Assist and Arrange**.

N.B. Use 5 A's independently and without mixing them. First finish assessing, and from the results of your assessment advise and correct misconceptions, if there are any. After you finish advising, check if patient agrees on the recommendations and treatment plan you proposed. If agreement is ensured, assist patient and arrange follow up visits. At any step, if the patient has question or doubt, explain and provide appropriate response in modest and understandable way.

a) ASSESS:

Assess the patient's/care giver's expectation for preparation (today's visit).

“What do you expect from this session? We will talk about your/ your child initiation of ART.”

While assessing **understanding of ARV therapy (ART)** use:

- **Specific questions** than general questions as it may overwhelm patients, answer yes or may not respond. E.g., instead of asking "Do you understand everything concerning antiretroviral therapy?" ask “What have you heard about antiretroviral therapy side effects?”
- **Open ended questions** than close ended questions like
“What are the benefits of ARV drugs?”
- At all times make sure you are non-judgmental, empathetic and active listener.
- Make the patient/care giver feel comfortable, not as if they are taking a test!

Questions that make patients/care givers explain in their own words are good to assess their understanding. If a patient/ care giver has misunderstood or forgotten some information, reassure the patient/care giver that this is normal, and explain once more.

Questions that can be asked to assess the patient's/care giver's understanding are:

Simple and specific questions

- ✓ What are the benefits of ART?
- ✓ Does ART cure patients from HIV?
- ✓ How long do you have to take ART?
- ✓ What is the effect of ART on the body's defense?
- ✓ Why is it important to come regularly to the health center/Hospital when you are on ART?
- ✓ What do you know about side effects of ART?
- ✓ Why is it important not to miss a dose when you take ART?
- ✓ What happens if you do not take ART correctly?
- ✓ Why is not good to combine ART with other drugs without consulting the health provider?

Assess interest in receiving therapy

Not all patients/ care givers are interested in receiving ART due to various reasons. Patients/ care givers may have other urgent preoccupations in their lives that make them want to postpone initiating therapy.

That is why it is necessary to assess in depth and to be sure that the patient/ child/ care giver understand the benefit of receiving ART.

b) ADVICE:

"I have some information about treatment for HIV/AIDS (or ART). Would you like to hear about it?"

Give the patient/care giver advice on the following topics. It is good to use drawings or flipchart to explain these topics.

- Explain in local language about HIV illness and expected progression.
- Opportunistic infections gradually become more serious, because HIV is attacking the body's defence.
- Explain that in children the progression of the disease is rapid and impairing their growth.

Give information about ARV therapy (ART):

- ARVs are life-saving drugs and that depends on starting early, taking the correct daily dose at the right time for lifetime.
- The pills do not cure HIV—they control replication; they delay or reduce the occurrence of OIs.
- The pills do not prevent transmission of HIV to others—you must still use condoms and practice safer sex. The therapy can never eradicate the virus from your body.

“After taking ART you will feel healthy but this does not mean you are healed, if you stop taking ART, you will become ill again and you may develop resistance.”

“If you take ART as recommended by your care provider you will be healthy, your CD4 cells will increase and viral load will decrease but the virus will NEVER be eradicated.”

- Possibility of side effects and drug interactions. (Please tell common and potentially serious side effects for the specific ART regimen the patient is taking including prophylaxis and OI treatments)
- Importance of disclosure of HIV+ status. (disclosure to partner and family member(s) and the importance of staged disclosure for children starting from early age)(see Paediatric psychosocial support section)
- Importance of testing partner and children (see family matrix tool).

- Drugs must not be shared with family or friends—patient must take the full dose.
- Importance of adjusting dosage for smaller children according to their changing body weights during follow up visits.
- Select appropriate drug formulations considering such factors such as taste and pill burden.
- Choose regimens similar to parents where it is appropriate.

N.B:

- **Do not overwhelm the patient/care giver with too much information** at once. The patient/care giver needs time to think about and digest some information before they will be able to concentrate on further information. That is why it is good to divide the advice over several visits, and indicate on the education side of the patient's/child's treatment Card the information that has been given already.
- When you give advice, it is necessary to evaluate the patient's/ care giver's **confidence and readiness to adopt the treatment**. If you notice the patient/ care giver is not paying attention anymore or does not make eye-contact with you anymore, it is good to ask if there is a reason for this. If you have the impression you have lost the patient's/ care giver's attention, it might mean the patient/ care giver is overwhelmed by the amount of information. Or it could mean the patient/ care giver is starting to think that all of this is too difficult for him.
- **Make sure you let the patient/ care giver to express their doubts**, ask any question or repeat what you have said. Also demonstrate on how to take or give ARV medications.

c) AGREE:

As treatment success depends on the optimum adherence, **determination of the patient** and agreement on treatment is the most important step.

Before initiating ARV therapy, establish that the patient/care giver is willing, motivated and agrees to treatment.

The patient is the one that must take the responsibility for taking the medication daily. The patient must be very motivated to take the medications, use a third person technique to motivate and tell how other patients improved with treatment and deteriorated because of not taking medications.

For children, both the decision to initiate and stay adherence to ART depends not only on the child but also the parents / care givers. Establishing willingness, motivation/ determination and agreement to start ART/treatment with the mother/ care taker (for younger children) or older children (≥ 10 years) before initiating ARV therapy, is important for success of treatment. Parents/ care givers are the most responsible person for the initiation & adherence of taking the ARV medication for children and must be very determined & motivated. As the children grow older, it is also important to make the children understand ART and build their motivation for good adherence.

Questions to check the patient's/ child's/ parent's/ care giver's agreement for medication:

"After hearing all the explanation and advice, how do you think you/your child could take this kind of treatment?"

"Do you agree to take your medications? Do you agree to give the medication to your child?"

"What difficulties do you anticipate that hinder you from taking your medication?"

"What difficulties do you anticipate that hinder you from giving the medication to your child?"

"What difficulties do you anticipate that hinder your child from taking his/ her medication?"

"Where will you keep the medications? How/ what time will you take it? What will you use to remind you?"

Besides getting answers to the above questions, it is good to look at some objective measures to check the motivation. Do not rely only on your personal feelings, studies have shown that the health care worker's impression does not always correspond with the real situation concerning motivation and adherence.

That is why it is useful to check:

- Has the patient/caregiver demonstrated the ability to keep appointments and to adhere to other medications?
Whether the patient comes on time on the appointment date (check book and patient chart) and take other treatments (for example CPT, TB treatment, IPT,FPT) correctly.
- Ask the patient to show you how to take ARV).

Does the patient have any history of non-adherence? TB treatment defaulter? This should not exclude this patient, but means that more work is required to prepare for ART adherence.

- **Has the patient disclosed his or her HIV status? If not, encourage the patient to do so. Disclosure to at least one person who can be the treatment buddy/ supporter is important.** Overall, supporting disclosure is important (see chapter 3 for ways to help patients with disclosure or ask a counselor to help). **Does the patient have a treatment supporter/ buddy?** Note that this might not be possible for everyone to bring their treatment supporter with them to clinic. For example, people who live far from the clinic and need to pay transport for two people might not be able to do so. So, just because the patient is unable to bring a treatment supporter is no reason to exclude them from ART.
- **Does the patient/caregiver want treatment and understand what treatment is for?** Willingness to start ART depends on how clearly the patient/ care giver understood the use and advantage of treatment. Check understanding and willingness and check if they have any concern/ worries.
- **Is the patient/ caregiver willing to come for the required clinic follow-up?** The patient shall be well aware of the regular follow ups visits to clinic for clinical care & to collect medications. Check the willingness & motivation of the patient/ caregiver to come for the required clinic follow-ups.

d) ASSIST

Explore what is needed to assist the **patient/caregiver** with ART:

"What problems might arise when you follow this plan? Like your work situation?"

"What questions do you have about this life- long treatment or how to follow this plan?"

"Do you need any help or advice on adjusting to the treatment plan to maintain good adherence? Adjusting specific follow up?"

Help the patient/ caregiver develop the resources/support/arrangements needed for adherence:

- **Ability to come/bring the child for required schedule of follow-up.** It is crucial to discuss this in detail. If the patient/care giver lives far away, it is necessary to ask how they will come. With a family member that has a car? Will the family member be available to

come regularly to the consultation, or if there is an unexpected complication? With public transport? Will you have money available to pay for public transport?

- **Home and work situation that permits taking medications regularly without stigma.** Some people might be away from home for more than 12 hours a day, and be surrounded by people all the time. It is important to check whether the patient will be able to cope with this situation. This means the patient always needs to carry some pills with him (in a pocket), and might need to find a place to take the drugs without others watching. Perhaps a watch with a discrete alarm is needed to remind them when it's time to take medication. It is important to help the patient figure out routines to set up before starting ART that fit regular pill-taking with the rest of their life.
- **Regular supply of free or affordable medication.** It is important to guarantee a continuous supply of drugs. This is a substantial commitment. Taking drugs for a limited time or with interruptions will create resistance. For patients buying their own (other than ARV) drugs, it is important to assess the viability of their financial resources.
- **Supportive family or friends.** Stigma and discrimination in the family may be a barrier to adherence. A family that knows about the diagnosis and are willing to remind the patient to take the drugs or to support the patient in moments where adherence is difficult can help support adherence.
- **ART adherence support group.** This can be important to support adherence for patients in the health facility. Community based organizations, or someone from the clinical team can set up such groups.
- **Treatment supporter—prepare this person:** This is a very important person for treatment success. The person supports and encourages the patient to be mindful about his/her medication and laboratory monitoring.

How to prepare the treatment supporter:

Have a meeting with the supporter before getting commitment to explain what is required (commitment, confidentiality, knowledge on HIV, ART and TB related needs and also emergency resource needs such as money, help with household, children, which can arise while on treatment).

- Educate on confidentiality,
- Remind patients to:

- Take the ARVs at the same time every day
- Attend the follow-up appointments,
- Remember / keep track of all important tests and results,
- Accompany patient to support group meeting if possible.
- Support/ Educate patient to prevent possible burn-out.
- Provide psychosocial support.
- Attend together with the patient the preparatory/ ART initiation visit
- Hold treatment supporters meeting at the health facility to deal with issues facing treatment supporter (burn out, patient not being adherent, barriers to treatment and adherence, etc.).
- Provide telephone number of the facility and assist supporters to access for consultation if a problem arises. Also take telephone number of supporters.
- Assist people living with HIV to identify someone from their families or friends or work colleagues as their treatment supporters and enlist.

e) ARRANGE:

When the patient/care giver is ready for ARV therapy, make a plan.

As early initiation is good for the patient, after preparation session, ART can be initiated as soon as possible including the same day.

If the patient/care giver needs another adherence preparation session, arrange a follow-up session; arrange an appointment with the ART support group.

Record the information you gave on the back of the Treatment Card.

After completion of adherence counseling session care givers should be clear with:

- Who will be responsible for administration of the drugs?
- What medication will be given?
- Timing and frequency of medication
- How to administer medication?

Discuss with the clinical team to recommend adherence support to the patient.

ART is a long-term commitment of both the clinical team and the patient.

EXERCISE 10-3

How would you respond to the following questions raised by a patient?

1. "How can I possibly remember to take a drug daily at the same time for life?"

2. "What is the purpose of taking such a complicated treatment if I will not be cured?"

3. "What if I go away for a few days and I forget to take the drugs with me?"

4. "My wife is HIV+ as well and she is not taking ART. How can I take it?"

5. "You are talking about side effects which might really bother me, how can I take drugs that make me feel so bad?"

6. "What if other people can see that I am taking ART?"

7. "What if people see me when I come to get my drugs at the clinic?"

8. "I have heard that in America, ART is not so complicated. They give one pill per week. Why do you want to give me pills to take daily?"

9. "You are saying that I need to tell at least one person that I am HIV+. I actually changed my mind about telling another person. I do not want anyone to know about my status or support me. I can do it all on my own."

10. "My wife—who will be my treatment supporter—does not have time to come to meet you since we have three children and one of them is sick."

Chapter summary:

- Adherence implies understanding, consent and partnership between the care giver and client.
- One of the principles of good chronic care is using the 5A's to prepare patients for ART adherence
- As treatment success depends on the optimum adherence, determination of the patient and agreement on treatment is the most important step.

Chapter 11: Prevention, Screening and Management of Opportunistic Infection

Chapter Description

This chapter describes opportunistic infections (OIs), their diagnosis and managements, and primary and secondary prophylaxis in PLHIV. In addition, this section also addresses management of cryptococcal meningitis, Leishmaniasis co-infection and STIs and screenings of different NCDs.

Enabling objectives:

By the end of this chapter, participants are expected to:

- Describe primary and secondary prophylaxis
- Discuss the management of respiratory problems in HIV infected individuals
- Discuss the management of GI problems in HIV infected individuals
- Discuss the management of CNS problems in HIV infected individuals
- Discuss the management of Skin problems in HIV infected individuals
- Discuss the management of Leishmaniasis co-infection and STIs
- Describe screening for different NCDs

Chapter outline:

- Introduction, General strategies to prevent OIs
- Co-trimoxazole preventive therapy
- Management of opportunistic diseases of the respiratory system
- Management of Gastrointestinal Opportunistic Diseases
- Management of Opportunistic Diseases of the Central Nervous System
- Cutaneous manifestations in HIV Positive individuals
- Prevention and management of STI
- Visceral Leishmaniasis and HIV co-infection
- Screening for Co-morbidities

Allocated time: 570 minutes

Learning activities

- Reading
- Case scenario,
- Discussion,
- Exercise,

11.1. Introduction on Opportunistic infections

Although ART is initiated for all HIV diagnosed patients as early as possible, some patients present for care and treatment at late clinical stages, therefore screening and management of OI is still critical. Opportunistic infections are the predominant causes of morbidity and mortality among HIV-infected patients. The common causative agents of the opportunistic infections in HIV are bacteria, fungi, viruses and protozoa. Main areas affected are the nervous, gastro-intestinal, respiratory systems and the skin. The level of immunity determines the occurrence and type of opportunistic infections. In general, milder infections, such as herpes zoster and other skin infections, occur early whereas serious life-threatening infections such as CNS toxoplasmosis and Cryptococci meningitis occur later with severe immune-suppression. Some life-threatening infections, such as pneumonia and TB, may occur early as well as later. When TB occurs later it is atypical, more disseminated and more extra pulmonary.

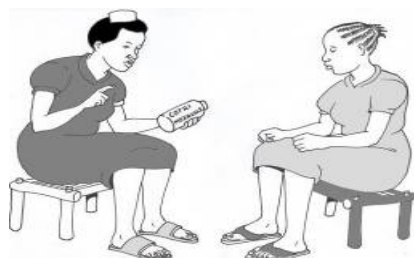
General strategies to prevent opportunistic infections are:

- Reduction of exposure (personal and environmental hygiene, safe sexual practices, ITN etc.,)
- Preventive therapies/ Chemoprophylaxis(primary/secondary),
- Immunization and
- Treatment (HAART)

For HIV chronic care and treatment, preventive therapies like Co-trimoxazole preventive therapy (CPT) and Tb preventive therapy (TPT) have significant role in reducing morbidities and mortalities among PLHIV.

11.2. Co-trimoxazole preventive therapy (CPT)

Case scenario:



Bekelu is 24 years old HIV positive woman. She was hospitalised last week with pneumonia. She is better now but concerned about getting ill again.

What is your approach?

Co-trimoxazole preventive therapy (CPT) should be implemented as an integral component of comprehensive HIV/AIDS care and treatment package. Existing recommendations cover initiation of CPT among adults, adolescents, pregnant women and children for prevention of *Pneumocystis pneumonia*, toxoplasmosis, bacterial infections & diarrhoea caused by *Isospora belli* or *Cyclospora* species, as well as benefits for malaria prophylaxis.

Table 11.1. CPT indication for primary prophylaxis

Age	Criteria for initiation	Criteria for discontinuation	Monitoring approach
HIV exposed infants	In all, starting at 6 weeks after birth irrespective of CD4 level	Until the risk of HIV transmission ends and HIV infection is excluded	Clinical at 3-monthly intervals with advice to report immediately if side effects appear.
HIV infected children < 5 year of age	In all	Continue until 5 years of age regardless of CD4% or clinical symptom	
Children ≥ 5 years of age, and Adults with HIV infection	Any WHO stage and CD4 count ≤ 350 cells/mm ³ Or WHO 3 or 4 irrespective of CD4 level	<i>Discontinued in those who are clinically stable (those individuals on ART for at least one year without any new WHO clinical stage 2, 3 or 4 events) with</i> – Evidence of immune recovery and/or viral suppression (CD4 count > 350 cells/mm ³ , with viral load suppression) or – Two consecutive CD4 count > 350 cells/mm ³ if no VL result	

Remarks:

- In addition to the criteria above the drug must be discontinued if the person develops Stevens-Johnson syndrome, severe liver disease, severe anemia, severe pancytopenia or negative HIV status in HIV exposed infants. Contraindications to co-trimoxazole preventive therapy: severe allergy to sulfa drugs (including Fansidar); severe liver disease, severe renal disease and glucose-6-phosphate dehydrogenase (G6PD) deficiency.
- Because of high prevalence of bacterial infectious diseases, co-trimoxazole should be given regardless of CD4 percentage or clinical stage for children under five years of age.
- Because of high prevalence of bacterial infections or malaria, co-trimoxazole prophylaxis may have to be continued longer for children.

Table 11.2. Dosage of Co-trimoxazole for adults, Adolescents, children and infants

Age(weight)	Suspicion(240mg/5ml)	Single strength tab (480 mg)	Double strength tab (960 mg)
Up to 6month (5KG)	2.5ml/day	1/4tab /day	-
6 months to 5 yr (5-15 Kg)	5ml /day	1/2 tab/day	-
6-14 yr (15-30 KG)	10ml/day	1 tab/day	½ tab/day
>14 yrs (>30 KG)		2tab/day	1 tab/day

Table 11.3. Adverse effects of CPT and management

Toxicity	Clinical description	Recommendation
Grade 1	Erythema , pruritus	Prescribe Anti-histamine and continue CPT & close Follow-up
Grade 2	Diffuse maculopapular rash, dry desquamation	Prescribe Anti-histamine and continue CPT & close Follow-up
Grade 3	Vesiculation, minor mucosal ulceration	STOP CPT ,manage .re-introduce after 2 weeks with observation (desensitize)
Grade 4	Exfoliative dermatitis, Steven-Johnson syndrome or erythema multiform,moist desquamation	STOP CPT NEVER RESTART CO-TRIMOXAZOL

NB: If the patient is sensitive to co-trimoxazole administration and there are no alternatives it possible to desensitize the patient under supervision (see table below). Look at photos 9Q (Stevens-Johnson syndrome) & R (Fixed dose Reaction) in the photo booklet.

Table 11.4. Desensitization protocol for mild to moderate co-trimoxazole hypersensitivity

Step	Dose	Remark
	Adult and adolescent	
Day1	2ml	Use co-trimoxazole oral suspension 40mg trimethoprine+200mg sulphamethoxazole per 5ml
Day2	4ml	
Day3	6ml	
Day4	8ml	
Day5	480mg tablet	
Day6	960mg tablet	

Dapsone as an alternative for CPT toxicity

In situations of severe allergy to co-trimoxazole or when desensitization is not successful, dapsone can be used instead of CTX if available. **It is primarily effective as prophylaxis against PCP but does not have the other prophylactic benefits of cotrimoxazole.** Dapsone will contribute to anaemia in most patients, and causes haemolytic anaemia in some patients, so patients should have a baseline Hgb before starting dapsone and Hgb monitored every 1-2 weeks for the first couple of months. Dapsone is not recommended during breastfeeding. When dapsone (as a substitute for CPT) is being used as PCP prophylaxis, it is only recommended for patients in WHO Stage 4 and/or absolute CD4 count < 200 cells/mm (Or For children CD4 % < 25% for children ≤5 years old), and should be discontinued once a patient achieves a sustained CD4 count of > 200 cells/mm (Or> 25% for children ≤5 years old) for at least 6 months.

Dose of Dapsone

Available as 25 mg and 100 mg tabs. Children: 2 mg/kg once daily (maximum dose: 100 mg) OR 4 mg/kg once weekly (maximum dose: 200 mg). Adults: 100 mg once daily.

EXERCISE 11.1

1. A 25 year-old woman comes to the clinic. She has been referred from the testing centre with a positive test for HIV. She has white patches in her mouth and weight loss. Does she need co-trimoxazole prophylaxis? How many pills will you give to the patient? What will you explain?

2. A 34 year-old man comes to the clinic asking for ART. He knew his HIV+ status two years back but was not on ART. He complains of seborrhoea and recurrent mouth ulcers. He had herpes zoster two years ago. How would you respond? What can be offered?

3. Five months old infant born to HIV positive mother brought to the clinic by his aunt. What can be done for this infant?

4. A 32 years old patient who was on ART and CPT for the last 18 months. The CD4 count at 12 and 18 months was 400 & 490 respectively. How do you manage?

Exercise 11.2. Yes/ No card sort for CPT. The facilitator will read cases and you will show a yes or No sign.

11.3. Management of Common Opportunistic diseases

11.3.1. Management of Opportunistic Diseases of the Respiratory System

Respiratory problems with HIV infection are caused by a number of infectious and non infectious etiologies, but the most common are pneumonia, tuberculosis, PCP and lymphoid interstitial pneumonitis (LIP) in children.

Case Scenario
Bisrat is a 6 years old HIV positive child who came to ART clinic complaining of fever and cough of four days duration. He also has difficulty in breathing. On examination he was having tachypnea and crepitation on right lower posterior lung field.

What are the possible causes of his symptoms?

How would you manage?

a. Bacterial pneumonia

This can occur in immuno-competent individuals but In HIV-infected patients, particularly those infected with *S. pneumonia*, incidence of bacteremia accompanying pneumonia is increased compared with that in individuals who are not HIV infected. Getting children vaccinated with PCV and pentavalent vaccines containing hemophilus influenza vaccines is very important to prevent the occurrence of pneumonia due to these bacteria. Bacterial pneumonia occurs during the whole spectrum of HIV disease, but tends to be more severe and recurrent as the CD4 counts drops significantly; in addition pneumonia can concomitantly present with sinusitis and/or bacteremia. If not treated promptly, extra pulmonary complications like empyema, meningitis, pericarditis, hepatitis and arthritis can follow. *Streptococcus pneumonia* and *Haemophilus influenza* are the most common etiologies of community acquired pneumonia. Typically the patient presents with sudden onset of cough, sputum production, chest pain, fever and/or shortness of breath.

Clinical Manifestations

Generally, the clinical manifestation is similar to HIV- uninfected individuals.

- Onset usually 3-5 days
- Cough, fever, chills rigors,
- Chest pain and pleurisy in older children and adults
- Tachypnea and decreased oxygen saturation indicate moderate to severe pneumonia
- May consider hospitalizing patients
- Egophony and pleural effusion on examination
- Grunting and chest in-drawing are severity signs in children younger than 5 years

Diagnosis:

- Predominantly clinical findings
- Radiology may assist in confirmation of diagnosis
- Leukocytosis may be seen in CBC

Treatment:

Adults:For non-severe pneumonia Amoxicillin 1000mg TID or erythromycin 500mgQID or doxycycline 100 mg BID for seven days. Avoid doxycycline in pregnancy. Alternative, Azithromycin 500 mg PO per day for three days, Clarithromycin 500 mg twice daily for seven days.

If not improving after three days and if patient is adherent to the antibiotic, review and consider switching to IV regimen, Ceftriaxone 1-2gm IV once per day plus erythromycin 500 mg oral or IV four times a day.

Children:For non-severe pneumonia, Amoxicillin 40mg/kg divided into three doses per day for seven days. In children and infants allergic to penicillin, use Erythromycin; for infants aged one month up to 2 years of age, erythromycin 125 mg four times daily for seven days; for age groups 2-8 years, erythromycin 250 mg four times per day for seven days; for age groups above 8 years, give erythromycin adult dose.

If not improving after three days and if patient is adherent to the antibiotic, review and consider switching to IV ceftriaxone 75-100mg per kg IV/IM once a day or equally divided doses twice a day for 7 to 10 days. Maximum dose is 2-4gm per day for seven days.

When the patient has presented with clinical evidence of severe pneumonia, admit for parenteral antibiotic treatment and supportive therapy or refer the patient if admission is impossible based on the following criteria.

Admission criteria for adults:

- tachypnea (RR > 30/minute),
- old age (> 65 years),
- cyanosis,
- hypotension, systolic blood pressure < 90mm Hg,
- multi-lobe involvement and
- altered mental status

Admission criteria for Children:

- Chest in-drawing,
- Grunting and
- Presence of danger signs

NB: All patients admitted to ward should be put on parenteral antibiotic treatment and supportive therapy or refer the patient if admission is impossible.

b. Pneumocystis Pneumonia

Pneumocystis pneumonia is caused by *Pneumocystis jiroveci* formerly known as *pneumocystis carinii* pneumonia (PCP), a ubiquitous organism that is classified as a fungus but also shares biologic characteristics with protozoa. The taxonomy of the organism have been changed; *pneumocystis carinii* now refers to the pneumocytes that infects rats, and *P.jiroveci* refer to the distinct species that infects humans. The abbreviation PCP is still used to designate *Pneumocystis pneumonia*. It commonly occurs when patients have significant immune suppression ($CD4 < 200 \text{ cells/mm}^3$ or $CD4 \text{ percentage} < 14\%$). The incidence of PCP has declined substantially with widespread use of prophylaxis and HART.

Clinical Manifestations

- Sub-acute onset of progressive dyspnea,
- Fever
- Non-productive cough, and
- Chest discomfort that worsens within days to weeks.
- In children highest incidence is seen between 2-6 months of age and is characterized by abrupt onset of fever, tachypnea, dyspnea and cyanosis.

Diagnosis:

- Mainly clinical when patients have severe and advanced immunosuppression ($CD4 < 200/\text{mm}^3$) in resource limited settings
- Presumptive diagnosis of PCP is based on clinical judgment and typical chest X-ray findings revealing a perihilar interstitial infiltration with tendency to spread outwards.
- Physical examination of the chest may be normal in mild cases and early presentation.
- Note that the chest X-ray can be normal in 20% of patients.
- Definitive diagnosis of PCP is based on demonstration of the organism from an induced sputum sample using special stains like Giemsa or methylene silver stains, but these tests are not routinely done in Ethiopia.

Treatment:

- Trimethoprim 15-25 mg/Kg and sulphamethoxazole 75-125mg/kg, three or four times daily for 21 days.
- If patient grows sicker, administration of oxygen is useful.
- In severely ill adults with marked respiratory distress prednisolone has to be given simultaneously;
 - o 40mg BID for the first five days then,
 - o 40 mg daily for the next 6 days and
 - o 20 mg daily until completion of intensive co-trimoxazole therapy.
- For severe cases of PCP in children provide prednisolone 2mg/kg per day for the first 7 - 10 days followed by a tapering regimen for the next 10 - 14 days.
- Secondary prophylaxis immediately after completion of the course of treatment with cotrimoxazole should be started.

Alternative regimens for mild to moderate cases of PCP include:

1. Clindamycin 600 mg qid plus primaquine 15 mg bid Or
2. Clindamycin 600 mg qid plus dapsone 100 mg daily

NB: Consider spontaneous pneumothorax in patients with sudden deterioration in clinical condition.

c. Lymphoid Interstitial Pneumonitis (LIP)

Epidemiology: LIP is one of the most common chronic lower respiratory conditions occurring in up to 25% of children with HIV/AIDS.

Clinical manifestations: Range from asymptomatic disease with isolated radiographic findings to bullous lung disease with pulmonary insufficiency. Symptomatic children present with insidious onset of tachypnea, cough, and mild to moderate hypoxemia with normal auscultatory findings or minimal rales or wheezing. Progressive disease is accompanied by digital clubbing and symptomatic hypoxemia. Associated physical findings include generalized lymphadenopathy, hepato-splenomegaly and parotid enlargement.

Diagnosis: Usually based on clinical exam findings. Diffuse bilateral reticulonodular infiltrate on X-ray with mediastinal lymphadenopathy. It is important to exclude tuberculosis and other infectious etiology.

Treatment: Provide symptomatic treatment (hydration, oxygen). Use antibiotics if there is a superimposed bacterial infection. Bronchodilators may be helpful in mild to moderate disease. Corticosteroids are usually reserved for children with significant hypoxemia and symptoms of pulmonary insufficiency. Give prednisolone 1 – 2 mg/kg/24 hrs for 6 – 8 weeks and then taper as tolerated.

d. Tuberculosis

TB is the most frequent life-threatening OI and a leading cause of death among HIV infected people. TB remains the leading cause of mortality among people living with HIV, despite substantial scale-up of ART, accounting for 30% of the AIDS-related deaths reported in 2019. TB increases HIV replication through the process of immune activation leading to increased viral load and this results in a more rapid progression of HIV disease. On the other hand, HIV increases susceptibility to be infected with M. Tuberculosis, the risk of progression to TB disease and the incidence and prevalence of TB. The lifetime risk of HIV positive individuals to develop TB is 50%, the annual risk is 10% .The World Health Organization (WHO) recent estimation indicates the risk of developing active TB disease is 18 times higher in people living with HIV (PLHIV). In Ethiopia, WHO estimation indicated that 5.3% of TB cases notified in PLHIV in 2019 . Moreover, a national level data report indicated that the prevalence of TB in PLHIV is 7.3%. Thus, it is essential for both TB and HIV control programs to synergize their joint efforts and intensify the implementation of TB/HIV collaborative activities to mitigate the dual burden of TB/HIV in populations at risk or affected by both diseases. The rationale for the integration is that tuberculosis and HIV prevention and control programs share mutual challenges of high impact of TB on HIV and vice versa. Therefore, the two programs must collaborate to provide better service for the co-infected patients.

Nationally Recommended TB/HIV Collaborative Activities

A. Strengthen the Mechanisms for integrated TB and HIV services delivery

- Strengthen the coordination mechanism for integrated TB/HIV services at all levels;

- Conduct surveillance to determine HIV burden among TB patients and TB burden among HIV patients;
- Carry out joint TB/HIV planning for integrated TB and HIV services delivery;
- Conduct monitoring and evaluation of collaborative TB/HIV activities.

B. The three I's (**I**ntensive case finding, **T**B Preventive Therapy and **I**nfection control) for HIV/TB co-infection.

- Intensify TB case finding and ensure quality TB treatment;
- Initiate TB prevention with earlier initiation of ART and Isoniazid preventive therapy (IPT) which is currently named as TB Preventive Therapy (TPT);
- Ensure Tuberculosis infection control in healthcare and congregate settings.

C. Decrease the burden of HIV among TB patients.

- Provide HIV testing and counseling to presumptive and confirmed TB patients;
- Provide HIV prevention services for presumptive and confirmed TB patients;
- Provide co-trimoxazole preventive therapy for HIV positive TB patients;
- Ensure HIV/AIDS prevention, treatment and care for HIV positive TB patients;
- Provide antiretroviral therapy for HIV positive TB patients.

The 3Is intervention for TB infection control is:

I. Intensified TB case finding and ensure quality TB treatment

Tuberculosis case finding should be intensified in all HIV testing and counseling services for HIV positive clients by using a set of simple questions for early identification of TB suspects. HIV positive clients coming through HCT services should be informed about the advantages of being screened for TB. Once informed about the risk of developing active TB, they should undergo screening for it. **Figure 11.1. Algorithm for initiating TPT in adults and adolescents ≥15 years living with HIV**

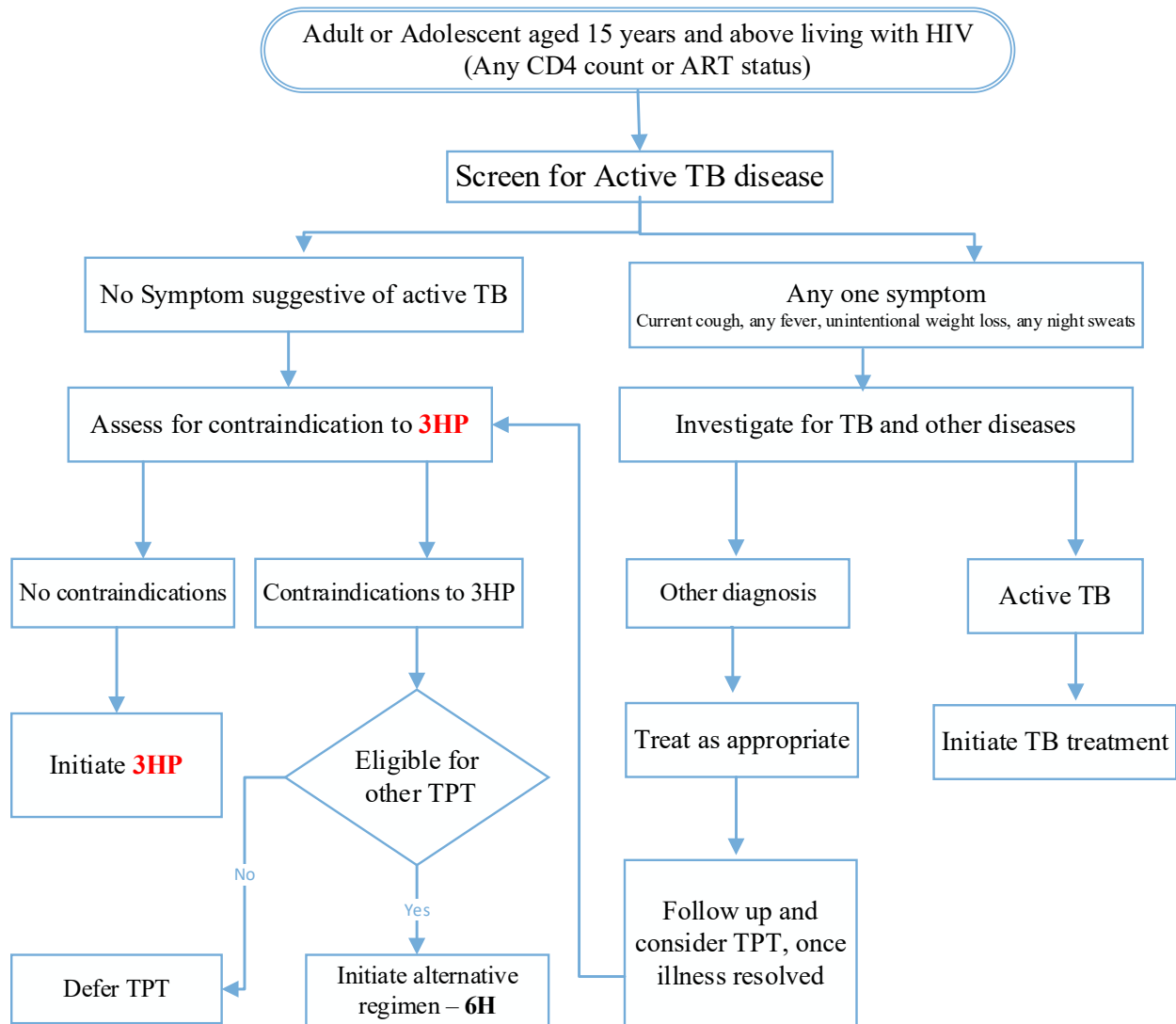


Figure 11.2: Algorithm for Tb screening and TPT for Children <15 years living with HIV and without household TB exposure

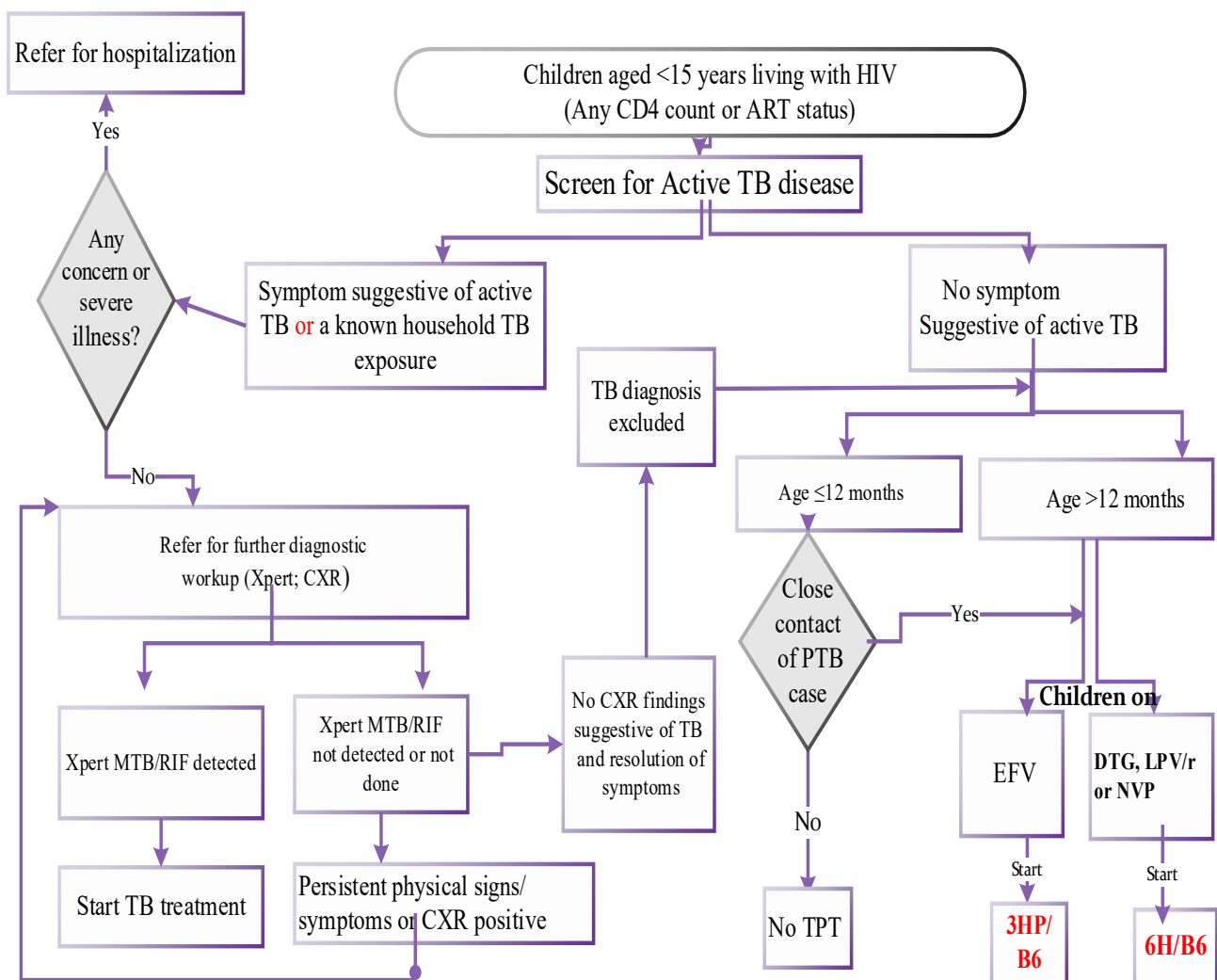
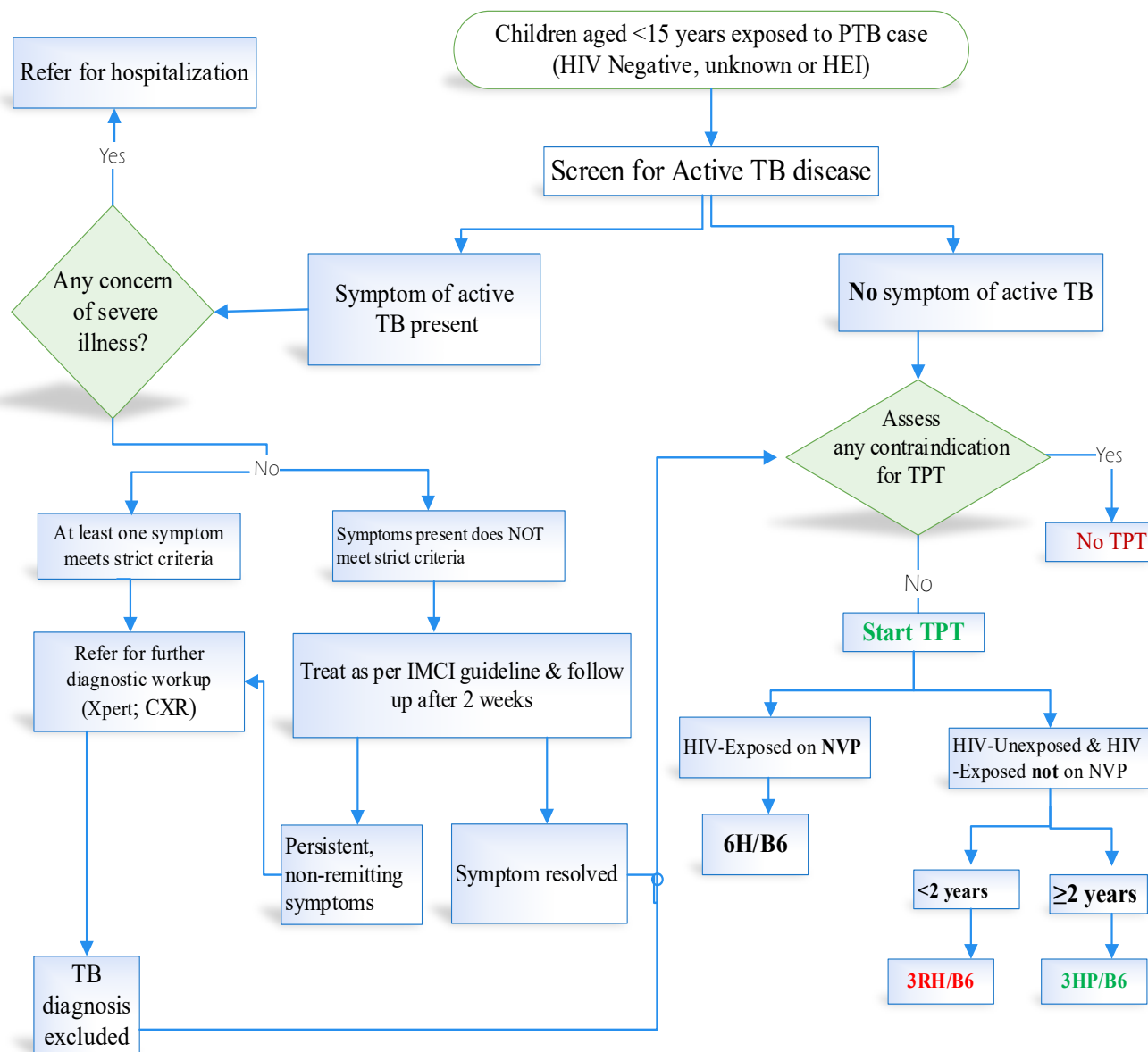


Figure 11.3: Algorithm for Tb screening and TPT HIV-exposed Children and HIV negative children and adolescents <15 years of age with household exposure to PTB case



Clinical manifestations of tuberculosis

- Cough, Weight loss, Fatigue, Malaise, Fever , Night sweats , Loss of appetite
- Poor weight gain and reduced playfulness in children
- Close contact with a known TB or chronically coughing patient –especially in young children,

Risk factors for TB disease

- Infection with HIV,
- Recent TB infection (<1 year),
- Malnutrition,
- Co-morbid conditions (Diabetes, malignancy etc)
- Age less than five years

Symptom based TB screening for all PLHIV

All clients living with HIV infection should be screened for active TB with **any one** of the following symptoms:

Adult and adolescents:

- Current cough
- Fever
- Weight loss or poor weight gain
- Night sweats

Children less than 10 years:

- **Current cough,**
- **Fever,**
- **Poor weight gain**
- **Close contact with a person with TB disease**

Clients with any one of the above symptoms should be investigated for active TB

Diagnosis of TB in HIV infected people

Diagnosis of TB is challenging in HIV positive individuals, especially when the stage of the disease is advanced. Standard TB diagnostic approaches and clinical algorithms should be followed to guide the diagnosis of TB in PLHIV. Although there have been significant improvements in the diagnosis and treatment of TB, there are still limitations in early detection and treatment of all forms of TB cases among PLHIV.

The conventional sputum microscopy is the cheapest and fastest method that is used to diagnose TB since 1882. However, the sensitivity of sputum smear microscopy for the diagnosis of TB is low in PLHIV due to poor quality sputum production and low bacillary concentration. Particularly in severely immunocompromised patients, the sensitivity of sputum smear microscopy is significantly reduced. Other challenges that make TB diagnosis difficult include non-specific clinical presentation of TB that is attributable to high prevalence of extra-pulmonary and disseminated forms of TB in individuals with advanced HIV disease.

Currently, rapid, and more accurate molecular technologies have been developed and available to diagnose TB in PLHIV. The diagnostic performance of these technologies has not been fully evaluated for the diagnosis of TB in PLHIV. Although rapid and accurate molecular assays have significantly reduced the gap in TB case detection among PLHIV, their accessibility continued to be a challenge, due to the infrastructure required and the cost of procurement and maintenance.

Eligible PLHIV for LF_LAM Test

WHO 2019 policy update on LF-LAM recommends the currently available urinary LAM assays have suboptimal sensitivity and are therefore not suitable as general diagnostic tests for TB. However, unlike traditional diagnostic methods, they demonstrate improved sensitivity for the diagnosis of TB among individuals coinfecting with HIV. The estimated sensitivity is even greater in patients with low CD4 cell counts.

Therefore, HIV-positive adults, adolescents and children with the following criteria are eligible for Urine LF_LAM test to assist in the diagnosis of active TB:

In inpatient settings

- With signs and symptoms of TB (pulmonary and/or extra pulmonary)
 - With advanced HIV disease^b or who are seriously ill ^a
 - Irrespective of signs and symptoms of TB, with a CD4 cell count of less than 200 cells/mm³
- a. “**Seriously ill**” is defined based on four danger signs: respiratory rate of more than 30/minute, temperature of more than 39 ° heart rate of more than 120/minute and unable to walk unaided.

- b. For adults, adolescents, and children aged 5 years or more, “**advanced HIV disease**” is defined as a CD4 cell count of less than 200 cells/mm³ or a WHO clinical stage 3 or 4 event at presentation for care. All children with HIV who are aged under 5 years should be considered as having advanced disease.

In outpatient settings

- With signs and symptoms of TB (pulmonary and/or extrapulmonary) or seriously ill
- Irrespective of signs and symptoms of TB and with a CD4 cell count of less than 100 cells/mm³

In outpatient settings, WHO recommends against using LF-LAM to assist in the diagnosis of active TB in HIV-positive adults, adolescents, and children:

- Without assessing TB symptoms
- Without TB symptoms and unknown CD4 cell count or without TB symptoms and CD4 cell count greater than or equal to 200 cells/mm³

Without TB symptoms and with a CD4 cell count of 100–200 cells/mm³

Remarks

- a. All patients with signs and symptoms of pulmonary TB who can produce sputum should have as their initial diagnostic test at least one sputum specimen submitted for Xpert® MTB/RIF (Ultra) assay. This also includes children and adolescents living with HIV who can provide a sputum sample.
- b. LF-LAM should be used as an add-on to clinical judgement in combination with other tests. It should not be used as a replacement or triage test. More details are given in Algorithms for LF-LAM use (Annex XX).

Note for using LF_LAM test for children less than 5 years:

Latest WHO guideline consider all HIV infected children age <5 years old to follow as advanced HIV disease , besides nationally we have a limited number of under 5 children on ART. Hence setting other criteria may limit the Urine LF_LAM utilization. Therefore LF_LAM test should be conducted for all HIV infected children under 5 years at least once as per below criteria:

- To offer LF_LAM for all HIV infected <5 years children backlog/currently on treatment once irrespective of clinical stage or CD count/CD 4 percentage
- All newly enrolled under 5 HIV infected children (at Enrollment) irrespective of clinical stage or CD4 count/CD4 percentage
- Once cleared the backlog, offer LF_LAM at any time when children presented with TB symptoms, or serious illness or advanced HIV stage or CD4 percentage less than 15%

Figure 11.4. Algorithm for LF-LAM testing to aid in the diagnosis of TB among PLHIV in outpatient settings

Adults, adolescents and children LHIV including: 1. All newly diagnosed HIV patients who are ART naive 2. HIV patients returning for care following a treatment interruption 3. HIV patients receiving ART regimen that is failing 4. HIV Patients presenting at the clinic and unwell

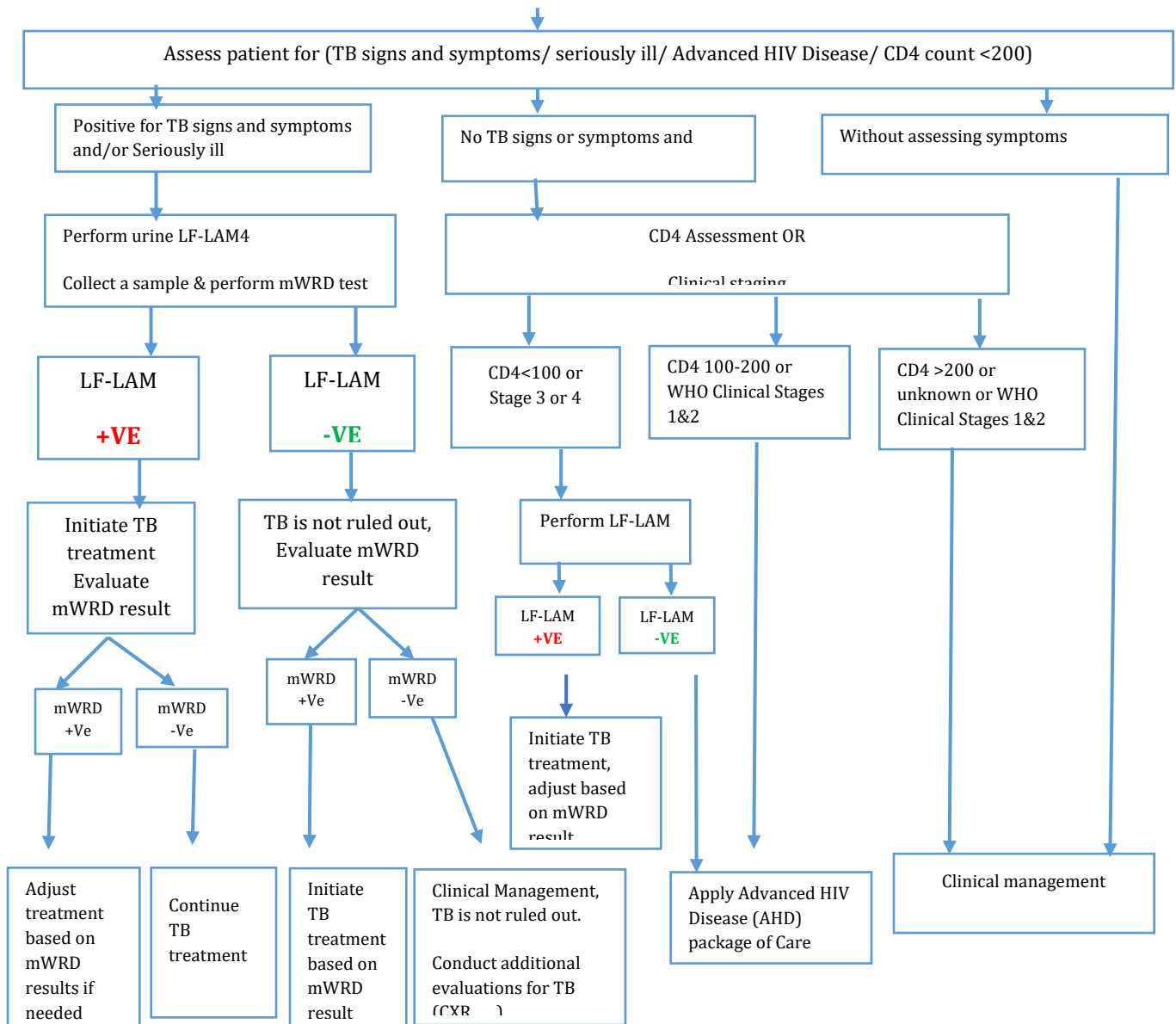
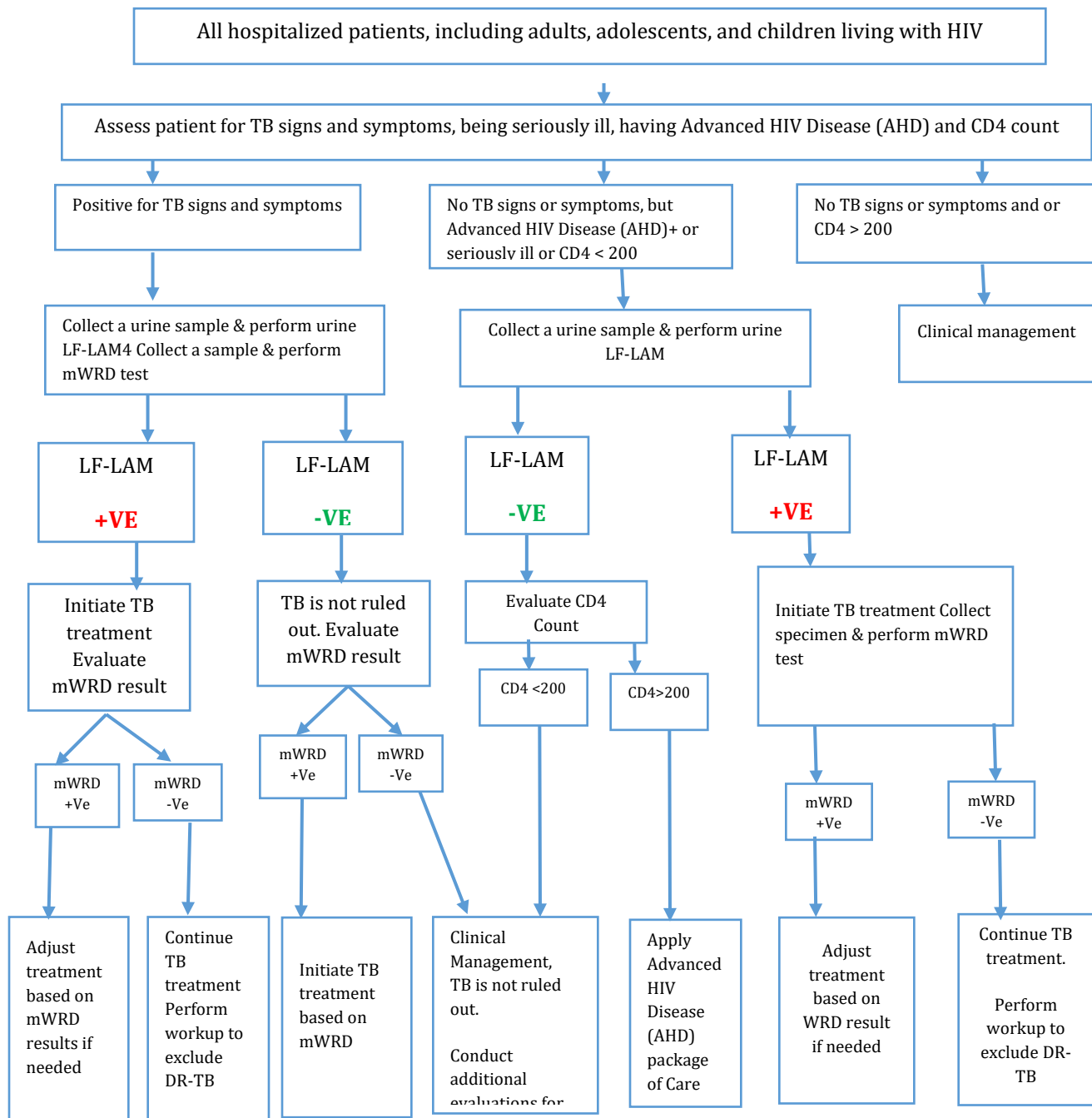


Figure 11.5. Algorithm for LF-LAM testing to aid in the diagnosis of TB among PLHIV in inpatient settings



To minimize the challenges that are associated to sputum based diagnostic tests, urine based rapid TB diagnostic tests are recommended to detect TB in PLHIV with Advanced HIV Disease. Alere lipoarabinomannan (LAM) assay is one of the urine based rapid diagnostic tests for TB detection in PLHIV. Evidence suggests the importance of LAM in the detection of TB at high TB/HIV burden settings. It is recommended for use as a simple point-of-care test to assist TB diagnosis in HIV positive adult patients with signs and symptoms of TB and with low CD4 cell counts. Based on the generalization of the data from adults, this recommendation is also used in children living with HIV who have signs and symptoms of TB. However, there is limited evidence on the specificity of LAM test in children. LAM test could decrease mortality through quicker diagnosis and early treatment commencement among PLHIV and severely sick.

XPert MTB/RIF Test (GeneXpert Test): The GeneXpert MTB/RIF system is a fully automated nested real time PCR system, which detects MTB complex DNA in sputum and other sample types (i.e pleural, lymph node aspirate or tissue, CSF, gastric fluid and tissue other than lymph node). It simultaneously identifies mutations in the rpoB gene, which are associated with rifampicin resistance. The assay detects M.TB and rifampicin resistance; conferring mutations using three specific primers and five unique molecular probes. It provides results on the same day and has minimal bio-safety requirements and training.

Xpert MTB/RIF test (GeneXpert Test) is recommended as an initial diagnostic test for all presumptive TB cases (individuals with TB symptoms) among HIV infected people.

- Xpert MTB/RIF should be used rather than conventional microscopy, culture and drug susceptibility testing (DST) as the initial diagnostic test in adults and children suspected of having HIV-associated TB or multidrug resistant TB.
- Xpert MTB/RIF should be used in preference to conventional microscopy and culture as the initial diagnostic test for cerebrospinal fluid specimens from patients suspected of having TB meningitis.
- Xpert MTB/RIF may be used as a replacement test for usual practice (including conventional microscopy, culture or histopathology) for testing specific non-respiratory specimens (lymph nodes and other tissues) from patients suspected of having extra pulmonary TB.

AFB microscopy: AFB Microscopy is indicated for HIV infected presumptive TB cases when access to XPert MTB/RIF test is limited. The recommendation for AFB microscopy is 2 spot tests with in 1 hour.

Chest radiography: Chest X-ray plays a significant role in shortening delays in diagnosis of TB in PLHIV. It can also be an important entry point to diagnose non-tubercular chest diseases, which are common among HIV positives.

Sputum culture: sputum culture is the gold standard for the diagnosis of tuberculosis in general. In patients with XPert negative results, sputum culture may be indicated as part of the diagnostic procedure for people living with HIV if clinical suspicion persists.

Diagnosis of extra-pulmonary tuberculosis in HIV positive

Extra-pulmonary tuberculosis is more HIV-related than pulmonary tuberculosis. The accurate diagnosis of extra-pulmonary tuberculosis is complex and difficult, particularly in peripheral

health facilities with limited diagnostic capacity. Therefore, it is important for healthcare workers to have high-index of suspicion and critically evaluate through clinical algorithms. For other sites, do organ specific investigations.

Histopathology: Pathology can play a complementary role in confirming the diagnosis of EPTB, such as tuberculous lymphadenitis, pleural effusion etc.

- ✓ Fine needle aspiration of the lymph nodes
- ✓ Effusions of the serous membranes can be aspirated
- ✓ Tissue biopsy (serous membranes, skin, endometrium and bronchial, pleural, gastric or liver tissue) can be taken, with or without surgery;

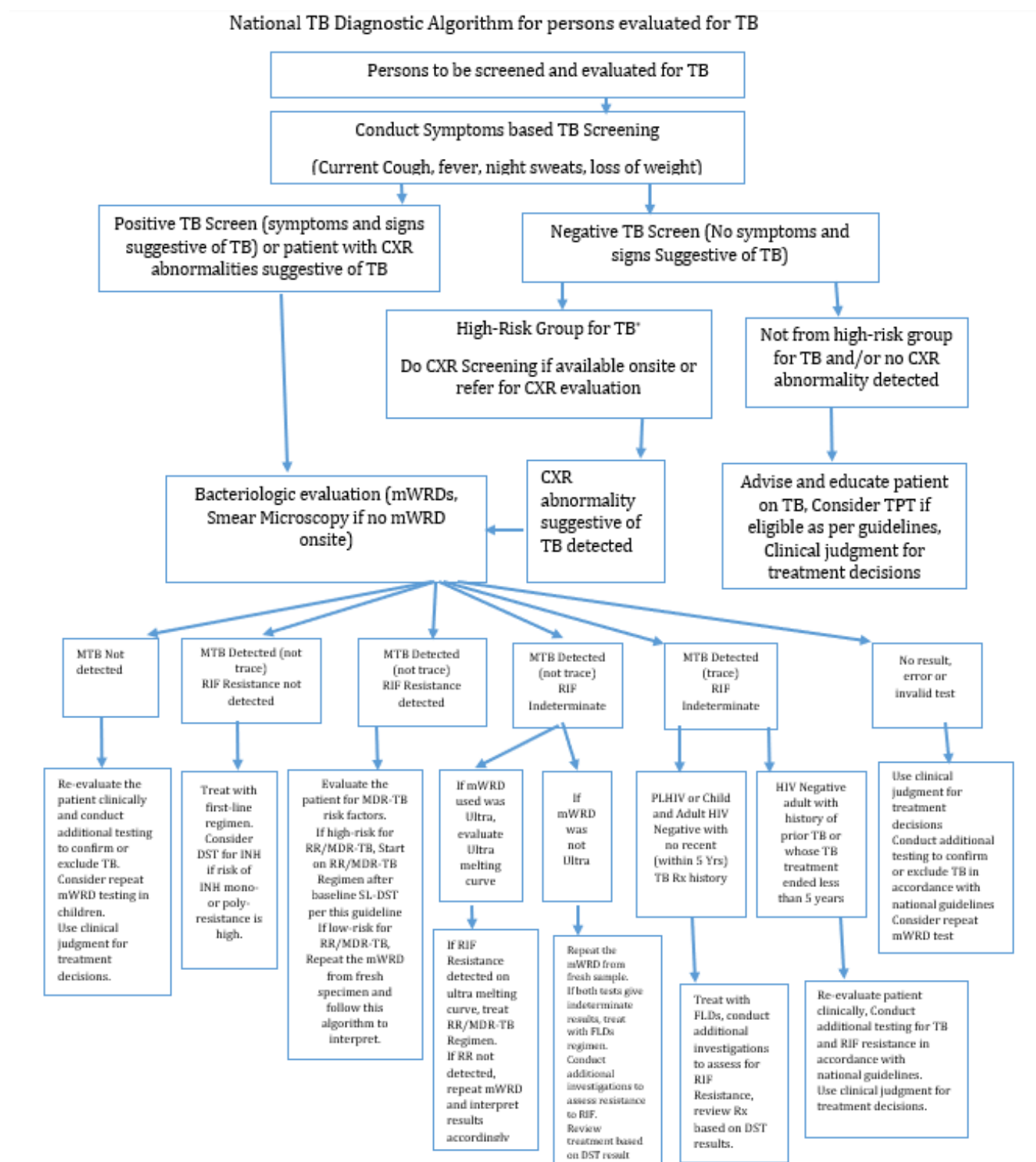


Figure 11.6. National TB and MDR TB diagnostic algorithm at health facility level

Standard TB Case Definition based on History of previous TB treatment

New patients: patients that have never been treated for TB or have taken anti-TB drugs for less than 1 month.

Relapse patients: patients that have previously been treated for TB were declared *cured* or *treatment completed* at the end of their most recent course of treatment, and is now diagnosed with a recurrent episode of TB (either a true relapse or a reinfection).

Treatment after failure: patients are those who have previously been treated for TB and whose *treatment failed* (: *smear positive results after fifth month during treatment*) at the end of their most recent course of treatment.

Treatment after loss to follow-up: patients have previously been treated for TB and were declared *lost to follow-up* at the end of their most recent course of treatment and is now diagnosed with TB.

Others: patients who have previously been treated for TB but whose outcome after their most recent course of treatment is unknown or undocumented, or patients that do not fit into any of the categories listed above.

Transfer in: A patient who is accepted to continue treatment in another reporting unit after starting treatment in another reporting unit.

Standard first line Anti-TB regimens

Table 11.5. Showing the standard first line Anti-TB treatment in Ethiopia

TB Patient type	Standard Regimen		Patient registration groups receiving the regimen
	Intensive phase	Continuation phase	
Drugsusceptible TB case (New and Previously treated)	2 (RHZE)	4 (RH)	▪ New TB patients ▪ Relapse ▪ Treatment after LTFU ▪ Treatment after failure of New regimen ▪ Others
	2(RHZE)	10 (RH)	▪ New patients with CNS TB(meningitis, tuberculoma) ▪ New TB patients involving vertebra and Osteoarticular space

* previously treated TB patients are treated with the same regimen as for New TB patients for six months after ruling out the presence of resistance at least for Rifampicin with rapid DST.

Table 11.6. Timing of ART initiation for adults and children with TB

New HIV positive Patients with signs and symptoms suggesting (presumptive tuberculosis cases)	Patients with tuberculosis found to be HIV positive	HIV positive patients taking ART diagnosed with TB
• Rapid ART initiation should be offered to all people living with HIV following a confirmed HIV diagnosis and clinical assessment and to people living with HIV with signs and symptoms suggesting TB. Except for central nervous system disease (meningitis), initiate ART while rapidly investigating for TB, with close follow-up within seven days to initiate TB treatment if TB is confirmed.	• ART should be started in all TB patients, including those with drug-resistant TB, irrespective of the CD4 count. • ART should be started as soon as possible within two weeks of initiating TB treatment, regardless of CD4 count, among all people living with HIV. (adult, adolescent and children), except when signs and symptoms of meningitis are present. • ART should be delayed at least four weeks (and initiated within eight weeks) after treatment for TB meningitis is initiated. Corticosteroids should be considered adjuvant treatment for TB meningitis. (Note. For people living with HIV with TB meningitis, immediate ART is associated with more severe adverse events compared with initiating ART two months after	• Start anti-TB • Modify ART regimen to avoid drug-drug interaction • Evaluate for treatment failure

	the start of TB treatment) • DTG should be used as the preferred drug in patients starting ART while on Anti-tuberculosis treatment but dose of DTG should be doubled (50 mg BID). • When second line is initiated LPV/r is preferable. Note that the dose of the ritonavir should be doubled to counter the effect of Rifampicin on the Protease inhibitors.	
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II. Tuberculosis Preventive Therapy

Latent tuberculosis infection (LTBI) is the state of persistent immune response to stimulation of mycobacterium tuberculosis antigens without evidence of clinically apparent active tuberculosis (TB).

Treatment of LTBI to prevent progression to active disease is one of the global key strategies to ending the TB epidemic. Increasingly, eligible targets and treatment options are expanding, with significant implications in the programmatic management of LTBI. TPT is the use of Isoniazid, rifapentine or other medications to sterilize latent TB infection. Screening for exclusion of active TB in HIV infected persons is the single most important step that should precede the decision to initiate TPT. Concerns regarding the development of INH resistance should not be a barrier to providing TPT.

For adults and adolescents living with HIV, TPT should be provided to those who are unlikely to have active TB based on appropriate clinical algorithm, irrespective of CD4 count, ART status, pregnancy status or history of treatment for prior episode of TB before three years. Children and infants less than 1 year of age should be provided preventive therapy only if they have a history of household contacts with pulmonary TB and active TB in the child has been excluded in investigation.

Clinicians should follow the algorithms for initiation and selection of regimen for the specific population groups eligible for TPT as indicated in section below.

Regimen and dosage for treatment of LTBI

Table 11.6.2. Regimen for treatment of latent TB infection.

Population group	Age group and ART regimen	Selection of TPT regimen	
		Preferred regimen	Alternative regimen
Persons living with HIV	Adults, adolescents, children, and infants of all ages taking a PI-based ART regimen	Daily isoniazid preventive treatment for 6 months (6H)	
	Children and adolescents aged <15 years taking a DTG-based ART regimen	Daily isoniazid preventive treatment for 6 months (6H)	
	Children and adolescents aged <15 years taking EFV-based ART regimen	Weekly isoniazid Plus rifapentine for 3 months (3HP*).	- Daily isoniazid preventive treatment for 6 months (6H) - Daily rifampicin Plus isoniazid for 3 months (3RH).
	Adolescents and adults living with HIV (≥ 15 years of age) taking non-PI based ART regimen	Weekly isoniazid Plus rifapentine for 3 months (3HP).	- Daily isoniazid preventive treatment for 6 months (6H)
HIV-negative persons	Infants and children <2 years of age)	Daily rifampicin Plus isoniazid for 3 months (3RH).	- Daily isoniazid preventive treatment for 6 months (6H)
	Eligible adolescents and children aged between ≥ 2 -14 years (refer to eligibility criteria specified above)	Weekly isoniazid Plus rifapentine for 3 months (3HP).	- Daily isoniazid preventive treatment for 6 months (6H) - 3RH will be used as alternative to 3HP
*3HP: should be taken with food to prevent GI upset; if patients are unable to swallow tablets (due to age or illness), the tablets can be crushed and added to a small amount of semi-solid food			

Drug-drug interactions with antiretroviral medicines and Management of adverse events (AEs)

Drug-drug interactions should be cautiously considered in using rifampicin and rifapentine in persons who are receiving antiretroviral treatment (ART). Both rifampicin and rifapentine should not be administered in persons who are receiving nevirapine or Protease Inhibitor (PI) based regimen. A three-month weekly rifapentine plus isoniazid can safely be used in patients receiving efavirenz-based antiretroviral regimen without the need for dose adjustment. Studies show that co-administration of rifapentine with raltegravir and dolutegravir based regimen is both safe and well tolerated in HIV infected adults and adolescents aged ≥ 15 and do not need any dose adjustment. TPT related adverse events (AEs), identified during treatment should be properly monitored, managed and reported per national recommendation, using health facilities reporting systems.

Individuals receiving TPT do not have active disease and therefore their risk for adverse events during treatment must be minimized. Moreover, the regimen can be withheld while an AEs is assessed, and there is time for the regimen to be recommenced and completed if safe to do so. Overall, 3HP is a safe and effective treatment for latent TB infection. Clinically significant drug reactions are rarely experienced by patients receiving 3HP, and even less commonly require discontinuation of treatment. Severe reactions are particularly rare. Nonetheless, healthcare workers should be familiar with the important drug reactions so that they can recognize rare occurrences and manage them appropriately.

Table 11.6.3.: Monitoring of TPT

Drug interactions	The most common drug reactions with 3HP are: • Liver toxicity (less common than for 6H) • Flu-like reactions (more common than for 6H) Drug reactions are usually mild and self-limiting, but occasionally they can be severe. Children usually tolerate 3HP very well and have much lower rates of drug reactions.
Baseline assessment	Active TB must be ruled out before commencing TPT. 3HP is currently not recommended in: • Pregnancy

	• Age <2years Information on baseline liver function is important in the following: • HIV+ (done as part of ART assessment but not mandatory) • Daily alcohol consumption • Liver disorders including viral hepatitis • Postpartum period (≤ 3 months after delivery) • Concomitant use of other hepatotoxic substances Individuals at higher risk of peripheral neuropathy should be offered vitamin B6 (pyridoxine) supplementation with 3HP; if B6 is not available, this should not delay starting a course of 3HP
Counselling for AEs	Red/orange discoloration of urine and other body fluids while receiving 3HP is normal and completely harmless. If patients experience any symptoms concerning for an AE: • Do not take any further doses of 3HP • Contact a healthcare provider for advice • Only continue receiving 3HP if advised to do so by a healthcare provider Individuals should be alert to the following symptoms: • Weakness, fatigue, loss of appetite, persistent nausea (early symptoms of hepatotoxicity) • Flu-like, or other acute symptoms appearing shortly after receiving a dose of 3HP • Symptoms of active TB (appendix1)

Adherence support and monitoring

Clients receiving TPT should be supported at home level, either by adherence case managers, HEWs or family supporter. They should have monthly scheduled follow up that is coordinated with other services, such as HIV care, child and maternal health services, as necessary. At each follow-up visit, the healthcare provider should:

- Educate clients and their families about the benefit of TPT, potential side effects, and importance of returning back to a health facility for new symptom/sign or any concern.
- Evaluate and counsel clients on importance of treatment adherence and completion. Case managers, adherence supporters, and HIV care and child health service providers should

support treatment adherence and monitoring, as part of comprehensive HIV care and child health services.

- Evaluate and routinely monitor drug side effects, including hepatitis, peripheral neuropathy or rash. Stop TPT if serious adverse effect is identified and manage the patient.
- Evaluate for signs and symptoms of active TB, other opportunistic infections (OIs) or diseases.
- Stop TPT, if active TB is diagnosed, which requires immediate start of anti-TB treatment

Contraindications for TPT

Individuals with any one or more of the following conditions should not receive TPT:

- Symptoms compatible with tuberculosis even if the diagnosis is not yet confirmed;
- Active hepatitis (chronic or acute);
- Regular and heavy alcohol consumption;
- Prior allergy or intolerance to medicine(s) in the regimen; and
- Symptoms of peripheral neuropathy

In addition, rifapentine is not currently indicated for children below 2 years, PLHIV receiving PI or NVP based ART regimens, and pregnant and breastfeeding women.

Patient management after treatment interruption:

TPT is said to be completed when a person completed a full course of treatment within the specified period. Completion of isoniazid-based TPT is defined as “completed” if a patient completed the full course of therapy within nine months period (i.e. the six months of doses should be finished in nine months). If a patient has interrupted TPT without medical personnel advice, the client should be traced by health extension workers (HEW) or through the index person or other family member who is already enrolled in care, and treatment must be resumed. It is important to identify barriers and support treatment adherence. However, concerns about

adherence should not be a barrier to the use of TPT. If the client discontinues isoniazid-based TPT for a period of less than three months: Resume the same course by adding for the missed doses at the end. If the client discontinues treatment for a period of more than three months, then re-initiate new course of treatment.

Completion for 3HP is defined when the patient took at least 11 doses of treatment in 16 weeks.

TB preventive treatment in Newborns

Once a pregnant woman with TB has been on treatment for TB for several weeks before delivery, it is less likely that the baby will become infected. The risk is highest if a mother is diagnosed at the time of delivery or shortly thereafter. If a pregnant woman is found to have pulmonary TB shortly before delivery, then the baby, and if possible, the placenta, should be investigated for evidence of congenital TB infection. If the result is positive, the newborn should be treated accordingly. If asymptomatic, the newborn should receive TB preventive treatment followed by BCG immunization. Breastfeeding can be safely continued during this period.

III. TB Infection control

People living with HIV are at high risk of acquiring TB in health care facilities and congregate settings. Each health care facility should have a TB infection control plan for the facility that includes administrative, environmental and personal protection measures to reduce the transmission of TB in health care and congregate settings and surveillance of TB disease among workers. Health care workers with HIV should be provided with ART and IPT if they are eligible.

Summary of recommendations for key actions for infection control:

- a. Administrative (facility-level infection control committee and protocols)**
 - A triage system to identify people suspected of having TB
 - Separate people with suspected or confirmed TB
 - Cough etiquette and respiratory hygiene
- b. Health workers and care providers**
 - Surveillance and information
 - Package of care for HIV-positive workers (ART and isoniazid preventive therapy)

- Protective equipment (particulate respirator masks that meet or exceed N95 standards)
- Relocation for health care workers living with HIV to a lower-risk area

c. Environmental

- Ventilation (mechanical)
- Ventilation (natural)

d. Personal

- Spend as much time as possible outside
- Cough etiquette
- Sleep alone while smear-positive

Avoid congregate settings and public transport while smear-positive.

Exercise 11.2

Read each cases and provide the right diagnosis and categorize documentation

	Case	Categorize the diagnosis
1	Patient had cough for 4 weeks. On the last visit sputum was sent. Two sputum samples are positive.	
2	PLHIV had cough for 3 weeks. The LF_LAM test done and positive however Xpert is negative	
3	Patient has had fever and is losing weight. You can find no cause for the weight loss other than HIV infection.	
4	Patient has no signs or symptoms of TB and is not taking TPT.	
5	Patient has completed three months of TB treatment.	
6	Patient returns for follow-up. Two weeks ago he complained of cough with yellow sputum. You gave him amoxicillin, and he is no longer coughing. Both sputums are negative.	
7	Patient was treated after being referred to the hospital for a large neck node and is now on TB treatment.	
8	Patient had TB last year and finished treatment 5 months ago. She has no cough, night sweats, or nodes.	
9	Patient has been put on INH prophylaxis and now is in the 3 rd	

	month of taking the prophylaxis. He continues to have no symptoms/signs of TB.	
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11.3.2. Management of Gastrointestinal Opportunistic Diseases

The GI OI diseases commonly manifest with diarrhea, nausea and vomiting, dysphagia and odynophagia among others. There are a number of opportunistic and pathogenic organisms causing GI disease in patients infected with HIV most common ones being isospora belli, cryptosporidium, shigella and salmonella, CMV etc. A scenario of multiple concurrent GI infections is fairly common. The general principle of managing GI OI is identifying and treating the specific offending agent providing supportive care to monitor situations such as fluid loss. A number of drugs can cause adverse effects that present with clinical manifestations referable to gastrointestinal symptoms which are similar to OIs of the GI, posing challenges in differential diagnosis.

a. Dysphagia and odynophagia

Dysphagia (difficulty in swallowing) and odynophagia (painful swallowing) are symptoms of oesophagitis occurring at advanced stages of AIDS.

They are usually caused by candida, HSV, CMV, and aphthous ulcers. In addition to be a sign of severe immunodeficiency, esophagitis seriously impairs the patient's nutritional status. Therefore, prompt diagnosis and treatment are mandatory to avert nutritional complications and inability to swallow prescribed medications. Children present with reluctance to eat, excessive salivation, or crying while feeding. If thrush is associated with dysphagia, odynophagia, and/or retrosternal pain, consider oesophageal candidiasis but this can also occur in the absence of oral thrush. Thrush or oropharyngeal candida is characterized by white, painless, plaque-like lesions on the buccal surface and/or tongue.

Diagnosis: is frequently made on clinical grounds, but when facilities are available, upper GI endoscopy with or without biopsy or contrast imaging may be done.

Treatment: Dysphagia and/or odynophagia are treated as oesophageal candida on clinical grounds, in particular when oropharyngeal candida is present. Patients are empirically treated

with Fluconazole in presumptive esophageal candida. If the response is unsatisfactory they are referred or investigated if facilities are available, to rule out other causes.

Drug of choice: Fluconazole 200 mg (6mg/kg/day in children) PO daily for 14-21 days alternatively, ketoconazole 200 mg (3-6mg/kg/day daily in children) twice daily for 4 weeks.

Risk of recurrence after completing treatment may be high. If the patient is on ART, s/he should be investigated for treatment failure. Take necessary precautions regarding drug interactions especially with ketoconazole. Patients may need hospital admission for supportive care till the esophageal symptoms improves and necessary long term treatments are started. If diagnosis suggests HSV esophagitis use acyclovir 400mg po five times for 14 to 21 days.

b. Diarrhea

Diarrhoea is defined as passing more than three loose or watery stools/day. It may be acute or chronic, persistent or intermittent. **Diarrhoea is among the most frequent symptoms of HIV disease.** Delay in treatment can result in fluid loss and hemodynamic instability. Chronic diarrhoea may also lead to nutritional deficiencies and wasting. Diarrhoea is caused by opportunistic or pathogenic organisms, such as viruses (including HIV), bacteria, protozoa, fungi, helminthes, non-infectious causes and drugs. (Diarrhoea occurs as an adverse reaction to a number of drugs).

Check the duration, volume, frequency, consistency of stools as well as any history of abdominal pain, tenesmus, nausea, vomiting, and presence of constitutional symptoms such as fever. Thorough physical examination is necessary to find out the state of hydration and the status of HIV disease.

Laboratory evaluation: Stool microscopy including modified acid fast stain
Stool culture when indicated (optional)

Management:

The most important first step is correction of fluid loss. Depending on the severity of dehydration, ORS or IV fluid therapy can be given. Patients with severe dehydration are admitted for intravenous fluid administration. In children zinc 20mg per day for 10-14 days (10mg per day for infants under 6months of age) should be added. If specific enteric pathogen is identified or strongly suspected on clinical grounds, it should be treated accordingly. Patients with bloody diarrhoea but repeatedly negative stool results: empirical treatment with

ciprofloxacin or norfloxacin (co-trimoxazole in children) can be given, especially when patient has constitutional symptoms such as fever.

Symptomatic treatment: In adults use anti-diarrhoeal agents Loperamide 4mg stat then 2mg after each bowel motion or Diphenoxylate 5mg QID. Necessary caution should be taken to avoid anti-diarrhoeal agents in bacterial or parasitic infectious colitis or enteritis, since toxic mega colon may occur. Patients with chronic diarrhoea develop nutritional deficiencies of variable severity; therefore proper nutritional assessment and support are helpful.

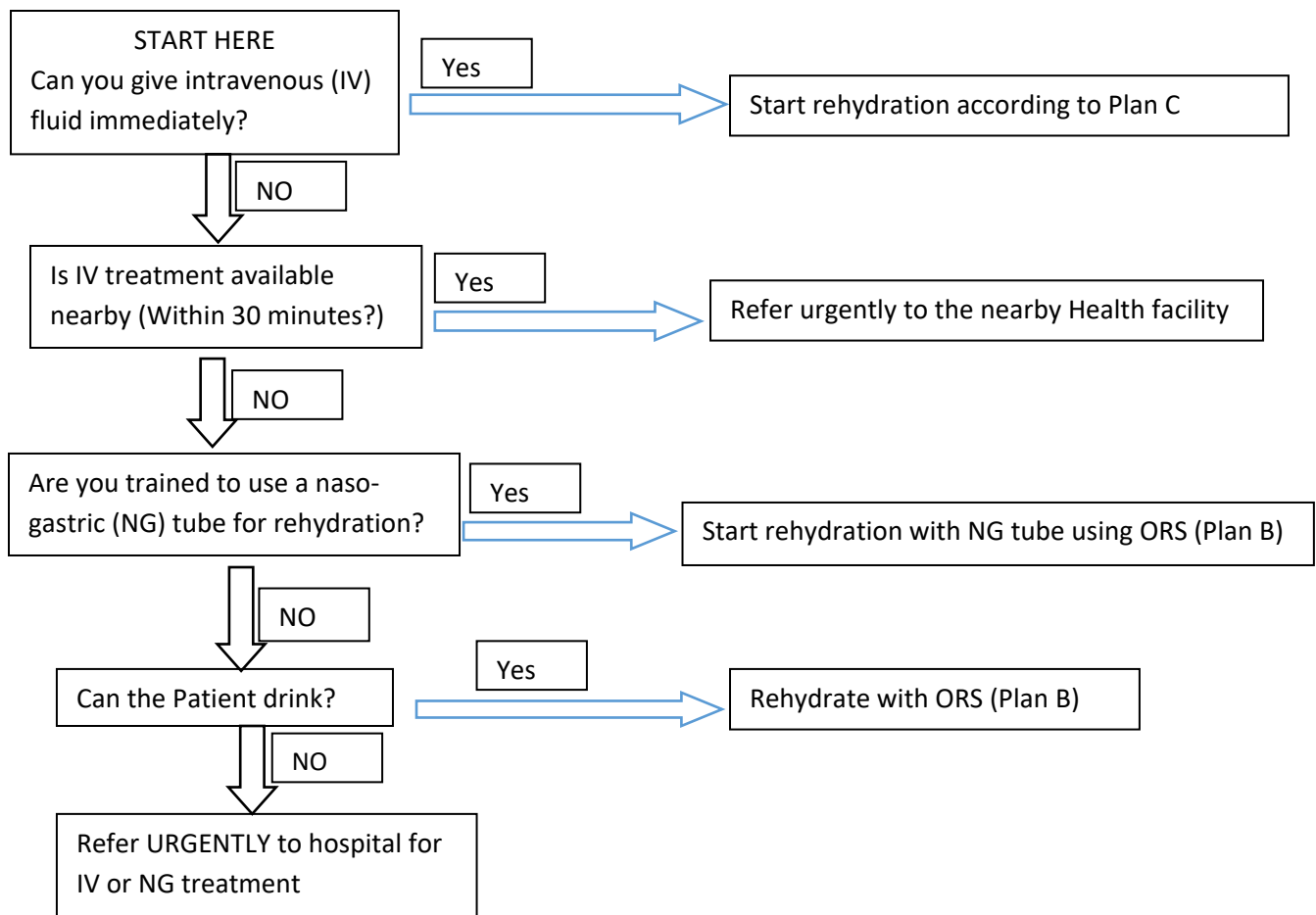


Figure 11.7. Decision algorithm for management of severe dehydration

Table 11.8a. Management of dehydration in children

Plan A: Treat Diarrhea at Home (For no Dehydration)	Plan B: Treat Some Dehydration with ORS	Plan C: Treat Severe Dehydration Quickly(See algorithm below)																								
Counsel the mother on the 4 Rules of Home Treatment: Give Extra Fluids, Give Zinc Supplements, Continue Feeding, When to Return 1. GIVE EXTRA FLUIDS (as much as the child will take) ☐ TELL THE MOTHER: - Breastfeed frequently and for longer at each feed. - If the child is exclusively breastfed, give ORS in addition to breast milk. - If the child is not exclusively breastfed, give one or more of the following: ORS solution, Food-based fluids (such as soup, rice water and yoghurt drinks), or clean water. It is especially important to give ORS at home when: -The child has been treated with Plan B or Plan C during this visit. - The child cannot return to a clinic if the diarrhea gets worse. ☐ TEACH THE MOTHER HOW TO MIX AND GIVE ORS. GIVE THE MOTHER 2 PACKETS OF ORS TO USE AT HOME. ☐ SHOW THE MOTHER HOW MUCH FLUID TO GIVE IN ADDITION TO THE USUAL FLUID INTAKE: Up to 2 years 50 to 100 ml after each loose stool 2 years or more100 to 200 ml after each loose stool Tell the mother to: - Give frequent small sips from a cup. - If the child vomits, wait 10 minutes. Then continue, but more slowly.	Give in clinic recommended amount of ORS over 4-hour period – DETERMINE AMOUNT OF ORS TO GIVE DURING FIRST 4 HOURS <table><tr><th>AGE</th><th>Up to 4 Months</th><th>4 months up to 12 months</th><th>12 months up to 2 years</th><th>2 years up to 5 years</th></tr><tr><td>Weight in kg</td><td>< 6 kg</td><td>6 – 10 kg</td><td>10 – 12 kg</td><td>12 – 19 kg</td></tr><tr><td>ORS in ml</td><td>200 - 400</td><td>400 - 700</td><td>700 - 900</td><td>900 -1400</td></tr></table> * Use the child’s age only when you do not know the weight. The approximate amount of ORS required (in ml) can also be calculated by multiplying the child’s weight (in kg) times 75 ☐ If the child wants more ORS than shown, give more. ☐ For infants under 6 months who are not breastfed, also give 100-200 ml clean water during this period. ☐ SHOW THE MOTHER HOW TO GIVE ORS SOLUTION: - Give frequent small sips from a cup. - If the child vomits, wait 10 minutes. Then continue, but more slowly. - Continue breastfeeding whenever the child wants. ☐ AFTER 4 HOURS: - Reassess the child and classify the child for dehydration.	AGE	Up to 4 Months	4 months up to 12 months	12 months up to 2 years	2 years up to 5 years	Weight in kg	< 6 kg	6 – 10 kg	10 – 12 kg	12 – 19 kg	ORS in ml	200 - 400	400 - 700	700 - 900	900 -1400	Start IV fluid immediately. If the child can drink, give ORS by mouth while the drip is set up. Give 100ml/kg Ringer’s Lactate Solution (or, if not available, normal saline), divided as follows: <table><tr><th>AGE</th><th>First give 30ml/kg in</th><th>Then give 70ml/kg in</th></tr><tr><td>Infants</td><td>1 hour*</td><td>5 hours</td></tr><tr><td>Children</td><td>30 minutes*</td><td>2 ½ hours</td></tr></table> Repeat once if radial pulse is still very weak or not detectable Reassess the child every 1-2 hours. If hydration status is not improving, give the IV drip more rapidly. Also give ORS (about 5 ml/kg/hour) as soon as the child can drink: usually after 3-4 hours (infants) or 1-2 hours (children). Reassess an infant after 6 hours and a child after 3 hours. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment. Refer URGENTLY to hospital for IV treatment. ☐ If the child can drink, provide the mother with ORS solution and show her how to give frequent sips during the trip ☐ Start rehydration by tube (or mouth) with ORS solution give 20 ml/kg/hour for 6 hours (total of 120 ml/kg) ☐ Reassess the child every 1-2 hours: - If there is repeated vomiting or increasing	AGE	First give 30ml/kg in	Then give 70ml/kg in	Infants	1 hour*	5 hours	Children	30 minutes*	2 ½ hours
AGE	Up to 4 Months	4 months up to 12 months	12 months up to 2 years	2 years up to 5 years																						
Weight in kg	< 6 kg	6 – 10 kg	10 – 12 kg	12 – 19 kg																						
ORS in ml	200 - 400	400 - 700	700 - 900	900 -1400																						
AGE	First give 30ml/kg in	Then give 70ml/kg in																								
Infants	1 hour*	5 hours																								
Children	30 minutes*	2 ½ hours																								

- Continue giving extra fluid until the Diarrhea stops. 2. GIVE ZINC SUPPLEMENTS TELL THE MOTHER HOW MUCH ZINC TO GIVE: Up to 6 months -1/2 tablet for 10 days 6 months or more -1 tablet for 10 days ☐ SHOW THE MOTHER HOW TO GIVE ZINC SUPPLEMENTS Infants- dissolve tablet in a small amount of expressed breast milk, ORS or clean water in a cup Older children- tablets can be chewed or dissolved in a small amount of clean water in a cup 3. CONTINUE FEEDING 4. WHEN TO RETURN } Advice mother	- Select the appropriate plan to continue Treatment. -Begin feeding the child in clinic. ☐ IF THE MOTHER MUST LEAVE BEFORE COMPLETING TREATMENT: -Show her how to prepare ORS solution at home. -Show her how much ORS to give to finish 4-hour treatment at home -Give her enough ORS packets to complete Rehydration. Also give her 2 packets as recommended in plan A Explain the 4 Rules of Home Treatment: 1. GIVE EXTRA FLUID 2. GIVE ZINC 3. CONTINUE FEEDING 4. WHEN TO RETURN See Plan A for recommended fluid and COUNSEL THE MOTHER	abdominal distension, give the fluid more slowly. - If hydration status is not improving after 3 hours, send the child for IV therapy. ☐ After 6 hours, reassess the child. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment. NOTE: ☐ If possible, observe the child at least 6 hours after rehydration to be sure.
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Table 11.8b Management of dehydration in adults and adolescents

Plan A: Treat Diarrhea at Home	Plan B: Treat Some Dehydration with ORS	Plan C: Treat Severe Dehydration Quickly(See algorithm below)															
Counsel the patient on the 3 Rules of Home Treatment: Drink extra fluid, continue eating, when to return. 1. Drink extra fluid (as much as the patient will take)—any fluid (except fluids with high sugar or alcohol) or ORS. – Drink at least 200-300 ml in addition to usual fluid intake after each loose stool. – If vomiting, continue to take small sips. Antiemetics are usually not necessary. – Continue drinking extra fluid until the diarrhea stops. It is especially important to provide ORS for use at home when: -The patient has been treated with Plan B or Plan C during this visit. -The patient cannot return to a clinic if the diarrhea gets worse. -The patient has persistent diarrhea or large volume stools. IF ORS is provided: TEACH THE PATIENT HOW TO MIX AND DRINK ORS. GIVE 2 PACKETS OF ORS TO USE AT HOME. 2. Continue eating. 3. Advise when to return.	Give in clinic recommended amount of ORS over 4-hour period. Determine amount of ORS to give during first 4 hours. Use the patient's age only when you do not know the weight. The approximate amount of ORS required (in ml) can also be calculated by multiplying the patient's weight (in kg) times 75. If the patient wants more ORS than shown, give more. <table border="1" data-bbox="741 683 1287 794"> <tr> <th>AGE*</th><th>5-14 years</th><th>15 years and older</th></tr> <tr> <th>WEIGHT</th><td>20- < 30 kg</td><td>30 kg or more</td></tr> <tr> <th>In ml</th><td>1000-2200</td><td>2200-4000</td></tr> </table> If the patient is weak, help him/ her take the ORS: - give frequent small sips from a cup. - if the patient vomits, wait 10 minutes then continue, but more slowly. -If patient wants more ORS than shown, give more. After 4 hours: - reassess the patient and classify for dehydration. - select the appropriate plan to continue treatment. - Begin feeding the patient in clinic. If the patient must leave before completing treatment: -show how to prepare ORS solution at home. -Show how much ORS to give to finish 4- hour treatment at home. -give enough ORS packets to complete rehydration. Also give 2 packets as recommended in Plan A. - explain the 3 rules of Home treatment: 1. Drink extra fluid 2. Continue eating 3. When to return	AGE*	5-14 years	15 years and older	WEIGHT	20- < 30 kg	30 kg or more	In ml	1000-2200	2200-4000	Start IV fluid immediately. If the patient can drink, give ORS by mouth while the drip is set up. give 100 ml/kg ringer's lactate solution (or, if not available, normal saline), divided as follows: <table border="1" data-bbox="1333 469 1873 534"> <tr> <th>Age</th><th>First Give 30ml/kg</th><th>First give 70ml/kg</th></tr> <tr> <td>>12mo.</td><td>In 30 minutes</td><td>In 2 ½ hours</td></tr> </table> *Repeat once if radial pulse is very weak or not detectable. • Reassess the patient every 1-2 hours. If hydration status is not improving, give the IV drip more rapidly. • Also give ORS (about 5 ml/kg/hour) as soon as the patient can drink: usually after 3-4 hours (infants) or 1-2 hours for children, adolescents, and adults. • Reassess an infant after 6 hours and older patient after 3 hours. Classify dehydration then choose the appropriate plan (A, B, or C) to continue treatment. Refer urgently to hospital for IV treatment. • If the patient can drink, provide the mother or family/friend with ORS solution and show how to give frequent sips during the trip Start rehydration by tube (or mouth) with ORS solution: give 20 ml/kg/hour for 6 hours (total of 120 ml/kg). • reassess the patient every 1-2 hours: - If there is repeated vomiting or increasing abdominal distension, give the fluid more slowly. - If hydration status is not improving after 3 hours, send the patient for IV therapy. • After 6 hours, reassess the patient. Classify dehydration. then choose the appropriate plan (A, B, or C) to continue treatment.	Age	First Give 30ml/kg	First give 70ml/kg	>12mo.	In 30 minutes	In 2 ½ hours
AGE*	5-14 years	15 years and older															
WEIGHT	20- < 30 kg	30 kg or more															
In ml	1000-2200	2200-4000															
Age	First Give 30ml/kg	First give 70ml/kg															
>12mo.	In 30 minutes	In 2 ½ hours															

Table 11.9. Diarrheal disease summary

Agent	CD4	Symptom	Diagnosis	Treatment
<i>E. histolytica</i>	Any	bloody stool, colitis	Stool microscopy	Metronidazole
<i>Giardia</i>	Any	Watery diarrhea	Stool microscopy	Metronidazole
<i>Strongyloides stercoralis</i>	Any	Watery diarrhea	Stool microscopy	Ivermectin or Albendazole
<i>Cryptosporidium</i>	<150	Watery diarrhea	Modified AFB	ART
<i>Isospora belli</i>	<100	Watery diarrhea	Modified AFB	TMP-SMX
<i>Microsporidium</i>	< 50	Watery diarrhea	Giemsa stain	Albendazole
CMV	<50	Watery/bloody diarrhea, colitis	Tissue biopsy	Ganciclovir

c. Peri-anal problems

A number of chronic or acute peri-anal problems commonly occur in patients with HIV disease, particularly in advanced stages of immunodeficiency. These include recurrent peri-anal abscesses, chronic peri-anal fistula, peri-anal herpes (severe, persistent and extensive), and peri-anal warts (sometimes large with obstructive problems). Patients with peri-anal problems frequently go to local healers and receive different kinds of local therapy that usually complicate the situation.

Treatment of peri-anal abscess in adolescents and adults: It is not difficult to make the clinical diagnosis of peri-anal abscess. All patients with acute or chronic peri-anal condition must be thoroughly evaluated and PR exam should be done routinely. Peri-anal abscess may extend depending on the immunological status of the patient; therefore early treatment is mandatory to avoid this and more serious morbidity. If patients require surgical incision, it should be done promptly on first visit, or referral made if the surgery is unavailable. Otherwise, broad spectrum antibiotics such as amoxicillin-clavulanic acid alternatively amoxicillin or ampicillin must be administered in sufficient dose for at least 10 days. Palliative care including Sitzz baths and analgesics are also important.

d. Peri-anal and/or genital herpes:

Latent or active infection with HSV I and II are common in the general population, and is usually mild in immune-competent persons. Severe cutaneous disease or visceral involvement is usually restricted to patients with advanced immunosuppression with a CD4 count $<100 \text{ cell/mm}^3$

The lesions become extensive, persistent, severe and sometimes bleeding. Unless thorough evaluation with regular inspection of genital and peri-anal areas is done, patients very often don't complain about genital lesions. The response to Acycovir is gratifying if it is done in sufficient dose (400mg 4 to 5 X/d) and sufficient duration (10 days to 2 weeks in moderately severe or severe cases). There is risk of recurrence with severe immunodeficiency. In such cases repeat treatment and put patient on chronic HIV care including ART. Herpetic oro-labial infection is treated the same way as ano-genital herpes.

The treatment of anal and genital warts is particularly frustrating when they are large. Unlike other opportunistic infections the response to ART is not satisfactory. Patients who have very well responded immunologically with ART continue to suffer from the warts. Depending on the size, cauterization, podophyllin treatment and surgical debulking, etc may be tried. Patients should be referred to where these services are available.

e. HCV/HIV and HBV/HIV co-infection management

HCV HIV co infection management

HIV patients are among high risk groups for HCV and should be given priority for screening. Therefore, all HIV patients should be screened and confirmation VL test should be done for HCV screened positives.

Anti-HCV rapid diagnostic test (RDT) or immunoassay (IA) can be used for screening and HCV RNA viral load test using either quantitative or qualitative PCR should be used to confirm chronic HCV infection.

All chronic HCV infected individuals should be treated to eradicate the virus and achieve cure so that complications can be avoided. Follow up quantitative or qualitative HCV RNA viral load is required to confirm if the patient has achieved Sustained virologic Response (SVR). This should be performed 12 weeks after the completion of therapy.

Treatment HCV in HIV infected individuals is not different from non HIV infected or HCV mono-infected. All combination of Directly Acting Anti-viral(DAA) including Sofosbuvir/Ledipasvir(SOF/LDV), Sofosbuvir/Ribavirin (SOF/RIB) and Sofosbuvir/ Daclatasvir (SOF/DCV) can safely be used. However attention should be given to drug-drug interaction interactions and shared side effects like head ache, fatigue and anemia.

According to the VH prevention and control guidelines the followings are treatment cited options (please consult the national VH prevention and control guideline for further reference):

- Sofosbuvir 400 mg oral once daily + Daclatasvir 60mg oral once daily for 12 weeks (dose of DCV be adjusted to 90 mg with Efavirenz and 30 mg with Atazanavir/r)
- Sofosbuvir 400mg oral once daily + Ledipasvir 90mg oral once daily for 12 weeks.
- For cirrhotic patient the treatment duration will be extended to 24 weeks for the above treatment options.

- Sofosbuvir 400mg oral once daily + Ribavirin 1000 mg (weight < 75kg) , 1200mg (weight ≥ 75Kg) twice on divided doses for 24 weeks.

It is recommended to thoroughly evaluate Chronic HCV infected person with cirrhosis and treatment duration be decided according to the genotype, type of drug selected and addition of ribavirin.

HBV/ HIV co infection management

HIV coinfection has been shown to have a profound impact on almost every aspect of the natural history of HBV infection and includes more rapid progression to cirrhosis and HCC, higher liver-related mortality, and decreased treatment response compared with persons without HIV coinfection. HIV/HBV-co-infected persons also demonstrated more rapid HIV disease progression compared to those who were HIV-infected alone, and had an impaired recovery of CD4 cells.

HIV patients are among the high risk groups for HBV and should be given priority for screening. i.e., all HIV patients should get screened for HBV and evaluated for chronic infection as per the national VH prevention and control guidelines. After test result, vaccination or treatment and care for reactive patients are recommended when resources permit.

Treatment options for patients with HIV –HBV Co-infection:

- ART should be initiated in all individuals with HIV regardless of WHO clinical stage or CD4count in the following situations:
 - Individuals coinfectd with HIV/ HBV with evidence of severe chronic liver disease. During HBV/HIV co-infection if treatment is indicated for HBV, combination antiretroviral therapy should be initiated with drugs containing TDF+3TC + EFV as a preferred regimen.
- For HIV positive patients who are co infected with HBV and qualify for treatment of both the diseases i.e. HIV and HBV, oral drug therapy is first line for these patients with at least 2 of the drugs having activity against HBV like combination of Tenofovir, lamuvidine and Efavirenz.
- The use of lamivudine as monotherapy in any of these diseases is contraindicated due to high YMDD resistance.

- When switching treatment in patients with HIV on ART failure, the regimen that will continue should have two of the drugs having activity against HBV.
- If tenofovir-associated renal toxicity occurs, the dose of tenofovir should be adjusted according to the renal clearance.
- If there is severe toxicity or contraindication to provide TDF in Hepatitis coinfecting patient, consult senior clinicians for other options

NB: For further reference please consult the national Viral Hepatitis prevention and control guidelines.

11.3.3. Management of Opportunistic Diseases of the Nervous System

Case Scenario: Zemen is a 34 year old HIV positive man who has disappeared from ART clinic 6 months back and came back with headache, fever, vomiting of 1 week duration. He became confused one day before he came to the hospital.

What are the possible causes?

How do you manage him?

Neurological manifestations of HIV can occur at any time from viral acquisition to the late stages of AIDS; they are varied and may affect any part of the nervous system including the brain, spinal cord, autonomous nervous system and the peripheral nerves. HIV affects the nervous system in 70-80% of infected patients. The effect may be due to direct effect of the virus, opportunistic infections and/or malignancies. For certain neurological manifestations a single aetiology is responsible while in others it is due to multiple causes.

Most life-threatening neurological complications of HIV occur during the severe immunodeficiency state and specific aetiological diagnosis in the Ethiopian setting is often a major challenge. Thus, this unit attempts to guide the management of common opportunistic infections and other treatable conditions in the nervous system.

Neurological conditions in HIV patients may be due to:

- HIV(HIV encephalopathy)
- OIs(Toxoplasmosis, Cryptococcal meningitis)

- Neurosyphilis
- Malignancies (primary CNS lymphoma) and
- Drugs, example EFV, the obsolete **d** drugs (ddi and D4T) etc

Diagnosis of neurological disorders in HIV in our setting depends on the history and standard neurological examinations. There can be single or multiple abnormal neurological findings in the same patient necessitating holistic neurological evaluation. Thus the examination should include assessment of:

- Mental status comprising cognitive function, orientation and memory.
- Cranial nerves
- Motor function including DTR
- Sensory function

a) *Toxoplasma gondii* Encephalitis

Toxoplasmic encephalitis (TE) is caused by the protozoan *Toxoplasma gondii*. Disease appears to occur almost exclusively because of reactivation of latent tissue cysts. Primary infection occasionally is associated with acute cerebral or disseminated disease. Sero-prevalence varies substantially in different communities; in Ethiopia, general prevalence is about 80%.

Clinical Manifestations

The most common clinical presentation of *T. gondii* infection is

- Focal encephalitis with headache, confusion, or motor weakness and fever.
- Patients may also present with non-focal manifestations, including only non-specific headache and psychiatric symptoms.
- Focal neurological abnormalities may be present on physical examination, and
- In the absence of treatment, disease progression results in seizures, stupor, and coma.

Diagnosis

HIV-infected patients with TE are almost uniformly seropositive for anti-toxoplasma immunoglobulin G (IgG) antibodies. **The absence of IgG antibody makes a diagnosis of toxoplasmosis unlikely but not impossible.** Anti-toxoplasma immunoglobulin M (IgM) antibodies usually are absent. Quantitative antibody titres are not useful for diagnosis. Definitive

diagnosis of CNS toxoplasmosis requires a compatible clinical syndrome; identification of one or more mass lesions by CT, MRI, or other radiographic testing; and detection of the organism in a clinical sample. In the absence of imaging support, empirical treatment is justified when patients present with focal neurological findings and the CD4 count is < 200 cells μ L. Failure to respond to conventional therapy, based on presumptive clinical diagnosis within a week or two of initiation of therapy, suggests the diagnosis to be unlikely. With empirical treatment for toxoplasmosis, nearly 90% of patients will demonstrate clinical improvement within days of starting therapy. Radiological evidence of improvement is usual after 14 days of treatment.

Treatment

First line treatment regimen for toxoplasma encephalitis in the Ethiopian context is

For Adults: **Trimethoprim/sulfamethoxazole 80/400**, oral, **4 tablets 12 hourly for 28 days**, followed by 2 tablets 12 hourly for 3 months in adults.

For children: 10mg of trimethoprim + 50mg of sulfamethoxazole/kg per dose every 12 hours for 28 days followed by maintenance therapy at 50% reduced dosage for three months.

Secondary prophylaxis: use co-trimoxazole 960mg daily for adults and See Table 10.2 for children

Alternative regimen

- I. **Sulfadiazine**, 1-2 gm p.o.q 6h for six weeks or 3 weeks after resolution of lesion

S/E: crystal urea, rash

C/I: severe liver, renal and haematological disorders; known hypersensitivity to Sulphonamides

Dosage/form: 500 mg tablets,

PLUS Pyrimethamine, Loading dose of 200 mg once, followed by:

Pyrimethamine 50-75 mg/day S/E: rash, fever and bone marrow depression (neutropenia and thrombocytopenia)

C/I: folate deficiency

Dosage/form: 25 mg tablets **PLUS**

Folinic acid (Leucovorin): 10-20 mg/d OR

II. Pyrimethamine and Folinic Acid (Leucovorin): (standard dose) PLUS Clindamycin: 600 mg q 6 hrs

S/E: toxicities include fever, rash, and nausea, diarrhoea (including pseudo-membranous colitis or diarrhoea related to *Clostridium difficile* toxin)

Adjunctive corticosteroids should be used for patients with radiographic evidence of midline shift, signs of critically elevated intracranial pressure or clinical deterioration within the first 48 hours of therapy. Dexamethasone (4 mg every six hours (0.15mg/kg/dose every 6 hours for children)) is usually chosen and is generally tapered over several days and discontinued as soon as possible.

Anticonvulsants should be administered to patients with a history of seizures, but should not be given routinely for prophylaxis to all patients with the presumed diagnosis of TE. Careful attention needs to be paid to any potential drug interactions.

b) Cryptococcal infection

Cryptococcal meningitis is one of the most important opportunistic infections and a major contributor to high mortality before and after ART is initiated. The main reasons for this high death rate include delayed presentation, together with poor availability and high cost of treatment. Furthermore, there are no standardized guidelines applicable to resource-limited settings for the diagnosis and management of cryptococcal disease. Most HIV-associated cryptococcal infections are caused by *Cryptococcus neoformans*.

Clinical manifestations of cryptococcal meningitis:

- Subacute meningitis or meningoencephalitis with fever, malaise, and headache.
- Neck stiffness and photophobia, occur in only one-quarter to one-third of patients.
- Some patients experience encephalopathic symptoms, such as lethargy, altered mentation, personality changes, and memory loss that are usually a result of increased intracranial pressure, thought to result from impaired cerebrospinal fluid (CSF) absorption.

Diagnosis

LP and CSF analysis:

- The opening pressure may be markedly elevated.
- CSF analysis

- Protein 30-150 mg/dl
- WBC 0-100 /mm³ (monocyte)
- Culture positive 95-100%
- Indian ink positive 60-80%
- Cryptococcal Ag > 95 % sensitive and specific

Management

Management of Cryptococcal management* requires hospitalization and evaluation by physician.

The management of **Cryptococcal meningitis** is **phased**:

Option A

The preferred regimen in Ethiopian, and it has 3 phases: induction, consolidation and maintenance.

Induction phase (2 weeks): High dose fluconazole:

Fluconazole 600 mg twice daily alone (children: 12mg/kg/day in 2 divided doses)

Consolidation phase (8 weeks): Fluconazole 800 mg/day (in children 12mg/kg/day)

Maintenance treatment (Secondary prophylaxis): Fluconazole 200 mg daily (in children, 6mg/kg/day)

Option B. (Alternative in Ethiopian situation)

Induction phase (2 weeks): Amphotericin B + Fluconazole:

Amphotericin 0.7-1 mg/kg/day + Fluconazole 800 mg/day

Consolidation phase (8 weeks): Fluconazole 400-800 mg/day

Maintenance treatment (Secondary prophylaxis): Fluconazole 200mg daily (in children, Fluconazole 6mg/kg/day).

Additional Points about Cryptococcal Meningitis management:

1. Management of elevated Intracranial pressure (ICP):

Management of increased ICP is critical as >90% of deaths in the first two weeks and 40% of deaths in weeks 3-10 are due to increased ICP. Failure to manage elevated ICP is the most common and most dangerous mistake in management (Since the ICP is non-communicating hydrocephalus there is no risk of CSF tapping within the recommended volume).

- Daily serial LP should be done to control increased ICP by drawing 20-30 ml of CSF based on patient's clinical response. Signs of ICP include headache, altered mental status, meningismus and changing in hearing or vision should be closely monitored, if possible opening pressure should be measured.
- There is no role for acetazolamide, mannitol, or corticosteroids to reduce intracranial pressure.

2. Discontinuation of maintenance treatment (secondary prophylaxis)

When patients are stable and adherent to ART and anti-fungal maintenance treatment for at least one year and have a CD4 cell count of greater than or equal to 200 cells/mm³ (two measurements six months apart).

3. Timing of ART initiation for patient with Cryptococcal meningitis

- Immediate ART initiation is not recommended in HIV-infected patients with cryptococcal meningitis due to the high risk of IRIS, which may be life-threatening.
- ART initiation should be deferred until there is evidence of a sustained clinical response to anti-fungal therapy, and
 - After 2-4 weeks of induction and consolidation treatment with amphotericin B-containing regimens combined with fluconazole, or
 - After 4-6 weeks of induction and consolidation treatment with high-dose oral fluconazole regimen

Poor prognostic signs of cryptococcal meningitis:

- Extra CNS manifestation (especially pulmonary)
- Altered mental status
- Low CSF WBC cell count less than 20 cells/ μ L

- High CSF cryptococcal antigen titer

Prevention of cryptococcal disease

- According to the pilot study conducted by MOH in collaboration with CHAI and ICAP-CU from June 2015 to July 2016, in 22 high case load facilities in all regions, the proportion of newly enrolled clients with CD4 count less than 100 was 25.88%. In the same study the prevalence of clients screened positive for cryptococcal antigenemia was high (9.9%).
- The use of routine serum or plasma CrAg screening in ART-naive adults followed by pre-emptive antifungal therapy if CrAg positive to reduce the development of cryptococcal disease, should be considered prior to ART initiation in:
 - a) Patients with CD4 count less than 100 cells/mm³; and
 - b) Where the population has a high prevalence (>3%) of cryptococcal antigenemia.
- The following algorithm diagram shows how to decide whether a patient needs prophylactic fluconazole treatment to CrAg positive and asymptomatic patients with CD4 count less than 100 cells/μl.

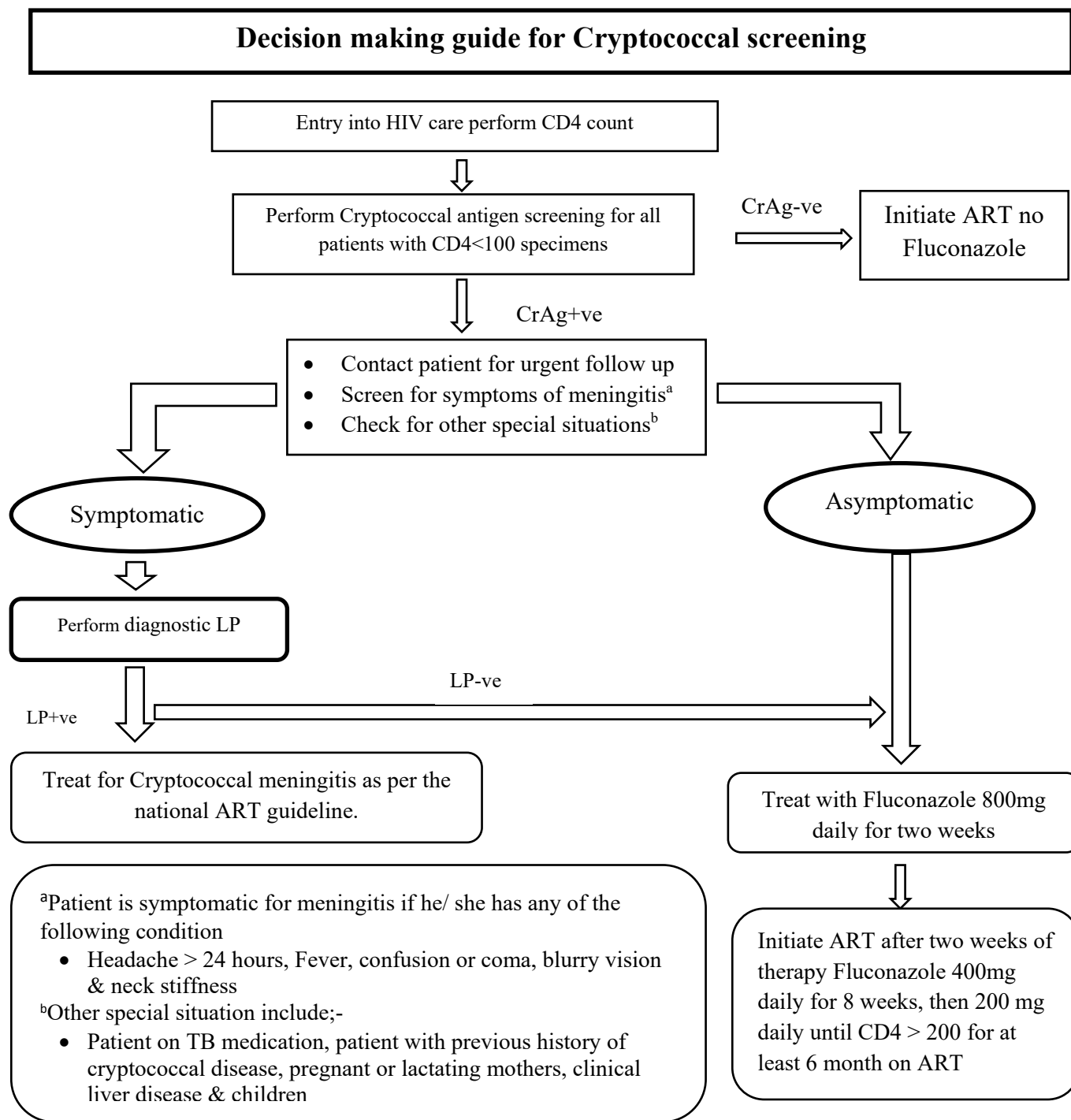


Figure 11.8.Decision-Making Guide for Cryptococcal Screening

c) Tuberculosis meningitis

Tuberculosis meningitis is one of the neurological manifestations in HIV infected patients. About 10% of AIDS patients present with TB will show signs of meningeal involvement. Symptoms include: fever, confusion, headache, meningismus, and focal neurological deficit (20%) especially cranial nerve palsy. Sometimes seizure and loss of consciousness is seen.

Diagnosis

LP- it is mostly safe and CSF analysis characteristically reveals

- Lymphocytosis (<500 cells/mm³)
- Low glucose
- High protein(100-500mg/dl)
- CSF geneXpert

AFB - seldom positive (10-40%)

- Examine the deposit after centrifugation of 10cc
- Examine at least for 30 minutes
- Repeat several CSF samples over several days

Suspect TB meningitis when there is

- Slow onset of neuro symptoms(over two weeks duration)
- Lymphocytosis on CSF
- Presence of abnormal chest x-ray
- Presence of abdominal lymph nodes

Treatment

- TB treatment according to the national protocol
- Delay ART for at least 4 weeks and initiate within 8weeks after treatment of TB meningitis is initiated
- Start prednisolone 1mg/kg for 2-4 wks then taper off over 4-8wks with Anti-TB for all patients

d) HIV Encephalopathy

It's the commonest CNS abnormality in HIV infected infants and children. It may be the initial manifestation of the disease or may present much later when severe immune

suppression occurs. With progression, marked apathy, spasticity, hyper-reflexia, and gait disturbance may occur, as well as loss of language, fine, and/or gross motor skills.

Depending on disease progression, two forms of HIV encephalopathy are observed; these are progressive and static encephalopathy.

Progressive Encephalopathy is the commonest form of encephalopathy; manifestations include:

- Loss or plateau of developmental milestones
- Cognitive deterioration
- Impaired brain growth resulting in acquired microcephaly
- Apathy, spasticity, hyperreflexia, gait disturbances, loss of language and other motor skills
- CT shows cerebral atrophy in 85% of children

Careful clinical evaluation is effective and cheap way for early identification of children with HIV encephalopathy.

Almost 10% of adult AIDs patients suffer from HIV encephalopathy (AIDS dementia complex) which progresses over months to years and manifested by cognitive problems (e.g. memory), impaired motor movements and behavioural problems (e.g. Apathy and social withdrawal).

NB: HIV encephalopathy is a diagnosis of exclusion.

HAART is the best treatment for HIV encephalopathy in both adults and children.

e) Peripheral Neuropathies

Peripheral neuropathies are among the most common causes of painful legs in HIV infection; they arise as a complication of HIV infection itself, of drug therapy, or of other metabolic or organ dysfunction or nutritional deficiencies.

Distal symmetrical sensory polyneuropathy is the most common presentation but mono-neuropathies can also occur. The neuropathies associated with HIV can be classified as:

- Primary, HIV-associated
- Secondary causes related to medications (INH), OIs or organ dysfunctions

Diagnosis: Peripheral neuropathy diagnosis in HIV-infected patients is based on the clinical picture presenting with pain, tingling sensations, paresthesia or numbness. Physical examination can reveal depressed or absent ankle reflex, decreased sensitivity to different modalities of sensation and in severe cases, difficulty in walking. The feet and sometimes the hands are involved in symmetrical distribution. The diagnosis can be supported by electro diagnostic studies including electromyography (EMG) and nerve conduction studies (NCS) when available. Blood tests are frequently obtained to exclude other causes of neuropathy. In most instances, however, diagnosis is almost always clinical.

Treatment of peripheral neuropathy:

- Avoid the offending agent if identified
- Remove other drugs associated with peripheral neuropathy
- Supplement vitamin intake for all patients including concomitant administration of pyridoxine with INH.
- Adjuvants for pain management (such as Amitriptylin, carbamazepin) indicated for patients with neuropathic pain and paresthesias.

Monitoring of events

- Recognize presence of peripheral neuropathy
- Assess severity at each clinical visit
- Avoid drugs causing neuropathy

Table 11.10 Summary of common HIV-associated neurological and related syndromes

Syndrome	Possibilities	Diagnosis
Painful legs (extremities)	Peripheral neuropathy, metabolic disorders or vascular insufficiency.	History, physical exam, Nerve Conduction Test
Difficulty walking	Musculoskeletal disorders, Spinal cord lesions, Myopathies, peripheral neuropathies, or some brain lesions.	Spinal X-ray, CT, MRI, CK, LDH LP when indicated
Severe headache +/-focal neurological deficit +/- seizure	Meningoencephalitis, Meningitis, CNS Toxoplasmosis, brain abscess, Tuberculoma, CNS lymphoma.	Imaging (CT, MRI) LP when indicated
Acute confusion	1. Meningo-encephalitis 2. Systemic infections like malaria, or septicaemia, 3. Metabolic disorders: –electrolyte abnormalities, hypoglycemia, renal or liver failure	Blood film, CBC, blood culture, LP (CSF analysis) RFT, LFT, RBS electrolytes if diarrhea or dehydration present
Chronic behavioral change	PML* AIDS dementia, Chronic meningitis.	Clinical CSF analysis

NB: Always exclude local causes.

*PML= Progressive Multifocal Leucoencephalopathy; has associated neurological deficits.

11.3.4. Cutaneous manifestations in HIV positive individuals

The skin is an organ frequently affected by OIs; early manifestations of HIV infections frequently occur in the skin. Different kinds of OIs, such as herpes zoster, and other viral, fungal and bacterial infections occur in the skin. Manifestations of adverse drug reactions and non-infectious conditions also occur in the skin. Some skin reactions to drugs such as Nevirapine may

be life-threatening. In most instances diagnoses of skin disorders with HIV disease are made on clinical grounds. Most skin disorders in HIV disease can be cured or ameliorated, but a few fail to improve even with good general clinical and immunological responses to ART.

Pruritis is the most common dermatologic symptom in HIV infected patients. It can be localized indicating primary skin lesion, or generalized that may or may not indicate primary skin lesions. In many patients pruritus may be severe and not amenable to available therapy. The most common skin conditions associated with pruritis in patients with AIDS include the following:

1. Papular pruritic eruption(PPE)
2. Excessive dryness of the skin (Xerosis cutis)
3. Eczemas like seborrheic dermatitis or contact dermatitis
4. Folliculitis that may include infections by *Staph.aureas* or hypersensitivity to insects.
5. Drug eruptions
6. Scabies
7. Intertrigo (Candida, tinea, herpes simplex)

In most patients, diagnosis can be established by examining the lesions. However, as immune deficiency advances it may be useful to use investigations such as biopsy to diagnose specific dermatosis or use staining and culture to diagnose specific infections.

a. Pruritic Papular Eruption

Pruritic Papular Eruption is common among HIV infected patients causing substantial morbidity in sub-Saharan Africa. Its prevalence ranges from 12-46% and it is uncommon in HIV negative patients. The pathogenesis is unknown but it may be related to hypersensitivity to arthropod bites. In extreme form, eosinophilia and eosinophilic infiltrates of the skin are present. Severity of rash often correlates with CD4 count.

Clinical manifestations

- Intensely pruritic discrete, firm papules with variable stages of development and predilection for extremities, though they can involve trunk and face.
- Pigmentation, scarring and nodules due to excoriation.

Treatment: Treat with topical steroid and oral antihistamines; however it is often refractory to treatment and hence short course prednisolone may be used. HAART is often effective.

b. Etiological Classification of Skin Disorders in HIV Disease

Skin disorders in HIV infected patients can occur due to infections, neoplasm, and hypersensitivity to foreign agents including drugs, or to unknown causes. Nevertheless, infections are commonly seen in clinical practice; refer to the following table:

Table 11.11 Common skin problems in HIV infected individuals

Infections	Disease	Clinical Presentations	Treatment	Remark
Bacterial	Cellulitis	Poorly defined Erythema. Pus and crust at the site plus signs of inflammation.	Amoxicillin 500 mg TID for 14 days or erythromycin 500 mg QID if allergic to penicillin.	Mostly encountered lower extremities and often unilateral.
	Impetigo	Erythematous small papules usually limited to few lesions coalescing in to crusted plaques.	Use topical antibiotics: use Amoxicillin for extensive disease.	Usually a superficial lesion.
	Carbuncle	Nodular Lesion with extensions to the deeper Structure. Signs of Inflammation present.	Use Cloxacillin 500mg qid for ten days.	Involves the trunk as well as extremities.
Viral	Herpes simplex	Painful vesicular lesion around mouth or genitalia. Recurrent and extensive, difficult to eradicate during advanced immune deficiency.	Acyclovir 400mg t.i.d for ten days. In children 20 mg/kg/dose 4X/d	If Chronic (> one month) patient will benefit from immediate ART initiation if not on ART
	<i>Herpes Zoster</i>	Painful and vesicular eruptions with dermatomal distribution. When healed, scar will remain.	Acyclovir 800mg 5X per day for seven days. Monitor renal function.	When it involves the eyes it is a medical emergency. Do not give Acyclovir* if duration is >72 hours.

	Warts / verrucae	Painless flat to raised warts over fingers or genitalia in advanced immune deficiency, they tend to be multiple and exophytic.	Podophyllin, Imiquimod, Cryotherapy Consult experts	Premalignant and risk for cervical cancer.
	Molluscum Contagiosum	Umbilicated and raised facial lesions that tend to be very big during immunodeficiency state.	May not require therapy;	Contagious
Parasitic infestation	Scabies	Pruritic lesions ranging from pinpointed erythematous papules involving interdigital and gluteal places to varying degrees of hyperkeratotic plaques associated with significant skin thickening and crusting.	BBL, lindane or permethrin to be applied to whole body. Ivermectin 200 microgram/kg stat orally.	Burrows are visible in mild infestations but in crust scabies may not be evident leading to misdiagnosis.
Fungus	Dermatophytosis	Superficial causing ringworm or athlete's foot	Topical antifungal for limited skin affected. Fluconazole for extensive lesion 100mg daily for ten days.	
	Thrush	White plaques on the buccal mucosa including the tongue that can be scraped off leaving red base. Can be associated with candida paronychia or intertrigo.	Miconazole gel 2% apply bid Fluconazole 100 mg daily for ten days for recurrent or oropharyngeal thrush.	

	Deep Fungal infection	Presentation varies from fungating nodules and tumors to ulcers and diffuse papulonodular disease	Disseminated Cryptococcus can be confused for Molluscum contagiosum. Treat with amphotericin induction and/or fluconazole maintenance.	
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*If patient has ophthalmic involvement refer to ophthalmic specialist.

Table 11.12 Skin manifestations related to Drug side effects

SIGNS	CLASSIFYAS	TREATMENT	HIV FEATURES
Generalized red, widespread with small bumps or blisters; or one or more dark skin areas (fixed drug reaction)	FIXED DRUG REACTION	Stop medications - Give oral antihistamine	Cotrimoxazole is a common cause
Generalized bullous lesion with skin sloughing, involving mucosal surfaces including the mouth and the eyes	STEVENS-JOHNSON SYNDROME	Stop medication - Refer URGENTLY to Hospital	a rare but fatal reaction to ARV's and co-trimoxazole

11.4. Prevention and management of STI

STIs are very common. The most widely known are gonorrhea, syphilis and HIV but there are more than 30 others. STIs can cause serious and permanent complications in infected people who are not treated in a timely and effective way. In people with untreated STIs, the complications and sequel can be devastating.

11.4.1. The link between STI and HIV/AIDS

Certain STIs facilitate the spread of HIV. In fact, the interrelationship between STI and HIV is more complex, in that:

- Certain STIs facilitate the transmission of HIV
- The presence of HIV can make people more susceptible to the acquisition of STIs
- The presence of HIV increases the severity of some STIs and their resistance to treatment.

Which STIs facilitate the transmission of HIV?

A person with open sores in the genital area is much more likely both to contract and to transmit HIV. Chancroid and syphilis are the main bacterial causes of sores: if promptly diagnosed and treated, these links can be reduced. Herpes genitalis is the most common cause by viruses.

Chlamydia, gonorrhea and trichomoniasis can also facilitate the transmission of HIV. This may be for one or both of two reasons:

- These non-ulcerative diseases stimulate the body's immune system to increase the number of white blood cells, which are both targets and sources of HIV.
- Genital inflammation associated with these STIs can cause microscopic cuts in genital tissues, creating potential sites where HIV can enter the body.

Management of STI in Ethiopia follows syndrome approach. A syndrome is simply a group of the symptoms a patient complains about and the clinical signs you observe during the exam. Implementing syndromic STI management will contribute to successful STI control.

STI control has important aims:

1. Interrupt the transmission of sexually transmitted infections.
2. Prevent development of diseases, complications and sequelae.
3. Reduce the risk of HIV infection.

STI treatment (antimicrobial) regimens are chosen to cover the major pathogens responsible for the syndromes in the specific geographic area. In order to make this determination, a laboratory analysis of the syndromes is made and the pathogens for each syndrome are identified. This means that, afterwards, the management of individual patients will not depend on laboratory investigation.

11.4.2. Components of syndromic management of STI

1. Drug treatment and follow-up
2. Partner notification and management
3. Health education and risk reduction

4. Condom provision and education
5. PITC
6. Abstinence from sex till all symptoms resolve
7. Recording and reporting

STI Management Flow-charts

- A decision box always has two exit paths (Yes/No)
- The action that need to be carried out tells you how to manage the case

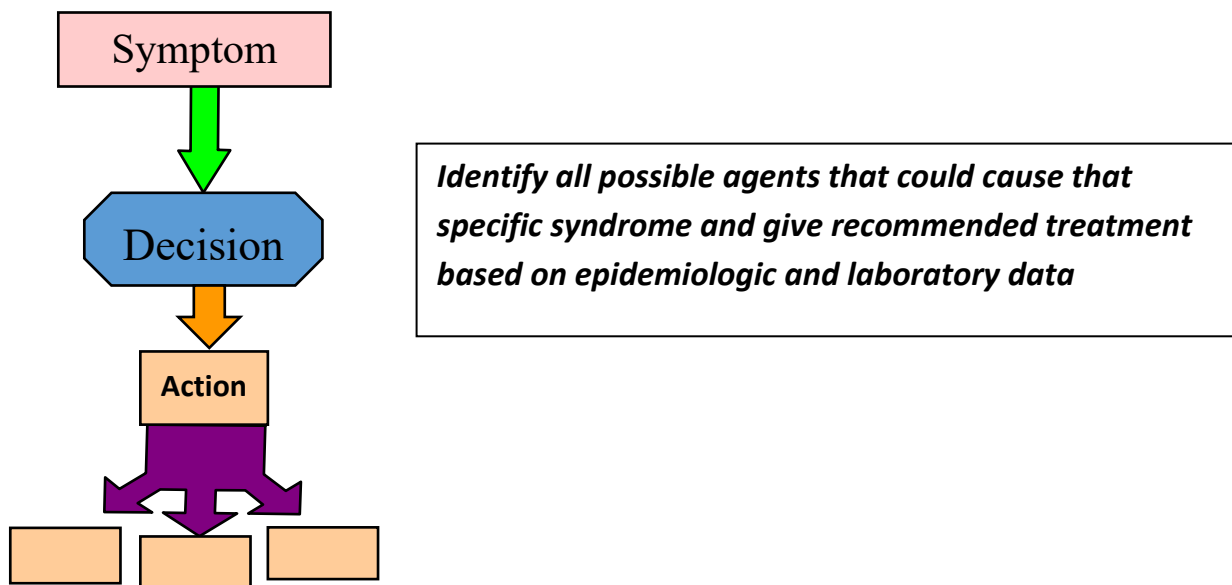


Figure 11.9. STI syndromic diagnosis approach

Table 11.13 Summary of Syndromic Management of Common STIs

Syndrome	Common causes	Sign and Symptom	Management
Urethral discharge Photo-7A	Nisseria gonorrhea, C.trachomatis, Mycoplasma genitalium, T.vaginalis, Ureaplasma urealyticum,	Burning sensation on urination, dysuria Urethral discharge Meatal excoriation	Ceftriaxone 250mg IM stat/ Spectinomycin 2 gm IM stat Plus Azithromycin 1gm po stat/ Doxycycline 100 mg po bid for 7 days/ Tetracycline 500 mg po qid for 7 days/ Erythromycin 500 mg po qid for 7 days in cases of contraindications for Tetracycline (children and pregnancy) Note: The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat
Vaginal Discharge with STI risk assessment positive (One or more of the following risks: age <25, new partner within the last three months, multiple partner within the last three months, Ever traded sex) Photo-6C	Gardnerella Vaginalis(Polymeric robial) Candida Albicans T.vaginalis Nisseria gonorrhea C.trachomatis	Vaginal discharge	Ceftriaxone 250mg IM stat/ Spectinomycin 2 gm IM stat Plus Azithromycin 1 gm po stat/Doxycycline 100mg po bid for 7 days Plus Metronidazole 500 mg bid for 7 days NB:If discharge is white or curd-like add Clotrimazole vaginal pessary 200 mg at bed time for 3 days Note: The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat plus Metronidazole 500 mg bid for 7days
Vaginal discharge with negative STI risk assessment	Bacterial vaginosis T. Vaginosis		Metronidazole 500mg po bid for 7 days Plus Clotrimazole 200mg vaginal tabs at bed time for 3 days If discharge is white or curd-like add Clotrimazole vaginal pessary 200 mg at bed time for 3 days

Lower abdominal pain	Neisseria gonorrhea, C.trachomatis, Mycoplasma genitalium, Bacteroides species, E. coli, H.influenza, Streptococcus,	Lower abdominal pain and Vaginal discharge Lower abdominal tenderness and cervical excitation tenderness	OPD Ceftriaxone 250 mg IM stat / Spectinomycin 2gm i.m stat Plus Azithromycin 1gm po stat/ Doxycycline 100 mg po b.i.d for 14 days Plus Metronidazole 500 mg po b.i.d for 14 days Note : The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat plus Metronidazole 500 mg bid for 14 days Inpatient Ceftriaxone 250 mg i.m/i.v/ Spectinomycin 2 gm i.m bid Plus Azithromycin 1gm po/ Doxycycline 100 mg po b.i.d for 14 days Plus Metronidazole 500 mg po b.i.d for 14 days Note: For inpatient PID, ceftriaxone, spectinomycin or azithromycin should continue for 24hrs after the patient remain clinically improved, after which doxycycline and metronidazole should continue for a total of 14 days
Scrotal Swelling Photo-7B	N.gonorrhea C.trachomatis	Scrotal swelling and pain	Ceftriaxone 250mg i.m stat/Spectinomycin 2gms i.m stat. Plus Azithromycin 1gm po stat/Doxycycline 100mg po bid for 7 days/ Tetracycline 500mg qid for 7 days Note :The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat

Genital Ulcer Photo-8A/8B/8C	Herpes simplex Virus (HSV1 and HSV2) T. Pallidum: H. Ducreyi: C. Trachomatis Serovars L1-L3: K. Granulomatis	Painful ulcer with or without vesicles	Vesicular: Acyclovir 400mg po tid for seven days OR Acyclovir 200mg five times daily for 10 days Non-vesicular Benzathine Penicillin 2.4 mIU IM stat / Doxycycline 100mg po bid for 14 days (for Penicillin allergy) Plus Ciprofloxacin 500mg po bid for 3 days/Erythromycin 500 mg po QID for 7 days plus Acyclovir 400mg po tid/10 days
Inguinal bubo Photo-8D	H. ducreyi C. Trachomatis Serovars L1-L3: LGV K. Granulomatis Treponema pallidum	Inguinal swelling fluctuating sometimes with discharge	Ciprofloxacin 500mg bid orally for 3 days Plus Doxycycline 100 mg bid orally for 14 days / Erythromycin 500mg po qid for 14 days If patient have genital ulcer, add Acyclovir 400mg tid orally for 10 days(or 200mg five times per day for 10 days) Note: surgical incisions are contraindicated; aspirate pus with hypodermic needle through the healthy skin
Neonatal conjunctivitis	N. gonorrhea C. trachomatis *S. pneumoniae *H. influenzae * S. aureus	Sticky eye discharge in the first 28 days after birth	Ceftriaxone 50mg/Kg) Max of 125mg IM stat Spectinomycin 25mg/kg IM stat (Maximum of 75mg) Plus Erythromycin 50mg/kg in four divided doses for 14 days

NB: For detailed guidance on management of specific sexually transmitted infection syndrome refer to the national syndromic STI management guideline

***These organisms are not sexually transmitted but included for the purpose of management.**

11.5. Visceral Leishmaniasis and HIV co-infection

Visceral leishmaniasis (VL) is a systemic parasitic illness, transmitted primarily by the phlebotomine sand fly from animal or human reservoirs. Visceral Leishmaniasis is endemic in Ethiopia, with patchy distribution in the southern and north-western lowlands. The causative parasite is *L. donovani*. VL has emerged as a major OI associated with HIV. In HIV patients, VL represents reactivation of latent infection with Leishmania parasite. HIV-1 increases the risk of developing leishmaniasis by more than 100 times and modifies the clinical spectrum of leishmaniasis by:

- Increased parasite load in peripheral blood and bone marrow
- Unusual localization of parasites (GI, lung)
- Low level of parasitic specific serum antibodies
- High rate of cutaneous dissemination and visceralization despite the species of leishmania involved
- High rate of treatment failure and recurrences (100% 12 months after treatment)

Clinical features: The cardinal signs of VL in patients with HIV infection are unexplained fever, splenomegaly and pancytopenia (anemia, leucopenia and thrombocytopenia). Presentation may not be typical. The bone marrow is packed with parasites but two-thirds of cases have no detectable anti Leishmania antibodies. CD4⁺ cell count in co-infected patients is usually <200 cells/ml.

Diagnosis:

- Parasitological diagnosis: Isolation of the organism from samples taken from reticulo-endothelial tissue and examined with Giemsa, Wright's or Leishmanial stain.
- Immunological diagnosis:
 - Antibody detection- rapid test **rk39 tests**,
 - Antigenic test-DAT freeze dried Antigen test- Confirmatory - negative or positive results,

Treatment: Liposomal Amphotericin B (Ambisome) 40mg/kg total, require longer treatment and more liable to relapse.

Treatment of relapsed patients: These are patients who are slower to respond and have a higher chance of further relapse and of becoming unresponsive to anti-monial drugs.

Treatment- Consider Referral to Referral/Specialized Hospitals

Same as above and see algorithm at **Annex 12** and national guideline for leishmaniasis

11.6. Screening for Co-morbidities

People living with HIV are at increased risk of developing a range of non-communicable diseases (NCDs), including cardiovascular disease, diabetes, chronic lung disease and some types of cancer. With effective ART, people living with HIV are also living longer and experiencing NCDs associated with ageing. Both HIV and NCDs require health systems that can deliver effective acute and chronic care and support adherence to treatment.

Chronic HIV care provides the opportunity for screening, monitoring and managing NCDs, especially through primary care. Integrating interventions such as nutrition assessment, dietary counseling and support, smoking cessation, promoting exercise, monitoring blood pressure and where available cholesterol as part of HIV care provide opportunities for reducing the risks of NCDs among people living with HIV. WHO has defined a package of essential NCDs (WHO PEN) interventions along with recommendations on screening for and treating NCDs.

People living with HIV have got increased risk of non-AIDs defining chronic diseases such as diabetes mellitus, cardiovascular illnesses, malignancies, chronic liver disease and chronic renal disease. With the advent of ART, people are living longer; hence they are at risk for age related diseases. Therefore, screening of PLHIV for these co-morbidities during every visit is a critical component of the care and treatment package.

11.6.1. Cervical cancer Screening

Cervical cancer screening is an important test in women living with HIV to prevent significant morbidity and mortality associated with HPV. Cervical cancer is a preventable disease and is curable if diagnosed and treated early. Women living with HIV have a higher risk of pre-cancer and invasive cervical cancer. The risk and persistence of HPV infection increases with low CD4 count and high HIV viral load. Cervical cancer screening leads to early detection of precancerous

and cancerous cervical lesions that will prevent serious morbidity and mortality. **Screening**

Frequency:

WLHIV should be screened every two years as part of routine care for HIV-positive women, regardless of whether they are on antiretroviral. Following abnormal results and/or treatment, repeat screening in one year. If follow-up screening is normal, return to screening every two years.

Screening methods:

Depending on the availability of resources, the following types of screening methods:

1. Visual inspection: with acetic acid (VIA) or Lugol's iodine (VILI)
2. HPV DNA test
3. Cytology: conventional (Pap smear) and liquid-based

Treatment of Precancerous Lesions:

- a. Cryotherapy
- b. Thermal ablation
- c. Loop Electrosurgical Excision Procedure (LEEP)
- d. Conization (Cone Biopsy)

Linkage and Integration of services:

Antiretroviral (ART) clinics should be closely linked with VIA clinics or sites providing cervical cancer prevention services, or ideally, provide the services themselves. (Refer to the national cervical cancer prevention guideline, 2021)

Healthcare delivery level by type of service and type of health care workers

Healthcare delivery level	Service provided	Type of health care workers
Tertiary hospital	VIA/cryotherapy/thermal ablation, PAP smear, colposcopy, LEEP biopsy, histopathology radiotherapy, radical surgery	Nurse, midwives, health officer, MD residents and specialists (gyn/obn, gyn-oncologists)

	chemotherapy and palliative care	
Regional/general hospitals	VIA/cryotherapy/thermal ablation, PAP smear, colposcopy LEEP biopsy histopathology, radical surgery and palliative care	Nurse, midwives, health officer, MD residents and specialists (gyn/obs, gyn-oncologists)
District hospitals /primary hospitals	VIA/cryotherapy/ thermal ablation, biopsy, palliative care	Nurse, midwives, health officer, MD
Health centers	VIA/cryotherapy/ thermal ablation, palliative care	Nurse, midwives, health officer

HIV and Mental Health Illnesses (MHI)

In general, around the world, mental health problems are more than twice as common among people living with HIV/AIDS compared to the general population. In Ethiopia, nearly half of the people living with HIV/AIDS were depressed or anxious, and the proportion increased to two-thirds among people co-infected with HIV and TB. Other studies have found that 30-40% of HIV+ pregnant women and new mothers in Ethiopia report symptoms of depression.

The co-occurrence of mental illness for individuals living with HIV significantly magnifies the burden of their disease. Mental illness has been associated with increased risk for HIV infection, and has been noted to occur at higher rates for HIV-infected individuals around the time of diagnosis or during the course of their illness. Further complicating the course of illness are the strong associations between the occurrence of psychiatric symptoms and poor adherence to antiretroviral treatments. Evidence suggests that treatment of co-morbid psychiatric conditions may improve adherence to HAART regimens, underlining the importance of recognition and treatment of psychiatric conditions.

Evidence further points to continued under-recognition and under treatment of these conditions among HIV-infected individuals. Factors contributing to under-recognition of psychiatric symptoms relate to:

- The overlap between the symptoms of depression and symptoms of HIV disease.
- The recognition of certain psychiatric disorders, specifically depression and anxiety, is the mistaken belief that these symptoms are expected in those diagnosed with HIV disease. Depressive or anxiety symptoms that are more severe or that persist beyond the period of initial discovery of HIV infection should prompt evaluation and treatment.
- The neuropsychiatric side effects associated with some ARVs. Zidovudine, didanosine, nevirapine, abacavir, and efavirenz have been associated with neuropsychiatric symptoms.

a. Why mental health care important for people with HIV/AIDS?

HIV infects the brain, and it can directly cause some mental health problems. When HIV infection results in compromised immunity it can lead to brain infections (toxoplasmosis, Cryptococcus, cytomegalovirus, and others) and tumours. The stresses of living with HIV/AIDS can also cause mental health problems, as can problems with poor nutrition, sleep, and chronic pain that are caused by, or made worse by, the illness. In addition to the trauma of learning about their diagnosis, people living with HIV face ongoing stigma, the burdens of ongoing medical treatment, and worry about their own health and that of friends and family members. In addition, some of the medicines used to treat HIV and the infections that come with it have mental health side effects.

Another reason why mental health problems are more common among people with HIV/AIDS is that having a mental health problem is one of the risk factors for becoming infected with HIV. People with mental health problems are more likely to be exploited by others and less able to negotiate safe sexual relationships with partners. They may be less likely to stay in the kind of steady, long-term relationships in which partners can protect each other from getting HIV. Some kinds of mental health problems (in particular substance abuse) make it less likely that people will take precautions (using condoms, avoiding impulsive sexual activity) to avoid getting infected.

b. Priority mental health disorders (WHO)

1. **Psychosis:** this is the collective name for a group of serious disorders characterized by changes in behavior (for example poor self-care, restlessness), strange thoughts or beliefs (for example believing that others wish to do the individual harm) and related dispositions.
2. **Mania:** a form of severe mental illness in which a person is excessively happy or irritable (experiences extreme mood swings), appears over-active and sleeps poorly. People with mania have poor reasoning skills (they have difficulty understanding what is good and what is bad), and display excessive self-confidence.
3. **Depression:** this is the most common priority disorder and is characterized by excessive sadness, loss of interest, lack of energy and related symptoms.
4. **Suicide:** refers to the intentional ending of one's own life.
5. **Abuse of alcohol and other substances:** this is the excessive use of these substances to the detriment of one's health.
6. **Childhood mental disorders:** these are different types of mental disorders related to childhood developments.
7. **Dementia:** is characterized by memory problems and broader problems with thinking and understanding.
8. **Epilepsy:** this is a chronic or longstanding condition caused by abnormal electrical conductions in the brain. In its most obvious form, it is characterized by episodic loss of consciousness and repetitive jerky movements of the body.

c. How to approach HIV clients with Mental Health Illnesses

Using the mental health screening tool, evaluate all PLHIV in care at enrollment and at every visit. If the client has yes for any of the screening points, link or refer the client for further psychiatric evaluation and support. See annex 24 for Mental Health Screening tool. Treat MHI clients according to the National Mental Health Training Manual.

11.7. Management of Advanced HIV Diseases

Definition of advanced HIV disease

For adults and adolescents, and children older than five years, advanced HIV disease is defined as CD4 cell count <200cells/mm³ or WHO stage 3 or 4 event.

All children younger than five years old with HIV are considered as having advanced HIV disease.

Diagnosis of Advanced HIV Disease:

Diagnosis of advanced HIV disease is done through CD4 testing of clients at base line for those initiating treatment,(re-engaging with care after a period of interruption for >28 days) and targeting those who have interrupted ART treatment and with persistently unsuppressed VL (>1000 copies per ml). In addition to CD4 testing and when CD4 testing is unavailable, a clinical diagnosis of WHO stage 3 or 4 can also be used to diagnose advanced HIV disease.

Package of care for PLHIV with advanced disease

In Ethiopia among adults (ages 15-64 years) living with HIV, 35.8% (45.6% of men and 31.1% of women) had immunosuppression with CD4 count less than 350 cells per microliter (µL). Whereas among adults who reported that they were unaware of their HIV-positive status, 22.0% had severe immunosuppression -a CD4 count less than 200 cells/µL (16.9% of men and 26.5% of women), (EPHIA 2017/2018). People with advanced HIV disease are at high risk of death, even after starting ART, with this risk increasing with decreasing CD4 cell count. The most common causes of death are TB, severe bacterial infections and cryptococcal meningitis.

A package of interventions including screening, treatment and/or prophylaxis for major opportunistic infections, rapid ART initiation and intensified adherence support interventions should be offered to everyone presenting with advanced HIV disease including those who are re-engaging with care after a period of interruption for >28 days. Baseline CD4 cell count testing

for all people living with HIV remains clinically important in order to identify those who have advanced HIV disease and who should be offered the package of care as presented in table 11.4.

Table 11.14 Summary of Components of care for people with advanced HIV disease

<i>Areas for the package</i>	<i>Intervention</i>	<i>CD4 cell count</i>	<i>Adults & Adolescents</i>	<i>Children</i>
<i>Diagnosis</i>	<i>Sputum Xpert ® MTB/ RIF as the first test for TB diagnosis among symptomatic PLHIV</i>	<i>Any</i>	<i>Yes</i>	<i>Yes</i>
	<i>LF-LAM for TB diagnosis among people with symptoms and signs of TB</i>	<i>≤100 cells/mm3 Or at any CD4 count if seriously ill</i>	<i>Yes</i>	<i>Yes</i>
	<i>Cryptococcal antigen screening</i>	<i>≤100 cells/mm3</i>	<i>Yes</i>	<i>No</i>
<i>Prophylaxis & preventive treatment</i>	<i>Co-trimoxazole prophylaxis</i>	<i>≤350 cells/ mm3 or clinical stage 3 or 4</i>	<i>Yes</i>	<i>Yes</i>
	<i>TB preventive treatment</i>	<i>Any</i>	<i>Yes</i>	<i>Yes^a</i>

	Fluconazole pre-emptive therapy for cryptococcal antigen– positive people without evidence of meningitis	<i><100 cells/ mm3</i>	<i>Yes</i>	<i>NA</i>
<i>ART initiation</i>	<i>Rapid ART initiation as mentioned above</i>	<i>Any</i>	<i>Yes</i>	<i>Yes</i>
	Defer initiation if clinical symptoms suggest TB or cryptococcal meningitis	Any	Yes	Yes
Adapted adherence support	Tailored counselling to ensure optimal adherence to the advanced disease package, including home visits if feasible	<200 cells/mm3	Yes	Yes

^a. For children younger than 12 months, only those with a history of TB contact should receive TB preventive treatment if the evaluation shows no TB disease.

Chapter summary:

- Opportunistic infections are the predominant causes of morbidity and mortality among HIV- infected patients.
- Main areas affected by HIV are the nervous, gastro-intestinal, respiratory systems and the skin.
- Co-trimoxazole preventive therapy (CPT) should be implemented as an integral component of a package of HIV-related services.
- HIV care settings should implement the WHO Three I's strategy: intensified TB case-finding, isoniazid preventive therapy (IPT) and infection control at all clinical encounters.
- Diarrhoea is among the most frequent symptoms of HIV disease. Delay in treatment can result in fluid loss and hemodynamic instability.
- Immediate ART initiation is not recommended in HIV-infected patients with Cryptococcal meningitis due to the high risk of IRIS, which may be life-threatening.
- Certain STIs facilitate the transmission of HIV.
- 30-40% of HIV+ pregnant women and new mothers in Ethiopia report symptoms of depression.

Chapter 12: Antiretroviral Therapy

Chapter Description:

This chapter describes the antiretroviral treatment for PLHIV and different types of combination for children and adults. In addition, it discusses the monitoring of ART.

Purpose: Explain preparation of patients for ART initiation and monitoring

Enabling Objective

By the end of this chapter participants will be able to:

- Demonstrate how to prepare a patient for ART initiation
- List combination of ARV drugs in the regimen,
- Understand the purpose and steps of adherence support and monitoring,
- Explain barriers and list different assessment methods/ tools for adherence monitoring,
- Describe common side effects of frequently prescribed ARV drugs,
- Describe how to monitor for treatment response, drug toxicity and treatment failure.
- Describe Enhanced adherence supports,
- Describe the way second line treatment failure identified and managed,
- Describe common modalities for differentiated service based on client's need,
- Describe how to ensure the adequate and continuous availability of quality-assured ARV pharmaceuticals,

Chapter Outline:

- Introduction on ART
- Preparation
- ART regimen
- Adherence support
- Monitoring
- Differentiated service delivery models

Allocated time: 695 minutes

Learning activities

- Reading
- Discussion,
- Demonstration,
- Exercise,
- Role play,
- Practice in skill station,

12.1 Antiretroviral therapy (ART)

ART is an abbreviation for Anti-Retroviral Therapy, the treatment of HIV infected individuals with anti-retroviral drugs. It is also called HAART (Highly Active Antiretroviral Treatment) to explain the use of at least three anti-retroviral drugs with different mechanism of actions for the treatment of HIV infection to have effective viral suppression.

The goal of ART is to suppress the replication and reduce the number of virus in the blood and increase the number of CD4 as much as possible and finally improve the general health of the client.

NB: Antiretroviral therapy suppress the viral replication to below detectable level, however the virus can never be eradicated completely from the body; hence the person should take the drugs lifelong, even if the symptoms have disappeared.

Since the virus cannot be eradicated, safer sex using condom should be practiced.

a. Benefits of ART:

- ❖ Significantly decreases morbidity and mortality
- ❖ Improves quality of life
- ❖ Decreased number of orphans
- ❖ Reduces mother-to-child transmission of HIV
- ❖ Increased awareness in the community: Increased number of people who accept HIV testing, hence increase in enrolment into treatment,
- ❖ Decreased stigma surrounding HIV infection,
- ❖ Increased motivation of health workers, they feel they can do more for HIV patients,
- ❖ Make treatment of opportunistic infections and providing palliative care less expensive,
- ❖ Improve the productivity of PLWHIV,

- ❖ Decreases new HIV infections in the community by large,

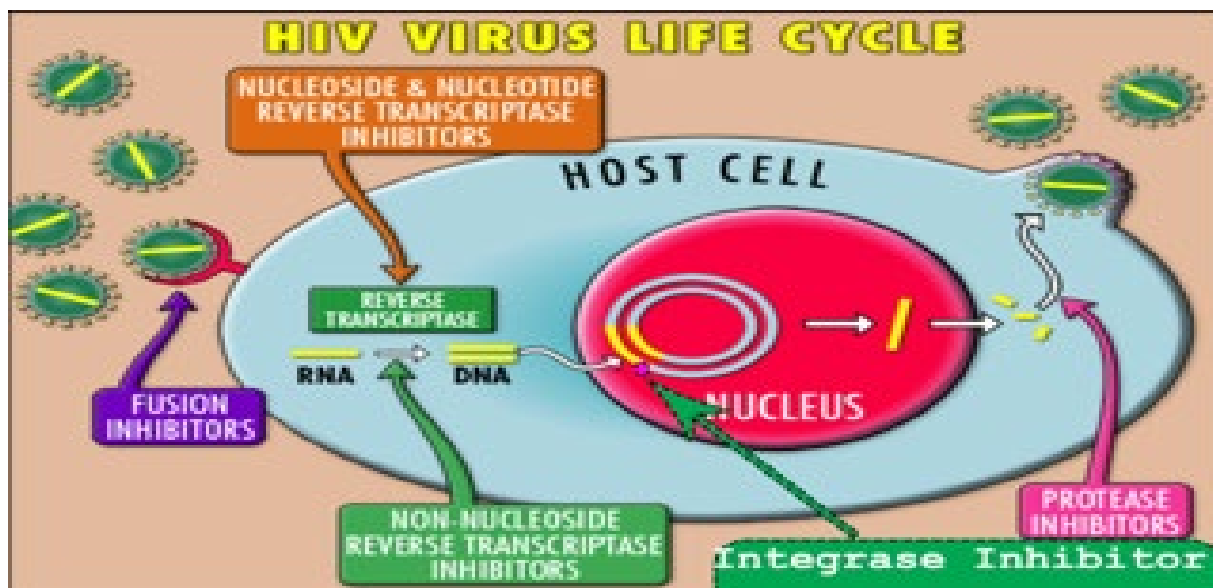


Figure 12.1. ARV drugs and their action sites on HIV virus

Table 12.1. Classes and dosages of Antiretroviral Drugs currently used in Ethiopia

Drug class/ drug		Adult Dose ^{a, b}
Nucleoside RTIs	Abacavir (ABC)	300 mg twice daily
	Lamivudine (3TC)	150 mg twice daily or 300 mg once daily
	Zidovudine (AZT/ ZDV)	300 mg twice daily.
Nucleotide RTI	Tenofovir (TDF)	300 mg once daily
Non-nucleoside RTIs	Efavirenz (EFV)	400 mg once daily or 600 mg once daily
Protease inhibitors	Lopinavir/ ritonavir (LPV/r)	400 mg/ 100 mg twice daily
	Atazanavir/ ritonavir (ATV/r)	300mg/100 oncedaily
	Darunavir/ ritonavir (DRV/r)	800 mg + 100 mg once daily ^c or 600 mg+ 100 mg twice daily ^d

Integrase strand transfer Inhibitors (INSTIs)	Dolutegravir (DTG)	50 mg once daily ^c
	Raltegravir (RAL)	400mg twice daily
^a These dosages are in common clinical use. The dosages in this table were selected on the best available clinical evidence. Dosages that can be given once daily or twice daily were preferred in order to enhance adherence to therapy. The doses listed are those for individuals with normal renal and hepatic function. Product-specific information should be consulted for dose adjustments that may be indicated with renal or hepatic dysfunction or for potential drug interactions with other HIV and non-HIV medications. ^b Zidovudine, lamivudine, dosages should be adjusted for the Creatinine clearance during renal insufficiency whereas Tenofovir should be avoided. ^c For individuals with no previous use of protease inhibitors ^d For individuals with previous use of protease inhibitors ^e In case of TB-HIV co-infection, the dose of DTG should be doubled, 50mg BID;		

The Pediatric ARV drug formulations and dosage depends on age and weight of the children. For the detail see Annex5 and Annex 6.

b. Why a Combination of 3 or more ARV drugs use?

- 1. It takes three or more drugs to have sustained viral suppression (low level of virus in the body).** HIV makes new copies of itself very rapidly. Everyday billions of new copies of HIV are made and many infected cells die. Giving a single drug might suppress viral replication for a short period of time but resistance to the drug develops soon. The same holds true to two drugs regimen and therefore giving two drugs alone for treatment is strongly discouraged. Whenever ART is given, it is administered as a minimum of three drugs combination referred as HAART.
- 2. Antiretroviral drugs from different drug groups attack the virus in different ways.** In the beginning of this chapter, we learned how different anti-HIV drugs attack HIV at different steps of the process of making copies of itself (first when entering the cell centre, and then when new copies want to leave the cell). Hitting two targets increases the chance of stopping HIV and protecting new cells from infection.

- 3. Combinations of anti-retroviral drugs may overcome or delay emergence of resistance.** Resistance is the ability of HIV to change its structure in ways that make ARV drugs less effective. HIV has to make only a single, small change to resist the effects of some drugs. For other drugs, HIV has to make several changes. When one drug is given by itself, sooner or later HIV makes the necessary changes to resist that drug. But two drugs are given together; it takes longer for HIV to make the changes necessary for resistance. When three drugs are given together, it takes even longer.

12.2 Preparing patients for ART

Before people start ART, it is important to have a detailed discussion with them about their willingness and readiness to initiate ART. The following issues should be addressed during the preparation:

- The benefits of ART, advantage of Rapid initiation over delayed initiation,
- Possible adverse effects of ARVs & OI medications,
- Information regarding lifelong treatment,
- The required follow-up and monitoring visits,
- Detailed examination and treatment of OIs,
- Proper adherence counseling and support,
- Counseling on effective FP planning choices,
- Counseling on disclosure and screening of family members,
- Education on safer sex

Requirements for initiation of ART:

1. HIV positive test result with written documentation,
2. Verification of HIV infection with re-testing at ART initiation point,
3. Ensure any opportunistic infection has been screened and addressed according to the standard guidelines,
4. Ensure that all adherence barriers are addressed,
5. Readiness of the patient for ARV therapy by ensuring:

- Patient understands benefits of treatment, possible side effects, adherence and lifelong commitment with regular follow up,
 - Patient expressed desire for treatment and actively involved in own care,
 - Treatment supporter arranged, if possible,
 - Patient has good adherence to care and other medications,
 - Barriers to adherence have been addressed such as highly unstable social situation, heavy alcohol use or other substances dependence, serious psychiatric illness or other severe comorbidities,
6. Patient fits criteria to be started on ART at the first-level facility (HCs, DICs, clinics):
1. the patient has no conditions requiring referral to the next level or evaluation by physician like:
 - Severe illness
 - Any condition in stage 4 with 2 exceptions: non-severe esophageal thrush or chronic Herpes simplex ulcerations,
 2. No jaundice or known liver problem,
 3. No Chronic illness such as diabetes mellitus, heart or kidney disease, etc.,
 4. No Prior ARV exposure other than for treatment or prophylaxis purpose (like sharing, substandard regimen,)
7. For patients with any of the above conditions that need referral to hospital or evaluation by physician, the condition will be managed and ART will be started at hospital.

Telling Patients on need of HIV retesting for verification: “You are going to start taking ART for life. Committing to lifelong treatment needs verifying of your result. So I will re-test you for verification, just like Bank verifies Checks.

NB: The retesting result should be documented on the retesting register (see annex 16)

EXERCISE 12.1

1. Define the abbreviation HAART: _____

2. How many different drugs do we need to combine in order to have a sustained viral suppression? _____
3. What are the two main goals of ART?

4. List requirements to initiate ART _____

12.3 When to Start ART?



Case scenario 1:

Abebech, a 32 years old woman, who has been diagnosed with HIV and referred to ART clinic where you are working. When she arrived, she has whitish patches on the tongue with difficulty of swallowing.

Q. What is the next step?



Case scenario 2:

Hailu is a 27 years old man, who is HIV positive and diagnosed with pulmonary tuberculosis. His CD4 count is 600cells/mm³

Q. What is the clinical stage?

Q. What is the next step?



Case scenario 3:

Zeberga is a 42 years old HIV positive man and came to the ART clinic referred from VCT. On history and examination he did not have any clinical findings. His CD4 count is 465cells/mm³.

Q. What is the clinical stage?

Q. What is the next step?



Case scenario 4:

Tolera is an 8 year old child tested HIV positive at paediatric OPD and came with his mother to the ART clinic. On clinical assessment there was no abnormal finding. His CD4 count is 900cells/mm³.

Q. What is the clinical stage?

Q. What is the next step?

Early treatment initiation is associated with improving or maintaining health, improving survival and reducing the incidence of Opportunistic infections and HIV prevention benefits at the family & community level.

12.3.1. When to start ART for adults and adolescents

All HIV positives are eligible for ART. The ideal time for ART initiation depends on the clinical condition and readiness of the client. Understanding of clients about HIV and the importance of life long treatment adherence need to be emphasized. All adherence barriers should be exhaustively assessed and addressed before considering ART initiation. For those HIV positive

clients, who understand the importance and benefitsof life long adherence and are ready for early initiation, **start ART as early as possible including same day.**

Rapid ART initiation is defined as initiation of ART same day or within seven days of HIV diagnosis if there are no contraindications. Rapid ART initiation should be offered to all people living with HIV following a confirmed HIV diagnosis, clinical assessment, and assessment of client readiness. ART initiation should be offered on the same day for people who are ready to start .The offer of rapid ART initiation, including sameday (initiating ART on the date of confirmed HIV diagnosis), increases the number of people starting ART, reduces mortality, and may further reduce both mother-to-childtransmissionand transmission to HIV-negative partners. This recommendation applies to all PLHIV at all age groups and is particularly important in people with very low CD4cell counts who have an increased risk of death.

History and clinical examination to evaluate for opportunistic infections shall be carried out before rapid ART initiation. People with advanced HIV disease should be given priority for assessment and initiation.

Table 12.2: Summary of recommendations on when to start ART in adults, adolescents, pregnant and breast feeding women and children

Category	Specific Recommendation
All Adults	Start ART rapidly, preferably same day,((i.e. on the date of confirmed HIV diagnosis) to all adults and adolescents who are ready and willing regardless of their WHO clinical stages and CD4 counts, except for those delaying ART initiation is recommended (TB meningitis and cryptococcal meningitis). diagnosis

and adolescents including pregnant and breast-feeding women	• Start ART in all TB patients living with HIV as soon as possible within 2 weeks following initiation of anti-TB treatment regardless of their CD4 count except when there is TB meningitis. If a patient has TB meningitis ,delay ART for at least 4 weeks and initiate within 8weeks after treatment of TB meningitis is initiated. • For women identified at labor and delivery, provide ART within the same hour of HIV diagnosis with brief counseling and provide detailed counseling on ARV and adherence after delivery • In addition antiretroviral therapy is recommended for all patients with HIV and drug- resistant TB (both MDR/XDR-TB) requiring second-line anti-tuberculosis treatment, irrespective of CD4 cell count, as early as possible (within the first two weeks) following initiation of anti-tuberculosis treatment. • ART should be delayed by 4-6 weeks of ART following initiation of treatment for cryptococcal meningitis. Earlier ART is associated with more severe adverse event and increased mortality with cryptococcal meningitis.
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12.3.2. When to start ART in children

- Start ART as early as possible to all children with a confirmed HIV diagnosis regardless of their WHO clinical stages and CD4 counts, **except when there is TB meningitis**. If a patient has TB meningitis ,delay ART for at least 4 weeks and initiate within 8weeks after treatment of TB meningitis is initiated. In addition, ART should be delayed by 4-6 weeks of ART following initiation of treatment for cryptococcal meningitis.As the level of maturity varies, children needs parental consent and support.
- For HIV infected infants diagnosed with the first DNA PCR result, initiate ART and take DBS specimen for confirmatory DNA PCR. Continue ART if the second DNA PCR confirms positive results; whereas if the second DNA PCR turns negative, without holding the ART, make the 3rd DNA PCR test.

12.4 What ART regimen to start with (First-line ART)?

12.4.1. First line ART for Adults and adolescents including pregnant and breast feeding women

It is recommended to use simplified, less toxic, and more effective with rapid onset of viral suppression and convenient regimens as fixed-dose combinations (FDC) for ART. The preferred first-line regimen for adults and adolescents (≥ 10 years of age or ≥ 30 kg body weight) including pregnant and breast-feeding women is TDF+ 3TC+DTG as a once-daily dose.

12.4.2. First-line ART for children:

The preferred first-line regimen for children > 4weeks and $\geq 3\text{kg}$ but less than 10 years is ABC+3TC+DTG. Availability of a new generic formulation of 10mg DTG in 2021 has created the opportunity to use DTG based regimen for children living with HIV (CLHIV) who are at least 4weeks of age and weight $\geq 3\text{kg}$ or more. For those who are not eligible for the preferred first line option, it is also important to provide alternative regimens ABC+ 3TC+LPV/r or AZT+3TC+DTG **or** as special circumstances ABC+3TC+EFV, AZT+3TC+EFV or AZT+3TC+LPV/

Children grow fast and increase their weight rapidly. It should be emphasized that progressive increase of pediatric ARV dosage that is adjusted to respective growth and weight gain along with age is crucial.

Table 12.3: Updated summary of first-line ART regimens for adults, adolescents, and children

Table. Summary table for what ART regimen to start with (first-line ART)

Population	Preferred first-line regimens	Alternative first-line regimens	Special circumstances ^a
Adolescents (10 to 19 years OR weight ≥ 30 kg), adults, pregnant, childbearing, and breast-feeding women including those with TB/ HIV- co infection	TDF+ 3TC+DTG ^b (FDC)	TDF + 3TC + EFV AZT + 3TC + DTG AZT + 3TC + EFV	AZT+ 3TC + ATV/r TDF+ 3TC+ ATV/r <i>ABC^c+3TC+DTG</i>
Children > 4weeks and ≥ 3 kg but less than 10 years	ABC + 3TC + DTG ^{b*}	ABC+ 3TC+LPV/r AZT+3TC+DTG	ABC+3TC+EFV ^d AZT+3TC+EFV AZT+3TC+LPV/r ^a

a. ABC or boosted PIs (ATV/r, LPV/r) can be used in special circumstances for those clients who could take neither DTG nor EFV due to contraindication and/or side effects.

^b In case of TB-HIV co-infection, the dose of DTG should be 50mg **BID**.

^{b*} In case of TB-HIV co-infection, the dose of DTG should be doubled depending on body weight of the child.

^c For PLHIV with renal insufficiency and anemia

^d EFV is for children 3 years and older.

EXERCISE 12.2

Identify the regimen code, and class of the drug

Regimen	Code		Class
	Adult&Adolescents (≥ 10 years old; OR ≥ 30 kgs body wt.)	Children > 4 weeks and ≥ 3 kg but less than 10 years	
TDF-3TC-DTG			
TDF-3TC-EFV			
ABC+3TC+DTG			
ABC+3TC+LPV/r			
AZT-3TC-EFV			
AZT+3TC+DTG			
ABC-3TC-EFV			

12.5 Adherence support for ART initiation and monitoring

Case scenario: Tulema must come regularly to the health facility. The nurse always checks about adherence. Tulema has never forgotten taking his drugs. He has been taking his medicines every day at the same time.



Whereas, Gemechis is a very busy man. He needs to travel a lot for his business. Sometimes, he is stressed because he has so many things on his mind. From time to time, he forgets the pills, especially when travelling, because he forgets to take them with him, or because he is in a meeting.

Q1. What would you do to help Gemechis?

Q2. What would you tell Gemechis to improve his adherence?

Q3. What is the key message we learn from the 2 cases?



12.5.1. Adherence Preparation for ART Initiation

The first time the patient goes home with ART is crucial. You must be sure all information is well understood and that all possible tools to assure adherence are provided. Explain and demonstrate how to open the ARV box.

It is important to explain carefully, using the Patient Treatment Card.

a) **ASSESS:**

Assess the patient's/care giver's goals for today's visit

"Today we will be initiating you/ your child on ART. Is there anything special about HIV/AIDS or ART you would like to address in today's session? Are there any concerns that you have which you would like to discuss?"

Check understanding of the information given before—make sure patient understands the illness, treatment, possible side effects and taking ARVs lifelong.

The patient has been given a large amount of information in preparing for ART. It is important to reassess the patient's understanding. Remember to use **specific questions** and to **make the patient feel comfortable**. You can say to the patient:

"I know that we have given you a lot of information about HIV/AIDS and ART. Let me ask you some questions to make sure that there is no confusion or misunderstanding as you start your treatment."

b) ADVISE:

Reinforce the information given before.

Advise on details of first-line regimen. Explain the purpose of and how to take each pill including role of parents / care givers for children. Provide and explain card summarizing treatment. Give patient a leaflet, brochure or other material available which corresponds to his/her treatment regimen. You can go through the materials together.

Make sure patient understands the importance of adherence. Remind the patient that ART is life saving and good outcome depends on taking them every day at the right time. Use the medication regimen and discuss which pills need to be taken at which specific time. Advise the patient not to miss any medication or timing. If there is missing, the patient can take the missed dose within 6 hours for medications taken on twice per day or 12 hours for medications taken once per day or 24 hours doses respectively. If these times passed, do not take, wait for the next dose.

Advise on diet. If the patient will be taking efavirenz, advise the patient not to take it with a fatty meal. Give examples of what s/he should avoid just before or after swallowing the pill.

Counsel to avoid alcohol use. These are important for adherence. It is important for the patient to realize that their regimen may not go well with alcohol:

- **Efavirenz has CNS side effects** which mean that it can make the patient sleepy or dizzy. Drinking alcohol may enhance these effects.
- **Efavirenz and NVP both can be toxic to the liver.** Alcohol is toxic to the liver. Taking both alcohol and one of these drugs can add up to cause liver damage. So it is important to reduce drinking alcohol to either very low risk levels or, if possible, to stop. There are materials that can help you counsel the patient on how to reduce or stop alcohol.
- For any medication, **high amounts of drinking can be a serious problem** because they may cause the patient to forget to take their medication.

Patients may find it difficult to stop socializing habits like binge of alcohol just to avoid stigmatization & discrimination. In such case it is good to advise on limiting alcohol intake to the level of low-risk drinking. **Explain side effects.** Warn about both the very common side

effects and how to manage them and symptoms that indicate the patient needs to come back urgently.

- **Prepare patient and treatment supporter to handle common side effects. Most side effects can be treated symptomatically.**
- **Explain which side effects are likely to improve with time (related to initiation of treatment) and their likely duration.** Again go through the Patient Treatment Card/job aid that corresponds to the regimen he/she is starting. Give the patient examples of the common side effects such as nausea or diarrhoea which occur often with the first line regimens, and tell the patient what he/she can do if these side effects occur.
- **Explain which are more serious and require urgent return to clinic.** Again use the specific Patient Treatment Card/job aid to go through the major side effects which can occur on the regimen the patient is taking.

Explain that patients can still transmit HIV infection while on ARV therapy. It is very important to still practice safer sex and other practices to prevent transmission.) Remind patients that they should still be using condoms with their partners to prevent transmitting the infection or to prevent getting re-infected (which can further impair immunity if re-infected with a different strain). It is important to repeat key information concerning HIV/AIDS and ART.

c) **AGREE:**

Make sure the patient/caregiver **agrees** to the regimen and is a true partner in the treatment plan. Ask if the patient is ready and willing? Ask if the patient is committed to start taking ARV medications lifelong?

Make sure patient/ caregiver **understands** that his or her life depends on taking the medicine every day. For children and patients who need caregiver for taking their medication, it is important to agree with caregiver/ parent on the regimen, dosing, timing, and being responsible for ensuring administration as agreed.

Agree on plan for support by treatment buddy and support groups.

Tell patient that you realize that taking the medications will be difficult and having a treatment buddy and attending the support groups can help the patient in managing this lifelong

commitment. The patient can learn adherence techniques and ask questions from others who are going through the same treatment.

d) **ASSIST:**

Develop (then reinforce on each visit) a concrete plan for the specific ARV regimen the patient is taking. Come up with a plan for how the patient will remember to take the medication before he leaves the clinic with his new ART drugs.

- **When to take?** Times for every 12 hour or every 24 hour dosing?
- **How to make it a habit?**
- **How to remember?** Provide and explain written schedule, pillbox, pill chart and other aids.

You can give a pill chart to the patient/caregiver. Check if the patient/caregiver understands correctly the information as written on the pill chart by asking to repeat the information.

For patients/caregivers who cannot read, it can be useful to indicate a colour on the pill bag (e.g. colour the bag with AZT red), and use the same colour on the pill chart. Use the sign of sun and moon to demonstrate morning and evening doses, and draw the adequate quantity of pills instead of writing the number.

Example of pill chart for patients who cannot read:

		
TDF/3TC/DTG(FDC) OR TDF/3TC/EFV(FDC)		●
AZT/3TC	●	●
AZT 300 mg	●	●
3TC 150 mg	●	●
EFV 200 mg		● ● ●
DTG 50mg/DTG 10mg		●
ABC/3TC	●	●
EFV 400/600 mg		●

If possible, provide a pillbox, so the patient can always check if they have missed a dose. Pillboxes are containers for storing medication with dividers for each day and each dose during the day. Make sure the patient understands the amount of daily medicines and the importance of refilling the pillboxes once a week.

Encourage the patient to use a pill beeper or alarm watch/cell-phone for both morning and evening doses. Beepers should be discrete to help the patient feel that confidentiality is not a risk.

You can provide a diary. This means the patient has a booklet where s/he ticks every time s/he takes the drugs. The disadvantage is that the patient can fill it out without taking the drugs, or that the patient can take the drugs but forget to tick it in the diary.

The availability of fixed dose combination ARVs has dramatically reduced pill burden and improved adherence.

Provide psychosocial support - Elicit concerns. Use empathy.

Encourage patient to join an ART adherence support group.

Provide the patient with specific information as to where and when they meet. ART adherence support group is a collection of peers/ PLHIV already taking ART and openly share their experience and challenges they face to get better adherence and overcome the challenges.

Arrange home visit, if feasible.

Really knowing your patient's home situation and helping him or her to make arrangements for adherence at home can be very helpful. The home visit also confirms where the patient lives so that if they fail to come from clinic, they can easily be followed up.

Home visits can be done in collaboration with the HEWs or community-based organizations providing support for ART.

Explore what is needed to assist the mother/caretaker with ART for her child:

"What problems might arise when you follow this plan?"

"What questions do you have about this treatment or how to follow this plan?"

Help the caregiver develop the resources/support/arrangements needed for adherence:

- Ability to bring the child for required schedule of follow- up
- Mentally clear, supportive and committed to give medication to the child,
- Home and work situation of mother that permits her giving medications regularly to the child without stigma
- Regular supply of free or affordable medication
- Supportive family or friends
- Disclosure to child and or family
- ART adherence support group

e) ARRANGE:

Arrange next follow-up visit in clinic, home visit if feasible, and next visit with responsible clinician (if required)

Agree on best way to access help between visits.

Make sure patients/ care giver for children understand where and when they will see health worker

Write it down for patient, if possible. Record what happened during this visit.

By the end of this visit, the health worker should have asked the following questions:

- ✓ Have I assessed if the patient/ care giver for child is ready for treatment?
- ✓ Is the regimen compatible with the patient's life-style?
- ✓ Have I reviewed common or potentially serious side effects and does the patient understand/ know what to do?
- ✓ Did I make the patient repeat dosing time and instructions?
- ✓ Does the patient know how to contact the health care team in case of problems or questions?
- ✓ Have I identified adherence barriers and made a plan on how to overcome them, in collaboration with the patient?
- ✓ Have I provided adherence education?
- ✓ Does the patient know the date of the next consultation?

When the caregiver and child are ready for initiating ARV therapy for the child, discussion with other clinical team members and make a plan is important.

Note that it may not be possible to prepare the caregiver and child for adherence on the same visit that you decide the child is medically eligible for ART. It usually takes at least 2 to 3 visits and the involvement of others on the clinical team and a treatment supporter.

If the caregiver needs another adherence preparation session, arrange a follow-up to reinforce key messages. Also arrange an appointment with the ART support group if the care giver wishes so.

After completion of adherence counseling session parents should be clear with:

- Who will be responsible for administration of the drugs to the child?
- What medication will be given?
- Timing and frequency of medication
- How to administer medication.

ROLE PLAY FOR CHILDREN:

PREPARATION FOR ADHERENCE TO ART

Martha is 18months old. Martha had an HIV antibody test which was positive and she has been classified as WHO stage 3. She was brought to the clinic today for a follow-up and was assessed to have: Oral thrush, and VERY LOW WEIGHT. The health clinical team has decided that she needs ART.

The health worker should counsel the mother on preparation for adherence to ART, using the 5A's. As the client is infant, the health worker should consider and emphasize the role and understanding of the caregiver.

The mother should try to behave as a real mother would in this situation.

The rest of the group should watch the role play and comment on the advice given and any difficulties faced by both the mother and the health worker during the consultation.

12.5.2. Monitor and Support Adherence

We have learned why it is necessary to take the drugs correctly, and not to miss a dose.

If the patient does not have good adherence, the virus might become resistant to the first regimen and we might have to change the initial ART regimen.

The second line regimen is often more expensive, and sometimes not so easy to take (larger amount of pills). It also often causes more side effects.

If the patient has to change several times to other regimens, it may happen that in the end there are no more possibilities to treat the patient because the virus became resistant to many types of drugs.

Fixed-dose combinations are the easiest for the patient to use. These will be provided whenever possible.

If the patient is given three separate pills to take twice daily, putting the three in a blister pack is helpful. You then need to explain to the patient to take all three at once.

a) ASSESS:

Do clinical review/evaluation (Chapter 7) and respond to any problems or change in status.

Ask about important factors that may interfere with adherence.

Possible factors/ barriers	Possible supports/ interventions/ solutions
Lack of trust to the health care worker or communication difficulties	• Try to find out why, and see if there are misunderstandings. Try not to take these things personally, and do not lose your temper. Focus on the practice and not on the person.
Literacy barriers	• Try to work with colours and pictures or symbols
Mental illness, especially depression or	• Treat depression, Refer if necessary.
alcohol abuse	• Counsel to stop alcohol.
Not understanding the disease and the treatment:	• Repeat the basic information, use visual aids.

Other beliefs of the patient may interfere:	• Instructions from some traditional healers or religions may make it difficult to accept instructions given by the classical health care worker.
Unstable living conditions, poor social support:	• Try to refer to social institutions or NGO to resolve these problems, refer to support group
Fear that when taking medications, other people will discover HIV status:	• Teach to pack the drugs you need for one day in a small bag, • and take them in the bathroom if you need to take them when visiting family or during work
Unwillingness to set long-term goals (person thinks only about the fact that NOW they have discomfort from the drugs, cannot see the benefit after many years):	• Repeated open discussion with the patient, start therapy only when the patient is ready to accept; do not look only at the medical criteria, do not lose patience.
Difficult access to health facility:	• Reimbursement of transportation fees if feasible. • Discuss possible referral to nearby ART Clinic.
Other immediate life-needs:	• Needs such as housing and food can be viewed as more pressing than taking the medications regularly, so do not ignore these things and try to find solutions
Barriers associated with the regimen	• Frequency of dosing, number of pills, food requirements/restrictions, complexity, and storage. • Check with doctor how regimen can be adapted
Barriers associated with side-effects:	• Individuals, especially those who did not have very severe symptoms of AIDS and who are now suffering side-effects from the ART, are likely to skip doses. Always listen carefully when patients mention side effects, and take it seriously. • Call for advice from doctor to see if a change in therapy can be made if side effects are interfering with

	adherence
Barriers associate with stigma and discrimination:	• Disclosure to family members, friends, • Get treatment buddy/ supporter,

There are many methods to assess adherence, but there is no single method that assess level of adherence perfectly.

a. Review the medications with the patient and their treatment buddy/ supporter. Determine whether there is any adherence problem.

Ask questions in a respectful and non-judgmental way. Ask in a way that makes it easier for the patient to be comfortable and truthful.

An individual's own report of his or her adherence (called a self-report) generally overestimates adherence. However, it can be more accurate if a good relationship with the provider established.

Usually, we ask the patient how many doses he forgot in the last three days, week and month, or we ask the most recent recall of a missed dose. However, keep in mind that patients may not be comfortable telling you the truth and they may tell you what they think you want to hear rather than what really happened.

You can use the following introduction:

"Most people with HIV have many pills to take at different times during the day. Many people find it difficult to always remember when to take the pills. It is important for me to understand how you are really doing with the medicine. Do not worry about telling me if you do not always take the dose. I just need to know what is really happening, not what you think I want to hear..."

Here are examples of other ways to assess adherence:

"Many patients have trouble taking their medications. What trouble do you have?"

"Can you tell me when and how you take each pill?"

"When is it most difficult for you to take the pills?"

"It is sometimes difficult to take the pills every day and on time. How many have you missed in the last three days?"

b. Use of self- monitoring tools like patient's medication diary, pill chart or tally sheet.

If any of these are used by the patient, ask to look at it.

c. Pill counting to check missed pills.

The provider can count the pills at the appointment to see how many doses are missed.

Disadvantages of pill count:

- Can lead patient to think/ perceive being audited/ controlled instead of being supported
- Can lead patients to deceive their care provider by manipulating the remaining pills to adjust expected adherence/ satisfy their provider's expectation, like throwing away missed pills before coming to the consultation
- Difficult to identify if the patient is sharing the drugs to other(s)
- Difficult to ensure if pills are taken at the right time

Estimating adherence based on the number of doses/ drugs missed in a certain period:

While calculating adherence using number of doses missed, always take the last one month.

Example: if you dispensed 60 doses (BID) at the last visit and the patient missed three doses, the adherence will be 95%.

$$(3/60) \times 100 = 5\%, \text{ as percentage missed adherence.}$$

If the patient did not miss any dose, then the adherence will be 100%.

Look at the following table to determine the level of adherence based on pills missed:

Table 12.5 Adherence level estimates based on doses missed

# of Missed doses out of 60 doses supplied for 1 month	% Missed	% Adherence	# of Missed doses out of 30 doses supplied for 1 month (FDC)	% Missed	% Adherence
1	2%	98%	1	3%	97%
2	3%	97%	2	7%	93%
3	5%	95%	3	10%	90%
4	7%	93%	4	13%	87%
5	8%	92%	5	17%	83%
6	10%	90%	6	20%	80%
7	12%	88%	7	23%	77%

8	13%	87%	8	27%	73%
9	15%	85%	9	30%	70%
10	17%	83%	10	33%	67%

Use your best judgment to estimate the percentage of doses the patients takes, as directed. Write your estimates of the patient's adherence ARV drugs (+co-trimoxazole prophylaxis) in the ARV drugs column of the HIV Care/ART follow-up card. Record the percentage or write G, F, or P:

G (GOOD) - equal to or **greater than 95%** adherence i.e., missing only 1 out of 30 doses or missing 2 from the 60 doses implies good adherence.

F (FAIR) - 85-94% adherence, i.e., missing 2-4 doses out of 30 doses or 4 to 9 doses from 60 doses.

P (POOR) - less than 85% adherence, i.e. missing ≥ 5 doses out of 30 doses or > 10 doses from 60 doses implies poor adherence.

Fair or poor adherence requires adherence counselling by health worker.

- **Ask the patient how many pills s/he has forgotten yesterday, last 3 days, and last month?**

It is often difficult for the patient to remember or estimate the real number of doses that were forgotten.

Asking the number of doses forgotten in the last three days makes it easier for the patient to recall forgotten doses.

Ask which dose missed, morning or evening? Any explanation?

- If adherence is **Fair** or **Poor**, the health care provider determines the reason by asking the following questions, records on the follow up form, and provides appropriate counselling based on the specific problem.
 1. Toxicity/Side effects
 2. Share with others
 3. Forgot
 4. Felt better
 5. Too ill
 6. Stigma, disclosure
 7. Drug stock out
 8. Lost/ran out of pills
 9. Delivery/travel problems
 10. Inability to pay
 11. Alcohol
 12. Depression
 13. Other

d. Other indicators of poor adherence are CLINICAL and BIOLOGICAL:

- i. Clinical indicators of poor adherence include recurrence/ or emergence of opportunistic infections and weight loss.
- ii. Biological indicators of poor adherence include the CD4 count going down or viral load going up. These can only detect adherence over many months. However, it may be helpful for the patient to know that their adherence to medication can be checked by a blood test.

b) ADVISE:

Reinforce the information given before.

Re-educate about ART and HIV/AIDS if you see that there is some confusion.

Give additional information that may help with adherence problem.

Explain any misconceptions the patient may have.

Taking the drugs does not interfere with effect of any religious activities or holy water. Continue taking ARV drugs as prescribed.

Advise on methods of coping with barriers:

- using alarm, placing the drug in specific place,
- putting extra pills/ drugs in hand bag, the hood of a car, at work or friend's place
- taking extra drugs with them if they go away
- assigning reminder from family member,
- collecting pills from nearby facility if run out of drugs/ or forgotten to take, ...

Advise on any suggested changes in the regimen (after consulting with clinician) if there is one. Usually side effects require only changing one drug, not stopping—consult with a doctor or medical officer if this is necessary).

c) AGREE:

Agree on any changes in Treatment Plan and solutions to adherence problems (if present).

Discuss the agreements you have reached and check for their commitment.

d) ASSIST:

Provide adherence support.

It is important to learn from previous difficulties or mistakes: discuss problems that occurred in adherence (not just ART, but also prophylaxis) and develop strategies to overcome them in the future.

Reinforce interventions which match the patient's needs and adherence problems, if present:

- Provide pill box if available, provide with format for tally sheet or drug diary, Link with supporting organizations which work on psycho-socio-economical supports.

Make sure the patient has:

- Plan to link taking medications with daily events such as meals.
- Any device or skill that he or she needs for proper adherence to treatment.
- Make sure the patient has the support he or she needs,
- **Get help from treatment supporter, other family members and friends or peers.** Encourage the patient to get friend(s) to help him or her remember difficult dose times, such as when they go out at night.

Help patient and treatment supporter to find solutions.

- ✓ Make sure your patient knows how to contact the clinic if he or she has difficulties taking the drugs
- ✓ Propose to arrange an appointment with someone who is already taking the same treatment (e.g. support group)
- ✓ Home visits might provide information about the living conditions of the patient, which might be helpful for the staff from the clinic to know. Home visits may sometimes encourage the patient.
- ✓ A 'treatment buddy' helps the person in adhering to his or her regimen by providing emotional support and helping the person by reminding support to them when to take the medication. The 'buddy' can be a friend or family member, HIV infected or not, and can also go to the consultation together with the patient. Remember to record their name and telephone number on the HIV Care/ART treatment card as Buddy.

If adherence problem is identified with a patient:

- Explain the problem/ situation to the patient,
- Get help! Engage more treatment team members.
- Link with home-based care for help and home visits, psycho-social support.

- Consult/ Seek help from higher level ART clinic provider/ adherence staff if regimen is too complicated or not tolerated or low adherence.

Refer to higher level if the identified barrier(s) need clinical management. If missing doses is repeated or adherence problem persisted, use special interventions (link to adherence supporters, make home visit, shorten visit appointment, or provide Enhanced Adherence Support (EAS), etc.).

All the time remind the patient

- ✓ THE DRUGS HAVE TO BE TAKEN EVERY DAY AT THE SAME TIME FOR LIFE!
- ✓ If a drug has to be taken two times per day, there should be a 12 hour interval between the two doses. For example, it is not good if you take the morning dose at 6:00AM and the evening dose at 09:00 PM, because the interval is more than 12 hours, and the concentration of drug in your body will be too low for several hours, which can create resistance.
- ✓ If you forget, take the dose within 6 and 12 hours for twice daily & daily regimen, respectively. After that, wait until the next dose.
- ✓ BUT DO NOT TAKE DOUBLE DOSE!

e) ARRANGE:

Record adherence estimate on patient's follow up card.

Arrange for refills.

Arrange for consented next follow-up visit at the health facility:

Make sure that the patient/ care giver for a child and supporter understand the follow-up plan and how to contact the clinic team if there is a problem. There is no major difference in approaches for monitoring treatment adherence in adults and children. All the above apply for both.

SOME TIPS AND TOOLS TO HELP YOUR PATIENT

Store the ARV drugs in a cool and dry place. Keep out from direct sunlight and away from reach of children. Storing medications in the car, behind refrigerator or in a warm place like kitchen table/ shelf due to fear of stigma or other reasons may compromise its effectiveness. A bottle of ARV drugs is usually provided with sachet of desiccants to protect

from moisture. Do not remove the desiccant from the bottle or if transferring the drugs to another bottle is important also take the desiccant as well.

Practice safer sex. As the patient will feel better with ART, yet the virus cannot be eradicated from the body, it is advisable to have sex with condom.

Initiate ART for patients who understand, get committed and motivated on the treatment.

The trained health care provider shall have interest and motivation to follow patients on ART.

For children, ARV dose shall be modified/ increased as their weight increase.

Patients on ART shall continue taking ART for life and be monitored regularly. Patients shall be advised to keep records of their daily medication in different forms like diary, tally or other convenience.

Never initiate patient on ART with medications which are not sustainably available. Initiate with national recommended regimen. If more than one members of a family are sick with HIV/ AIDS, all efforts should be made to treat all members of the family, in order to avoid sharing drugs.

Priority shall be given to Pregnant/ lactating women and under 5 children in all service packages. (Initiation, VL monitoring)

SOMETHING TO THINK ABOUT...

Some of the patients who are now feeling very well under ART were previously so sick that their only thought was how to survive, or how to arrange things before they die. *When they get well, the patient living condition may change and emotional challenges may arise. For example: Starting a new relationship? Establishing a family? Having children?*

These are important issues that every health worker should be aware of.

Role play with EPT:

A 56 years old HIV/AIDS patient who was diagnosed 9 years ago has got a history of stage IV disease and has started ART with TDF/3TC/DTG three Weeks. He describes having nausea initially, but it has resolved. In discussing the adherence with the patient, he tells you that he/she sometimes forgets to take his medication, but only when he goes out to drink with his friends. Also figures it did not matter because when he remembers just doubles the next dose. (Patient treatment support is his spouse).

12.6 Monitoring ART



Case scenario on monitoring the response to

ART: Kabsay was started with TDF-3TC-DTG. After a month of treatment his condition deteriorated with worsening of cough, high grade fever and enlarged multiple tender cervical lymph nodes.

- What is the most likely diagnosis?
- What are the differential diagnoses?
- What are the next steps?

12.6.1. Clinical and Laboratory monitoring before and after initiating ART

Standardized clinical assessment of patients and, when available immunological, are recommended at baseline for assessment of clinical condition and decision making. Before ART initiation and thereafter patient readiness and adherence to therapy are always assessed and necessary support provided. Opportunistic infections including TB, IRIS, and co morbidities shall always be looked for and managed.

Table 12.6. Baseline and follow-up assessment

Baseline assessment, week 0 (First encounter)	
Objective: to conduct initial assessment.	
Assess	Act
• Check/Assess if retesting for verification is done and documented. • Assess: socio-economic status, any HIV related illnesses in the past, advanced HIV disease, symptom screen for TB, other OI, co-morbidities, pregnancy, past and current medication. • Determine WHO staging. • Clinical records: fill intake form, follow up form and registers (positive tracking, pre ART or ART registers). • Counselling and education: adherence, treatment readiness, disclosure, and address adherence barriers. • BP and BMI	• Do retesting for verification if not done • Determine treatment readiness. • Start CPT and TPT if clinically indicated • Consider provision of prophylaxis for other OIs (e.g. Fluconazole) • Treat OIs. • Manage co-morbidities • Start ART for those who are ready and have no adherence barriers; and give appointment to return after two weeks. If not, give appointment to return within one week. For those who still defer ART continue to counsel on benefit of rapid ART initiation and start them on ART as early as possible. (N.B. continue follow up counselling sessions until the patient is initiated on ART). • Patients who are not initiated on ART due to meningitis (TB/cryptococcus), (Refer section xx) • Continue ART for transfer-ins. • Screen for STI • Screen for cervical cancer for women between 15-49 • Refer if necessary.
2nd visit: 2 weeks after initiation	

Objective: To determine toxicity/intolerance, adherence, and IRIS	
• Clinical assessment for: IRIS, toxicity etc. • Assess and support adherence, addressing adherence barriers. • Provide counseling and education including prevention of HIV transmission. • Lab tests if necessary. • Support disclosure if not done. • BP and BMI	• Manage toxicity as indicated. • Treat OI if diagnosed. • Give appointment to return in 2 weeks. • Screen for cervical cancer for women between 15-49 if not done • Screen for STI • Refer if necessary.
3rdvisit: 4 weeks after initiation	
Objective: Same as second visit	
• Same as 3rdvisit. • Hgb if patient is on AZT. • BP and BMI	• Refill ART and other medicines as necessary for one month. • Treatment of OI and comorbidities if identified. • Manage drug toxicity and intolerance. • Assess and provide adherence support and patient education including HIV prevention. • Do other lab tests as indicated • Screen for STI • Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 4 weeks.
4thvisit- 8 weeks after initiation	
Objective: same as 3 rd visit	

• Same as 4th visit. • BMI and BP	• Refill ART and other drugs as necessary for 1 month. • Treatment of OI and co-morbidities. • Manage toxicity and intolerance. • Provide adherence support and patient education including HIV prevention. • Screen for STI • Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 4 weeks.
5thvisit: 12 weeks after initiation Objective: Same as 4 th visit.	
• Same as 5thvisit. • BP and BMI	• Refill ART and other drugs as necessary for 1 month. • Treatment of OI and other co-morbidities. • Manage toxicity and intolerance. • Assess and provide adherence support and patient education including HIV prevention. • Screen for STI • Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 4 weeks
6thvisit: 16weeks after initiation Objective: Same as 5 th visit	
• Same as 6thvisit. • BP and BMI	• Refill ART and other drugs as necessary for 1 month. • Treatment of OI and other co-morbidities. • Manage toxicity and intolerance. • Assess and provide adherence support and patient education including HIV prevention. • Screen for STI

	• Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 4 weeks.
7thvisit: 20 weeks after initiation Objective: Same as 6 th visit	
• Same as 6thvisit. • BP and BMI	• Refill ART and other drugs as necessary for 1 month. • Assess and provide adherence support and patient education including HIV prevention. • Treat OI and other co-morbidities. • Manage toxicity and intolerance. • Screen for STI • Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 4 weeks.
8thvisit: 24 weeks after initiation • Objective: Same as 7th visit	
• Same as 6thvisit. • BP and BMI	• Assess and provide adherence support and patient education including HIV prevention. • Determine viral load. • Refill ART and other drugs as necessary for 3 months. • Treat OI and other co-morbidities. • Manage toxicity and intolerance. • Screen for STI • Screen for cervical cancer for women between 15-49 if not done • Refer if necessary. • Appointment to return after 3months.

Note:

- Newly started patients will be appointed every two weeks during the first month of treatment and every 4weeks (every month) then after until 24 weeks of treatment. After the 24th week of initiation of antiretroviral therapy patients will be scheduled to return every twelve weeks.
- OIs prophylaxis, treatment for co-morbidities and SRH services should be provided in the context of/integrated with ART multi month dispensing.
- Patients should be encouraged to come at any time if they have concerns and can be seen out of the above schedule whenever necessary.

For follow up schedule during COVID19 pandemic, refer the national interim guidance

Table 12.7: Recommended tests for HIV treatment monitoring and approaches to screening for co-infections and non-communicable diseases

Management	Recommended	Desirable (if feasible)
Baseline/ART initiation	• CBC including hemoglobin ^d • CD4 cell count. • Cryptococcus antigen if CD4 cell count ≤ 100 cells/mm³. • FBS • Gene Xpert, LF-LAM if eligible	• HBV (HBsAg) serology. ^a • HCV serology. • Pregnancy test. • HPV-DNA if available • Lab investigations for major non communicable chronic diseases and comorbidities ^b • Pregnancy test • Serum creatinine and estimated glomerular filtration rate (eGFR) for starting TDF^c
Receiving ART	• HIV viral load at 6 and 12 months after initiating ART and every 12 months thereafter • Viral load testing for pregnant mothers: ○ Newly diagnosed mothers: after 3 months followed by every six months until the MTCT risk ends, then routinely every 12 months. ○ For those who are already on ART and their VL test done before 6 months, do the VL soon after pregnancy is known; then continue every	• Serum creatinine and eGFR for TDF^c • Pregnancy test, especially for women of childbearing age not receiving family planning. • CD4 cell count if indicated • HPV-DNA if available

	six months until the MTCT risk ends, then routinely every 12 months . ○ For all pregnant women, regardless of ART initiation timing: conduct viral load testing at 34–36 weeks of gestation (or at the latest at delivery) to identify women who may be at risk of treatment failure and/or may deliver infants at higher risk of perinatal transmission ○ For all breastfeeding women, regardless of when ART was initiated: conduct a viral load test three months after delivery and every six months thereafter to detect viremic episodes during the postnatal period ● FBS at 6 months after ART and every 12 months annually.	
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^a*If feasible, HBsAg testing should be performed at baseline to identify people with HIV and HBV coinfection and who should therefore initiate TDF-containing ART.*

^b*Consider assessing for the presence of chronic conditions that can influence ART management, such as hypertension, other cardiovascular diseases, diabetes and TB according to the WHO Package of Essential NCD interventions (PEN), mental health Gap Action Program (mhGAP) or*

national standard protocols

^c *Monitoring may include a range of tests, including serum creatinine and estimated glomerular filtration rate (eGFR), serum phosphate and urine dipsticks for proteinuria and glycosuria.*

eGFR = $140 - \text{age (years)} \times \text{body weight (kg)} / (72 \times \text{serum Cr in mg/dL})$ for male and **eGFR** = $140 - \text{age (years)} \times \text{body weight (kg)} \times 0.85 / (72 \times \text{serum Cr in mg/dL})$ for female.

^d *Among children and adults with a high risk of adverse events associated with AZT (low CD4 or low BMI).*

^e *Among people with a high risk of adverse events associated with TDF: underlying renal disease, older age group, low body mass index (BMI), diabetes, hypertension and concomitant use of a boosted PI or potential nephrotoxic drugs.*

N.B. If the recommended baseline tests are not available, it should not delay ART initiation

12.6.2. What to expect in the first months of ART?

Although taking ART is a lifelong commitment, the first six months of therapy are especially important. There will be complex reactions during this period. Some complications are commonest when the people starting ART already have advanced HIV disease with severe immunodeficiency and existing co-infections and/or co-morbidities, severely low hemoglobin, low body mass index and very low CD4 counts or are severely malnourished.

Scenarios to look for in the first few months after initiation:

- Clinical and immunological improvement and virological suppression
- Opportunistic infections and/or immune reconstitution inflammatory syndrome (IRIS) may develop
- Early adverse drug reactions, such as drug hypersensitivity, especially in the first three months of ART.
- Death rates are also highest in the first three months of ART.

a. CD4 recovery

In most adults and children, CD4 cell counts rise when ART is initiated and immune recovery starts. Generally, this increase occurs during the first year of treatment, and then continues to rise further during the second year. However, severe immunosuppression may persist in some

individuals who do not experience a significant rise in CD4 cell count with treatment, especially those with a very low CD4 cell count when initiating ART. Failure to achieve some CD4 recovery should alert the health care provider to potential adherence problems or primary non-response to ART, and consideration should be given to continue prophylaxis for opportunistic infections such as co-trimoxazole preventive therapy.

b. Immune reconstitution inflammatory syndrome (IRIS)

IRIS is a spectrum of clinical signs and symptoms thought to be associated with immune recovery brought about by a response to ART. It occurs among 10–30% of the people initiating ART, usually within the first 4–8 weeks after initiating therapy. It may present in two different ways: **paradoxical IRIS**, when an opportunistic infection or tumor diagnosed before ART initially responds to treatment but then deteriorates after ART starts; or **unmasking IRIS**, in which initiating ART triggers disease that is not clinically apparent before ART. IRIS should be considered only when the presentation cannot be explained by a new infection, expected course of a known infection or drug toxicity.

The clinical spectrum is diverse, and IRIS has been reported for many different infections, tumors and non-infectious conditions. The most serious and life-threatening forms of paradoxical IRIS are for TB, *Cryptococcus*'s, Kaposi's sarcoma and hepatitis.

BCG vaccine-associated IRIS (localized and systemic) may occur in some HIV infected infants. IRIS is generally self-limiting, and interruption of ART is rarely indicated, but people may need to be reassured in the face of protracted symptoms to prevent discontinuation of or poor adherence to ART.

Through evaluation to identify disseminated opportunistic infections or tumors in patients with low CD4+ cell count (<50 cells/mm³) is very important before initiating ART. Another risk factor for IRIS is shorter duration of therapy for opportunistic infections before ART initiation. .

The most important steps to reduce the development of IRIS include:

- Earlier HIV diagnosis and initiation of ART before a decline to below 200 CD4 cells/mm³;
- Improved screening for opportunistic infections before ART, especially TB and *Cryptococcus*; and
- Optimal management of opportunistic infections before initiating ART.

Timing of ART in people with opportunistic infections requires balancing a greater risk of IRIS after early initiation against continuing high mortality if ART is delayed.

Most or all of the following features should be present in order to make the diagnosis:

- A low pretreatment CD4 count (often less than 100 cells/ μ L) except in tuberculosis. IRIS secondary to preexisting M. tuberculosis infection may occur in individuals with CD4 counts >200.
- A positive immunological response to ART.
- The absence of evidence of drug-resistant infection, bacterial super infection, drug allergy or other adverse drug reactions, patient noncompliance, or reduced drug levels due to drug-drug interactions or malabsorption after appropriate evaluation for the clinical presentation.
- The presence of clinical manifestations consistent with an inflammatory condition and
- A temporal association between HAART initiation and the onset of clinical features of illness- usually within the first 6 months.

Management of IRIS

- Patients should generally be treated for the underlying opportunistic infection as soon as possible
- Continuation of ART when IRIS occurs

Role of anti-inflammatory agents:

Anti-inflammatory agents may be particularly helpful in the setting of obstructive mass lesions (e.g. expanding cervical lymph node).

Use of anti-inflammatory agents, particularly corticosteroids, must be weighed against potential risks and side effects.

When a decision is reached to treat with corticosteroids, initiate therapy with prednisone at a dose of 1 mg/kg/day (maximal dose 60 to 80 mg) followed by a rapid taper over a 10 to 14-day period. IRIS in closed spaces (e.g. CNS OIs) should be managed promptly or referred to appropriate center to avert significant morbidity and mortality.

IRIS is not indicative of treatment failure or drug side effect. It is a transient phenomenon and is not a reason to stop ART or change regimen. The OIs should be treated using standard guidelines and in critically sick patients' short course of corticosteroid might be indicated to control severe symptoms.

12.6.3. Re-engaging to care after ART interruption:

Interruption in treatment is defined as client has missed appointment for greater than 28 days from the last appointment date. People who interrupted their treatment need to be re-evaluated for possible adherence barriers and advanced clinical conditions.

For re engaged clients after missing appointment for greater than 28 days (interrupted treatment), the following essential elements should be part of the care.

- Exhaust all possible adherence barriers and provide an ongoing Enhanced Adherence Counseling (EAC).
- Resume the same (previous) ART regimen they used before interruption.
- People interrupting treatment on NNRTI –containing regimen are at higher risk of drug resistance and should restart ART using a DTG-containing regimen if not contraindicated to DTG.
- Maintain close or frequent follow-up schedule.
- Monitor the viral load closely and determine the VL at 3 and 6 months after resuming ART. If the viral load is ≥ 1000 copies/ml in the two consecutive measurements, switch to second or third line regimen and continue adherence support
- Those with low viral load level, (50 –1000 copies/ml) need to be provided with enhanced adherence counselling and viral load testing after 3 months to promote more viral suppression. (see fig XXX)

*For clients who missed their appointments less than 28 days, assess for the cause and strengthen adherence support. In addition, ensure the clients are on optimized regimen (DTG based treatment for those who are on first line).

12.7. ARV drug toxicities: Monitoring and Management

Session Objective

- Describe common side effects of frequently prescribed ARV drugs,
- Describe how to monitor for treatment response, drug toxicity and treatment failure.
- Demonstrate how to report ADR/ or fill ADR format

Adverse effects have been reported with the use of all ARVs. Toxicities contribute to about 25% of patients discontinuing therapy in the first year. Moreover, 25% of patients also do not adhere to their regimens due to toxicity.

Operational Terminologies

Side effect -Any unintended effect occurring at doses normally used in human and is related to the pharmacological properties of the drug.

Adverse reaction -A noxious and unintended effect which occurs in doses normally used in human and may not be related to its pharmacological properties.

ARV toxicities -Indicate both the side effects and ADRS (not related to over dose of drugs)

Case studies for Monitoring and management of Toxicities



Case 1:

Zeberga was started on TDF-3TC-EFV. After two weeks he came complaining of night mares, bad dreams and headache.

- What is happening in this patient?
- What is the next step?



Case 2: Muktar is a 38 year-old man who has been on (ZDV+3TC) +DTG combination tablets for 5 months. Today he came to your ART clinic. When you asked him how he is doing with treatment, he reports that he has become very tired to the extent of couldn't get out of bed in the last two weeks. He also has felt nauseated on and off ever since he started the medications. He believes these changes are a result of his antiretroviral medications; therefore, he has begun missing doses.

- What is the problem with Muktar?

- What drug(s) might be responsible for his fatigue?
- What additional information do you need?
- How severe (grade of severity) is Muktar's problem?
- How do you manage?

Most common toxicities associated with ARVs appear within the first 3 to 6 months of ART initiation, therefore it is important to be vigilant and check for these toxicities in the follow up visits. What should be done?

- Establish whether the adverse event is due to ARV drugs, other drugs, drug-drug interaction or clinical illnesses including new OIs.
- Try to identify the responsible ARV drug.
- Assess the severity using toxicity grading matrix (see annex 8)

12.7.1. What are the common types of toxicities?

Common types of toxicities are classified on how soon they appearance along with medication and degree of severity. There are three types of side effects base on time of onset.

a. TYPE ONE TOXICITY:

Early side effects that are uncomfortable for the patient, but not dangerous.

(a) Some side effects are very **common**, but do not cause serious danger to the health of the patient. Because they are very common, it is important to inform the patient about them and provide simple advice on what to do if they occur. They include symptoms such as nausea, headache, dizziness, diarrhea, feeling tired, and muscle pain. Usually they occur when treatment begins and improve within two to four weeks. The patient should be reassured that this will go away after some weeks. For example: efavirenz can cause strange dreams and nightmares, mood changes, dizziness, and loss of concentration often within the first 2 weeks after initiation of treatment. If the patient does not understand these side effects and does not have a plan of how to cope with them, adherence will be poor then HIV resistance will occur and treatment outcome will not be good.

The list of possible side effects which are not dangerous is long. Telling the patient about all of them could be discouraging and confusing. For each drug (and each drug regimen), we need to

teach what the very common side effects are that we should prepare the patient to deal with. We also need to know how to provide clinical management when the patient seeks care, because the symptoms may persist or become severe. Sometimes even these common side effects can be more severe if they persist for a long time or if the patient has an especially severe reaction. It should be made clear to the patient that, after trying home management of the symptoms, if the symptoms persist or become worse, the patient should seek care.

(b) Less common and not serious side effects. It is not necessary (or advisable) to warn patients about these side effects. For example: AZT may cause blue nails. You should be able to explain the patient that this is only of cosmetic importance.

b. SECOND TYPE: Early and potentially serious side effects

These require emergency consultation. The patient needs to be warned about these potential side effects. For some, the patients need to seek care urgently if they occur. These shall be marked on the Patient Treatment Card.

Examples:

- Pallor (anemia—can occur with AZT)
- Yellow eyes due to sick liver (acute hepatitis—can occur with NVP or EFV)
- Severe abdominal pain
- Rash. Although many new rashes that occur on NVP or EFV are mild, they can be serious, so it is important for the patient to be advised to show them to the health worker as soon as possible.

c. THIRD TYPE: Side effects occurring later during treatment

They tend to occur after the patient has been taking ARV drugs for several months or even years. The most common one is an abnormal distribution of body fat—fat gain on the abdomen, breasts, shoulders, neck (sometimes with fat lumps under the skin), as well as fat loss from legs, arms, buttocks and face (lipodystrophy). Some serious side effects can occur late e.g. lactic acidosis.

Table 12.8 Types of toxicities associated with first, second and third line ARV drugs

ARV Drug	Major types of toxicity	Risk factors	Suggested management	Remark
ABC	Hypersensitivity reaction.		Substitute with TDF or AZT	
ATV/r	Electrocardiographic Abnormalities (PR and QRS interval prolongation).	Pre-existing conduction system disease. concomitant use of other drugs that may prolong the PR interval	Use with caution in people with preexisting conduction disease or who are on concomitant drugs they may prolong the PR or QRS interval	
	Indirect hyperbilirubinemia (clinical jaundice)	Underlying hepatic disease, HBV and HCV co-infection. Concomitant use of hepatotoxic drugs.	This phenomenon is clinically benign; however, can be associated with social and personal discomfort and in some circumstances with stigma. Substitute with LPV/r only if adherence compromised.	Strongly advice the client not to seek traditional healers and herbal medications
	Renal stone	History of renal stone.	Substitute with LPV/r.	
AZT	Anemia, neutropenia	Baseline anemia or neutropenia CD4 count ≤ 200 cells/mm ³ .	Avoid use of AZT for people with HIV and severe anemia at baseline (hemoglobin <7.0 g/dl) as first-line therapy. Substitute with TDF or	

			ABC.	
DRV/r	Hepatotoxicity	Underlying hepatic disease; Coinfection with HBV or HCV; Concomitant use of hepatotoxic drugs.	Substitute with ATV/r or LPV/r. When it is used in third-line ART, limited options are available.	
	Severe skin and hypersensitivity reactions.	Sulfonamide allergy.	For hypersensitivity reactions, substitute with another therapeutic class.	
DTG*	Hepatotoxicity Hypersensitivity reactions	Coinfection with Hepatitis B or C , Liver disease.	Substitute another therapeutic class: EFV or boosted PIs	
	Insomnia Body weight gain or obesity	Older than 60 years Low CD4 or high viral load Female African ethnicity	Consider morning dose or substitute EFV, boosted PI Monitor body weight and promote anti-obesity measures (such as diet and physical exercise). If significant increase despite measures, consider substituting EFV or boosted PI	
LPV/r	Electrocardiographic abnormalities (PR and QT interval prolongation, torsade's de pointes).	People with pre-existing conduction system disease; concomitant use of other drugs that may prolong the PR interval, Congenital long QT	Use with caution in people with pre-existing conduction disease or those on concomitant drugs that may prolong the PR or QRS intervals	

		syndrome, Hypokalemia Concomitant use of drugs that may prolong the QT interval		
	Hepatotoxicity	Underlying hepatic disease HBV and HCV co- infection Concomitant use of hepatotoxic drug	If LPV/r is used in children, substitute with DTG or EFV according to age. ATV can be used for children older than 6 years. If LPV/r is used in second- line ART for adults, use ATV/r. If boosted PIs are contraindicated and the person has failed an NNRTI based first line ART use DTG or consult specialist	
	Pancreatitis	Advanced HIV disease, Alcohol	Substitute another therapeutic class (INSTIs)	
	Dyslipidemia	Cardiovascular risk factors such as obesity and diabetes	Substitute another therapeutic class (INSTIs)	
	Diarrhea	Risk factor unknown	Substitute with ATV/r or DTG	
NVP (For HEIs proph	There are rare reports of Hypersensitivity and hepatotoxicity		Use AZT only	

ylaxis)				
EFV	Persistent central nervous system toxicity (such as abnormal dreams, depression or mental confusion) ²	Depression or other mental disorder (previous or at baseline) Daytime dosing	DTG or boosted PI	
	Hepatotoxicity	Underlying hepatic disease – HBV and HCV co-infection Concomitant use of hepatotoxic drug		
	Convulsions	History of seizure		
	Severe skin Hypersensitivity reaction.	Risk factors unknown.	Use DTG or boosted PI	
	Gynecomastia	Risk factor unknown.	Use DTG or boosted PI	
TDF	Chronic kidney disease, acute renal injury, Fanconi syndrome.	Underlying renal disease; older age; BMI <18.5 (or body weight <50 kg); untreated diabetes mellitus; untreated hypertension; Concomitant use of nephrotoxic drugs or boosted PI.	Substitute with AZT or ABC. Do not initiate TDF at eGFR <50 mL/min, uncontrolled hypertension, untreated diabetes, or presence of renal failure. It is recommended to	

² Most CNS side effects will improve within 2-4 weeks after initiation

	Decreases in bone mineral Density.	History of osteomalacia and pathological fracture; risk factors for osteoporosis or bone loss.	monitor growth in children taking TDF containing regimen.	
	Lactic acidosis or severe hepatomegaly with steatosis.	Prolonged exposure to nucleoside analogues; Obesity.		

NB:

- *For those patients with HBV and HIV co-infection suffering from TDF toxicity, consult or refer national viral hepatitis guideline.*
- *The clinical manifestations due to hypersensitivity reactions for some ARVs may be confused with IRIS*
- **Hyper glycemia has been reported from different sites in Ethiopia and small-scale studies from other countries.*

Drug substitutions for ARV drug toxicity

Drug regimen or single agent substitutions may be required for drug toxicity and to avoid drug interactions.

Clinical considerations

- Delaying substitutions or switches when there are severe adverse drug effects may cause harm and may affect adherence, leading to drug resistance and treatment failure.
- When drug interruptions are required, it is important to consider the various half-lives of ARV drugs. For example, when a NNRTI needs to be discontinued, a staggered approach should be used by prolonging the use of the NRTI backbone for two weeks except life threatening conditions (grade 4 conditions) where you have to discontinue all ARV drugs.

12.7.2. Monitoring for specific drug toxicity

a. TDF

TDF nephrotoxicity is characterized by proximal tubular cell dysfunction that may be associated with acute kidney injury or chronic kidney disease. It is advisable for high-risk people (those who are older or have underlying renal disease, long-term diabetes or uncontrolled hypertension concomitant with boosted PIs or nephrotoxic drugs) to detect and limit further progression of renal impairment. A significant uncertainty remains around how best to monitor TDF-related bone toxicity among children however it is recommended to monitor growth in children taking TDF containing regimen.

Clinical recommendations

- Laboratory monitoring is not mandatory to initiate treatment with TDF.
- Routine blood pressure monitoring may be used to assess for hypertension.
- Urine dipsticks may be used to detect glycosuria or severe TDF nephrotoxicity in individuals without diabetes using TDF-containing regimens.
- If the creatinin test is routinely available, use the estimated glomerular filtration rate at baseline before initiating TDF regimens.
- Do not initiate TDF when the estimated glomerular filtration rate is <50 ml/min, or in long term diabetes, uncontrolled hypertension and renal failure.
- Monitor growth in children using TDF.

b. AZT

AZT is associated with a risk of hematological toxicity, and measuring hemoglobin is recommended before initiating ART and monthly at least for the next three months. People with HIV with severe anemia at baseline (hemoglobin <7.0 g/dl) should avoid AZT as first-line therapy.

c. NVP

Currently NVP is no more recommended for the treatment of HIV except for infant prophylaxis.

d. EFV

The main type of toxicity of EFV is central nervous system side effects, which typically resolve after a few weeks. However, in some cases, they can persist for months or not resolve at all. Patients for whom the CNS side effect persisted, it is advisable to substitute with EFV400 or DTG whichever is convenient.

e. DTG

The main type of toxicity are mild or moderate nausea, headache and diarrhea; abnormal liver function, particularly in patients with HBV and HCV coinfection; potentially serious hypersensitivity reaction; neuropsychiatric symptoms, insomnia, body weight gain or obesity. It is better to check for HBV & HCV co-infections at initiation or when reaction happens. Most side effects of DTG are limited to toxicity level Grade-1 or Grade-2 that doesn't prevent the use of DTG. Insomnia and weight gain are more common in patients older than 60 years, low CD4 or high viral load, female and African ethnicity.

Clinical Grading of ARV toxicity

Grade 1 (Mild reaction): are bothersome but do not require changes in therapy

Grade 2 (Moderate reaction): consider continuation of ART as long as feasible. If the patient does not improve in symptomatic therapy, consider single-drug substitution.

Grade 3 (Severe reaction): Substitute offending drug without stopping ART. Closely monitor using laboratory and clinical parameters.

Grade 4 (Severe life-threatening reaction): Immediately discontinue all ARV drugs, manage the medical event with symptomatic and supportive therapy, and reintroduce ARV drugs using a modified regimen (i.e. with an ARV substitution for the offending drug) when the patient is stabilized. Life-threatening toxicity includes severe hepatitis, pancreatitis, lactic acidosis or Steven-Johnson syndrome.

Table 12.9 Clinical Grading of toxicities

Item	Grade 1 Mild toxicity	Grade 2 Moderate toxicity	Grade 3 Severe toxicity	Grade 4: Severe life- threatening toxicity
Peripheral neuropathy	• Transient or mild discomfort, no limitation of activity • No medical intervention/treatment	• Moderate limitation of activity, some assistance might be needed • Non-narcotic analgesia required	• Marked limitation in activity, some assistance usually required, medical intervention/therapy required, hospitalization possible • Severe discomfort	• Life-threatening, extreme limitation in of activity, significant assistance required, significant medical intervention/ therapy required, hospitalization/ hospice care

	required		and/or severe impairment (decrease or loss of sensation up to knees or wrists) narcotic analgesia required	• Incapacitating or not responsive to narcotic analgesia • Sensory loss involves limbs and trunk
Cutaneous/Rash/Dermatitis*	Erythema, pruritus	Diffuse, maculopapular rash or dry desquamation	Vesiculation or moist desquamation or ulceration*	Erythema multiform or suspected SJS or Toxic Epidermal Necrolysis (TEN)
Diarrhea	3-4 loose stools a day or mild diarrhea lasting less than one week	5-7 loose stool a day or diarrhea lasting more than one week	Bloody diarrhoea or over 7 loose stools a day or needing IV treatment or feeling dizzy when standing	Hospitalization required (possible also for grade 3)
Fatigue	Normal activity reduced by less than 25%	Normal activity reduced by reduced by 25-50%	Normal activity reduced by over 50%. Cannot work	Unable to care for yourself
Nausea	Mild or transient reasonable food intake	Moderate discomfort or intake decreased for less than 3 days	Severe discomfort or intake decreased for minimal food intake for more than 3 days	Hospitalization required
Vomiting	2-3 episodes a day or mild vomiting for less than one week	4-5 episodes a day or mild vomiting for more than one week	Severe vomiting of all food and fluids over 24 hours or needing IV treatment or feeling dizzy when standing	Hospitalization for IV treatment (possibly also for grade 3)
Mood disturbance	Mild anxiety, able to continue daily tasks	Moderate anxiety/disturbance, interfering with ability to work, etc	Severe mood changes requiring medical treatment. Unable to work	Acute psychosis, suicidal thoughts

Management	Continue ARV Provide careful clinical monitoring Consider change of a single drug if condition worsens	substitute responsible drug	Stop ARV and consult experienced physician
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Table 12.10. Laboratory Grading of Adverse Events in Adults and adolescents (ACTG)

Laboratory Test Abnormalities				
Item	Grade 1 toxicity	Grade 2 toxicity	Grade 3 toxicity	Grade 4 toxicity
Haemoglobin	8.0-9.4 g/dL	7.0-7.9 g/dL	6.5-6.9 g/dL	<6.5 g/dL
Absolute Neutrophil Count	1,000-1,500 mm ³	750-990 mm ³	500-749 mm ³	<500 mm ³
Platelets	-75,000-99,000	50,000-74,999	20,000-49,999 mm ³	<20,000
Creatine Kinase, High	3 to < 6 x ULN	6 to < 10x ULN	10 to < 20 x ULN	≥ 20 x ULN
Creatinine, High <i>**Report only one</i>	1.1 to 1.3 x ULN	> 1.3 to 1.8 x ULN OR** Increase to 1.3 to < 1.5 x participant's base line	> 1.8 to < 3.5 x ULN OR** Increase to 1.5 to < 2.0 x participant's baseline	= 3.5 x ULN OR** Increase of = 2.0 x participant's baseline
Creatinine Clearance ¹ or eGFR, Low <i>**Report only one</i>	NA	< 90 to 60 ml/ min or ml/ min / 1.73 m ² OR** 10 to < 30% decrease from participant's base line	< 60 to 30 ml/min or ml/ min / 1.73 m ² OR** 30 to < 50% decrease from participant's base line	< 30 ml/min or ml/ min / 1.73 m ² OR** = 50% decrease from participant's baseline or dialysis needed
ALT	1.25-2.5 X upper normal limit (ULN)	2.5-5 X upper normal limit	5.0-10 X upper normal limit	10 X upper normal limit
Bilirubin	1-1.5X ULN	1.5-2.5 X ULN	2.5-5 x upper limits of normal (ULN)	>5 x upper limits of normal
Amylase/lipase	1-1.5X ULN	1.5-2 X ULN	2-5 x upper limits of normal	>5x upper limits of normal

Triglycerides *	200-399mg/dL	400-750 mg/dL	751-1200mg/dL	>1200mg/dL
Cholesterol *	1.0 –1.3 X Upper normal limit	1.3-1.6 X Upper normal limit	1.6-2.0 X Upper normal limit	2.0 X Upper normal limit
MANAGEMENT	Continue ARV Repeat test 2 weeks after initial test and reassess		substitute responsible drug	Stop ARV and consult experience physician
	Lipid imbalances could be managed with diet, exercise and pharmacologically with the use of fibrates. ALWAYS SEEK EXPERT ADVICE IN CASE OF DOUBT			

* Additional clinical indications to change ARVs due to grade 3 toxicity (See Table 12.9 for substitution)

¹Use the applicable formula (i.e., Cockcroft-Gault in mL/min or Schwartz, MDRD, CKD-Epi in mL/min/1.73m²). Sites should choose the method defined in their study and when not specified, use the method most relevant to the study population.

**Reminder: Choose the method that selects for the higher grade.

Nausea: Severe discomfort or minimal intake for >3 days

Vomiting: Severe vomiting of all foods/fluids in 24 hours, orthostatic hypotension, or necessity for IV therapy

Diarrhoea: severe diarrhoea, with orthostatic hypotension, or necessity for IV therapy

Headache: Severe or requires narcotic therapy

Allergic reaction: Angioedema or anaphylaxis

Fatigue: Normal activity reduced >50 %

** For a patient on nevirapine, rash with mucosal involvement or associated with fever and/or systemic symptoms, and/or derangement in liver functions should be treated as Grade 4 toxicity.(N.B., NVP is phasing out and it will not be in use anymore)All antiretroviral drugs should be stopped immediately.

- Primary care providers should consult with more experienced physician before recommencing ART after a serious adverse effect,
- Never re-challenge with nevirapine,

Do not substitute EFV for nevirapine after G4 toxicity but when there are no alternatives to substituting NVP it may be possible to use EFV cautiously. Triple NRTI may be used.

12.7.3. Management of Common ARV Drug Toxicities

1. Management of Gastrointestinal (GI) related Side Effects

Gastrointestinal (GI) problems are the most common side effects of almost all ARVs drugs. These include **nausea, vomiting, diarrhea, abdominal discomfort, loss of appetite**. They are mainly associated with **NRTIs and most PIs** especially during initiation of ART. Patients should be informed that these side effects usually resolve after four to six weeks of treatment.

- Non-pharmacologic management includes
 - Taking AZT with meal reduces the risk of nausea and vomiting
 - Ginger, peppermint or chamomile teas or sweets may also be helpful, as well as frequent small meals.
 - Care should be taken with fatty foods and dairy products.
 - Coffee, smoking, alcohol, aspirin and very spicy foods should be avoided if possible
- Pharmacologic management
 - For symptomatic treatment, metoclopramide is useful.
 - Dimenhydrinate or ondansetron can also be used.

Diarrhea: Diarrhea occurs frequently with AZT and all PIs, particularly with lopinavir and nelfinavir. Other causes such as gastrointestinal infections or lactose intolerance should be excluded.

- Non-pharmacologic management
 - The priority is to treat dehydration and loss of electrolytes.
 - Difficult to digest foodstuffs (particularly those rich in fats or glucose) should be avoided and those that are easy to digest (e.g. potatoes, rice, noodles) are recommended.
 - Avoid spicy, fatty, starchy, or processed foods, caffeine, alcohol, dairy products, and foods that give you gas,
- Pharmacologic management
 - PI-associated diarrhea can also be managed by calcium, taken as calcium carbonate, at a dosage of 500 mg bid (taken 2 hours apart from ARVs)

- Symptomatic treatment consists of loperamide (initially 2 - 4 mg, followed by 2 mg, up to a maximum of 16 mg daily).

2. Management of Hepatotoxicity

Hepatotoxicity is common and is associated with most antiretroviral agents. Severe hepatotoxicity occurs in up to 6% of patients on ART, but liver failure is rare. It is commonly associated with NRTIs (D4T and DDI), **NNRTIs** (NVP > efavirenz) and **PIs** (Full-dose ritonavir). The patients may present with signs or symptoms of hepatitis (anorexia, malaise, jaundice, nausea, vomiting, bilirubinemia, hepatomegaly, and hepatic tenderness). Other constitutional symptoms may include fever, arthralgia, fatigue, and other findings or generalized organ dysfunction.

Who are at higher risk for ARV- induced hepatotoxicity?

- Baseline elevated serum aminotransferases,
- Chronic hepatitis B or C co-infection,
- Concomitant hepatotoxic medication,
- History of alcohol abuse,
- Protease inhibitor therapy,
- Thrombocytopenia and renal insufficiency.
- Female gender and baseline High CD4 count (NVP induced hepatotoxicity)
 - Women with CD4 count >250 cells/dl
 - Men with CD4 count >400 cells/dl

When is hepatotoxicity expected?

- **NNRTIs** especially nevirapine often cause hypersensitivity reaction within the first 12 weeks.
- **NRTIs** lead to hepatic steatosis; which occurs after more than 6 months on treatment.
- **PIs** cause hepatotoxicity at any stage of treatment
- **INSTI (Coinfection with hepatitis B or C or Liver disease, specially DTG)**

Management of Atazanavir induced Hepatotoxicity

Atazanavir causes hyperbilirubinemia due to inhibition of hepatic enzyme UDP-glucuronosyl transferase. However the levels of bilirubin return to normal following discontinuation of the

drug. If bilirubin is mildly elevated (< 3X ULN) and the serum liver enzyme levels are normal, treatment change is not mandatory. If the bilirubin is constantly elevated, the drug should be discontinued.

3. Monitoring and management of Renal problems (nephrotoxicity)

Tenofovir use has been associated with renal insufficiency, particularly in older patients with underlying renal disease, long-term diabetes or uncontrolled hypertension. Renal toxicity occurs after some months, rarely at the beginning of therapy. Other risk factors include a relatively high TDF exposure, low body weight, and co-administration of nephrotoxic drugs.

- Do not initiate TDF when the estimated GFR <50 ml/min, or in long-term diabetes, uncontrolled hypertension and renal failure.

Estimated GFR can be calculated using this formula:

$$eGFR = [140 - \text{age (years)} \times \text{body weight (kg)}] / (72 \times \text{serum Cr in mg/dL}) \text{ for male and}$$

$$eGFR = [140 - \text{age (years)} \times \text{body weight (kg)} \times 0.85] / (72 \times \text{serum Cr in mg/dL}) \text{ for female.}$$

- Renal function tests including serum creatinine, urea or Blood urea nitrogen, creatinine clearance (CrCl), proteinuria, glycosuria, blood and urine phosphate should be checked.
- In case of renal dysfunction, especially in patients with low body weight, avoid tenofovir if possible, or make dose adjustment based on CrCl.

4. Management of Neurological Side Effect

Management of *Peripheral neuropathy*: NRTIs are the agents most frequently associated. Patients complain of paresthesia, tingling, numbness and pain in their hands and feet. Risk factors for polyneuropathy include advanced AIDS, Vitamin B12 deficiency, alcohol abuse, diabetes mellitus, malnutrition and treatment with other neurotoxic drugs, e.g. 6H (Less in 3HP).

Symptoms frequently improve within the first two months following discontinuation of the drugs but are not always fully reversible.

- Non-Pharmacologic management
 - ✓ Tight shoes or long periods of standing or walking should be avoided and
 - ✓ Cold showers may relieve pain before going to bed.
- Pharmacologic (Symptomatic) treatment includes:
 - Acetaminophen (paracetamol), Carbamazepine, Amitriptyline, Gabapentine

and Opioids

- Vitamin B supplementation can help to improve peripheral neuropathy faster.

Management of Central Nervous System Toxicities: CNS toxicities such as dizziness, insomnia, nightmares, mood fluctuations, depression, depersonalization, paranoid delusions, confusion and suicidal ideation are mainly associated with EFV in first days and weeks of treatment. They can occur rarely with 3TC and ABC.

- Reassure the patient that they will go away with continued EFV therapy
- Avoid administration of EFV with fatty meal
- Discontinuation of therapy becomes necessary in only 3 % of patients. If the CNS side effects persist for more than two to four weeks,
 - The dose of efavirenz can be divided into a 400 mg at night and a 200 mg in morning.
 - Lorazepam can diminish the CNS side effects, and haloperidol can be given for panic attacks and nightmares. But lorazepam and haloperidol should be restricted to severe cases, because of their side effects and addictive potency.

5. Management of AZT Induced Hematological toxicities

AZT is myelo-suppressive causing neutropenia and macrocytic anemia. Anemia generally occurs during the first 6 months of AZT therapy and may be completely asymptomatic or symptomatic with fatigue or dyspnea. Risk factors include advanced HIV infection, pre-existing myelosuppression, chemotherapy or co-medication with other myelotoxic drugs such as pyrimethamine, amphotericin B, ribavirin, and interferon.

- Zidovudine should be discontinued in severe cases and a **blood transfusion** may be necessary.
- Change AZT to less myelotoxic drugs
- If there are no options to change, **erythropoietin is an option**, but should be avoided as a long-term option if possible, due to the associated high costs.
- Due to **drug-induced neutropenia**, despite viral suppression, the CD4+ T-cell count may remain low after an initial rise.
 - Change the treatment to less myelotoxic ARVs such as TDF, ABC & 3TC, most of

the PI and all NNRTIs.

- AZT should be avoided.

6. Management of Hypersensitivity Reactions

Although virtually any drug can cause a hypersensitivity syndrome, it is common with

- **NRTIs:** abacavir (5-8%),
- **NNRTI:** Nevirapine (15 to 20 %, discontinuation in 5 to 10 %) and less frequently on efavirenz therapy, where only 2 % of the patients discontinue the drug.
- **PIs:** Atazanavir (6 % in patients and is usually mild) and tipranavir

Management of NNRTIs induced HSR

Approximately 50 % of NNRTI allergies resolve despite continuation of therapy. Treatment should be discontinued immediately in cases with:

- mucous membrane involvement, blisters and exfoliation
- Hepatic dysfunction (transaminases > 5 times ULN) or fever > 39°C.

Management of ABC induced HSR

ABC induced HSR typically presents with a combination of symptoms including **fever** (which is almost always present), **constitutional symptoms** (eg, malaise, dizziness, and headache), and **gastrointestinal disturbances** (eg, nausea, vomiting, diarrhea). Respiratory symptoms (eg, dyspnea, cough, sore throat) occurred in approximately one-third of patients. Rash is often a late symptom that is absent in up to 30% of patients. Most HSR (90%) occur within the first 6 weeks and the median onset is 7-8 days.

- Exclusion of Human leukocyte antigen type B (HLA-B*5701) individuals from ABC treatment could largely prevent HSR.
- If ABC is discontinued in time, the HSR is completely reversible within a few days.
- Following discontinuation of ABC, further supportive treatment includes, intravenous hydration and possibly steroids therapy
- **NEVER re-challenge ABC again.** Re-challenge to abacavir after an initial HSR can result in a more rapid, severe, and potentially life-threatening anaphylactic reaction
- Treatment with abacavir requires detailed counseling (and documentation) on the possible occurrence and symptoms of the HSR.

- It is difficult to differentiate HSR due to ABC or NNRTIs. Therefore, synchronous initiation of ABC and NNRTIs is generally avoided if possible.

7. Management of Mitochondrial Toxicity

The clinical presentation of mitochondrial toxicity depends on the target organ that is involved. Mitochondrial toxicity is the major cause of NRTIs induced myopathy, neuropathy, lipoatrophy, and lactic acidosis.

NRTI induced Lactic Acidosis: Lactic acidosis occurs less often with zidovudine, abacavir and lamivudine. Lactic acidosis usually follows a minimum of six months of treatment. Nowadays lactic acidosis is not a clinical problem since d-drugs (d4t, ddI) are no longer in use. If incase it occurs, management of hyperlactatemia and lactic acidosis involves discontinuation of NRTI treatment and initiating supportive treatment such as correction of the acidosis. Mortality of patients with lactate levels above 10 mmol/l is approximately 80%. Agents used for treatment of congenital mitochondrial disorders may hasten recovery (thiamine, riboflavin, coenzyme Q, L-carnitine)

NRTIs induced Myopathy: Symptoms of myopathy included proximal muscle weakness, tenderness, and myalgia which are mainly associated with AZT. The majority of patients has elevated serum creatinine phosphokinase (CK) levels and may occasionally have elevated lactate levels

If NRTI myopathy is suspected the drug should be discontinued and alternate drug initiated. Following discontinuation of NRTIs, CK normalized in about 4 weeks and clinical complaints resolves in about 3 months

8. Management of Insulin Resistance

Insulin resistance is mostly associated with PIs (40%). Hyperglycemia (3-17%), new cases of diabetes mellitus (1%) and worsening of preexisting diabetes has also been reported. Therefore, patients receiving PIs should be advised about the warning signs of hyperglycemia, such as excessive thirst, excessive urination, and excessive appetite. Hyperglycemia resolves in some but not all patients after the discontinuation of PI based therapy.

Switching from PI-based regimens often allows improvement however, most experts, would continue ART with supportive therapy (oral hypoglycemic drugs or insulin) in the absence of severe diabetes.

9. Management of Lipid Abnormalities

It has been linked to treatment with all the **PIs** (higher with ritonavir). Rises in cholesterol and triglycerides may put someone at increased risk of heart disease particularly if they smoke or overweight or have high blood pressure. Lipid abnormalities are generally **treated with fibrates and/or statins**. However, beware of drug interactions between ART, statins, and fibrates that may lead to risk of myositis.

10. Management of Bone disorders

Avascular necrosis: Avascular necrosis is death of bone tissue due to a lack of blood supply. Avascular necrosis most often affects the head of the thighbone (femur), causing hip pain. Risk factors for avascular necrosis are alcohol abuse, hyperlipidemia, steroid treatment, hypercoagulability, hemoglobinopathy, trauma, nicotine abuse and chronic pancreatitis.

Identify risk factors and take measure to eliminate. Once the diagnosis is confirmed, refer patients to an orthopedic surgeon.

- Physiotherapy, rest and sometimes surgery are recommended.
- Bisphosphonate medications, such as alendronate and
- Nonsteroidal anti-inflammatory drugs (e.g. ibuprofen) are the treatment of choice for analgesia.

Osteopenia/Osteoporosis: Treatment with PIs (boosted) and NRTIs lower bone density in patients in addition to HIV infection itself. Other factors such as malnutrition diminished fat tissues, steroid treatment, hypogonadism, and immobilization can induce osteopenia/osteoporosis. Osteopenia and osteoporosis are often asymptomatic. Thus, it is reasonable to screen patients with numerous risk factors for osteopaenia with Dual energy X-ray absorptionometry.

- Preventive measures (such as physical exercise, sufficient ingestion of calcium and vitamin D) and elimination of risk factors (such as alcohol, tobacco and poor diet) are warranted.

- Osteopenia should be treated with vitamin D daily and a calcium-rich diet or calcium tablets with a dose of 1200 mg/day.
- In cases with osteoporosis, Bisphosphonates such as alendronate treatment for 48 weeks increased bone mass density
 - ✓ Tablets should be taken on an empty stomach 30 min before breakfast, and an upright position should be maintained for at least 30 min.
 - ✓ No calcium should be taken on this day.
- Since testosterone suppresses osteoclasts, hypogonadism should be treated.

12.7.4. Pharmacovigilance and Adverse event reporting

12.7.4.1. Pharmacovigilance

Pharmacovigilance is the science and activities relating to detecting, assessing, understanding, and preventing adverse effects or any other possible drug-related problems. Pharmacovigilance is a key component of comprehensive patient care and the safe use of medicines. Failure to monitor, understand and manage these events can result in poor adherence and treatment failure and can reduce confidence in antiretroviral therapy among both people living with HIV and care providers. Pharmacovigilance is required not only for long-term antiretroviral therapy but also for the antiretroviral drugs used for PMTCT, PrEP and PEP.

The main pharmacovigilance methods include spontaneous reporting, cohort event monitoring and targeted spontaneous reporting.

- Spontaneous reporting system is the most widely used system. Health professionals and pharmaceutical manufacturers voluntarily submit suspected adverse drug reactions to the national regulatory authority. A serious limitation here is underreporting.
- Cohort event monitoring is a prospective observational cohort study of adverse events associated with one or more medicines. All adverse events occurring to a person taking antiretroviral drugs are collected regardless of the causality or relationship with the antiretroviral drugs.
- Targeted spontaneous reporting is a method that builds on the principles of both spontaneous reporting and cohort event monitoring. Targeted spontaneous reporting enables focus on a specific drug of interest (such as DGT, Tenofovir), a specific

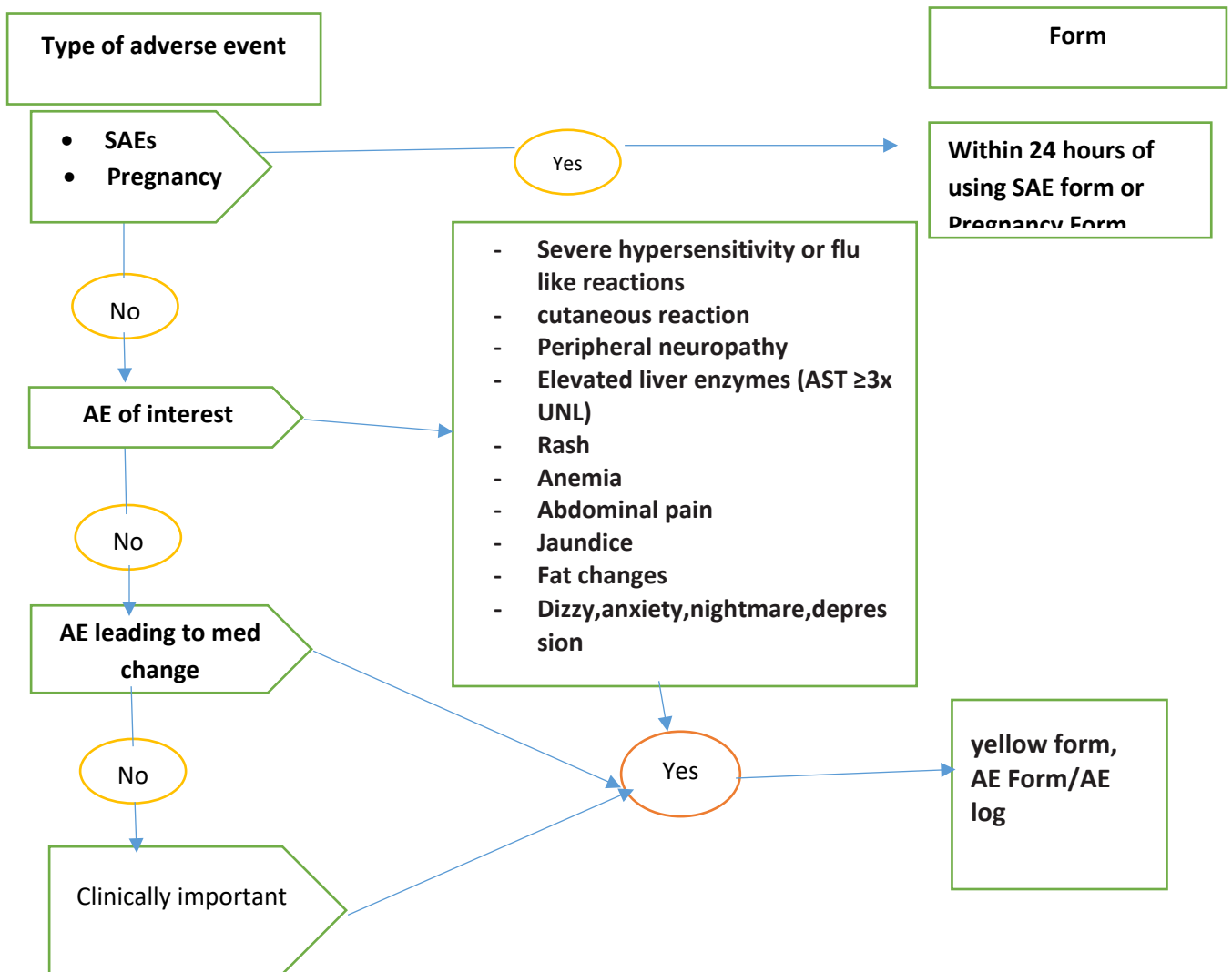
population of interest (such as women, children) or a specific adverse drug reaction (such as anemia).

- **12.7.2. Reporting of adverse events**

The national pharmacovigilance guideline (2014) indicated adverse drug events shall be reported to the responsible authority body, Ethiopian Food and Drug Authority (EFDA). The adverse events that need to be reported includes medicine-related injuries, with at least a reasonable possibility to be caused by the direct pharmacological mechanism of a medicine, an individual's particular vulnerability, drug interactions, unexpected therapeutic ineffectiveness (e.g. resulting from drug interactions, product quality, problems or antimicrobial resistance), medication errors, and product quality defects.

In line with this, adverse events associated with the TPT drugs should be reported using either the standardized adverse event reporting format (yellow form) or electronically on e-reporting of ADR available on the apps or website of EFDA (www.fmhaca.gov.et) by health care providers and public health programs.

Figure 12.3. General Reporting approach and Timeline for active DSM



Diagnosis and Management of Treatment failure

Session Objective

- Explain treatment success and failure;
- Discuss how to suspect/identify, work up and manage treatment failure cases
- Describe second line and third line regimens
- Describe Enhanced adherence supports for identification & management of suspected treatment failure cases,



Case Scenario 1: Abebech has been taking her treatment irregularly for the past two years. She presented with severe headache and stiff neck. With suspicion of meningitis, LP was done and the CSF analysis revealed Indian ink positive stains.

- What is wrong with Abebech?
- How would you manage her?

Case scenario 2: Hailu has been taking his medication correctly for the past five years. In the past two years, after his wife died he started taking alcohol and started to forget taking the medications regularly. His viral load in the last two years has been TND, TND, 480, 10,000..

- What is going on with this patient?
- What is the next step?

12.8.1. Identifying Treatment failure

Monitoring individuals receiving ART is important to ensure the success of treatment, to identify adherence problems and determine whether and which ART regimens should be switched in case of treatment failure. The value of viral load testing as a more sensitive and earlier indicator of treatment failure is increasingly recognized. Viral load testing should be done after initiation of ART at 6 months, 12 months and every 12 months then after in order to detect treatment failure

proactively. Viral load testing should be used aside from the routine testing whenever there is clinical or immunologic suspicion of treatment failure. The viral load request form should be filled completely before sending it to the designated laboratory (annex 15) and the request and returned result should be documented on the viral load register (Annex 18).

Implementation considerations for treatment monitoring of pregnant and breastfeeding women

Whenever possible, use same-day point-of-care testing for viral load testing of pregnant and breastfeeding women to expedite the return of results and clinical decision-making. If this is not available, viral load specimens and results for pregnant and breastfeeding women should be given priority across the laboratory referral process (including specimen collection, testing and return of results).

For all pregnant women, regardless of ART initiation timing:

Conduct viral load testing at 34–36 weeks of gestation (or at the latest at delivery) to identify women who may be at risk of treatment failure and/or may deliver infants at higher risk of perinatal transmission.

In addition:

a) For pregnant women receiving ART before conception:

Conduct a viral load test at the first antenatal care visit (or when first presenting) to identify women at increased risk of in utero transmission.

b) For pregnant women starting ART during pregnancy:

Conduct a viral load by three months after ART initiation to ensure that there has been rapid viral suppression. If viral load testing is expected to be undertaken near the planned viral load at 34–36 weeks of gestation, the first viral load test can be delayed until weeks 34–36 of gestation

For all breastfeeding women, regardless of when ART was initiated:

Conduct viral load test three months after delivery and every six months thereafter to detect viremic episodes during the postnatal period.

Table 12.11: Definitions of virological, immunological and clinical failure for the decision to switch ART regimens

Failure	Definition	Remark
Virologic failure	Viral load above 1000 copies/mL based on two consecutive viral load measurements in 3 months apart, with adherence support following the first viral load test	This is the earliest sign of failure before manifesting any of the clinical or immunological failure. An individual must be taking ART for at least 6 months before it can be determined that a regimen has failed. VL testing should not be done when there is an acute infection/fever as there can be blips of VL. See guidance below.
Immunologic failure	Adults and adolescents CD4 count at or below 250 cells/mm ³ following clinical failure Or Persistent CD4 levels below 100 cells/mm ³ Children <5 yrs: Persistent CD4 levels below 200 cells/mm ³ or <10%	Without concomitant or recent infection to cause a transient decline in the CD4 cell count. Persistent is to mean at least 2 CD4 measurements below the threshold. Current WHO clinical and immunological criteria have low sensitivity and positive predictive value for identifying individuals with virologic failure.
Clinical failure	Adults and adolescents New or recurrent clinical event indicating severe immunodeficiency (WHO clinical stage 4 condition and certain WHO clinical stage 3 conditions (pulmonary TB and severe bacterial infections) may also indicate treatment failure) after 6 months of effective treatment Children New or recurrent clinical event indicating advanced or severe immunodeficiency (WHO clinical stage 3 and 4 clinical condition with exception of TB) after 6 months of effective treatment.	It is the last presentation that comes after immunological & virological failures. The condition must be differentiated from immune reconstitution inflammatory syndrome occurring after initiating ART.
Note^a If serial (two or more) CD4 values are available, the rate of decline should be taken into consideration.		

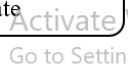
Attention:

1. Viral suppression: is a viral load that is undetectable, equal to or less than 50 copies/ml.

2. Low-level viraemia: is one or more viral load results that are detectable (more than 50 copies/ml) but equal to or less than 1000 copies/ml.

3. Virological failure Viral load above 1000 copies/ml based on two consecutive viral load measurements in 3 months, apart with enhanced adherence support following the first viral load test.

Treatment failure threshold should remain at 1000 copies/ml. Viral suppression or undetectability, however, are defined as viral load equal to or less than 50 copies/ml. PLHIVs should be supported with adherence counselling to achieve viral suppression. However, treatment failure should be considered for those with a repeat viral load result >1000 copies/ml after three months of EAS following the first viral load. Those with low-level viraemia at the first viral load test (50–1000 copies/ml) need to be provided with enhanced adherence support (EAS) and repeat viral load test after 3 months to promote viral suppression. If viral load is still 50-1000 copies/ml, maintain ARV drug regimen and continue viral load test every six months. In addition, continue the routine follow up support and link to community-based adherence support services.

^a. Unsuppressed viral load results should be immediately communicated

^b Conduct same-day testing using point-of-care viral load testing for a repeat viral load test, where available, to expedite the return of results. If not available, viral load specimens and results for a repeat viral load should be given priority across the laboratory referral process (including specimen collection, testing and return of results).

12.8.2.Enhanced Adherence Support (EAS)

Enhanced adherence support is important for patients that have unsuppressed VL, persisted or new immunosuppression, developing new OI or have multiple adherence barriers. It shall be systematic and with documenting the interventions provided during the EAS period. VL shall be repeated 3 months after effective EAS.

Table 12.13: Summary of components of Enhanced Adherence Support(EAS)

Enhanced adherence support sessions overview	
First Session (When the viral load result >50 is received; Day 0)	- Review cognitive, behavioral, emotional, and socio-economic barriers to adherence: - Treatment literacy - Medications: dosage, timing, storage - Side effects - Motivation - Mental health screening (screen for depression and other common mental problems using national Mental Health Assessment tool) <u>Action:</u> - Discuss risk reduction (e.g. for substance abuse) - Discuss patient's support systems - Assist patient to develop adherence plan to address the identified issues. - Referrals and networking when necessary
Second Session (30days after the first session)	- Review adherence plan from the first session and discuss any challenge. - Identify other possible gaps and emerging issues. <u>Action</u> - Assist patient to modify the adherence plan to address the identified issues - Referrals and networking when necessary.

Third Session (60 days after the first session)	- Review adherence plan from the first and second session and discuss any challenges. - Identify other possible gaps and emerging issues. <u>Action</u> - Assist patient to modify the adherence plan to address the identified issues. - Decision on repeat VL based on current adherence: - If the adherence is good, plan repeat VL testing after a month and explain possible ways forward, emphasizing the role of the patient and the health facility in terms of strengthening adherence. - If adherence challenges persist, link to appropriate client centered care.
Fourth Session (90 days from the first session)	- Take the second VL sample. - All efforts should be employed to get the VL result as soon as possible. - As soon as the VL result is received, discuss the result with the client. - Actions:: - If VL is < 50 copies/ml, maintain the current regimen and encourage adherence - If VL is 50-1000 copies/ml, continue the current regimen and the monthly EAS for the next 3months then repeat the viral load testing. If the low-level viremia (50–1000 copies/ml) persists, m In addition, continue the routine follow up support and link to community-based adherence support services. If VL is > 1000 copies/ml, switch to appropriate regimen (second or third line).

12.8.3. What ART regimen to switch to (second-line ART):

Using a **boosted PI + 2 NRTI** combinations is recommended as the preferred strategy for second-line ART for adults, adolescents and also for children when NNRTI-containing regimens were used in first-line ART. Two NRTI + DTG can be used as second line regimen if it is not

used in the first line. In children using a PI-based regimen for first-line ART, switching to NNRTI or maintaining the PI regimen is recommended according with age. Two NRTI + DTG can be used as second line regimen if it is not used in the first line.

Guidance for Changing ARV Regimens for treatment failure:

- *Ensure diagnosis of treatment failure to avoid premature switching.*
- *Assess level of adherence and address barriers*
- *Address Drug interactions issues*
- *Do not add one drug to a failing regimen!*
- *Consider resistance & cross resistance*
- *Get advice from experienced clinicians*
- *Use at least two new drugs in the new regimen, preferably one new drug class*

Secondline ART for children (including adolescents)

- For children starting first-line ART with an NNRTI or DTG based regimen, PI-based regimens remain the recommended choice for second-line therapy. LPV/r is the preferred boosted PI option, but ATV/r may be considered if more appropriate formulations become available for appropriate age. After failure of a first-line LPV/r-based regimen, children can be switched to DTG based regimen as per table below.
- After failure of a first-line regimen of ABC or TDF + 3TC, the preferred NRTI backbone option for second-line ART is AZT + 3TC and failure of a first-line regimen is containing AZT + 3TC, the preferred NRTI backbone option for second-line ART is ABC or TDF + 3TC.

Table12.14:Summary of preferred second-line regimens for adults,adolescents and children

Population	1 st line regimens		2 nd line regimens
Adults and adolescents 10	Preferred	TDF + 3TC + DTG	AZT+3TC+ATV/r or LPV/r

years & older	Alternative	TDF+3TC+EFV	AZT+3TC+DTG or ATV/r or LPV/r
		AZT+3TC+ EFV	TDF+3TC+DTG or ATV/r or LPV/r
		AZT+3TC+DTG	TDF+3TC+ ATV/r or LPV/r
	Special Circumstances	ABC+3TC+EFV	AZT+3TC+ DTG or ATV/r or LPV/r
		ABC+3TC+DTG	AZT+3TC+ATV/r or LPV/r
		TDF+3TC+ATVr or LPV/r	AZT+3TC+DTG
		AZT+3TC+ATVr or LPV/r	TDF+3TC+DTG
	Preferred	ABC+3TC+DTG	AZT+3TC+LPV/r
Children 4weeks to 10 years and >3kg	Alternative	ABC+3TC+LPV/r	AZT+3TC +DTG
		AZT+3TC+DTG	ABC+3TC+LPV/r
	Special circumstances	ABC+3TC+EFV	AZT+3TC+DTG
		AZT+3TC+EFV	ABC+3TC+DTG or LPV/r
		AZT+3TC+LPV/r	ABC+3TC+DTG or EFV

c In PI-experienced patients, the recommended DRV/r dose should be 600mg/100 mg twice daily; refer the weight band for pediatrics

d DRV/r should not be used in children younger than three years of age.

e DTG based 3rd line following use of INSTI must be administered with DTG BD.

Second line ART service in health centers

Patients who have treatment failure on first line ARV regimens in health centers can be switched to second line ART in the same facility based on the following criteria, also patients who were switched to second line ART in hospitals can be transferred to selected health centers for follow up

Eligible health centers to provide second line ART service

Second line ART service will be made available at health centers that have a patient load of 200 or more. That is a reasonably adequate patient load which could give rise to viable number of prospective clients who might need to be switched to a second line ART.

Eligible patients for second line ART service at health centers

The following criteria should be used to sort out treatment failure patients who will be switched to second line ART or will have follow up at the health centers:

- a. Patients who fulfill the criteria of treatment failure and with no signs of clinical failure
- b. Patients who are initiated on second line ART at hospital and became stable with no signs of OI or drug toxicity and transferred out to health centers having second line ART.

Identification and Management of Second Line Treatment Failure

Session Objective

- Describe the characteristics used to identify treatment failure for second line regimen
- Describe the way second line treatment failure managed

Patients who are on second line regimen and have high viral load level (>1000 copies/ml) after 6 months of treatment need to go through the algorithm (as described for first line treatment failure) with enhanced adherence support. As the third line regimen is only salvage therapy, regimen change shall be done cautiously. *Consider genotyping before constructing a third line regimen when accessible, however switching to third line should not be delayed while waiting for the result, if the TAT is prolonged.* Once patients are confirmed to have second line failure, they should be referred to hospitals providing third line treatment.

Management of Second Line Treatment Failure

When second line regimen failure is suspected, the issue should be discussed with experienced clinician in HIV care and treatment before switching to third line regimen. Failure to treatment while on a second line PI regimen is often due to non-adherence. The Health care provider shall be directly involved in the adherence assessment and provision of Enhanced Adherence support, and this effort shall not be delegated to the adherence counselor alone. (*See adherence section for detailed steps of adherence counseling for patients with adherence problems*).

Before switching to third line regimen, ensure the following considerations:

- Two consecutive viral load measurements $> 1,000$ copies/ml 3 months apart.
- The patient shall stay on the second line regimen (switched PI) for a minimum of 6 months with good adherence before considering the First viral load measurement.
- Enhanced adherence support shall be instituted, and the patient shall be followed every month for 3 months.
- The repeat VL test shall be done after 3 months of enhanced adherence support.

The approach in switching to third line follows the guiding principles used when switching to second line regimen. The table below shows the regimens to be selected for third line considering

the previous exposure during 1st line and 2nd line regimen.

Table 12.16: Summary of sequencing options for third-line ART regimens depending on first and second line experiences in adults, adolescents, pregnant women and children

Population	1 st line regimens		2 nd line regimens	3 rd line regimens ^{a,b}
Adults and adolescents 10 years & older	Preferred	TDF + 3TC + DTG	AZT+3TC+ATV/r or LPV/r	DRV/r ^c +TDF+3TC+EFV DRV/r+TDF+3TC+DTG ^e
	Alternative	TDF+3TC+EFV	AZT+3TC+DTG or ATV/r or LPV/r	DRV/r+TDF+3TC +DTG ^f
		AZT+3TC+ EFV	TDF+3TC+DTG or ATV/r or LPV/r	DRV/r+AZT+3TC+DTG
		AZT+3TC+DTG	TDF+3TC+ ATV/r or LPV/r	DRV/r+AZT+3TC +EFV
	Special Circumstances	ABC+3TC+EFV	AZT+3TC+ DTG or ATV/r or LPV/r	DRV/r+TDF+3TC +DTG
		ABC+3TC+DTG	AZT+3TC+ATV/r or LPV/r	DRV/r+TDF+3TC +EFV
		TDF+3TC+ATV/r or LPV/r	AZT+3TC+DTG	DRV/r+TDF+3TC +EFV
		AZT+3TC+ATV/r or LPV/r	TDF+3TC+DTG	DRV/r+AZT+3TC +EFV
Children 4weeks to 10 years and >3kg	Preferred	ABC+3TC+DTG	AZT+3TC+LPV/r	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r ^d +DTG ^g +ABC+3TC
	Alternative	ABC+3TC+LPV/r	AZT+3TC +DTG	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +ABC+3TC+ EFV
		AZT+3TC+DTG	ABC+3TC+LPV/r	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +AZT+3TC+EFV or DRV/r+DTG+AZT+3TC

	Special circumstances	ABC+3TC+EFV	AZT+3TC+DTG	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +ABC+3TC+EFV
		AZT+3TC+EFV	ABC+3TC+DTG or LPV/r	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r +AZT+3TC+EFV or DRV/r+DTG+AZT+3TC
		AZT+3TC+LPV/r	ABC+3TC+DTG or EFV	For those <3 years, maintain second line regimens till they become 3years. For children 3-10 Years, switch to DRV/r+AZT+3TC+EFV or DRV/r+DTG+AZT+3TC

a Consider genotyping before constructing a third line regimen when accessible, however switching to third line should not be delayed while waiting for the result, if the TAT is prolonged.

b. When constructing third line regimens for special circumstances in the absence of genotyping please consult senior experts on HIV

c In PI-experienced patients, the recommended DRV/r dose should be 600mg/100 mg twice daily; refer the weight band for pediatrics

d DRV/r should not be used in children younger than three years of age.

e DTG based 3rd line following use of INSTI must be administered with DTG BD.

Table 12.17.Recommended dose of third line ARV drugs for adults and adolescents

Generic Name	Dose
Proteases inhibitors (PIs)	
Darunavir + ritonavir (DRV/r)	800 mg + 100 mg once daily ^a or 600 mg + 100 mg twice daily ^b
Integrase strand transfer inhibitors (INSTIs)	
Dolutegravir (DTG)	Dolutegravir (DTG) 50 mg once daily

^aFor individuals with no previous use of protease inhibitors.

^b For individuals with previous use of protease inhibitors.

Table 12.18. Recommended strength of Third line ARV drugs for children

Drugs	Strength of tablets (mg) or oral liquid (mg/ml)
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DRV ^a	Tablet 600 mg
	Tablet 150 mg
	Tablet 75 mg
DRV ^a	Solution 100mg/ml

Note: Doses for children based on their weight and age. See Pediatric dosage at Annex 6 for detail.

For programmatic implementation purpose, implementation third line treatment is limited to selected high load hospitals.

Exercise 12.4

1. What is a second-line regimen?

2. Identify if these drugs are used for first line or second-line therapy and for adults or children.
(use thick mark(✓) to show your answers)

	Regimen	First line	Second line	Third line	Adult	Paediatrics
1	ABC-3TC-DTG					
2	TDF-3TC-DTG					
3	TDF-3TC-EFV					
4	AZT-3TC-EFV					
5	TDF-3TC-ATV/r					
6	AZT-3TC-ATV/r					
7	TDF-3TC-LPV/r					
8	AZT-3TC-LPV/r					
9	ABC-3TC-LPV/r					
10	ABC-3TC-EFV					
11	DRV/r+TDF+3TC+EFV					
12	DRV/r+TDF+3TC+DTG					
13	DRV/r+AZT+3TC+DTG					
14	DRV/r+AZT+3TC +EFV					

3. What is Enhanced Adherence support?

4. What is a third line regimen?

5. Is TDF-3TC-ATV/r used as a first-line or as a second-line regimen?

6. When do we use DTG?

7. What happens if we do not take the combination of several antiretroviral drugs?

8. What do you think is the most common problem/ cause of treatment failure?

9. How do you follow a patient suspected with treatment failure?

10. List down the needed intervention and materials for EAC.

12.7 ARV Drug Interactions

Session Objective

- Describe common drug interactions that may occur in patients taking ARVs
- Describe effects of ARVs on efficacy of ARVs and other co-administered medications

Care must be taken for drug interactions when patients are on ARV therapy. As you prescribe medications, watch for possible combinations of drugs that may cause drug interactions. Also, notice whether the patient's condition has changed that might be related to a drug reaction.

There are three ways ARV drugs interact with other drugs:

1. The effects of the drugs on the kidneys and liver

Besides ART drugs, patients might need to take other drugs. The drugs have an effect on how hard the kidneys and liver work to clear the body of the drugs.

Some drugs can make kidneys and liver work harder, and others can make kidneys or liver become 'lazy'.

If a patient takes a drug that makes liver work harder, the amount of antiretroviral drugs in the blood will decrease faster. As a result, the amount of antiretroviral drugs will be insufficient, which might create resistance to the treatment.

If liver starts to work slowly due to the use of another drug, the amount of antiretroviral drug in the blood might be too high. In this case, the patient might develop toxic blood levels leading to side effects.

2. The toxic effects and side effects of the drugs

The risk for side effects increases if two or more drugs that have similar side effects are given together.

3. The way the drugs stop the virus

A drug can reinforce the action of another drug, or block its action, making it ineffective. Different categories of antiretroviral drugs have different ways of stopping the virus from replicating. Drugs in the national first-line regimens are known to work well together. Used in the correct combination, the right drugs will **reinforce each other's effect**. That is why it is

important to stick with the first-line regimen combinations as recommended in national guidelines. Some ARV drugs work in **exactly the same way** to stop the virus. Since they work in exactly the same way, on the same spot in the CD4 cell, there is no enough space to work well, and the drugs will not reinforce each other, but **block each other**.

Whenever prescribing ARVs, check these points:

- Are there any combinations of drugs in the prescription that could cause a drug reaction?
- Are there ARVs that could affect effectiveness of hormonal contraceptives?
- Check whether the patient is taking other drugs or herbal medications.

Table 12.19. Key ARV drug interactions and suggested management

ARV drugs	Key interactions	Effect	Suggested management
AZT	Ribavirin and pegylated interferonalpha-2a		Substitute with TDF
EFV	Bedaquiline		Avoid the combination
	Amodiaquine, DHA/piperaquine		Use an alternative antimalarial agent or substitute EFV for DTG
	Artemisinin or lumefantrine		Use an alternative antimalarial agent or substitute EFV for DTG Risk of QT prolongation with ATV/r and LPV/r
	Methadone		Adjust the methadone dose as appropriate
	Hormonal contraceptives		Use alternative or additional contraceptive methods
	Amlodipine		Adjust the amlodipine dose as appropriate
	Simvastatin and atorvastatin		Adjust the statin dose as appropriate

	Low-dose dexamethasone (COVID-19)		Double dose of dexamethasone
Boosted PI (ATV/r, DRV/r, LPV/r)	Rifampicin		Adjust the PI dose or substitute with DTG
	Lovastatin and simvastatin	Increase concentration	Use an alternative dyslipidemia agent (for example pravastatin)
	Halofantrine and lumefantrine		Use an alternative antimalarial agent
	Estrogen-based hormonal contraception		Use alternative or additional contraceptive methods
	Methadone and Buprenorphine		Adjust methadone and buprenorphine doses as appropriate
	Astemizole and terfenadine		Use alternative antihistamine agent
	Rifapentine	Reduce serum level	Do not provide 3HP with protease inhibitors and consider 6H in this case
TDF	nephrotoxic drugs [e.g. aminoglycosides, amphotericin B, ganciclovir, pentamidine, vancomycin or interleukin-2]	Exacerbate nephrotoxicity	Avoid concurrent use
	ritonavir boosted PIs		Closely monitor renal function

	Ledipasvir- or velpatasvir-containing regimens		Monitor for TDF-associated adverse effects, including renal dysfunction, particularly when TDF is coprescribed with boosted HIV PIs
	Lithium		TDF: monitor renal function closely
DTG	Carbamazepine, Phenobarbital, and phenytoin		Use alternative anticonvulsant agent, (such as valproic acid or gabapentin) or if not possible substitute DTG with EFV and for children below 3 years substitute with boosted PIs
	Polyvalent Cation products containing Mg, Al, Fe, Ca, and Zn	Absorption of DTG is affected/reduced	Use DTG at least 2 hours before or at least 6 hours after supplements containing polyvalent cations, including but not limited to – Fe-, Ca-, Mg-, or Zn-multivitamin supplements; mineral supplements, cation containing laxatives and Al-, Ca- or Mg-containing antacids. Monitor for virological efficacy.

	<i>Rifampicin,</i>	Increases metabolism of DTG, and hence reduces concentration of DTG in the blood.	DTG 50mg BID. For pediatrics DTG BID by weight band. Continue with twice daily dosing of DTG in children for 2 weeks after use of rifampicin has ended based on their weight band
	<i>Rifapentine</i>		No evidence that changes of dose of rifapentine or DTG is needed to achieve adequate exposures of DTG
	<i>Metformin</i>	DTG increases metformin levels, which may lead to hypoglycemia.	Start metformin at lowest dose and titrate based on glycemic control. Monitor for adverse events of metformin Or adjust metformin dosing if already on metformin When starting or stopping DTG in patients on metformin dose adjustment of metformin may be necessary to maintain optimal glycemic control and/or minimize adverse events of metformin.

Table 4.10: Drug-drug interactions (ARVs and anti-TB drugs for treatment of DR-TB).

Clinical Condition	Responsible ARV drug/s	Responsible anti TB drug/s	Recommendations
Bone marrow suppression	AZT	Lzd, R, H	Monitor blood counts regularly. Replace AZT if bone marrow suppression occurs. Consider suspension of Lzd. Also consider cotrimoxazole, if patient is taking.
Hepatotoxicity	EFV protease inhibitors, NRTIs	H, R, E, Z, Bdq, PAS, Eto/ Pto, FQ	When severe, stop both the ART and TB medications, and restart the TB medications first. Consider cotrimoxazole, if patient is taking. Also rule out viral hepatitis (Hepatitis A, B, C & Cytomegalovirus(CMV)).
Renal toxicity	TDF	Amino-glycosides, amikacin	TDF may cause renal injury. If possible, avoid TDF in patients receiving aminoglycosides or amikacin. If TDF is absolutely indicated, serum creatinine and electrolytes should be monitored (at least every two weeks). Even without the concurrent use of TDF, PLHIV have increased risk of renal toxicity secondary to aminoglycosides and amikacin. In the presence of renal insufficiency, ARV and anti-TB medications need to have their doses adjusted.

Central nervous system (CNS) toxicity	EFV	Cs, H, Eto/Pto, FQ	EFV has a high rate of CNS side effects in the first 2–3 weeks of use, but typically self-limited and resolves. At present, there are limited data on the use of EFV with Cs; concurrent use is the accepted practice with frequent monitoring for central nervous system toxicity. Psychosis can occur with Cs, but is rare with EFV alone; other causes should always be ruled out.
QTc Prolongation	Protease inhibitors (PIs) LPV/r	Bdq, Dlm, Mfx, Gfx, Cfz, Lfx, Ofx	PIs may result QTc prolongation. The additive effects of combining ART with known second-line anti-TB drugs on QTc prolongation is not known. Close monitoring is indicated.
Dysglycaemia (disturbed blood sugar regulation)	Protease inhibitors (PIs)	Gfx, Eto/Pto	Protease inhibitors tend to ^[1] _{SEP} cause insulin resistance and hyperglycemia. Eto/Pto tends to make insulin control in diabetics more difficult, and can result in hypoglycemia and poor glucose regulation.
For patients who are co-infected with HIV and MDR/XDR-TB, there is limited information on interactions of ARV drugs with new drugs such as bedaquiline and delamanid. Concomitant use of EFV and PIs with bedaquiline may interfere with drug concentrations and require close clinical monitoring; alternative ARV options should be considered, if possible. N.B. For details of ARVs and anti-TB drugs for treatment of DR-TB interactions, please refer the latest TB guideline			

a. Effects of ARVs on Hormonal Contraceptives

In general, women taking ARVs are eligible for all hormonal contraceptive methods. But special consideration is necessary for women using some **hormonal methods** (i.e. especially etonogesterel implants) to use condom additionally. Advise to use condom in addition to hormonal contraceptives for women living with HIV using EFV or ATV/r or LPV/r or DRV/r. There is no significant interaction of ARV's with DMPA. DTG does not have significant interaction with hormonal contraceptives.

***If the patient is a woman of childbearing age (15-49 years), check the HIV Care/ART Card to see if she is pregnant. If not indicated, ask: *Do you think you might be pregnant?* Write the response in the place provided on the card. If you see drugs that can cause a drug reaction or should not be taken if the woman is pregnant, call the clinician to ask for an alternative drug prescription.**

b. Effects of food on ARVs:

Food can also have an influence on how well the drugs can be dissolved and absorbed. So far, Efavirenz is the only ARV affected by food type. For, the presence of **fatty food** increases the absorption rate and subsequently increases the side effect of **efavirenz (EFV)**. Hence, it is important to advise the patient to avoid eating fatty meals 3 hours before taking EFV containing regimen and 3 hours after. For the other four ARV drugs in the first-line regimens, there are **no** food restrictions.

c. Drug Interactions of ARVs with herbal medications:

In the Ethiopian context, the interactions of herbal medications and ARVs are unknown. Therefore, the concurrent use of herbal medications and ART should be discouraged. It is important to ask your patients if they are taking herbal medications.

Exercise 12.5

1. For each of the ARVs given in the table below, circle the drugs which could have a drug interaction or other difficulty when given together. There may be more than one interactions.

1. DTG+3TC+TDF • Rifampicin • Rifapentine • Multivitamins containing Mg, Al, Fe, Ca, or Zn • Phobarbitone	2.DTG+3TC+TDF • Metformin • Carbamazepine • Antacids containing Mg or Al • Implant
3.TDF-3TC-EFV • oral contraceptive • diazepam	4.TDF-3TC-EFV • cotrimoxazole • rifampicin • phenobarbital
5.AZT-3TC-EFV • Diazepam • Aciclovir • Ganciclovir	6.AZT-3TC-DTG • Implant • Depo/ injectable contraceptive • Phenytoin
7.AZT-3TC-LPV/r or ATV/r or DRV/r • Rifampicin • Rifapentine • Co-trimoxazole • Oral contraceptive • Methadone • Statins	8.TDF-3TC-EFV • Rifampicin • Oral contraceptive • Implant
9.AZT-3TC-EFV • Rifampicin • Ketoconazole • Phenytoin	10.AZT– 3TC- NVP • Co-trimoxazole • Oral contraceptive

2. What advice would you give to HIV positive woman on ART who is taking oral contraceptive?

3. If a patient is on 3HP or treated for tuberculosis with rifampicin, which ARV drug should be avoided?

For what reason? _____

What ARV drugs can be taken by the patient? _____

4. If a patient is taking TDF-3TC-EFV, what other advice would you give the patient? _____

Section summary:

- Some ARVs (EFV, PIs and NVP) reduce efficacy of hormonal contraceptives that would lead to un-intended pregnancy. ART providers should provide condom in addition to hormonal contraceptives for their clients who are using hormonal contraceptives. Advise patient to use condom in addition to hormonal contraceptives for clients using EFV or NVP or ATV/r or LPV/r or DRV/r.
- The goal of ART is to suppress viral replication and increase the number of CD4 as much as possible and finally improve the general wellbeing/ health of the client.
- Before people start ART, it is important to have a detailed discussion with them about their willingness and readiness to initiate ART.
- In the appointment spacing care model, stable clients will be appointed every six months for clinical visit and medication refill after complete 1 year on ART.
- The value of viral load testing as a more sensitive and early indicator of treatment failure is increasingly recognized.
- If the patient does not have good adherence, the virus might become resistant to the first regimen and we might have to change the initial ART regimen.
- Most common toxicities associated with ARVs appear within the first 3-6 months of ART initiation, therefore it is wise to be vigilant and check for these toxicities in the follow up visits.
- Enhanced Adherence support is important for patients that have uncontrolled VL.
- Dual contraceptive using condom is recommended for women using hormonal contraceptives for FP (implants).

12.8 DIFFERENTIATED HIV SERVICE DELIVERY

Session Objective

- Describe common modalities for differentiated service based on client's need.
- Describe patient classification model for differentiated service delivery models,
- Describe the criteria for implementation of appointment spacing model,
- Describe other differentiated service delivery models,

“Differentiated care is a client-centered approach that simplifies and adapts HIV services across the cascade to reflect the preference and expectations of various groups of people living with HIV (PLHIV) while reducing unnecessary burdens on the health system.”

The central driver to adapting service provision is the client's needs. Differentiated care applies across the HIV continuum to all three of the 90-90-90 targets. Differentiated care includes models of testing people unaware of their HIV status to treatment care to viral suppression of HIV clients enrolled in care. Based on the level of stability of the clients, DSD models can be based in the facility based or outside in the community.

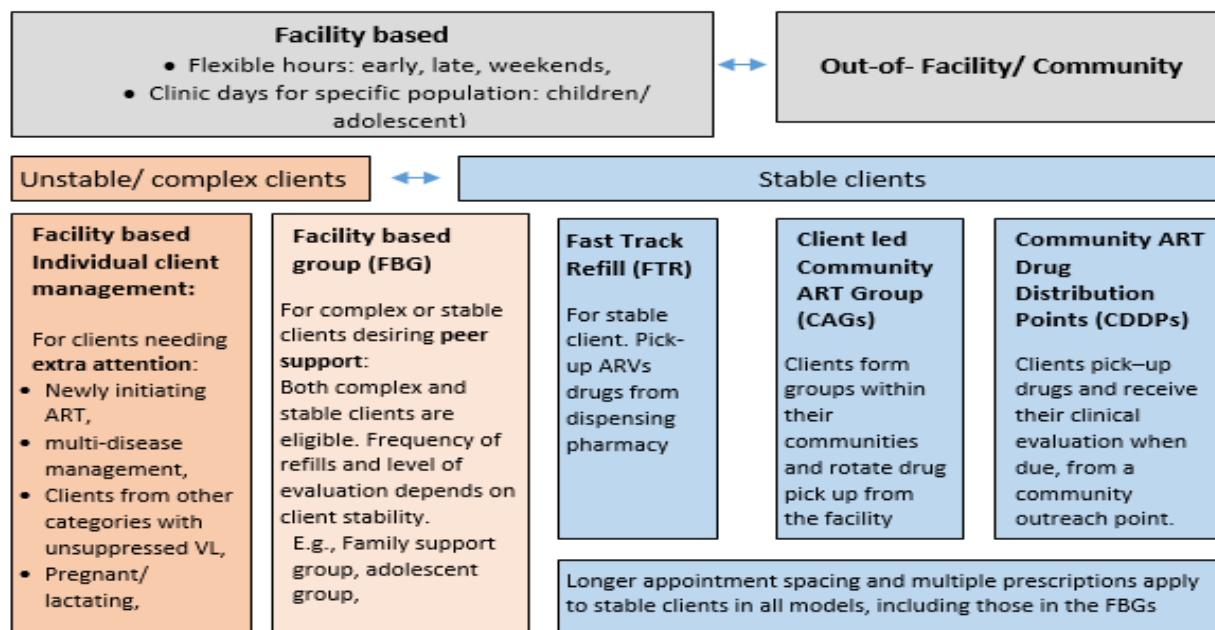


Figure 12.3. Generic Differentiated service delivery model (DSD)

To accommodate the growing number of stable individuals on ART and improve the retention in care and health outcomes, innovative models of differentiated ART service delivery have been developed. The WHO differentiated service models are

- i. Appointment spacing Model (ASM) for clinical and drug refill visits,
- ii. Peer educator-led ART refill groups,
- iii. Community ART distribution points, and
- iv. Patient-led community ART groups

All these approaches are found to reduce the burden for patients (reduced time and cost of travel to clinic and less income loss) and the health system (reduced clinic attendance), while maintaining high retention in care (more than 90% retained in care across multiple time points).

So far Ethiopia has endorsed these models and on the way of piloting additional modalities:

1. Appointment spacing model of differentiated HIV service delivery (ASM)

In this care model, stable clients will be appointed every six months for clinical visit and medication refill. For those clients who complained of storage for 6 months there will be a Multi-

month prescribing (MMP) letting them collect their ARVs at 3rd month without need of going through clinical visits.

Clients in this care model also need to get additional supports like arrangement of treatment supporter at home level among their family members, arrangement of adherence reminders like alarm and education on how to maintain the drug quality at home level. Clients need to be counseled and encouraged to disclose their status and to participate in peer support groups or to be a member of PLHIV associations. Those clients who disclosed their status will get adherence support from PLHIV associations via home visits or telephone follow up by community level volunteer adherence supporters in addition to the support by the case managers at facility level. As the number of clinical visits are limited to twice per year, the clinical visits will be intensified that include thorough clinical evaluation, assessment of adherence and viral load monitoring.

Customized and standardized checklists and formats have been prepared to facilitate and monitor the implementation in addition to the already available standardized forms and registers. After appropriate classification based on the suggested criteria, clients will be informed and need to give verbal consent on the frequency and location of the service delivery. Eligibility for each category and adherence will be monitored continuously according to a standardized assessment tool. Through continuous assessment and based on their current needs, patients may move between the categories or groups over the course of their care and treatment.

- Those stable clients (category 4) will be given appointment every six months for clinical follow up and medication refill with the health care providers (Nurse/HO/Physician). In addition, these clients need additional supports like:
 - Enhanced counseling to disclose to their family members and at least one treatment supporter will be arranged for each client (among their own family members).
- Ensure clients are supported on innovative and simple ways regarding Medication safety techniques to maintain the quality of the drugs over the six months' period.
- Clients included in the appointment spacing need to be linked to adherence case managers and adherence supporters to get additional counseling and support.

- Adherence case managers and adherence supporters will link the clients included in the appointment spacing to community based adherence counseling and support based on consent.
- Couple clients will be counseled to have the follow up visits alternatively and each of them will pick the medication of the other partner during their visit.
- All clients on appointment spacing model will be assessed at every visit for possible reclassification as their care need will be changed over time.

Patient classification based on their care needs in the Ethiopian context:

Category 1a: People, who present when, well, potentially with higher CD4 cell counts.

Category 1b: Pregnant/Breast feeding women with no other complications, **rapidly growing children** (0–5 years old) and **adolescents** with no advanced disease or other complications.

Category 2: People with advanced disease are defined as those presenting to care with a CD4 count below 200cells/mm³ or WHO disease stages 3 and 4.

Category 3: Those who are already on ART but need careful monitoring to ensure timely action as required (For example treatment failure suspects, patients with other chronic comorbidities, patients with identified adherence barriers).

Category 4: Stable individuals are defined as those who have received ART for at least one year and have no adverse drug reactions that require regular monitoring and have good understanding of lifelong adherence and evidence of treatment success. Evidence of treatment success can be seen using:

- Two consecutive viral load measurements below 1000 copies/mL with no current illnesses, excluding children, adolescents, pregnant and lactating women. **OR**
- In the absence of viral load monitoring, rising CD4 cell counts or CD4 counts above 200cells/mm³, an objective adherence measure either from the clients self-report or by doing pill count, can be used to indicate treatment success.

Table 12.20. Framework for appointment spacing model of HIV service delivery in Ethiopia

Patient group/category	Minimum care package/services	Location of Service delivery	Provider of service	Frequency of service delivery	Additional support required
Category 1a: People, who present when well, potentially with higher CD4 cell counts,	• Clinical and lab evaluation • OI screening and prophylaxis if needed • ART initiation • additional and targeted adherence and retention support in order to commit to lifelong ART	HC or Hospital based on consensus	Prescriber nurse/HO/Physician	Biweekly during the 1 st month, monthly during month 2 to 6, Every three months then after, Assess for possible change of category after 12 months	Intensive adherence support during the first six months as they feel well and may be reluctant
Category 1b: Pregnant/Breast feeding women with no other complications, rapidly growing children (0–5 years old) and adolescents	• Clinical and lab evaluation • OI screening and prophylaxis if needed • ART initiation if not already started • additional and targeted adherence and retention support in order to commit to lifelong ART, • <i>Rapidly growing Children and adolescents may need dose adjustment based on their weight</i>	HC or Hospital based on consensus	Prescriber nurse/HO/Physician	If client newly initiated Biweekly during the 1 st month, monthly during month 2 to 6, Every three months then after	Intensive adherence support.

category 2: People presenting to care with advanced disease	• Rapid initiation of ART (after appropriate base line evaluation) • Systematic screening for OIs; • Provision of IPT and CPT; • Intensive follow-up and monitoring • Lab monitoring as per the guideline	Hospital	Prescriber nurse/HO/Physician	As per the national treatment guideline and may be assessed for reclassification after stabilization	Additional adherence counseling and support by case managers
Category 3: Those who are already on ART but need careful monitoring to ensure timely action as required	• Clinical care and lab monitoring, • additional adherence support • Timely switch to second-line ART regimen in case of treatment failure	Hospital	Experienced Nurse/HO/Physician	Monthly and change of category if stabilized or issue settled	Provide enhanced adherence counseling and support if there is high viral load
Category 4: Stable individuals who are already on ART for ≥ 1 year excluding children, adolescents, pregnant and lactating women.	• Clinical review and medication refill • Adherence assessment and support • Adherence counseling to Re-enforce the adherence practice • Lab monitoring as per the guideline • <i>Treatment supporter at home level,</i>	HC or Hospital	Nurses	Every six months for clinical follow up and drug refill. If the client is not willing to be included in the six monthly follow up, a pharmacy drug refill will be arranged every other three months	Additional adherence counseling and support by case managers (at the facility) and community adherence groups, encourage disclosure, involve treatment/adherence supporters at home

The formats used for the implementation of the appointment spacing - the individual client classification tool and the client enrollment register - are attached under Annexes 13 & 14.

2. Urban Health Extension professional managed community ART refill groups (UHEP_CAG)

In this model, stable clients receive their ART refills in a group established at their nearby

community. Health care professionals are responsible for the management of the provision of ART drug supply, care and support to groups of stable clients.

Six to 10 stable clients meet for 30-60 minutes at convenient place in the community and get their pre-packed ARVs refilled by healthcare professional. In the pilot program, the urban health extension professional (UHEP) are trained to manage the group providing the pre-packed ARVs and performing a brief symptom screen and referral wherever necessary, every 3 months (alternating with the 6 month clinical appointment at ART clinic). There is also a peer support as necessary.

3. Fast Track ARV Drugs Refill Model,

This is one of the facility based Differentiated Service Delivery Models of HIV care where stable clients make clinical visit once every six months but collect their medication every three months from Pharmacy.

Another models for unstable and stable clients that need special support at ART clinic are **Peer Support Differentiated Service Delivery Model** and a **viremia clinic** with support group for clients with high viral load results. Adolescents refill their ART & clinical visit together with psychosocial support groups in the adolescent DSD Model.

Section summary:

- For ART is a life long treatment for HIV, clients need a differentiated service that fit to their routine life,
- Differentiated HIV service in the form of appointment spacing and establishing community level treatment group for PLHIV both alleviate the economic burden on clients and workload on health care facilities,
- Stable clients can have their clinical visit to ART clinic twice a year and collect their medicines every 6 month,
- People living with HIV can have community level ART treatment provision with distribution of pre-packed ART among members of peer group,

12.9 Pharmaceuticals Supply Management in HIV

The objective of pharmaceuticals supply management system is to ensure adequate and continuous availability of the most safe, effective, quality-assured ARV pharmaceuticals at service delivery sites, in the right quantity, at the lowest possible cost and in a timely manner.

The Ethiopian Pharmaceutical Supply Agency (EPSA) is responsible for quantification, procurement, storage and distribution of pharmaceuticals to health facilities. Health facilities obtain ARVs and related commodities through the Integrated Pharmaceutical Logistics System (IPLS), a single reporting and distribution system of the country.

1. Selection of ARV Drugs and Related Supplies:

Selection of ARV products is done at national level considering their safety, efficacy, quality and cost. Accordingly, the National Essential Medicines List (EML) is updated whenever new ARVs are included.

2. Quantification:

National quantification of ARV pharmaceuticals is conducted by EPSA in collaboration with FMOH and other stakeholders and is done every two years which is revised every year based on changes in guidelines at national level and the data obtained from ART sites. It is important to have detail report on number of clients currently on ART, disaggregated by regimen and patient type, attrition rate, and other relevant data.

3. Procurement:

Generally, the procurement of ARVs and related pharmaceuticals is executed by EPSA. The procurement process follows the national and international procurement regulation, which is a competitive approach through international bidding.

4. Storage and Distribution:

Proper storage of ARVs, including refrigeration, is critical to maintain the quality of the drugs and related supplies. Central EPSA will deliver products to its Hubs; subsequently the hubs distribute the pharmaceuticals to health facilities every two months based on orders placed by the

health facilities to EPSA hubs. Health facilities are expected to follow consumptions and keep record regularly. They get ARV drugs and related supplies if and only if they submit their report and order, using the national Report and Requisition Form (RRF) with accurate data and in a timely manner.

Treatment for second-line failures will be provided in selected hospitals where third line regimen ARV drugs are available. As the total numbers of patients on 3rd line are low and provided with specially arranged follow up in these hospitals, EPSA hubs will provide third line drugs to these hospitals per their request on the IRRF.

5. Inventory Control:

Inventory control system is mandatory to maintain an appropriate stock level of products at all level of the supply chain to avoid shortage and overstock. IPLS dictates the health facilities to keep a minimum & a maximum of 2 & 4 months' stock, respectively. Hence, health facilities shall closely monitor their stock status to avoid shortage and unnecessary expiry of ARV drugs and related supplies using bin cards.

At health facility level, ARV utilization by clients/ consumption data will be captured by different recording and documentation mechanisms, namely, patient information sheet (PIS)/ yellow card, ARV drugs dispensing daily register, monthly ARV drugs dispensing and consumption summary form, and Intra-facility report and resupply form (IFRR).

By performing all activities properly, the national ARV drugs supply management system will be strengthened.

6. Rational medicine use:

Rational medicine use (RMU) dictates that patients must receive medicines appropriate for their clinical needs, in doses that meet their individual requirements, for an adequate period of time and at the lowest cost to them. RMU implies promotion of rational prescribing and ensuring good dispensing & counselling practice for appropriate medicine use and adherence by the patient.

The ART pharmacy service as part of the continuum of chronic care promotes the RMU through implementing effective pharmaceutical care. ARVs and OI medications provided for HIV patients

are recorded in the patient information sheet and the ART pharmacy provider provides counselling on potential side effects (ADR), drug-drug and drug-food interactions, allergies, contraindications, proper storage and timing of ARV administration.

A multidisciplinary approach of addressing client's issues and the effective communication of information among MDT members optimize RMU in the facility.

7. Quality Assurance:

Ethiopian Food and Drug Authority (EFDA) is responsible in leading the quality control of pharmaceuticals. Quality of ARV drugs & related supplies from time of reception to dispensing to the patient shall be assured at the facility level through using visual inspection, documenting complaints from patients, implementing a First-Expiry-First-Out (FEFO) system, and removing expired/ damaged medicines from shelf, and maintaining good storage system.

Chapter 13: Psychosocial Support and Paediatric HIV Disclosure

Chapter Description:

This chapter describes the psychological and sociological needs of children and the importance of disclosure. In addition, it discusses the steps and processes for disclosure and supports.

Purpose: Understand the importance of psychosocial needs and process of disclosing HIV status to HIV infected children and adolescents.

Enabling Objectives:

By the end of this chapter, the participant will be able to:

- Explain psychosocial needs of children and adolescent.
- Illustrate how to communicate with children and adolescent.
- Identify children and adolescent with psychosocial problems.
- Provide psychosocial support to children and adolescent.
- Describe advantages and disadvantages of disclosure.
- Describe the process of disclosure.
- Provide effective disclosure counseling for parents or caregivers.

Outline:

- Introduction to psychosocial support
- Psychosocial needs of children and adolescents
- Disclosure of HIV status to children and adolescents,

Allocated time: 130 minutes

Learning activities

- Reading
- Discussion,
- Role play,
- Exercise,

13.1. Introduction to Psychosocial support

Psycho: the mind (unique feeling, emotions, thoughts, understanding, attitudes and beliefs of an individual).

Social: external/interpersonal relationships and what goes on, in the environment. It includes social attitudes, values (culture) and influence exerted by family, peers, school and community.

Psychosocial: is the dynamic relationship between social and psychological experiences where the effect of one continually influences the other.

Children and adolescents in our society undergo psychosocial stress which is often unnoticed by adults and caregivers. They do not have an opportunity to express themselves freely because in our culture children are seen but not heard. A diagnosis of HIV infection in a child usually implies that other family members too may be living with HIV, although family members who have not yet been tested for HIV may not be aware of their status. For this reason, the HIV-infected child cannot be treated in isolation; therefore, care of the HIV-infected child must be family centered as well as child-focused.

Counselling, psychological and social support are integral components of the holistic approach to caring for an HIV-infected child. This is a continuous process that usually begins at the first point of contact in the healthcare delivery system and continues through non-medical sector support services. Psychosocial issues must be addressed from the perspectives of the child, the caregiver and family, the healthcare provider, as well as other systems around the child that include schools.

Clinical presentation of a child with psychosocial problems

A child at risk of developing psychosocial problems will present with the following:

- Emotional disorders, e.g. anxiety, depression, personality changes that may include mood swings, poor interpersonal relationships and poor impulse control.
- Behavioral disorders, e.g. delinquency, disobedience, aggressiveness, disruptive and/or restlessness.
- Psychological disorders, e.g. depression, schizophrenia, and other forms of mental illness.

Issues to address in providing psychological and social support to children infected/affected by HIV:

1. Issues from an infected or affected child's perspective

HIV-infected children and/or uninfected children from HIV infected family members have to deal with many psychosocial issues, including:

- Dealing with his/her own chronic ill health, pain and discomfort.
- Being different from other children.
- Watching a parent battle chronic and/or terminal illness, and sometimes even assuming a care-giver role for sick parents.
- Bereavement and its consequences, such as separation from close family members or change in socio-economic circumstances (having to leave school, having to do without basic necessities, having to get a job to make a living, having to look after younger siblings, or being homeless).
- Asking questions that are not adequately answered or getting evasive answers.
- Having to take drugs daily.

2. Issues from the caregiver's perspective

Regardless of how a child became infected, parents experience some degree of guilt. In sub-Saharan Africa where mother-to-child transmission is the main mode of HIV transmission in children, the mother, who is invariably the primary caregiver, is also HIV-infected. However, the primary caregiver may be someone other than the mother of the child. Psychological issues that caregivers will need to deal with include:

- Dealing with his or her own HIV diagnosis (in cases where the caregiver is one or both

parents of the child).

- Dealing with the child's illness and the related feelings of guilt, anger and hopelessness.
- Deciding whether to share or not and what to share with the spouse, child, relatives, neighbors, or school authorities.
- Fear of disclosure.
- Reproductive desires and decisions in the face of HIV infection.
- Time away from work (for frequent visits to health care facilities) and implications for job security, and family earnings.
- Concern about who will take care of the children in case of the caregiver's death.
- Fear of his or her own death.

3. Issues from the healthcare service provider's perspective

Healthcare service providers for HIV-infected children often find it challenging to address the children's psychological needs. These Challenges include:

- Inadequate knowledge and skills to communicate effectively.
- Inadequate knowledge of information that is appropriate for children at different developmental ages.
- Lack of enough time to develop and nurture a relationship designed to make a child 'open up' and share his/her feelings.
- Unavailability or lack of knowledge of referral options.
- Limited resource materials to guide health care providers, cultural and traditional factors of 'talking to' children but 'not with' children.
- Limited knowledge of the needs of children.

13.2. Psychosocial needs of children and adolescents

All children need care, attention, security, love, nurturing, play, acceptance, a supportive environment, and specific help to overcome their individual problems.

When children lose someone they love, they need simple and age appropriate information about what has happened. They need to be listened by someone who is prepared to answer the same

questions several times. Most importantly, they need reassurance that they will be taken care of and loved.

Communicating with children about care and treatment

Communicating effectively with children requires skills in listening, observing, and understanding their messages and responding appropriately. At least one person who is familiar with and normally cares for the child should be present. This is true for all children, especially younger ones who often find it difficult to trust and communicate with someone they do not know well, or are not familiar with.

The different ways (media) of communicating with children include:

- Make-believe play - also known as **pretend play**, is a loosely structured form of play that generally includes role-play, object substitution and nonliteral behavior.
- Using stories and asking children to tell their stories.
- Drawing pictures.
- Music, dance and drama.
- The children writing about their own experiences.

Tips of communication with children:

- Get the child's attention before speaking.
- Communicate on the same level as the child.
- Speak as if you mean it.
- Make requests simple.
- Use more positive direction than negative.
- Talk with - not at – children.
- Keep lines of communication open by listening attentively when the child talks to you.
- Use kind words and actions to encourage and support the child.

Communicating with adolescents

Adolescents are going through dramatic biological and emotional changes that affect their self-confidence, relationships, social skills, and general thinking. Adolescents may feel fearful, embarrassed, or uncomfortable around health workers. Most adolescents have concerns about

confidentiality and health workers need to reassure them that will practice confidentiality. We can establish rapport by treating everyone equally, being genuine, not criticizing their appearance, using words and language they can understand, keeping their secrets confidential, reassuring them and let them know you are there to help.

Healthcare service providers and caregivers should be careful to avoid criticizing children and adolescent, for criticism may inhibit free expression. Acceptance attitude and treating each as an individual is very important while communicating and working with children and adolescent. Children will continue to ask questions about a topic even after explanations have been given to them; this holds for questions about disease as well. They deserve to be given correct answers, appropriate to their ability to understand and comprehend. **A general rule is never to tell lies to children and adolescents; this may lead to loss of trust when they eventually find out the truth.**

On-going psychosocial support for HIV-infected children and adolescent in care

HIV-infected children and adolescent need support to remain in care. The support may take the form of:

- Attending peer support group activities, where they share their experiences and support each other in coping with living with HIV. This is usually done through music, dance and drama, sports, testimonies and other activities carried out in support group meetings. Vocational training and skills building can be provided in these groups, especially to adolescents out of school.
- Building life skills: they can be trained in life skills, such as developing self-awareness and having high self-esteem, coping with emotions, communication, assertiveness and negotiation, and appropriate decision making.
- Utilizing mobile phone technology to aid communication with children and adolescent: where possible important messages can be passed on to them through short message services (SMS) e.g. reminders to take their medications, reminders about clinic appointments, and others.

Bereavement counseling for children

When children lose a family member, attention must be paid to helping them and their families to move through this time with the least amount of suffering and as much support and dignity as possible. Open communication about what is happening should be encouraged among the children themselves, their parents, and health workers. Children and adolescent in a family require continued counseling and psychological support after the death of a loved one. Parents and caregivers also need support for their emotional reaction toward a terminally ill child and adolescent.

Children whose parents are terminally ill are affected in many ways and have a wide range of problems and needs:

- Psychological distress.
- Anxiety about their security and safety.
- Lack of parental nurturing.
- Lack of basic social needs.
- Loss of inheritance.
- Need to work.
- Less education and skills.
- Mental health needs.
- Emergency and long-term child care.
- Bereavement and grief counseling.

Responding to a dying parent's needs will also address many of the child's immediate concerns, including reassurance that the child will receive care when the parent is no longer available. To appropriately respond to the needs of a child whose parent is dying, it is important to understand how child developmental stages affect children's perception of death and dying (see Table 13.1).

Table 13.1.Children's perceptions of death and possible interventions

Age	Perception of death	How to help the child
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1–3 years	Equate death with sleeping and expect people who have died to eventually wake up. Fear separation from parent/caregiver	Keep the child's daily routine as unchanged as possible. Make time each day to hold, talk to, and comfort the child.
3–4 years	Children this age do not accept death as final and think of it as a temporary separation. Children may believe that they are in some way responsible for the death because of powerful imaginations (magical thinking). Some may believe that perhaps, if they 'wish hard enough, the dead person will come back'.	Explain clearly why the person died: 'died because she was not well. It had nothing to do with you, with something that you did or didn't do'
5–8 years	Begin to accept death as final and view it as separation from loved ones. They have a great fear of a sick parent dying, and of being abandoned. They worry about their own death	Reassure the child that minor illnesses and injuries can be treated. Reassure the child that it is okay to cry, to feel angry, sad, or frightened when someone dies. Reassure them that they are not responsible for the death
8–10 years	Children learn that all living things must die; they begin to feel sorrow and loss. Interest in the mystery of death grows	Answer questions as fully as possible. Do not discourage normal curiosity about death. Acknowledge the child's feelings. Allow the child to cry and talk about the loss
9–11 years	React strongly to death. Interested in what happens after death Death is accepted as a part of life	Answer questions fully Acknowledge and explore the child's feelings Interventions may include: talking about memories; writing a daily journal; drawing pictures of how he or she feels; prayer; compiling a picture album of the loved one

The uninfected sibling of an HIV-infected and sick child may, similarly, have unmet psychosocial needs as a result of the continuous attention given to the sick child. Secrecy and lack of communication may deter the child from asking questions.

Resentment may occur because of feelings of deprivation and exclusion. It is important for the family to set time aside for this sibling or siblings and to communicate what is happening, within the limit of each individual child's developmental stage and understanding.

13.3. Disclosure of HIV Status to Children and Adolescents

Disclosure is a process that involves ongoing discussions about the disease as the child matures cognitively, emotionally and sexually. Disclosure process is optimal when process involves caregiver/parent and child, with parent/caregiver initiating/leading the process with counselor/HCW providing a supportive role.

Testing a child for HIV and informing the status is a sensitive issue. Many health care providers may find it difficult to reveal HIV status. The benefits of disclosure however significantly outweigh the drawbacks. Children often want and ask to know what is wrong about their health. The process of informing HIV status to a child is an ongoing process rather than a single event. All families are unique and there is no fixed rule as to when and how to inform a child about HIV test result. Taking the child's, family and community situation into account, disclosure has to be individualized.

Information should be given to a child in a manner that he/she understands and at a pace that he/she can cope with. The parent or caregiver has the prime responsibility for informing the child of the result. The healthcare worker should support the parent or caregiver in disclosing HIV status to a child. Information provided should be truthful, coherent, and consistent and in language that is understandable and age-appropriate. Some caregivers are very protective of their children and need help on the importance of informing the child of their HIV status. This process is often made easier if it is adapted to a child's needs, expectations and requests. As children

grow, they undergo both physical and psychological changes. In general, children and adolescents of different ages will have different emotional needs, fears and expected behaviors that need to be considered in the process.

Exercise 13.1

List the advantage of disclosure and disadvantages of non-disclosure of HIV status to a child.

Benefit of Disclosure

Disclosure is important for children for the disclosed child. A disclosed child who knows about his status and the treatment he is taking will have the following benefits. She/ He will:

- Feel relieved at learning the cause of his/her illness.
- Help to stop self-blaming as cause of the sickness.
- Feel more in control.
- Have greater open involvement in medical care decisions.
- Avoid situation of accidental disclosure (e.g. child overhears caregiver discussing).
- Decrease disruptive behaviors.
- Improve adherence to medication.
- Improve social functioning and school performance.
- Be more willing to access health education, social support, peer support.
- Communicate with respect, interact with children and build trust.

Disclosure is also important for the **parents/ caregivers**, as the disclosed caregivers:

- Relieve the stress and anxiety that accompanies secrecy and deception.
- Have opportunity to develop trusting relationship and open communication with child.
- Have improved cooperation from a child.
- Able to cope with feeling of helplessness and hopelessness by sharing positive messages with their children.
- Have easier access to other services, care and support.

Disadvantage of non-disclosure: keeping the status of the child secret and not telling the child

his/ her problem and the reason for treatment has deleterious effect on the child psychosocial and academic development and interaction. Some of the effects are:

- Self-blame: the child blames him/herself for everything,
- Wrong belief: Maintains false understanding of one's illness like the belief of being cured.
- Child's imagination creates unnecessary worry.
- Parents may use forceful means to ensure child takes medication.
- Start engaging with sexual relationships without knowing status.
- Not able to enquire or access support they need.
- Discovering own HIV status inadvertently without any preparation or in a traumatic way.
- Loss of trust or failure to maintain or rebuild trust with parents/ caregiver in future.

Barriers to Disclosure

Whenever we consider disclosure to children, it is important to consider the following conditions like beliefs and feelings, which practically hindered the process.

- Fear that the infected child will **inappropriately disclose** his/her HIV status, especially in families in which the diagnosis remains closely guarded (fear of shared stigma).
- **Fear** of stigma, rejection, and loss of support by the family/community.
- **Desire to protect** the child from worrying about his/her future.
- The possibility that the burden of learning of his/her HIV status will lead to depression or other mental health issues on the child.
- Feelings of **guilt and shame** may prevent HIV-infected caregivers from disclosing their own infection to their child.

Process for Disclosure

Disclosure is not an event or a one-time conversation. It is a process that takes time and constant communication in an age-appropriate manner. The best way for a child to learn about his/her HIV status is through age-appropriate information shared by a loving and trusted caretaker. Disclosure is generally started when the child is about 6 years old. Disclosure to children should never happen casually, inadvertently or in the heat of anger or conflict. A child's maturity and cognitive

capacity varies and is not only dependent on age. It is important to tailor the discussion to the child's cognitive level and to the child's personal and individual situation. It is important to assess readiness of the entire family for disclosure and address potential barriers to disclosure.

NB: It is recommended to use the disclosure tool “Why am I taking Medicines” as an aid during the disclosure process. (*Refer Pediatrics psychosocial care training material, Ethiopia*)

Commonly used Terminologies in Disclosure:

- **Full Disclosure** to a person who is HIV positive means telling him/her that they have HIV. It can also include the reason for taking medicines. Example: You have HIV and we are giving you ARVs/medicines to help your body fight the virus.
- **Partial Disclosure** to a person who is HIV positive means sharing information about his/her illness but not using the word “HIV”. Example: Your body is sick and we are giving you medicines to help keep your body healthy.
- **Complete Non-Disclosure** to a person who is HIV positive means not telling a person anything about their condition. Example: A HCW may assess the person's health status and give him/ her medications without explaining anything about what the medicines are for or why the person must take them.
- **Deception** is when a person who is HIV positive, is intentionally told something untrue. Example: You are taking medicines because you have TB.

Assessing Readiness for Disclosure

Consider the following issues while assessing readiness of caregiver and the child for disclosure.

The Child

- Is the child symptomatic? Taking medications?
- How old is the child?
- Is the child living with a sick parent or family member?
- Is the child asking questions about HIV?
- Does the child appear distressed, anxious or worried?
- Is the child sexually active and at risk of contracting or spreading HIV?

The parent or caregiver

- Has the parent or caregiver been tested for HIV?
- Is the parent or caregiver infected? Symptomatic/taking medication?
- Is the adult ill? Is s/he in need of help from children in the household?
- Is the infected adult an important attachment figure for the child?

The family or household

- Are there any adults in the household with HIV infection? Who is aware?
- Are other children in the household HIV-infected? Who is aware?
- How many family members are taking HIV-related medications?
- Is the family unit cohesive, or characterized by separations and/or conflicts?

The community

- Are testing and treatment generally available in the community?
- Are there people in the community who are open about their own HIV status?
Does the child know anyone in the community who is open about his/her HIV status?
- How strong is the stigma surrounding HIV in the community?
- Are there risks to the family (isolation, discrimination) if inadvertent disclosure occurs?
- Are there resources within the community for children – a youth group and or trusted adults they can talk to?

Source: Columbia Paediatric Clinical Manual 2005

Once the decision has been to disclose to the child, it is important to understand that the topic will have to be visited over and over again. It is important to give a clear message and listen actively; take cues from the child and avoid lecturing; the emphasis should be on asking directly and indirectly and listening. This involves preparation, education, planning and follow-up (see table below).

Process of assisting families for disclosure
Preparation • Why disclose now? • What do you want to communicate to your child? • What will be the most difficult questions for you to answer when your child knows his/her HIV status? • How will this information affect the relationship between you? • Acknowledge difficulty of disclosure and affirm motivation to begin process.
Education • How to explain HIV transmission to a child. • Anticipate questions and responses from child. • Post disclosure expectations.
Planning • When and where? • Who will be there? • What will you say? • Plans after disclosure.
Follow-up • School and family functioning. • Monitor medical treatment adherence. • Disclosure to peers and others. • Support group counselling. • Continue to reinforce the positives.

It is also important to discuss benefits of disclosure which have both short and long term impact on the family. Most children without cognitive deficits have the capacity to understand the diagnosis and concepts about immune systems and health. Whenever possible, disclosure should occur at a time when the child is clinically and emotionally stable, and the caregiver is ready to disclose. Preferably, disclosure should not take place during an acute illness or family crisis and should not coincide with other events such as birthdays, holidays or graduations. Disclosure can provide an opportunity for a family to share important information and to focus on health, communication, and relationships. The disclosure process should not be rushed; however, the timing of disclosure becomes more pressing as the child nears adolescence, a time when the potential to engage in sexual and substance-using behavior may emerge.

Disclosure plan in collaboration with parents/caregiver

Preparing caregivers to disclose HIV status to the infected child may take several months or even years. Many caregivers with HIV infection have difficulty accepting the illness, either their own or their child's, and struggle with whether or not to disclose the diagnosis of HIV infection. Clinicians should respect caregivers' reasons for fearing or resisting disclosure and should attempt to understand the barriers to disclose. Validation of caregivers' concerns can foster a partnership and prevent the development of an oppositional relationship between the members of the healthcare team and caregivers.

Partial disclosure may be an effective strategy to use to help caregivers who are not ready to use the terms "HIV" or "AIDS". Partial disclosure involves teaching a child about his/her body and how the immune system works before using the actual words "HIV" or "AIDS".

Stages in Pediatrics HIV disclosure

Stage 1:

This is for children around the age of six. If they are symptomatic, they want to know what will happen to them. They do not need to be informed of their diagnosis but the illness must be discussed with them. At this stage, start providing partial disclosure. Communicate with the child as follows:

- You are taking medicines to keep you healthy.
- You have body soldiers that keep you healthy.
- Your medicines increase the number of body soldiers and keep them strong so you can stay healthy.
- As long as your body soldiers are strong and you have many, you can do what you want to do in life.
- You need to take your medicines in the right time (in the evening or morning and evening) when your mother/ caregiver gives them to you.

Stage 2:

Once the child understood the above messages or if the child is school age (7-13years), move on to stage two and then to stage three. Communicate with the child as follow:

- Your body soldiers became weak because something was attacking them (“a germ”).
- If you take your medicines every day, you keep the “germ” asleep, so it can’t attack your body soldiers.
- Body soldiers can then increase in number and stay strong to keep you healthy.
- If you don’t take your medicines the “germ” could wake up and start attacking your soldiers.
- May consider asking the child if he/she would like to know how strong their body soldiers are. Check the file carefully first:
 - If CD4 trend is up, show child that their own “body soldier” numbers are going up and stronger because they have taken their medicines.
 - If CD4 trend not up, re-enforce previous messages (while looking into possible causes).
 - If viral load is suppressed, show child that the “germ” inside their body is under control because they are taking their medication properly.
- Building on the story, introduce the concept of resistance:
 - “If you forget your medicine and the “germ” wakes up too often, he can become “tricky”; then your medicines may not work to keep him asleep.”

Stage 3

Make sure the child and caregiver are ready for final disclosure. Some may have difficulty coping with disclosure information leading to changes in behavior (acting out in school, i.e. fights, low grades, truancy, anger, crying fits, and no expression of emotion). Others may have concerns that

other children in the school or community will make fun of them. Encourage them to ask questions; do not be disappointed if they do not react in the manner you expected.

- Introducing the words HIV, CD4 and viral load:
 - Only proceed if child and caregiver are ready.
 - Anticipate how the child might react.
- In presence of caregiver:
 - As usual, ask child why taking medicines; congratulate for what he/she has learned.
 - Ask child if he/she wants to know the other names for the soldiers and the bad guy.
 - “Germ” is a virus called HIV.
 - “Body soldiers” are called CD4 cells.
- Ask child what he/she has heard about HIV and correct any misconceptions.
- Choose words that avoid assigning blame, e.g. if child asks where he/she got HIV.
 - “Some children are born with the virus, and we think that is what happened to you.”
 - **NOT** “Your mother gave it to you.”
- Put new information back in the context of what the child has already learned.
 - “As long as you keep taking your medicines well, keep the germ asleep and the soldiers strong, you can do everything you want to do in life.”
- Continue to check understanding at each visit.

***N.B:** Full disclosure can be provided to most children over 10 years of age. Each steps of disclosure should be documented on the patient’s chart.*

Adolescents 14 years and older

Adolescents should know of their HIV status. They should be fully informed in order for them to appreciate consequences for many aspects of their health, including sexual behavior and treatment decisions. Be supportive and non-judgmental.

Post –disclosure assessments

Disclosure is a process that does not end with telling an HIV-infected child the name of their illness or diagnosis. After the HIV diagnosis has been disclosed to the infected child, follow-up visits are needed to monitor the child and family’s understanding of the illness and their

emotional and psychological adjustment.

Once the diagnosis has been explained to a child, it needs to be reinforced or regularly discussed as the child develops because many children will not have understood the full implications of the disease or diagnosis at the time of disclosure. For example, preadolescent children can cognitively understand the concepts about the virus but may be less likely to think of the future implications, such as transmission risks and safe sexual practices. As the child ages and matures, he/she will slowly understand and integrate the implications of the diagnosis into his/her life. Children's perception of self, health, illness, and death evolve as they mature through different developmental stages.

Some children who learn of their HIV status may experience guilt and shame and may isolate themselves as a result of the stigma and secrecy surrounding the disease. Changes in behavior and school functioning may occur in these children and may be symptoms of depression. Patients and families who have a difficult adjustment to HIV disclosure without progress over time should be referred for mental health services and additional support. In young adolescents it will be important to discuss about modes of HIV transmission, sexuality and reproductive health.

Exercise 13.2:

Group exercise; your facilitator will lead this group exercise

1. Do you disclose HIV status to adolescents in your facility?
2. Is adolescent peer group support available at your facility?
3. If no, what can be done to establish adolescent peer group support at your facility?
4. What activities can be done during adolescent peer group meeting?

Role play on disclosure

Adanech is a 32 year old married woman who has been on ART for the last 12 years. She has a 6 year old boy named Eyob, who has been on ART since his birth. Adanech didn't inform her child the reason why he is taking medications because of the fear of stigma and discrimination of her child by his peers. Her child sometimes asks her about his medication and she always tells him that she will explain in the future. She feels sad about her son's status and for not able to tell the truth to him.

- (a) Role play on how to prepare family for disclosure as a health provider.
- (b) Role play on starting to provide partial disclosure to the child.
- (c) Role play on stage two of HIV disclosure to the child.
- (d) Role play on stage three of HIV disclosure to the child.

Discuss your observations based on the role plays.

NB: *For detailed guidance on psychosocial support and Paediatric HIV disclosure, refer to the national psychosocial support and paediatric HIV disclosure training manual.*

Chapter summary:

- Disclosure of HIV status should be guided by the needs of the child and care givers.
- Disclosure should be an ongoing discussion about health and treatment.
- Age, developmental stage and health status of the child should be taken into account in considering disclosing HIV status.
- Disclosure does not end with telling an HIV-infected child the name of their

Chapter 14: Nutrition and HIV/AIDS

Chapter Description: This chapter describes the relationship of nutrition and HIV disease progress and entails the methods of assessment and management for malnutrition. The different standards for measuring nutritional problems will be also discussed.

Purpose: Learn how to conduct nutritional assessment and manage based on classification.

Enabling objectives

At the end of this chapter, participants will be able to:

- Describe the effect of HIV and AIDS on nutrition.
- Determine additional energy requirement for PLHIV of different age and disease stages.
- Assess and classify nutritional status of adults and children using appropriate anthropometric measurements.
- Counsel and manage patients with under nutrition based on appropriate care plan.
- Describe the preferred infant feeding options in the context of maternal HIV infection in Ethiopia.

Outline:

- Introduction
- Nutritional assessment
- Management of malnutrition for PLHIV
- Infant Feeding in the context of Maternal HIV Infection

Allocated time: 120 minutes

Learning activities

- Reading
- Demonstration,
- Exercise,

14.1. Introduction

The insufficient, excessive or imbalanced consumption of micronutrients and macronutrients leads to a state of malnutrition resulting in a number of nutritional disorders.

Under nutrition = Inadequate amount, combination and timeliness of energy or nutrients to carry out needed physiological functions. Under nutrition impairs growth, leads to wasting and stunting and ultimately death due to infectious and metabolic complication.

Over nutrition = More energy and nutrients than the body needs over time. Patients with over nutrition need to be counseled on dietary and life style management.

In the developing world, because of the high prevalence of under nutrition, malnutrition often denotes under nutrition and the associated complications.

Effects of HIV on Nutrition (HIV and Malnutrition)

Under nutrition is very common in people living with HIV. HIV can lead to under nutrition by multiple mechanisms. HIV affects nutrition in three, sometimes overlapping, ways:

- Reduces amount of food consumed.
- Interferes with the digestion and absorption of nutrients.
- Alters metabolism of nutrients.

HIV is associated with reduced intake of food/nutrients because of various factors:

- a) Cognitive impairment and/or depression results in reduced motivation and ability to access and prepare foods.
- b) Family instability or poverty leads to reduced access to food.
- c) HIV, OI or medicines induce anorexia and nausea and OIs of mouth and esophagus bring about painful swallowing.

HIV also affects nutritional status by interfering with the digestion and absorption of nutrients; this occurs due to diarrheal diseases and damages to intestinal mucosa.

HIV is associated with altered metabolism of food/nutrients. PLHIV require more energy than non-infected individuals of the same status in order to meet the elevated nutritional needs that result from the infections and changes in metabolism caused by HIV. WHO does not recommend

increasing protein, fat, or micronutrient intake over the recommended dietary allowances (RDA) since there is no increment in requirement of the nutritional types.

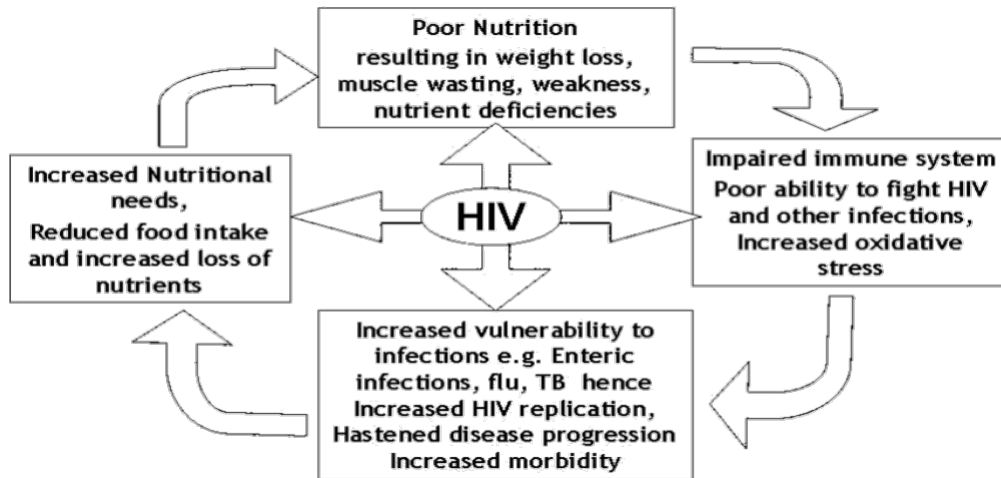


Figure 14.1. The vicious cycle of malnutrition and HIV

Table 14.1. Average daily energy requirements in calories (Source: WHO, 1993)

	Status/Age	Energy Need (kcal)
Men	Average active	2,430
Women	Average active	2,170
	Pregnant	2,460
	Lactating	2,570
Children	6–11 months old	730
	12–23 months old	1,250
	24–59 months old	1,500
	5–10 years old	1,800
Boys	10–14 years old	2,360
	15–18 years old	2,800
Girls	10–14 years old	2,040
	15–18 years old	2,100

Table 14.2.Additional energy requirement for PLHIV

Category		Additional energy requirement in %
Adult	Asymptomatic	10%
	Symptomatic	20% - 30%
Children	Asymptomatic	10%
	Symptomatic losing weight	50% -100%
	Symptomatic not losing weight	20% - 30%

Exercise 14.1

1. Kebede a four years old boy who is HIV positive but not symptomatic. Calculate his daily energy requirement in kilocalories and recommend local foods to meet his energy requirement.
2. Wro Kibe, is age 37 years old symptomatic HIV positive lactating mother. Calculate her daily energy requirement in kilocalories and recommend local foods to meet her energy requirements.

*Refer **Annex 3 and 4** for available local foods and their energy content.

The components of nutritional assessment and counselling (NACS) are:

1. Assessment:
 - Anthropometry (BMI, MUAC, Wt/Ht, Growth monitoring)
 - Biochemical
 - Clinical
 - Dietary
2. Counseling:
 - Clinical (including adherence)
 - Dietary (including WASH & food safety)
 - Psychosocial
 - Referral to social services including, economic strengthening, livelihood and food security support

3. Commodity Support:

- Therapeutic/supplementary food support
- Safe water treatment
- Multi-micronutrient supplements

Foods to avoid for PLHIV:

- Raw egg s and raw/ undercooked meat.
- Water that is not boiled or juice that is made from water that is not boiled.
- Alcohol.
- “Junk” foods such as chips, biscuits, and sweets with little nutritional value.
- Foods that aggravate symptoms related to diarrhea, nausea/vomiting, bloating, loss of appetite, and mouth sores.

14.2. Nutritional Assessment in Adolescents and Adults

Weight of HIV patients should be regularly measured and recorded in every visit. For adults, height should be recorded once at entry however adolescents may require repeated measurement.

14.2.1. Body Mass Index (BMI)

BMI is a number calculated from a person’s weight and height. BMI is a reliable indicator of body fatness for people. BMI can be considered an alternative for direct measures of body fat. Additionally, BMI is an inexpensive and easy-to-perform method of screening for weight categories that may lead to health problems. BMI can’t be used for pregnant and lactating mother rather MUAC can be used for nutritional assessment.

BMI cut offs are also not accurate in pregnant women or adults with edema. For these groups, MUAC can be an effective indicator of nutritional status. MUAC can also be problematic for individuals with changes in body composition due to ART, e.g. lipoatrophy.

BMI is calculated as the weight of the client in kilograms divided by the square of the height in meters.

$$\text{BMI} = \frac{\text{Weight in kg}}{\text{Height in m}^2}$$

Table 14.3. Classification of nutritional status of Adults based on BMI:

BMI < 16 kg/m ²	=	Severely malnourished
BMI = 16 - 16.99 kg/m ²	=	Moderately malnourished
BMI = 17 - 18.49 kg/m ²	=	Mildly malnourished
BMI = 18.5 - 24.99 kg/m ²	=	Normal weight
BMI = 25 - 29.99 kg/m ²	=	Overweight
BMI > 30 kg/m ²	=	Obese

BMI-for –Age: In children between the ages of 5 – 17 years, the computed BMI shall be compared against a BMI- for- Age reference chart under Annex1, to decide whether the computed BMI indicates malnutrition or not, as children are in a dynamic growth process and will show varying weight and height at different ages.

Table 14.4. Classification of nutritional status of children 5-17 years using BMI-for-age

Children Aged 5 – 17 Year, BMI –for –Age	Nutritional Status
< -3 SD	Severely malnourished
-2 to -3 SD	Moderately malnourished
-1 to -2 SD	Mildly malnourished
> -1 SD	Normal weight
<i>Source:</i> WHO 2006	

14.2.2. Measuring and Classifying MUAC

To measure MUAC:

1. Keep your work at eye level. Sit down when possible. Very young children can be held by their mother during this procedure. Ask the mother to remove clothing that may cover the child's left arm.
2. Calculate the midpoint of the child's left upper arm by first locating the tip of the child's shoulder with your fingertips.
3. Place the tape at zero, which is indicated by two arrows, on the tip of the shoulder and pull the tape straight down past the tip of the elbow. Read the number at the tip of the elbow to

the nearest centimeter. Divide this number by two to estimate the midpoint. A piece of string can be used for this purpose. Mark the midpoint with a pen on the arm. Straighten the child's arm and wrap the tape around the arm at the midpoint.

Make sure the numbers are right side up. Make sure the tape is flat around the skin.

4. Inspect the tension of the tape on the child's arm. Make sure the tape has the proper tension and is not too tight or too loose.
5. When the tape is in the correct position on the arm with correct tension, read and call out the measurement to the nearest 0.1cm.
6. Immediately record the measurement.

Please refer to Annex 2 for pictorial MUAC measurement technique.

Table 14.5.Nutritional classification in relation to MUAC measurement

MUAC	Classification
Children 6 - 11 months old: <11 cm Children 12 - 59 months old: <11 cm Children 5 - 9 years old: <13.5 cm Children 10 - 14 years old: <16 cm Adult: < 18 cm Pregnant and lactating: <19 cm	Severe malnutrition
Infants 6 - 11 months old: 11 - 12 cm Children 12 - 59 months old: 11 - 13 cm Children 5 - 9 years old: 13.5 - 14.5 cm Children 10 - 14 years old: 16 - 18 cm Adults: 18 - 21 cm Pregnant and lactating: 19 - 23 cm	Moderate malnutrition

Infants 6 - 11 month old : > 12 cm Children 12 - 59 months old : > 13 cm Children 5 - 9 years old : > 14.5 cm Children 10 - 14 years old: > 18 cm Adults: > 21 cm Pregnant and lactating: >23 cm	Normal
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Patients with malnutrition should be managed according to national guideline (See national guideline for HIV/AIDS and nutrition, FMOH 2008).

14.2.3. Nutrition Education and Counseling for PLHIV

Nutrition counseling is an interactive ongoing process between a counselor and client. All PLHIV should receive appropriate individual nutrition counseling based on their nutrition assessment results. The nutrition counseling should interpret information generated during assessment, understand client preferences, constraints and options, and plan a feasible course of actions that supports healthy dietary practices and referral for additional services. Good counseling can result in positive changes in nutrition related behavior and help to improve the quality of life of PLHIV.

In addition to treatment of malnutrition, comprehensive nutrition counseling should advocate for improved diet through consumption of locally available foods and link individuals to appropriate support. Nutrition counseling helps identify client preferences, barriers to behavior change, and possible solutions to overcome those barriers. With such information, the client and care provider can jointly plan a feasible course of action to support desired nutrition practices. The care provider may use job aids to select appropriate messages and guide counseling sessions.

Table 14.6.Critical nutrition practices for PLHIV

1. See a health care provider for periodic nutrition assessment (especially weight).
2. Increase energy intake by eating a variety of foods, especially energy-rich locally available foods and eating more frequent meals, especially if sick.
3. Maintain high levels of hygiene and sanitation.
4. Drink plenty of clean and safe (boiled or treated) water.
5. Maintain a healthy lifestyle by avoiding unprotected sex, alcohol, tobacco, sodas, and other colored and sweetened drinks and do physical activity (exercise).
6. Seek early treatment for infections and manage symptoms through diet.
7. Take medicines as advised by the health worker and manage food and drug interactions or side effects.

14.3. Management of Malnutrition in PLHIV

Nutrition management of PLHIV depends on a classification of the nutritional status of the patient, age of the patient, other medical conditions and the available therapeutic or supplementary food products. The management includes treatment of malnutrition by prescribing ready-to-use therapeutic foods, supplementary foods and locally available diversified energy and nutrient dense foods for rehabilitation from malnutrition based on nutrition care plans.

Nutrition care plans are interventions determined based on PLHIV clients' nutritional status and health conditions that affect their nutritional needs and absorption/utilization.

There are three nutrition care plans for treatment of malnutrition in PLHIV:

Nutritional Care Plan C: Clients classified as Severe Acute Malnutrition (SAM) with medical complications and/or failed appetite test and those classified as Moderate Acute Malnutrition (MAM) with medical complication and/or failed appetite test will be managed at inpatient/stabilization center according to the national SAM management guideline. Therefore, should be referred to those facilities as identified.

Clients classified as SAM without medical complication and passed appetite test will be managed at Outpatient Therapeutic Program (OTP) or ART unit with Ready-to-Use-Therapeutic Food (RUTF) - Plumpynut.

Dose and duration of prescription for outpatient management:

Adults: 4 sachets per day; maximum for 3 months. Then, transition to care plan B.

Children: dosage based on their weight (Kg) as indicated on the national guideline.

Nutritional Care Plan B: Clients classified with MAM and without medical complication and those transitioned from care plan C will be managed with Ready-to-Use Supplementary Food (RUSF)-Plumpy Sup based on the availability of supply. Otherwise, stress on nutritional counseling on the need to have additional energy from the locally available foods.

Dose and duration of prescription: Adults: 2 sachets per day; maximum for 3 months.

Nutritional Care Plan A: Educate and counsel on the 7 critical nutrition practice (Table 14.5) during the subsequent follow-up sessions.

Table 14.7. Criteria for classification of acute malnutrition among young Infants 0 to 6 months

ASK	LOOK andFEEL	Signs	Classify	Treat
Ask for signsof medical complications: Is the child: Too weakto suckleeffectively? Vomiting everything? Does the child haveRecent weightloss or failure togain weight?	1.Check for presence of oedema of both feet (or sacrum) · Does the child have oedema? 2.Check the weight and Length . Is the weight-for-length less than- 3zscore? 3.Check for signs of medical complications: Hypothermia: axillary temp <35 °C or rectal < 35.5° C, Fever ≥ 37.5°C Ineffective feeding (attachment, positioning and suckling) directly observed for 15 to 20 minutes Any medical or social issue needing detailed assessment or intensive support Severe anemia (severe palmar pallor), Jaundice Convulsions, Very Weak, lethargic or unconscious Pneumonia or fast breathing	WFL <-3Z score, OR Visible wasting OR Oedema of both feet And presence of any one of these complications	Complicated Severe Acute malnutrition	Admit Inpatient Management (CARE PLAN C-Inpatient)
		WFL < -3Z score AND no complications OR WFL ≥ -3Z to < -2Z score, AND No edema of both feet	Uncomplicated Severe Acute Malnutrition or Moderate Acute Malnutrition	Course of broad-spectrum oral antibiotic, such as amoxicillin . Detailed assessment of underlying cause(s) of malnutrition and tailored action to address these . Plot and appraise growth chart for monitoring progress
		If WFL ≥ -2Z score AND No edema of both feet	No Acute Malnutrition	Congratulate and Counsel the mother on infant and child feeding/care (CARE PLAN A)

Table 14.8. Criteria for classification of acute malnutrition among children 6 to 59 months of age

ASK	LOOK AND FEEL	Signs	Classify	Treat
1. Ask for signs of medical complications 2. Is the child: Too weak to suckle effectively? Vomiting every thing? 3. Does the child have: Recent weight loss or failure to gain weight?	4. Check for presence of oedema of both feet (and sacrum): Does the child have oedema? If yes, is it generalized involving upper arms and face? 5. Check the weight and height: Is the weight-for-height less than -3z-score? 6. Check MUAC: Is the MUAC < 11 cm? 7. Check for signs of medical complications: • Hypothermia: axillary temp < 35°C or rectal < 35.5°C • Fever $\geq 38.5^{\circ}\text{C}$ • Pneumonia/severe pneumonia • Shock • Dehydration (watery diarrhoea with recent sunken eye balls.) • Hypoglycemia • Severe anemia (severe palmar pallor) • Jaundice • Dermatitis +++ • Corneal clouding or ulceration • Measles (now or with eye/mouth complications) • convulsions • Very Weak, lethargic or unconscious	WFL/H < -3Z score OR MUAC < 11 cm OR oedema of both feet (+, ++), PLUS • Any of the medical complications or • Failed appetite test OR +++ oedema, OR - Marasmic Kwashiorkor (WFL/H < -3Z with edema, OR MUAC < 11.5 cm with edema)	Complicated Severe Acute malnutrition	Admit for in-patient management (CARE PLAN C-Inpatient)
		WFL/H < -3Z score OR MUAC < 11 cm OR Oedema of both feet (+, ++) AND No medical complication AND passes appetite test	Uncomplicated Severe Acute Malnutrition	Manage in OTP using the OTP protocol (CARE PLAN C-outpatient)
		WFL/H $\geq$ -3Z to < -2Z score OR MUAC 11 cm to < 12.5 cm AND No oedema on both feet	Moderate Acute Malnutrition	Admit to TSFP if available, Counsel on infant and child feeding/care (CARE PLAN B)
		If WFL/H $\geq$ -2Z score OR MUAC $\geq$ 12.5 cm AND No oedema on both feet	No Acute Malnutrition	Congratulate and Counsel the mother on infant and child feeding/care (CARE PLAN A)

Table 14.9.Criteria for classification of acute malnutrition among children 5 to 18 years of age

ASK	LOOK AND FEEL				Signs	Classify	Treat	
Ask for signs of medical complications: 1. Is the child vomiting everything? Does the child have Blood in stool? Diarrhoea>14d? Bleeding tendencies? History of recent sunken eyeball? Coughing for more than 21 days Active TB or on treatment for it? Convulsion (more	1	Check for presence of oedema of both feet (orsacrum) • Does the child have oedema? • If yes, is it generalized involving upper arms and face?			WFL/H < -3Z score OR MUAC in severe category OR BMI for age <-3SD OR oedema of both feet (+, ++), PLUS Any one of the medical complications , or Failed appetite test OR+++ oedema,OR Marasmic Kwashiorkor (WFL/H< -3Z score with Oedema,OR MUAC in severe category with Oedema)	Complicated Severe Acute Malnutrition	Admit for in-patient management (CARE PLAN C-Inpatient)	
	2.	• Is there no other cause for oedema? Check the weight and height:						
	3.	• Is the weight-for-height less than-3z-score? Check BMI for age:						
	4	Does he have low BMI for his age? Check MUAC:						
		Age (years)	MUAC severe (cm)	MUAC moderate (cm)				MUAC normal (cm)
		Children 5 to 9	<13	13 - <14				≥14
	Children 10 to 14	<16	16 - <18	≥18				
	Children 15 to 17.9	<17	17 - <19	≥19				

than one or prolonged for >15 min)		Check for signs of medical Complications: Hypothermia :axillary temp $<35^{\circ}\text{C}$ or rectal $<35.5^{\circ}\text{C}$ Fever $\geq 38.5^{\circ}\text{C}$ Pneumonia/Severe pneumonia Shock Dehydration (watery diarrhoea with recent sunken eye ball) Hypoglycemia Severe anaemia (severe palmar pallor) Jaundice Dermatitis +++ Corneal clouding or ulceration Measles(now or eye/mouth complications) Very weak, lethargic or unconscious	$< -3\text{Z}$ score OR MUAC in severe category OR BMI for age $< -3\text{SD}$ OR Oedema on both feet (+, ++) AND No medical complication AND passes appetite test	Uncomplicated Severe Acute Malnutrition	Manage in OTP using the OTP protocol (CARE PLAN C-outpatient)
			WFL/H $\geq -3\text{Z}$ to $< -2\text{Z}$ score OR MUAC in moderate category OR BMI for age -3 to $< -2\text{SD}$ AND No oedema on both feet	Moderate Acute Malnutrition	Admit to TSFP if available Counsel on feeding and (CARE PLAN B)
			If WFL/H $\geq -2\text{Z}$ score OR MUAC in normal category OR BMI for age $> -2\text{SD}$ AND No Edema on both feet	No Acute Malnutrition	Congratulate and Counsel the mother on feeding and care (CARE PLAN A)

Table 14.10. Criteria for classification of acute malnutrition among adults 18 years or older

ASK	LOOK AND FEEL	Signs	Classify	Treat
1. Has the client lost weight unintentionally during the past month?	1. Check for presence of oedema of both feet (or sacrum): • Does the client have oedema?	BMI <16 OR MUAC <18cm for PLWs <19 cm OR oedema on both feet without clear cut other causes, OR FOR HIV Positive client: BMI ≥16 and <17.5 OR MUAC ≥18 and <21 cm PLUS Any one of the medical complications	Complicated Severe Acute Malnutrition	Admit for in-patient management (CARE PLAN C-Inpatient)
2. Has the client had: • Active TB or on treatment for TB? • Diarrhoea for >14 days? • Other chronic OIs or malignancies? (e.g. oesophageal infection, mouth soar or oral thrush)	• If yes, is it generalized involving upper arms and face? • Is there no clear cut other causes? 3. Measure weight and height • Is the BMI <16? 3. Check MUAC • Is MUAC <18 cm? • Is MUAC <19 cm for PLW?	BMI <16 OR MUAC <18cm for PLWs <19 cm OR Oedema on both feet without clear cut other cause, AND No medical complications	Uncomplicated Severe Acute Malnutrition	Manage in OTP using the OTP protocol (CARE PLAN C-outpatient)
3. Has the client had noticeable changes in his/her body composition, Specifically his/her fat distribution? • Thinning of limbs and face, • Change in fat distribution on the limbs, breast, stomach region, back or shoulders?	4. Examine for conditions that cause secondary malnutrition (see above and in ``ASK`` part 5. Check for signs of medical complications: • Severe anemia • Severe dehydration • Active TB • Pneumonia • Shock following: • Jaundice	BMI ≥16 and <17.5 OR MUAC ≥18 and <21 cm For PLW: MUAC ≥19 and <23 cm OR For HIV positive client: Confirmed (>5 per cent weight loss since last visit) or reported weight loss (e.g. loose clothing) Regardless of BMI or MUAC: Any of the following: Chronic lung disease, TB, persistent diarrhea, other chronic opportunistic Infections (OI) or malignancy AND No Oedema on both feet	Moderate Acute Malnutrition	Admit to TSFP if available Counsel on Critical Nutrition Practice (CNP) (CARE PLAN B)

4. Has the client experienced the following? • Nausea and/ or vomiting • Persistent fatigue • Poor appetite	Dermatosis +++ Very weak, lethargic or unconscious	BMI >17.5 OR MUAC >21 cm (for pregnant and lactating mothers >23 cm) AND No Oedema on both feet	No Acute Malnutrition	Congratulate and Counsel on CNP (CARE PLAN A)
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14.4. Infant Feeding in the context of Maternal HIV Infection

Breastfeeding has unsurpassed advantages over any forms of infant feedings to date. In resource limited countries, breastfeeding is a key component of child survival interventions. Infant mortality is higher in settings where infants are not breastfed and breastfeed for short periods. Breastfeeding accounts for 30-40% of perinatal transmission in populations where breastfeeding is practiced for up to 2 years of age. However, replacement feeding if not instituted properly is associated with increased risk of morbidity and mortality at a young age in low resource settings. Exclusive Breast Feeding (EBF) during the first 6 months of life is associated with decreased transmission of HIV and improved child survival compared to non-exclusive breast feeding. EBF for as long as possible until 6 months is recommended as the only feasible and the safest option for infant feeding amongst HIV infected women. At 6 months, complementary foods should be introduced in order to sustain normal growth. Most Ethiopian children will continue to benefit from breastfeeding until 12-18 months of age. Healthcare workers should be trained to provide infant feeding counseling to all mothers in general and to HIV infected mothers in particular to support them exclusively breastfeed.

The primary goals of infant feeding counseling and support are:

- Improve child survival by actively supporting exclusive breastfeeding for the first 6 months of life.
- Decrease HIV transmission via breastfeeding by treating pregnant and lactating HIV infected women with ART.

14.4.1. Factors that increase risk of HIV transmission during breast feeding

Maternal:

- Advanced HIV/AIDS (low CD4 count and high viral load).
- Contracting new HIV infection while breast feeding.
- Poor maternal nutritional status (low BMI and anemia).
- Breast problems (cracked nipples, mastitis and breast abscess).

Infant:

- Mixed feeding.
- Lesions in infant's mouth and intestine.
- Prematurity.

14.4.2. Recommended infant feeding option

For the first six months of infant:

- Exclusive breast feeding is the recommended option.
- Prompt treatment of breast problems (cracked nipples, mastitis, breast abscess), and lesions in the infants' mouth.
- Avoid mixed feeding during the first 6 months since this is associated with increased risk of HIV transmission.
- Avoid early cessation of breastfeeding as it is associated with increased risk of death from diarrheal illnesses, malnutrition and pneumonia.
- Ensure that mother is adherent to ART.

After 6 months:

- Introduce complementary diet at 6 months of age.
- Continue breast feeding until 12-18 months of age.
- Breast feeding should only be stopped once a nutritionally adequate and safe diet without breast-milk can be provided.

In developing countries like Ethiopia, replacement feeding is usually associated with malnutrition and gastroenteritis and is not a recommended infant feeding option. Once the mother is on ART and adherent to therapy the risk of HIV transmission with breast feeding is very low.

14.4.3. Infant feeding counseling

First six months of life:

- Initiate breastfeeding within one hour of birth.
- Encourage mother to exclusively breastfeed for the first 6 months of life.
- Breastfeed 8-12 times per day based on demand.
- Empty one breast before the other.
- Demonstrate how to practice safe breast feeding - proper positioning and attachment.
- Irrespective of early infant PCR result, encouraged mother to continue breast feeding.
- Infants who are determined to be HIV infected should continue to breast feed according to recommendations for the general population.
- Provide follow-up counseling and growth monitoring for first 2 years of life.
- Ensure that the mother is adherent to ART.

Signs of good positioning:

- Infant's neck is straight or bent slightly back.
- Infant's body is turned towards the mother.
- Infant's body is close to the mother.
- Infant's whole body is supported.

Signs of good attachment:

- Chin touching breast.
- Mouth wide open.
- Lower lip turned outward.
- More areola visible above than below the mouth.

Feeding from 6 months to 24 months

- Ensure appropriate complementary feeding is initiated at 6 months of age to all infants whether they have been on exclusive breastfeeding or exclusive replacement feeding.
- Encourage continued breastfeeding.
- Promote good hygiene, proper food safety and handling.
- Infants should continue to receive growth monitoring and nutritional advice.

- Teach mothers to provide suitable complementary foods made from nutrient rich foods, emphasizing the use of locally available energy dense foods and increased use of fruits and vegetable.
- For infant 6-12 months give complementary diet 3 times per day and for children more than 12 months give complementary diet 5 times per day.
- Infants should be weaned only when a supply of animal or formula milk can be provided.

Feeding orphans

Abandoned children or maternal orphans require special consideration. Such infants should be given appropriate exclusive replacement feeding preferably infant formula for the first 6 months.

Advise to be provided to care giver of an orphan

- Use commercial infant formula, for the first six months of life.
- Home modified animal milk should only be used if commercial infant formula is not available or could not be afforded.
- Teach care giver how to prepare the replacement feeding and provide intensive counseling on hygienic preparation of formula at each visit.
- Avoid bottle feeding, in order to avoid the risk of diarrhea and malnutrition.
- Ensure close follow-up to monitor growth and nutritional status monthly to prevent malnutrition and gastroenteritis during the first 2 years of life. Introduce complementary diet at six months of age.

14.4.4. Feeding HIV Infected Children in the Context of Maternal HIV Infection

Children with confirmed HIV infection, but still asymptomatic, should increase their energy intake by 10% to maintain growth. HIV-infected children who are experiencing weight loss need to have energy intake increased by 50-100% and symptomatic children but without weight loss need 20-30% additional energy. Children should receive vitamin A and other micronutrient supplements according to current WHO or national recommendations.

Study the following table for clinical situations when the nutrition of an HIV-infected child is affected.

The table suggests what additional action you should take for these children.

Table 14.11. Feeding the HIV infected child in specific circumstances

Clinical situation that may impair the nutrition of HIV-infected children	Consequence	What action should you take?
Recurrent or chronic infection	Increased metabolic needs Significantly higher caloric demands	Offer feeds more frequently than before: If the child is breastfeeding breastfeed at least 8 times in 24 hours If the child is on complementary foods offer small meals at least 5 times a day. Increase the energy value of these feeds by adding, for example oil / margarine / ground nuts
Intestinal infections	Increased nutrient requirements Impaired absorption and loss of appetite may decrease food intake	Follow the same feeding recommendations for the child with recurrent or chronic infection Treat for worms if the child has not been treated during the previous 6 months .Give Vitamin A if the child has not been treated during the past 6 months
Oral or esophageal thrush	Potential pain with swallowing may result in decreased oral intake primarily for solids, but also for liquids	Offer foods that have been mashed up Avoid spicy foods Paracetamol half an hour before feeds may be helpful in extreme cases
Persistent diarrhea caused by cryptosporidia or other parasites	Impaired absorption of nutrients	Follow the feeding recommendations for the child with recurrent or chronic infection (above); the child with intestinal infections (above) and the child with persistent diarrhea
Nausea and vomiting as a result of ARV drugs	Reduced intake and impaired absorption	Encourage small frequent fluids and give food that the child likes Let the child eat before medication

It is important to identify local foods that are available and affordable and to advise the mother on how to increase the energy content of foods. Always advise the mother to continue feeding and continue giving fluids during any illness.

EXERCISE 14.2

In this exercise you will answer questions about the feeding recommendations that you have learnt about in this module.

1. Write a "T" by the statements that are True. Write an "F" by the statements that are False.

- A. ___ Children should be given fewer feeds during illness.
- B. ___ A 3-month-old HIV positive child should be exclusively breastfed.
- C. ___ A 2-week-old child of unknown HIV status, born to an HIV infected mother should never be breastfed.
- D. ___ A breast-feeding child born to an HIV positive woman can continue breast feeding for as long as 12-18 months.
- E. ___ A 5-month-old child whose mother is HIV negative should be breastfed as often as he wants, day and night.
- F. ___ A 9-month-old child who is HIV positive on virological tests should continue breastfeeding.
- G. ___ All breast feeding HIV positive women transmit HIV to their infants.

2. When should other foods be added to the diet of a child born to an HIV positive mother? What foods should be added and with what frequency? _____

Chapter summary

- HIV affects nutrition by reducing amount of food consumed, interfering with the digestion and absorption of nutrients, and altering metabolism of nutrients.
- HIV and OIs increase catabolism. Therefore, additional energy required.
- BMI is an inexpensive and easy-to-perform method to assess nutritional status of PLHIV.
- Primary goal of infant feeding counselling and support is to improve child survival and decrease HIV transmission.

Chapter 15: Palliative Care

Chapter Description:

This chapter is intended to describe the palliative service needed for PLHIV while on ART. It describes different ways of assessing pain in adults and children using different standard scales and the WHO Pain management cascade.

Purpose: To understand the components of palliative care and enable participants to manage pain in patients with HIV.

Enabling Objectives

By the end of this chapter, the participant will be able to:

- Define palliative care and its role in the management of HIV.
- Describe components of palliative care.
- Illustrate pain management in adults and children.
- Discuss challenges of palliative care in the Ethiopian setting.

Chapter outline:

- Introduction to palliative care
- Components of palliative care in patients with HIV/AIDS
- Challenges of palliative care

Allocated time: 50 minutes

Learning activities

- Brainstorming,
- Reading,
- Discussion,

Activity

What is Palliative Care?

Palliative care is interventions that improve the quality of lives of patients and their families facing problems associated with life-threatening chronic illnesses such as AIDS, diabetics and cancer. It is also important for those with curable illnesses who may have symptoms for many months before they get cured. It is prevention and relief of sufferings, pains and other physical problems as well as psychosocial and spiritual issues.

Palliative care is an integral part of a comprehensive HIV/AIDS care and support framework. In the framework of a continuum of care from the time of HIV diagnosis until the end of life; and regards dying as a normal process and affirms life. It also offers support to help the patient and family cope during the patient's illness and in the bereavement period.

The palliative care needs of patients increase with time, particularly in a situation where the underlying disease is getting worse rather than better. In areas where patients present late for a medical care the need for palliative care is immense.

With good treatment and support, palliative care can help many patients live comfortably with disease for many years. For those who do have advanced disease in a terminal phase, palliative care focuses on promoting quality of life by providing good symptom management. This can help patients continue to function and enjoy life at home for as long as possible.

Why is palliative care important in HIV/AIDS care?

- Chronic disease that requires lifelong treatment.
- Presence of adverse drug reaction and toxicity.
- The need for holistic approach (psychosocial, spiritual).
- Helps to improve adherence, retention and quality of life.
- Contribute for effective and well-functioning health system.

Palliative care is an important component of care for any medical condition. Palliative care includes the management of symptoms such as fatigue, dyspnea, and neuropathic pain, and treatment of drug side effects such as nausea, vomiting, and diarrhea. Palliative care also addresses psychosocial needs,

for example depression and therefore by addressing these issues, the role of palliative care may extend beyond the individual to reach the community by reducing the emergence of drug resistance.

15.1. Components of palliative care

1. Symptomatic management
2. Preventive care
3. Psychosocial and spiritual support
4. End of life care

15.1.1. Symptomatic management

ART care providers should assess and manage common symptoms associated with HIV/AIDS and treatment side effects. Most common symptoms are listed below:

Table 15.1. Common symptoms and ARV side effects

Common symptoms from OIs	Common ARV side effects & complications
- Pain - Dyspnea - GI problems - Skin and mouth problems - Fever - Neurological disorders - Anxiety, fatigue etc	- Nausea and vomiting - Diarrhea - Peripheral neuropathy etc.

N.B: For further reading on symptom management refer to national palliative care guideline.

Pain

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage. This highlights that pain is not just a physical sensation but an emotional experience too. In simple words pain is what patient says, hurts, which gives emphasis on the patient's experience.

The concept of total pain, refers to the global nature of pain perception not only as a physical ailment but that it has a psychological, spiritual and social consequences. Pain is classified as acute vs chronic and nociceptive vs neuropathic.

1. Acute versus chronic pain

Acute pain is due to a definable acute disease or injury, duration limited to healing of tissue in days or weeks, accompanied by anxiety and sympathetic over-activity (sweating, tachycardia, tachypnea etc.). It has the physiologic purpose of protecting the body from further injury or facilitating healing. Chronic pain on the other hand persists months beyond the usual course of an acute disease reasonable time for an injury to heal, or is associated with a chronic pathological process which causes continuous pain or pain which recurs at intervals for months or years. It is accompanied by psychological problems such as depression but no sympathetic response due to adaptation of the sympathetic nervous system.

2. Neuropathic versus Nociceptive Pain

Neuropathic pain is caused by damage to the central or peripheral nervous system. Neuropathic pain can be caused by injury, compression or infiltration of a nerve. Examples include post herpetic neuralgia or sciatic pain resulting from prolapsed intervertebral disc. Injured nerves react abnormally to stimuli or discharge spontaneously as they become hyper-excitabile. Often neuropathic pain is described as a burning, tingling or stinging sensation or a shooting electric shock-like sensation – ‘pins and needles’. On the other hand, nociceptive pain arises from noxious stimuli, potential or actual injury of somatic or visceral tissues of the body.

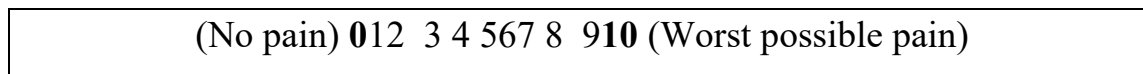
Assessment of pain

Pain should be considered as the **5th vital sign**. A proper assessment of pain is essential for successful treatment. Pains could be unrelated to or only indirectly related to the primary disease and they would need different modalities of treatment.

Grading of pain

Numerical Scale:

It has 0 at one end meaning **no pain** and 10 at the other end meaning **worst imaginable pain**.



Grading of pain in children *faces pain scale* and five finger score:

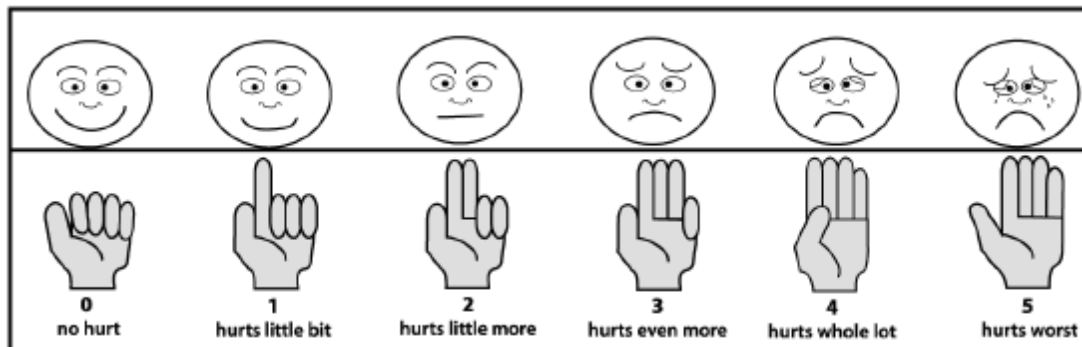


Figure 15.1.Grading of pain in children

Management of Chronic Pain

- Pain should be treated whenever possible, consider pharmacologic precautions for all analgesic drugs.
- **‘By the mouth’**: Oral medications should only be abandoned if the patient is unable to take or retain them.
- **‘By the clock’**: Analgesics should be given *‘by the clock’*, i.e. at regular intervals and strict schedule.
- **‘By the ladder’**: the two/three step analgesic ladder (two step for pediatric, three step for adults) which is meant for mild, moderate and severe pain is to be followed.
- **‘For the individual’**: the right dose of an analgesic is the dose that relieves the pain.
- **‘Attention to detail’**: it is essential to monitor the patient’s response to the treatment to ensure that the patient obtains maximum benefit with as few adverse effects as possible.

- **‘Combination therapy’**: using two or more analgesia of different mechanism of action is evidence-based practice.

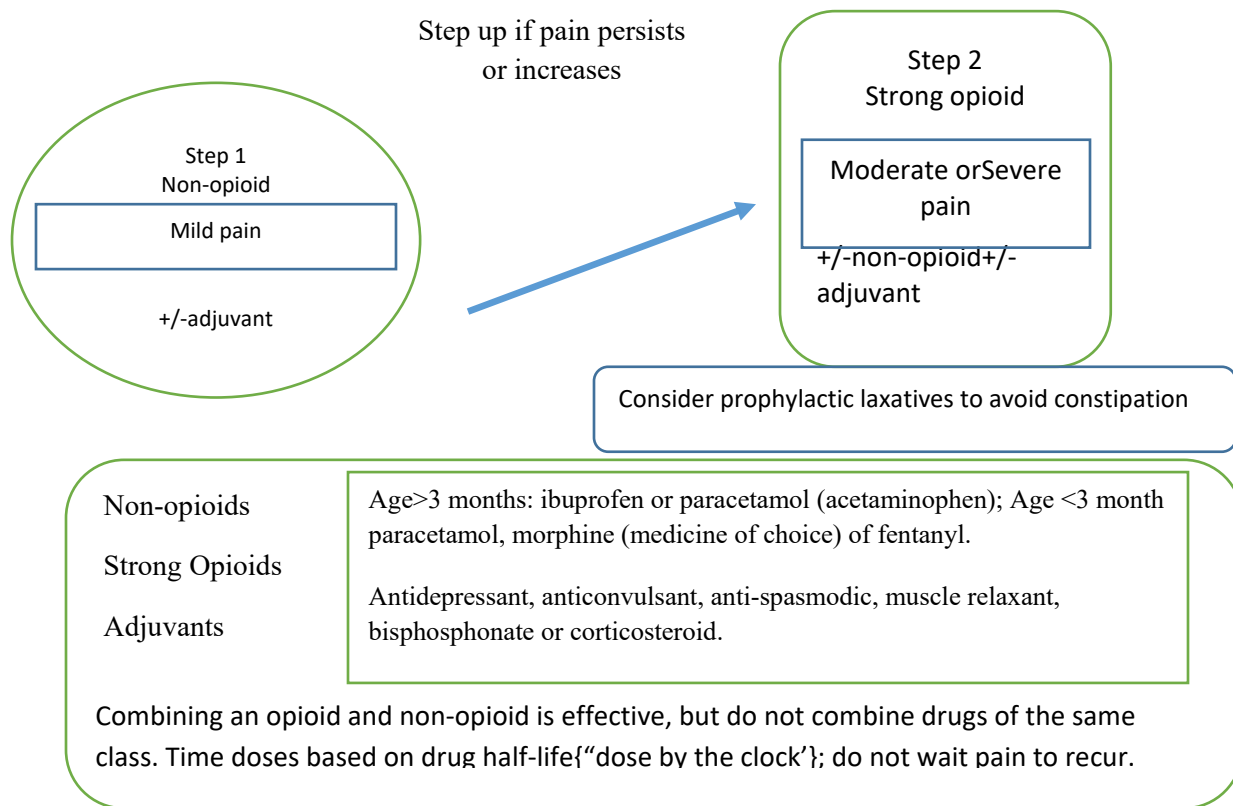


Figure 15.2 Two-Steps analgesics ladder

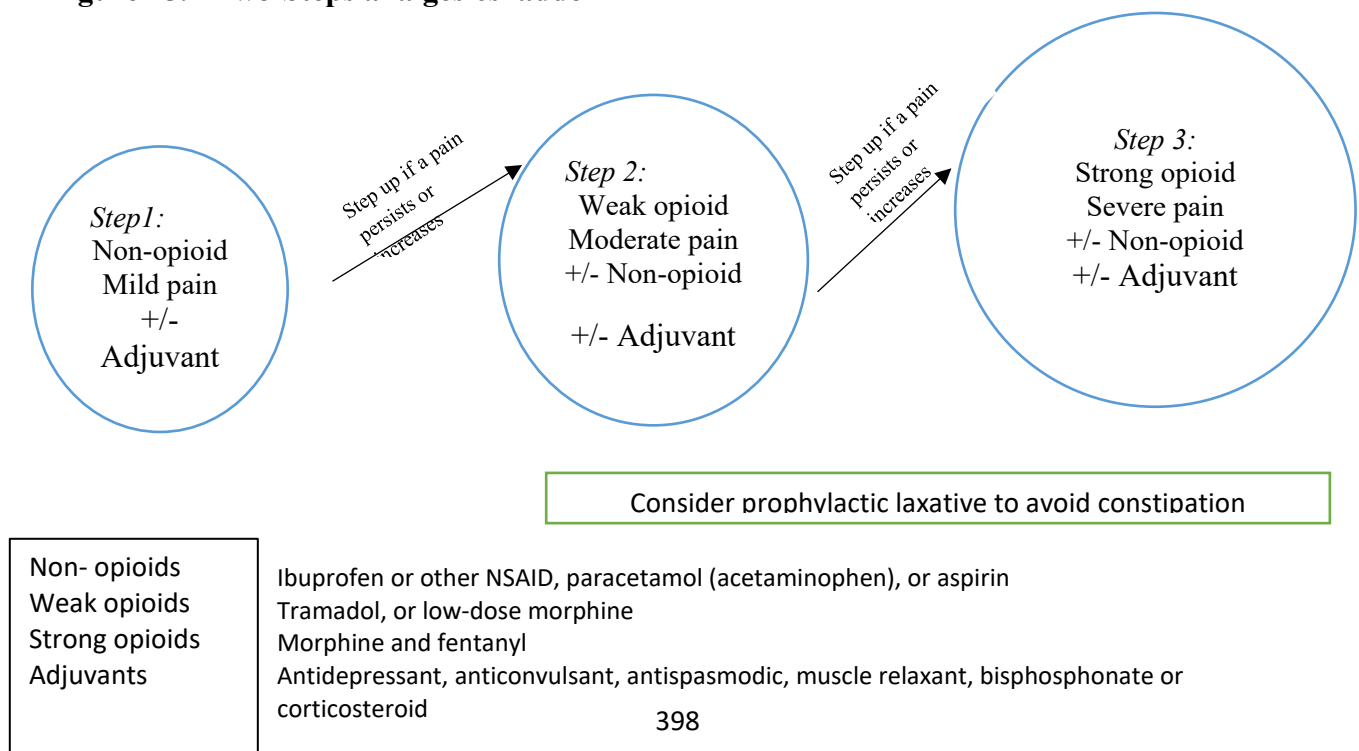


Table 15.2. Summary of analgesic drugs use for Step-by-Step

	Usual Adult Dose	Pediatric Dose
Step1 Acetaminophen	PO 500-1000 mg q 6 hr	PO 10-15 mg/kg /dose q 4-6hr (max. dose 90 mg/kg/24 hr.)
Step1 Aspirin	PO 300-600 mg q 4-6 hr	Not recommended
Step1 Diclofenac	-PO 50 mg q 8 hrs -Suppository 100 mg once/day (extended release) - IM 75mg 12 hourly	Children: 25mg suppository eg. 2-3mg/kg in 2-4 divided doses (max 200mg/day)
Step1 Ibuprofen	400 mg q 4-6 hr PO	5-10 mg/kg/dose q 6-8 hr (max. dose 40 mg/kg/24 hr)
Step1 Indomethacin	-PO 50-200 mg/daily divided 8-12hrly -suppository 100mg 12 hrly	Not recommended for children
Step 2-Codeine	-PO 30-60 mg q 4 hr Max. dose 240 mg/d-IM 30-60 mg q 4 hr	Not recommended for children
Step 2 – Tramadol	-PO 50-100 mg q 6 hrs Max 400 mg/day -IM/IV 50-100 mg q 4-6 hrs. Max. 400 mg/day	Not recommended for children
Step 3- Morphine ^a	2.5-5mg every 4 hours (dose can be increased by 1.5 or doubled after 24 hours if pain persists)	• < 1 year 0.1mg/kg Q6hrs • 1-12yrs 0.2-0.4 mg/kg Q4hrs • >12 yrs. use adult dose

^aReduce morphine once pain is controlled; if used for weeks reduce it gradually to avoid withdrawal symptoms.

Non-Pharmacological Management of Pain

Palliative care includes many non-pharmacological ways to manage ‘total’ pain. These therapies address the physical, psychological, social and spiritual dimensions of pain. The following are a few holistic non-pharmacological ways of treating pain.

Anything that enhances quality of life can, in turn, relieve pain.

- Listening and empathy.
- Counseling – provides emotional support and practical suggestions.
- Companionship and accompaniment – can help ease pain and increase comfort.
- Activities such as favorite music/ games/ gardening/ memory book provides meaning & distraction.
- Spiritual/pastoral support and prayer – provides comfort, meaning and hope.
- Positioning – enhances comfort and relieves pressure areas.
- Bathing, grooming, and other care measures – enhances comfort and self-respect.
- Exercise – improves mobility, circulation and skin integrity.
- Massage, therapeutic touch.
- Traditional therapies that are beneficial, healing and comforting.
- Heat/cool applied locally – can reduce swelling and help relaxation.
- Treatments such as radiotherapy can reduce inflammation, pain and tumor size.

Note: Please refer to national palliative care guideline, February, 2016, for further reading on pain management.

15.1.2. Preventive care

In Ethiopia as well as in other countries, palliative care also includes preventive care aimed to keep people healthy as long as possible. This is consistent with the new definition of palliative care that rejects the notion of terminal care alone and embraces the concept of care from the time of diagnosis and continuing to end of life and death. Below are the components of preventive care package. For the details of some of the components, please refer in the appropriate sections.

- Active TB screening
- CTX prophylaxis
- Malaria prevention
- Positive living and prevention health and dignity
- Nutritional counseling and supplementation
- Safe water supply

Improved water, sanitation and hygiene is very important in PLHIV because PLHIV have a weak body defense, drinking contaminated water will give PLHIV diarrhea accelerate progression of HIV, and may lead to death.

Prevention of water and food contamination

Hand washing:

- Use soap, ash, or sand for hands washing.
- Wash hands at critical times: before eating, before cooking meals, after using the latrine, after cleaning a baby or an adult's bottom or cleaning the potty, and after taking care of a sick person including PLHIV.

Safe drinking water:

- Drink water treated with Wuha Agar.
- Drink water stored in container with cover.
- Draw drinking water from covered containers like jerrican, pot or other container without contaminating it through contact with fingers or unclean materials/utensils.
- Pour drinking water into a clean cup or glass.
- Always keep cups or glasses used for drinking clean and upside down.
- Store the drinking water container in a safe place out of the reach of animals and children.

Safe handling and disposal of feces for mobile people

- Use a latrine and/or potty during day and night time.
- Help sick and weak people to use a latrine and/or potty during day and night time; use a device (rope, pole) to help sick people use the latrine if needed.
- Dispose of the feces from the potty in the latrine.
- Wash potty with water and ash/sand after disposing of the feces.
- Wash hands with water and soap after caring for a bed bound person with diarrhea.

Safe handling and disposal of feces for bedridden people

- Cover the bed with plastic sheet.
- Wash feces stained bed sheets, cloth, and plastic sheet with water and soap.
- Wash hands with water and soap after caring for a bed bound person with diarrhea.

Care for women with menstruation

- Wear gloves to handle blood stained materials.
- Wash blood stained materials (bed sheets, clothes, plastic sheeting) with water and soap and dry in the sun.

- | |
|---|
| <ul style="list-style-type: none">• Wash hands with water and soap after caring for a bed bound HIV+ woman with menses. |
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15.1.3. Psychosocial and spiritual support

In addition to curative and preventive care, palliative care also includes psychosocial and spiritual care for patients and their family members/caretakers throughout the continuum of care. Psychosocial and spiritual supports are important part of the inter-disciplinary team within palliative care, offering holistic service to patients and families. These are potentially a universal service and supports clients facing long-term situations involving life-limiting illness, dying, death, grief, and bereavement. Spiritual care involves being a compassionate presence to patients even as they suffer. It recognizes that emotional and spiritual healing can take place even though a physical cure is impossible.

Consider the following in psychosocial and spiritual support

- Provide counseling for parents and family members on positive living, dealing with stigma, adhering to care and treatment.
- Map and create linkage with community and faith-based organizations that provide financial, spiritual, psychosocial and food support.
- Helping the patient and family prepare for death.
- Support at the end of life and death.

15.1.4. End of life care

End of life care is about the total care of a person with an advanced incurable illness and does not just equate with dying. End of life care provides physical, mental and emotional comfort, as well as social support, to people who are living with and dying of advanced illness.

Health care providers have a key role to play in helping patients and their families prepare for and deal with death. Facility-based staff should work with community health workers and spiritual leaders to ensure that quality end of life care are provided.

Consider the following while providing end of life care

- Be honest, attend to emotional responses and spiritual needs in the context of social and cultural customs in developmentally appropriate manner.
- Prepare for death and advanced care planning patient and family / care takers.
- Provide comfort near end of life, including pain management.
- Prepare the body for burial.
- Help family and friends deal with loss and grief.

15.2. Challenges of palliative care

The challenges include late disease presentation, inadequate diagnostic facilities and assessment skills, poor availability of drugs to alleviate pain particularly **opioids** due to regulatory issues, pricing obstacles and ignorance as well as false beliefs about its use. Symptom management is more effectively accomplished when started early; however, because of limited access to care, and long travel distances, palliative care in Ethiopia may be challenging. Moreover, inadequate medical equipment and lack of trained personnel make accurate assessment of palliative care needs very difficult.

Chapter summary:

- Components of palliative care are symptomatic management, preventive care, psychosocial and spiritual support, and end of life care.
- Improved water, sanitation and hygiene is very important in PLHIV.
- ART service providers should regularly assess patients for the presence of pain and manage according to WHO step-ladder treatment criteria.
- Oral medications should only be abandoned if the patient is unable to take or retain them.
- Analgesics should be given ‘by the clock’, i.e. at regular intervals and strict schedule.
- End of life care provides physical, mental and emotional comfort, as well as social support, to people who are living with and dying of advanced illness.

Chapter 16: Standard Precautions, Infection Prevention, HIV pre-exposure prophylaxis and Post-Exposure Prophylaxis

Chapter Description:

This chapter is intended to describe what standard precaution is meant and for. Pre-exposure prophylaxis and post exposure prophylaxis are described with respective minimum criteria and management steps in this section.

Purpose: To identify Standard Precautions and Post-Exposure Prophylaxis procedures to use when working with HIV patients and providers.

Enabling Objectives:

By the end of this chapter, participants will be able to:

- Describe the basic principles and procedures of standard precautions.
- Describe the principle and application of HIV pre-exposure prophylaxis (PrEP)
- Describe the principles of HIV post-exposure prophylaxis (PEP)
- List the management steps of occupational exposure and sexual assault.
- Describe measures taken to maximize the effectiveness of PEP.

Outline:

- Standard precaution (SP)
- Pre-Exposure prophylaxis (PrEP)
- Post-Exposure Prophylaxis (PEP)

Allocated time: 105 minutes

Learning activities

- Reading
- Discussion,
- Exercise with case study,

16.1. Standard Precautions

Standard precaution is a set of standards of infection control developed to prevent exposure and transmission of blood-borne pathogens (HIV, HBV, and HCV). It should be implemented and practiced at all times by all health care providers and caregivers in all settings (hospital, clinic, community settings, and patient homes).

Examples:

- **Physical barrier:** personal protecting equipment (gloves, face masks, goggles, gowns, plastic or rubber aprons, and drapes).
- **Physical disinfection:** high level disinfection (by boiling or steaming and sterilization by autoclaving or dry heat ovens).
- **Chemical disinfection:** antiseptics (alcohol-based antiseptic agents) and high-level disinfectants (chlorine and glutaraldehyde).

Standard precautions were developed because it is not possible to identify all patients with blood borne pathogens. Most patient care does not involve any risk of HIV transmission. Therefore, routine HIV testing of all health care workers or patients is NOT recommended.

These precautions are designed to provide for the safe handling of infectious material including **amniotic fluid, spinal fluid, exudates, transudates, serum, semen, vaginal fluids, blood, bloody body fluids**, etc. As part of this, barriers were developed like gloves, gowns, masks, goggles to protect from splash or spatter.

Safety involves not just patient contact, but the management of the environment in the patient care setting. These procedures are designed to keep all health care workers safe and to protect the public health against infectious waste that could be of risk.

Standard precautions include the following:

- Increased attention for the correct handling of sharps and all contaminated materials.
- Safe disposal of sharps can be made easily. Homemade containers with an open top, firm sides and of durable materials can be used as sharp containers for used needles, scalpels and suture materials.
- Safe disposal of waste contaminated with blood or body fluids.

- Safe disposal of waste bandages, dressings, linens or materials contaminated with blood or body fluids must be handled with gloved hands and placed in containers for safe disposal.
- Hand washing with soap and water before and after all procedures. The single most important step that the healthcare worker can take to ensure the safety of his patients and himself.
- Use of protective barriers, such as gloves, gowns, masks, goggles when in direct contact with potentially infected body fluids. These protective barriers have served to protect healthcare workers from occupational exposure to infectious material.
- With standard precautions, everyone is considered infectious. It is impossible to tell ahead of time who is infected and who is not infected. Gloves provide an important barrier between infectious material and the healthcare provider. Using gloves does not mean that you don't have to wash your hands. You must wash before and after gloving and between patients.

Proper disinfection of instruments and other contaminated equipment in an effort to make standard procedures routine, more efforts were made to prevent bi-directional infections by proper disinfection of instruments and contaminated equipment.

Standard precautions have been associated with a reduction of needle stick injuries and occupational injury.

16.2. Pre – Exposure Prophylaxis (PrEP)

Oral Pre – Exposure Prophylaxis (PrEP): is the use of ARV drugs by people who are not infected with HIV to block the acquisition of HIV. It is an evidence-based HIV risk-reduction intervention that is offered to all people at **substantial risk of acquiring HIV**.

Substantial risk of HIV infection: is provisionally defined as an incidence of HIV higher than 3 per 100 person-years in the absence of pre-exposure prophylaxis (PrEP).

Target population: FSW and HIV negative partners of sero-discordant couples

16.2.1. Eligibility and exclusion criteria

Table – 16.1: Eligibility criteria for PrEP

HIV Negative FSWs	HIV Negative Partners of Sero- Discordant Couples
• HIV negative using a rapid antibody test on the day of PrEP initiation, • Self- identifying, • Substantial risk of HIV infection (any ONE of the following WHO criteria) in the past 6 months: ○ Have sex without a condom with more than one partner, ○ History of a new STI, ○ Use of PEP for sexual exposure for GBV, ○ Has a known HIV positive sexual partner who is not on ART/ on ART for less than six months • No contraindications to PrEP medicines (TDF/3TC) • Age ≥ 18 years, • Willing to give consent for and use PrEP as prescribed, including periodic HIV testing, safety laboratory testing and enhanced adherence counselling sessions. • Willing to provide detailed contact information for tracing.	• HIV negative using a rapid antibody test on the day of PrEP initiation • Substantial risk of HIV infection (any ONE of the following in the past six months): ○ Has a known HIV positive sexual partner(s) who is not on ART or on ART for less than six months and not yet achieved viral suppression or ○ If the HIV negative partner has doubts about the effectiveness of the partner's treatment or has other partners besides the HIV-positive partner on treatment, or ○ If there is history of partner violence and controlling behaviour. ○ Refuses to report risk category but still requests PrEP, • No contraindications to PrEP medicines (TDF/3TC) • Willing to give consent for and use PrEP as prescribed, including periodic HIV testing, safety laboratory testing and enhanced adherence counselling sessions. • Willing to provide detailed contact information for tracing.

The Exclusion Criteria and Contraindications for PrEP

- HIV infection (rule out HIV infection by testing using the national algorithm at the same day of PrEP initiation).
- Signs/symptoms of acute HIV infection, with probable recent exposure to HIV.
- Estimated creatinine clearance of less than 60 ml/min (if known).
- Allergy or contraindication to any medicine in the PrEP regimen (TDF or 3TC).
- Presence of Hepatitis B infection (clients with HBsAg Positive test result).

- On MDR-TB treatment.
- Significantly mobile (not planning to be in that town for the next 6 months) persons that will not be able to attend visits as prescribed, that is clients that will not be in a region where PrEP can be provided at the next visit or clients that do not have contact information.
- Age below 18 years
- Unwillingness to use PrEP as prescribed

16.2.2. Follow – Up and Monitoring Of PrEP

Currently the recommended regimen is TDF/3TC FDC, 1 tablet, orally daily.

Prescription Intervals:

For HIV negative partners of sero - discordant couples, the initial dose will be for one month. Once the HIV test at 1 month is still negative, provide supply for 3 months and continue the follow up every 3 months with repeat HIV testing.

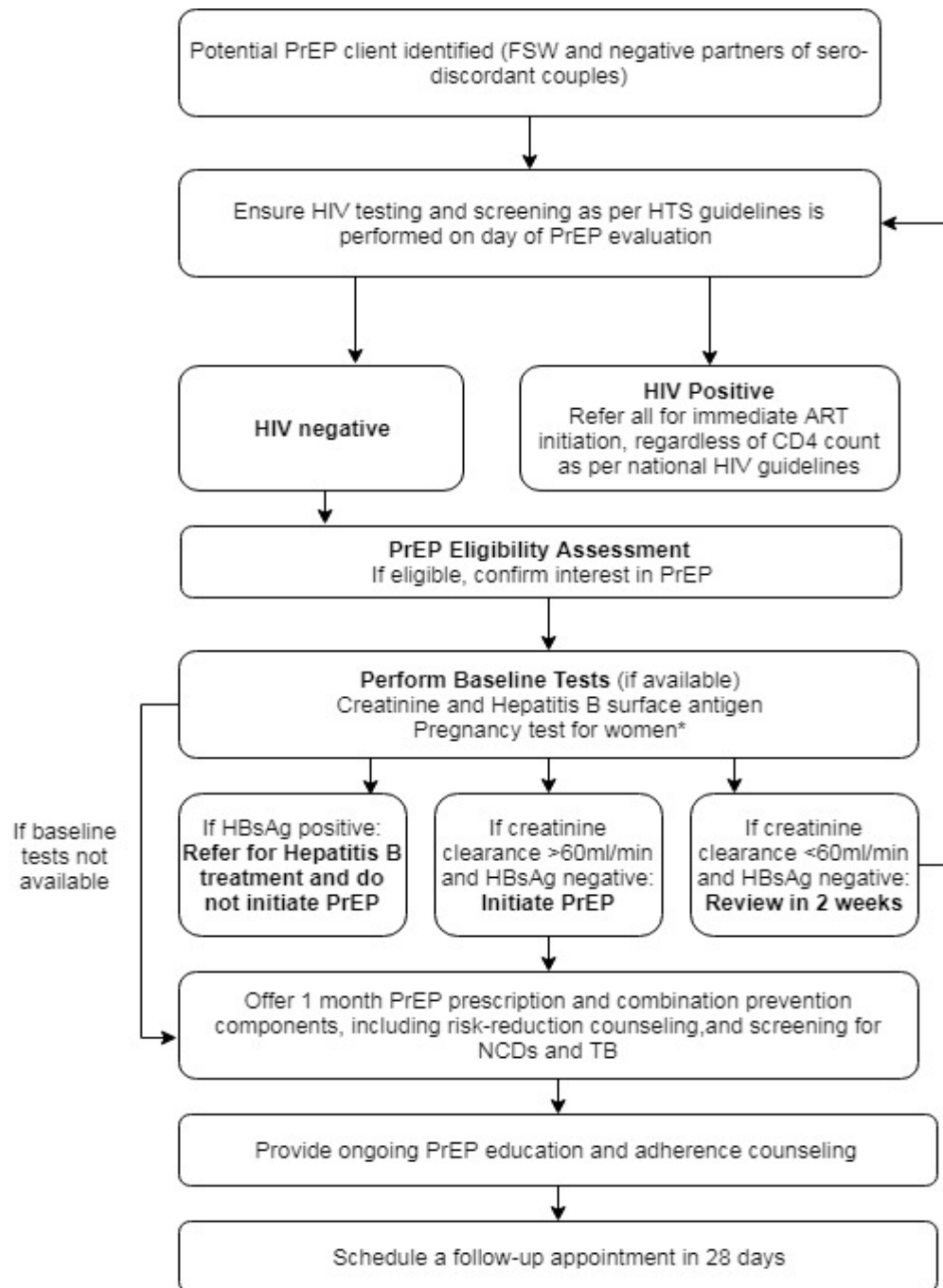
For HIV negative Female sex workers, monthly clinic visits for the first 3 months is important to address their changing routines and multiple health needs. During each visit, every other component of combination HIV prevention package has to be offered to PrEP users. HIV testing will be repeated at one month and at 3rd month and will be continued 3 monthly.

Initial Follow up

At Initial visit, before initiation, there are lab tests performed to ensure there is no contraindication for PrEP.

Table 16.2: Necessary Lab Tests at Initial Follow –Up Visits

Investigation/ intervention	Rationale
HIV test	To assess HIV infection status To complete a symptom checklist for possible acute HIV infection
Serum creatinine	To identify pre-existing renal disease, Calculate creatinine clearance using age, sex and weight. If estimated creatinine clearance is less than 60 ml/min, defer PrEP
Hepatitis B surface antigen	If negative, provide PrEP. If positive, refer for further testing and assessment for hepatitis B treatment.
Screening for sexually transmitted infection (STI)	To diagnose and treat STI (syndromic or diagnostic STI testing, depending on local guidelines).
Pregnancy testing	To guide antenatal care, contraceptive and safer conception counselling, and to assess risk of mother to child transmission. Pregnancy is not a contraindication for PrEP use (see section below).
Counseling	To assess whether the client is at substantial risk of HIV. To discuss prevention needs and provide condoms and lubricants. To discuss desire for PrEP and willingness to take PrEP. To develop a plan for effective PrEP use, sexual and reproductive health. To assess fertility intentions and offer contraception or safer conception counselling. To assess intimate partner violence and gender-based violence. To assess substance use and mental health issues.



*If client is pregnant, offer PrEP services through PMTCT clinic

Figure 16.1. PrEP Screening and Initiation Algorithm

Table 16.3: Follow – up after initiation of PrEP

Service	Interval
Confirmation of HIV-negative status	Every 3 months. Consider also testing at 1 month
Address side-effects	Every visit.
Brief adherence counselling	Every visit.
Estimated creatinine clearance	Every 6 months. Consider more frequently if there is a history of conditions affecting the kidney, such as diabetes or hypertension; consider less frequently if age is less than 45, baseline estimated creatinine clearance more than 90 ml/min, and weight > 55 kg.
Screening for sexually transmitted infection (STI)	To diagnose and treat STI (syndromic or diagnostic STI testing, depending on local guidelines). Every visit.
Counseling	To assess adherence and provide counselling regarding effective PrEP use (adherence), To assess whether the client is still at substantial risk of HIV. To discuss prevention needs and provide condoms and lubricants. To assess fertility intentions and offer contraception or safer conception counselling. To assess intimate partner violence and gender based violence. To assess substance use and mental health issues.

Initial follow up will be after a month of initiation and every three months then after.

16.2.3. When should PrEP be stopped or discontinued?

PrEP should be stopped if the client:

- Has a positive HIV test result, then start ART,
- Develops renal disease (refer to the national guideline on comprehensive HIV care and treatment)
- Has become non - adherent to PrEP (i.e. as per the national definition for adherence, poor adherence, non – adherence).

- If the client misses 5 or more doses during the month, Enhanced Adherence Counseling (EAC) will be offered 2 times (once per each visit monthly). If the client keeps on missing doses, and has poor adherence on three consecutive visits, providers may consider discontinuing PrEP.
- Does not need or want PrEP,
- No longer meets the eligibility criteria, , index or partner is no more in substantial risk for HIV infection, i.e., sustained viral suppression of HIV partner,
- If there are safety concerns, where the risks of PrEP use outweigh potential benefits.

After discontinuation of PrEP, there should have to be follow - up testing for HIV after a month in order to confirm that the client is still HIV negative.

16.3. Post-Exposure Prophylaxis (PEP)

PEP is the use of therapeutic agents to prevent infection following exposure to a pathogen. In the case of HIV, PEP is short-term antiretroviral treatment to reduce the likelihood of HIV infection after potential exposure. The types of exposures include **per-cutaneous** (needle stick injury), **splash, bite, suspected sexual assaults**. For health-care workers, PEP commonly is considered for exposures to HIV and Hepatitis B.

The least expensive way to deal with a disease is via prevention. The implementation of standard precautions with appropriate training and monitoring is the number one priority.

Although standard precautions will decrease the need for PEP, there are “accidents” and unanticipated occupational exposure. PEP will be made available free of charge for accidental occupational exposures for health workers and emergency personnel as well as for rape victims.

Table 16.4.Risk of viral transmission with sharp injury from infected source.

Source	Risk (%)
HBsAg ⁺	30
HCV	1.8
HIV	0.3

The risk of transmission of HIV is much less than that of Hepatitis B by a factor of 100. These transmission rates apply to occupation risk due to sharp injury. However, risk could vary depending on severity of injury and viral load in the source patient.

16.3.1. Occupational Exposure to HIV

Most injuries in the healthcare workplace are due to contaminated sharp injuries. Body fluid in contact with mucous membranes is an important risk. Procedures such as gynecological exams, spinal taps, labor and delivery, surgery and line placement and others can place the health worker at risk. Transmission is estimated to be about one in three hundred for each exposure independent of any other exposure.

16.3.2. Risk factors for Sero-conversion

- There is increased risk for occupational HIV exposure when the exposure is deep as opposed to superficial.
- When the device in question had blood visible from the source person, there is increased risk of transmission.
- Risks are significantly increased if the source patient (SP) has advanced disease, such as the stage of illness, extent of viremia (viral load) circulating free viruses as opposed to cell bound virus and the existence of ARV in the source person. Basically, this has to do with the amount of virus that is available for the initial inoculum.
- If the person was injured by a device that was in an artery or vein also increases the exposure risk.
- Type of exposure or route of exposure is important. Hollow bore needles are more likely than solid needles to cause sero-conversion (**hollow bore**-IV needles, injectable; **solid**-suture material with needle). Per-cutaneous route is therefore an important route for transmission. The mucous membrane route of exposure is often much overlooked. This is also a route of exposure but less than per-cutaneous route. Cutaneous exposure is much less likely to transmit HIV.
- Glove use - 50% decrease in volume of blood transmitted.

16.3.3. Steps needed from the time of injury and contact to blood, and body fluid through the initiation of PEP if it is warranted

- Percutaneous injury or injury to non-intact skin :
 - Wash the exposed site with soap and water as soon as possible, without scrubbing.
 - Avoid using antiseptics.

- Allow free bleeding but don not squeeze the wound.
- Exposed mucous membranes:
 - Irrigate copiously with clean water or saline.
- After a splash contacts with unbroken skin, do the following:
 - Wash the area immediately.
 - If, running water is not available, clean the area with a gel or other hand rub solution or whatever is customarily available.
 - Do not use strong disinfectants.
- After a splash contacts with the eye, do the following:
 - Irrigate the exposed eye immediately with water or normal saline.
 - Sit in a chair, tilt the head back and have a colleague gently pour water or normal saline over the eye pulling the eyelids up and down to make sure the eye is cleaned thoroughly.
 - If contact lenses are worn, leave these in place while irrigating the eye as they form a barrier over the eye and will help protect it. Once the eye has been cleaned, remove the contact lenses and clean them in the normal manner. This will make them safe to wear again.
 - Do not use soap or disinfectant on the eye.
- After a splash contacts with the mouth, do the following:
 - Spit the fluid out immediately.
 - Rinse the mouth thoroughly using water or saline, and spit again. Repeat this process several times.
 - Do not use soap or disinfectant in the mouth.
- Report the exposure: To the PEP focal person or the ART physician or nurse in the facility immediately. The PEP focal person or the ART clinicians needs to do:
 - a)** Clinical evaluation, counseling and testing of the exposed person and complete the exposure reporting form.
 - b)** Do risk assessment and determine the Exposure Code (EC) and source HIV status code (HIV SC) (use PEP algorithm).
 - c)** Using the EC and HIV SC determine whether PEP is warranted for the exposed HCW (Use PEP algorithm).

- d) If the HCW is warranted to take PEP: Chose appropriate PEP regimen, counsel about the ARVs and prescribe (use PEP algorithm).
- e) Document properly on the PEP follow-up register.
- f) Appoint the exposed person and follow.
- g) Do all follow up HIV testing at 6 weeks, 12 weeks and 24 weeks, and manage accordingly.
- **Testing source:** rapid test is done after counseling and consent has been secured. If the source patient is negative, there is no need of further assessment of the exposed health care worker. If the result is positive, the health care worker needs to be tested.
- **Testing of health care worker:** HIV serology should be performed immediately after exposure. If the result is positive, there is no need for PEP, but if negative you should administer PEP and then repeat serology at 6 weeks, 3 months, and 6 months.
- Following HIV exposure there is a need for psychosocial support, safe sexual practice and close monitoring for adherence and toxicity of ARVs.

Note: Provision of PEP is an emergency and the set up should be organized to provide the drugs within hours (preferably the 1st 2hours).

- Remember to initiate PEP immediately after exposure until test result confirms the HIV status of the victim.
- Stop PEP if the health worker is positive for HIV antibodies.

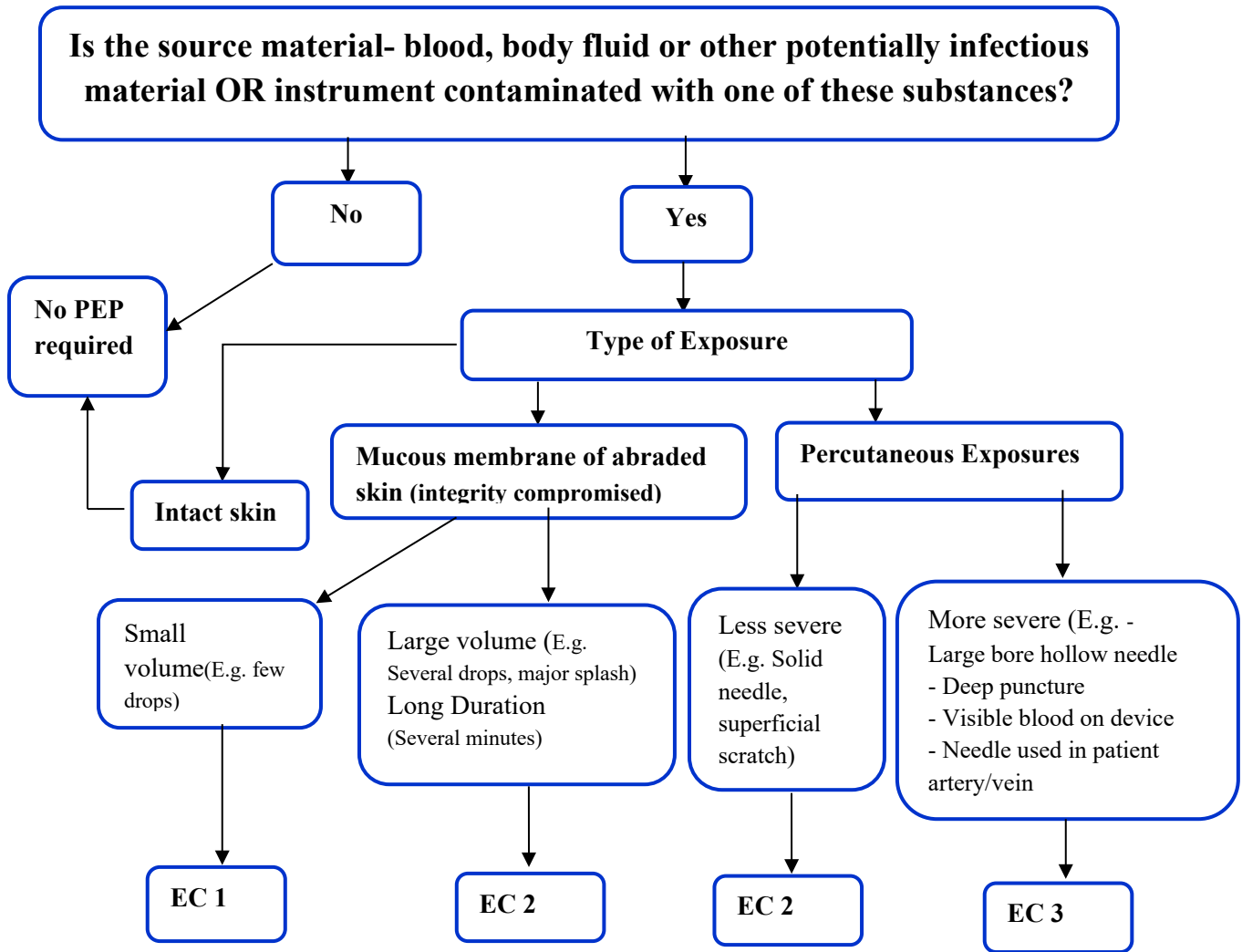


Figure 16.2. Algorithm to determine the severity of exposure (Exposure Code): Risk assessment of exposure

Table 16.5.Interpretation of exposure code (Severity of Exposure)

	Exposure Code	Type of exposure
1	EC 1	Is a minor mucocutaneous exposure to small volume of blood for short period (few seconds to minutes)
2	EC 2	Is a major mucocutaneous exposure to large volume of blood for longer duration (Several minutes) or Mild percutaneous exposure (with solid needle or superficial scratch or injury).
3	EC 3	Sever percutaneous exposure (Large bore hollow needle, deep puncture, visible blood on devise, needle used in patient artery/vein).

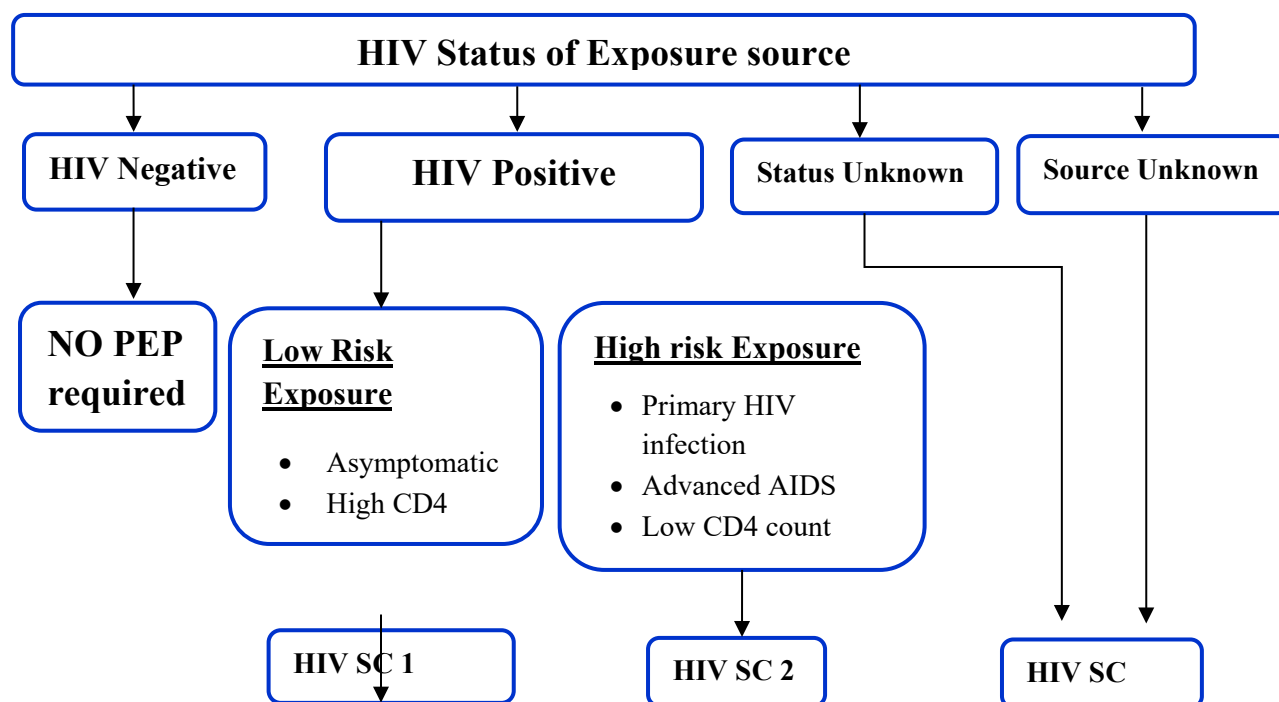


Figure 16.3.Assessing the HIV status and condition of the exposure source (HIV Status Code)

Table 16.6.Interpretation of the HIV status of the source patient

	HIV source code (SC)	The HIV status and severity of the illness in the source patients
1	HIV SC 1	The source patient is HIV Positive but is asymptomatic and has reasonably good immune status.
2	HIV SC 2	The source patient is HIV Positive and is symptomatic, may have AIDS or has other evidence of advanced illness (Low CD4 or High viral load)
3	HIV SC unknown	The HIV status of the source patients is unknown (either the patient has refused HIV testing or died or discharged before HIV testing) or the source patient is unknown (e/g Unlabeled blood sample in a laboratory)

16.3.4. Recommendation of PEP based on Risk assessment

Table 16.7.Recommended HIV post exposure prophylaxis for percutaneous injuries and Mucous membrane or non-intact skin exposure

Status code	Exposure code		
	EC 1	EC 2	EC 3
SC 1	Basic 2 drug PEP	Basic 2 drug PEP	Expanded 3 drug PEP
SC 2	Basic 2 drug PEP	Expanded 3 drug PEP	Expanded 3 drug PEP
SC unknown	Consider basic 2-drugs PEP		
HIV negative	No PEP warranted	No PEP warranted	No PEP warranted

16.3.5. Timing and duration of PEP

- To be effective prophylaxis has to commence as soon as possible (within 1 to 2 hours, post-exposure). Don't consider PEP beyond 72 hours post exposure.
- Prophylaxis is to be given for 28 days.

Table 16.8. Recommended ARV drug regimens for PEP

ARV drug regimen	Dose	Frequency	Duration
2-Drug Regimen:	<i>TDF 300mg</i>	<i>Once daily</i>	<i>28 days</i>
<i>Tenofovir (TDF) + Lamivudine(3TC)</i>	<i>3TC 300mg</i>	<i>OR</i>	
<i>or</i>	<i>AZT 300mg</i>	<i>12 hourly</i>	
<i>Zidovudine (AZT) + Lamivudine (3TC)</i>			
3-Drug Regimen:	<i>Triple FDC</i>		
<i>Preferred:</i>	<i>(TDF 300mg, 3TC 300mg,DTG50mg)</i>	<i>Once daily</i>	
<i>Adults & older children:</i>	<i>AZT 300mg+3TC 150mg+DTG50mg</i>		
<i>Tenofovir+Lamivudine+Daltugravir (TDF+3TC+DTG) FDC</i>			
<i>Children>6yrs:</i>			
<i>Zidovudin+Lamivudine+Dalutegravir (AZT+3TC+DTG)</i>	<i>LPV/r400mg/100mg</i>	<i>12hourly</i>	<i>28 days</i>
<i>Children <6yrs: AZT+3TC+LPV/r</i>	<i>ATV/r300mg/100mg</i>		
<i>Alternative:</i>		<i>Once daily</i>	<i>28 days</i>
<i>Adult: Tenofovir+Lamivudine +Efavirenz/lopinavir/Atazanavir (TDF+3TC+EFV/LPV/r/ AZT/r)</i>			

Important Note:

A) Do not use Nevirapine containing regimen for PEP, as the risk of toxicity is higher

B) For drug experienced source patient, start the available PEP soon and refer for consultation.

As this is an emergency, PEP should be started as soon as possible within 1 hour and then after refer or consult the person to an experienced physician on HIV.

16.3.6. Prevention of HIV transmission after sexual assault:

- Any person/child presenting to a health facility after potentially exposed to HIV after sexual assault should be counseled by the examining health care worker about the potential risks of HIV infection.
- Source should be considered unknown sero-status.
- Parents/guardian of traumatized children should be counseled and informed on the risk of HIV infection after sexual assault.
- **The following points should be covered in the counselling sexual assault victims:**
 - a) The risk of transmission is not known, but it exists.
 - b) It is important to know the victim's HIV status prior to using any antiretroviral treatment.
 - c) It is the patient's choice to have immediate HIV testing or, if she/he prefers, this could be delayed for not more than 72 hours after incident (**management guidelines on sexual assault provides for a 3-day starter pack for those who prefer not to test immediately, or those that are not ready to receive results immediately**). However, encourage the patient for the importance of testing.
 - d) PEP is not recommended after 72 hour following sexual assault. Accordingly, the victims should be counseled about the possible risk of infection and the possibility of transmitting infection during sero-conversion.
- It is strongly recommended that the implementation of post-rape prophylaxis to be monitored and evaluated carefully.
- Other post assault service need to be provided:

- Emergency Contraceptives or IUCD.
- Client education on safer sex practice.
- Long term psychosocial support
- Police and legal support.
- Screening for conventional STIs screening and prophylaxis/ treatment.
- HBV vaccination.
- TAT if there is physical injury (need based) check for allergy/hypersensitivity.
- Testing for HIV at 6 weeks, 12 weeks(3 month) and 24 weeks(6 month).

Table 16.9.Recommendation for HIV prophylaxis after sexual assault.

PEP is recommended -During significant risk of exposure immediately within 1 hour and within 72 hours and -Exposure of vagina and/or rectum to semen and -Source likely to be infected.
PEP is not recommended – Delay more than 72 hours from time of exposure. – Following condom leak or tear.
Recommended regimen TDF+3TC+DTG for 28 days. Alternatively, TDF +3TC+EFV OR Kaletra OR Boosted Atazanavir,

16.3.7. Follow-up of client Exposed to HIV

a) Post exposure Testing

- A **client** who are taking PEP should be followed in the adult ART clinic.
- They should be instructed to return at 6 weeks, 3 months and 6 months post sexual assault for voluntary counselling and HIV testing.

b) Monitoring and Management of PEP Toxicity

- Exposed **clients** should be reassessed within 3-5 days for medication tolerability and toxicity. If further details about the source become available, a risk assessment re-evaluation may also be appropriate.
- **Clients** taking PEP should be monitored for drug toxicity by testing at baseline and again 2 weeks after starting PEP. The scope of testing should be based on medical conditions in the exposed person and the toxicity of drugs included in the PEP regimen.
- Minimally, lab monitoring for toxicity should include a complete blood count and liver function tests.

If toxicity is noted, modification of the regimen should be considered.

EXERCISE 16.1

Case Study 1

27 years old female nurse presents to OPD for evaluation of needle stick injury 2 days ago from a diabetic lancet. Source patient (SP): 35 year old male known HIV+ (wasn't known at the time).

Discussion:

- Would you offer her PEP?
- What do we need to know about PEP in order to manage her exposure?
- What is her risk for contracting HIV?
- Are there factors that might affect this risk?
- How effective is PEP?
- Is it too late to start PEP?
- What are the drawbacks of starting PEP?
- Which regimen(s) should be considered?
- What follow-up should be arranged?

Case Study 2

24 year old dental technician splashed in the eye during dental procedure 3 hours ago

Source patient: 33 year male, co-infected with HIV and HCV.

- What else do you need to know?

- Which Fluids are potentially Infectious for HIV? - Blood?, Saliva?, Sweat?, Feces?, Spinal fluid?, Pleural fluid?, Pus?, Urine?

Chapter summary

- Standard precautions should be implemented and practiced by all healthcare providers.
- The most important infection control method is hand washing.
- Proper handling of sharps is critical for reducing occupational exposure to blood borne pathogens.
- Risk of HIV seroconversion after occupational exposure varies depending on source patient and exposure circumstance.
- When indicated, PEP should be employed immediately (within hours).
- A set up should be in place to administer immediate PEP including testing for HIV in health care facilities.
- Clients taking PEP should be put under follow up in ART clinic.
- PrEP is useful prevention modality for FSWs with substantial risk and HIV negative discordant partners until the positive partner has sustained viral suppression,

Chapter 17: Team Building and Service Delivery

Chapter Description:

This chapter describes about team work to deliver effective chronic HIV care with final goal of viral suppression to individuals and control of HIV transmission. The roles and responsibilities of multi-disciplinary team members the teaming process for HIV care and treatment are described.

Purpose: Understand importance of team building in delivering HIV prevention, care and treatment service

Enabling objectives:

By the end of this chapter, participants will be able to:

- Explain the composition of MDT at different levels of facility
- Describe the function of MDT
- Describe what constitutes minimum clinical team for chronic HIV care
- Explain how the clinic should be organized to implement the care and treatment service
- List on the clinical and recording tasks of each team member for optimal clinic functioning
- Understand common problems after the National Consolidated HIV Care and treatment training and how to help the clinical team solve these problems

Outline:

- Team building,
- Multidisciplinary team
- Sequence of care,
- Problem solving

Allocated time: 160 minutes

Learning activities

- Brainstorming,
- Reading
- Discussion,

17.1. Team Building

A "clinical team" is a team of health workers that collaborate in the clinical care of a patient. Each member of the clinical team may have different roles and be concerned with different aspects of the care of the patient, but they discuss each case and decide together on the treatment plan. Several clinical teams may be situated in one health facility. But not all members of the clinical team have to be located in the same facility, as in the case of a doctor/medical officer who is part of the clinical team at several small health centers. The clinical team members are comprised of:

- Medical Doctor
- Health officer
- Pharmacist /Druggist
- Laboratory technologist/technician
- Nurse/ lay counselor
- Case manager and Adherence supporters/ mother supporters,
- Data clerk/ health information technician (HIT)

17.2.Members of Clinical/ Multi-disciplinary Team (MDT)

- **ART clinicians:** could be medical doctors, health officer or nurses who are trained to manage and follow ART clients. They could be involved in mentoring and supervision of other clinicians. Senior clinicians (MD) will be consulted in management of complicated cases and will be responsible for substitution and switch to second/ third line ART regimens.
- **Laboratory staffs:** are laboratory technologist or technicians that are responsible to conduct HIV/AIDS related laboratory testing, coordinate sample collection, transport and result delivery in health facilities, conduct laboratory mentorship and lead the HIV/AIDS related laboratory tests.
- **Pharmacy service provider:** they manage logistics of HIV related pharmaceutical care: select, request, receive, store and dispense ARV drugs, provide patient counseling and adherence support, maintain patient information.

- **PMTCT service provider:** are trained to manage and follow HIV positive pregnant and lactating mothers with their exposed infants at ANC, labor and delivery. They could be involved in mentoring and supervision of other clinicians.

Trained non-clinical staffs (PLHIV volunteers)

- **Adherence case managers:** they do adherence counseling, support index case family testing, screen for mental illness and link to treatment, facilitate referral linkage with community resources, patient tracking and education and support
- **Adherence supporters:** they facilitate intra and inter facility referral linkage, track patient appointment and conduct tracing for lost patients.
- **Mother mentors:** they provide education, adherence counseling and support for pregnant and lactating WLHIV mothers. They are also involved in tracking of mothers and HEI.
- **Data clerk/HITs:** they work on registration and documentation of HIV service; they update/fill Pre-ART, ART registers and update ART data base. They also report through monthly report cohort analysis reporting forms.

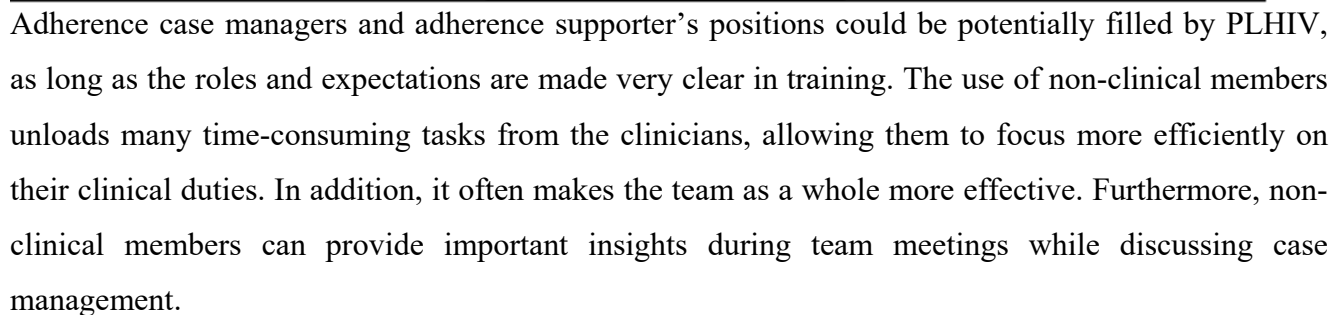
NB. Other clinical staffs such as medical directors (CCO) and head nurse/ metron/ Outpatient directorate director could be involved in MDT based on the complexity of the facility.

Key stake holders working in the community could also be involved in MDT meetings. The minimal clinical team for HIV care and ART depends on the category of health facility. In most settings, a doctor is needed to supervise more than one clinical team, with each team having a nurse and health officer. ART nurses supervise and provide support to one or two adherence case managers and adherence supporters. At a large hospital, for example, there may be several clinical teams under the supervision of one doctor. In a region with several health centers, their respective clinical teams may be under the supervision of one doctor who rotates regularly through the health centers and is in contact through distance communication.

The overall roles of clinical team/MDT:

- a. Lead the overall HIV care and treatment activities in the facility.
- b. Discuss challenging cases and new treatment updates.
- c. Provide solution for logistics and admin challenges in the clinic.
- d. Monitor the overall activities of HIV service.
- e. Ensure reports are sent as per the standard.

Many of the 11 steps in the *Sequence of Care* are best done by the non-clinicians on the clinical team. These steps are shown on the left side of the diagram below.



The first step in the *Sequence of Care* is triage. He or she must determine the reason for the visit, take an interval history and facilitate contact with clinicians. There are patients coming for unscheduled and scheduled chronic HIV care visits. .

It is important to keep an appointment book in the triage area. All chronic HIV care patients need to be given follow-up dates. If all the chronic HIV care patients are arriving in the morning and slowing down the clinic, appointment times (early morning, late morning, early afternoon, late afternoon), can be given to spread them out over the day. Refer **Patient flow in HIV chronic care** Figure 6.1 in Chapter 6,

Understanding of the patient flow in the HIV chronic care is important to get clear picture on reorganizing and supervising the implementation of HIV prevention, care and treatment in the specified health facility.

Exercise 17.1. Patient flow

Do this exercise in groups. Represent all relevant offices of the health care unit HIV service provided. Draw a map on a flipchart sheet of the area of the facility that will be used for HIV Care. The relevant rooms should be included. Draw the route of patient flow on the map. Represent all services delivered with integration.

And discuss the place where adherence support and patient education provided, patient triage, patient monitoring, patient data management.

17.4. Problem solving and addressing gaps:

As the continuum of HIV chronic care involves different service providers and units, consorted monitoring effort is important to supervise the implementation and to address whenever problem happens or gaps observed. To mention some of the challenges that happen in the continuum of HIV care:

- Provision of HTS for the right clients (case identification),
- Linkage of HIV positives identified to HIV care and treatment,
- Enrollment to care and initiation of treatment,
- Linkage to laboratory monitoring service, specimen collection and result delivery,
- Supply of ARV drugs and other supplies,

- Other medical needs of PLHIV other than HIV/ AIDS, like admission and X-rays,
- Documentation and Report compilation

In conclusion, effective HIV program is the result of a strong and a well-coordinated team work.

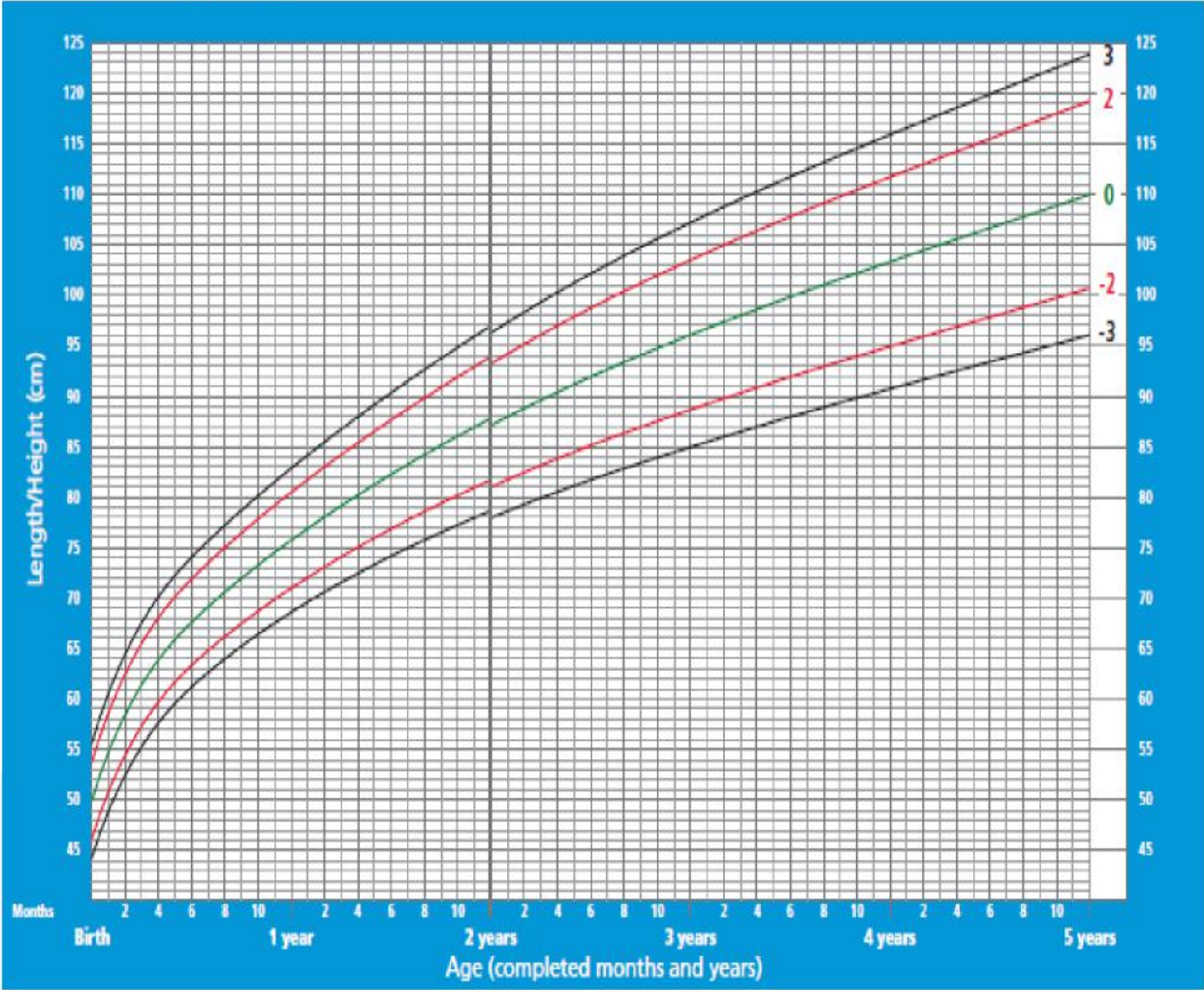
Chapter summary:

- Roles of Clinical Team/MDT
 - Lead the overall HIV care and treatment activities in the facility
 - Discuss challenging cases and new treatment updates
 - Provide solution for logistics and admin challenges in the clinic
 - Monitor the overall activities of HIV service
 - Ensure reports are sent as the per the standard
- The use of non-clinician members unloads many time-consuming tasks from the clinicians, allowing them to focus more efficiently on their clinical duties.

Annex 1: Growth Curves and Charts

Length/height-for-age BOYS

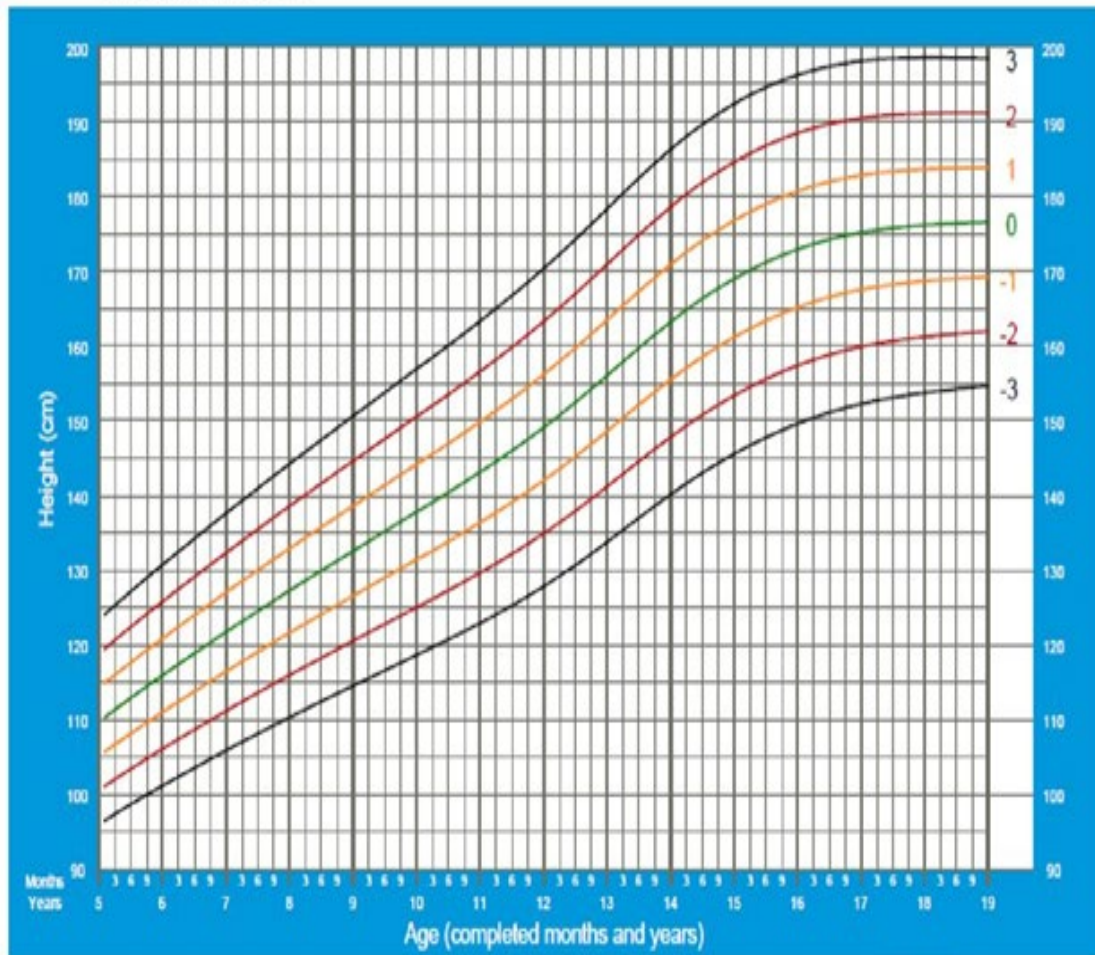
Birth to 5 years (z-scores)



WHO Child Growth Standards

Height-for-age BOYS

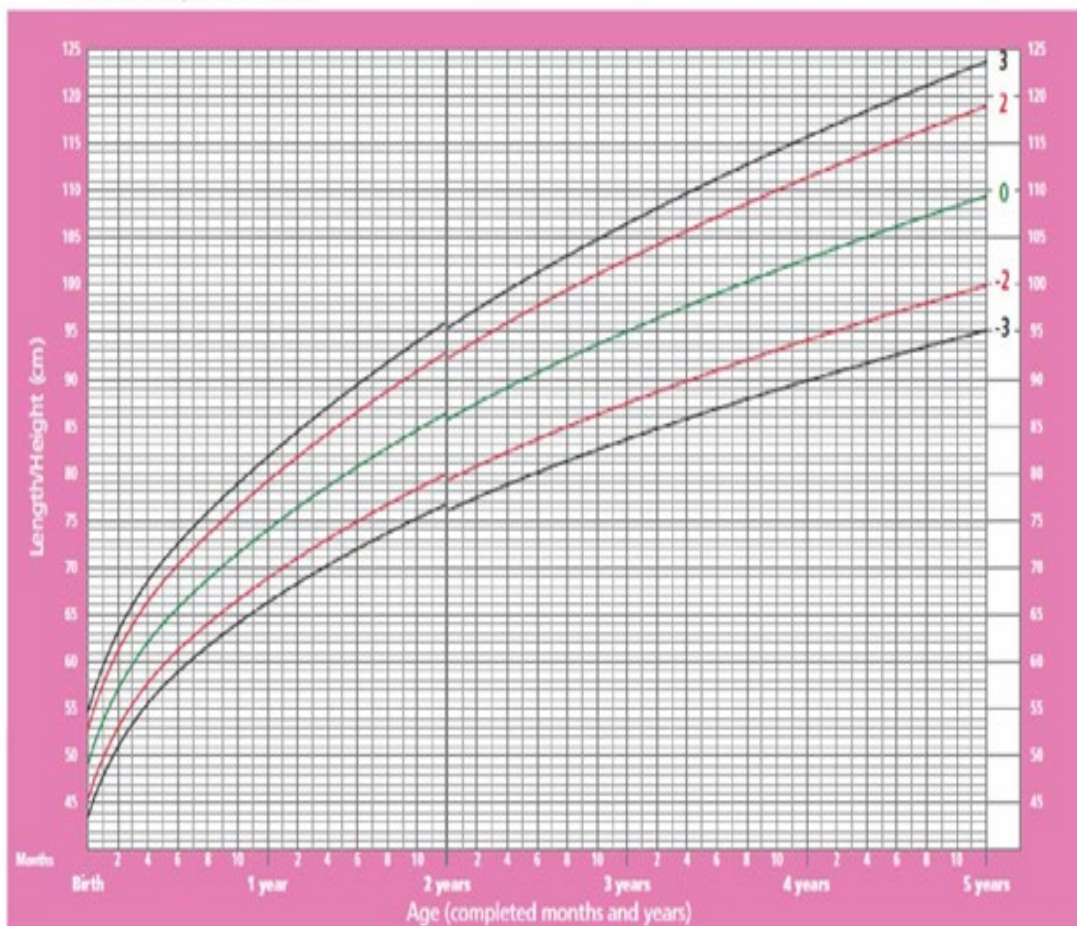
5 to 19 years (z-scores)



2007 WHO Reference

Length/height-for-age GIRLS

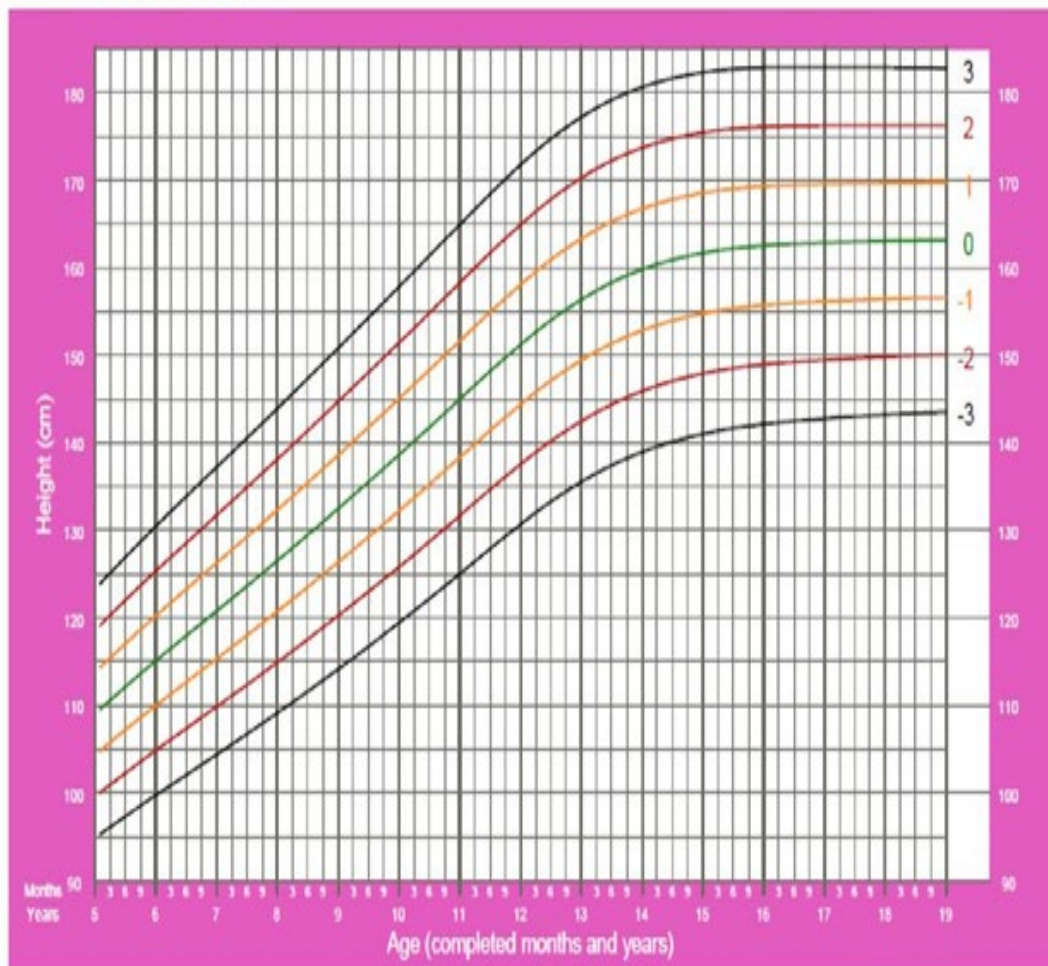
Birth to 5 years (z-scores)



WHO Child Growth Standards

Height-for-age GIRLS

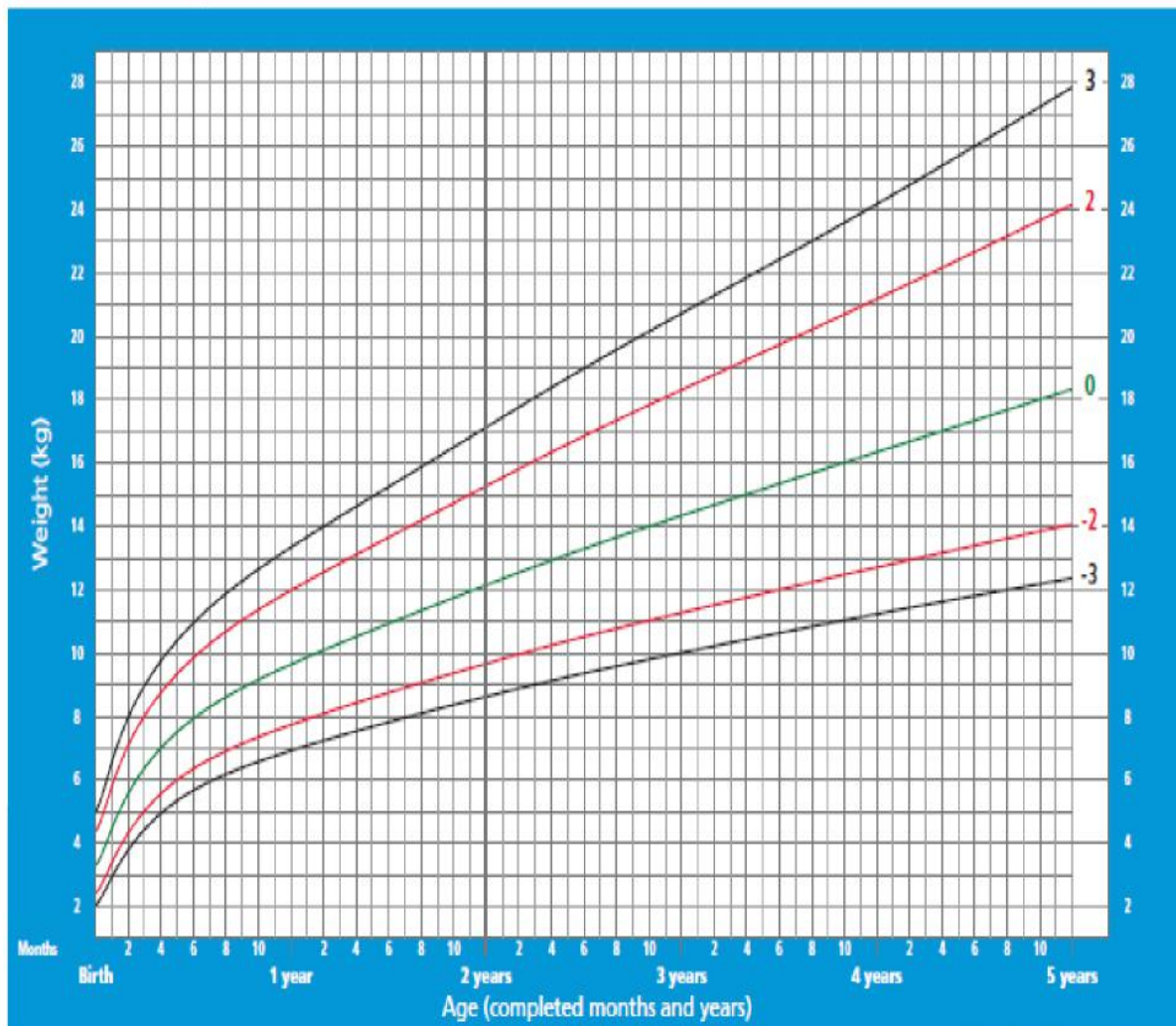
5 to 19 years (z-scores)



2007 WHO Reference

Weight-for-age BOYS

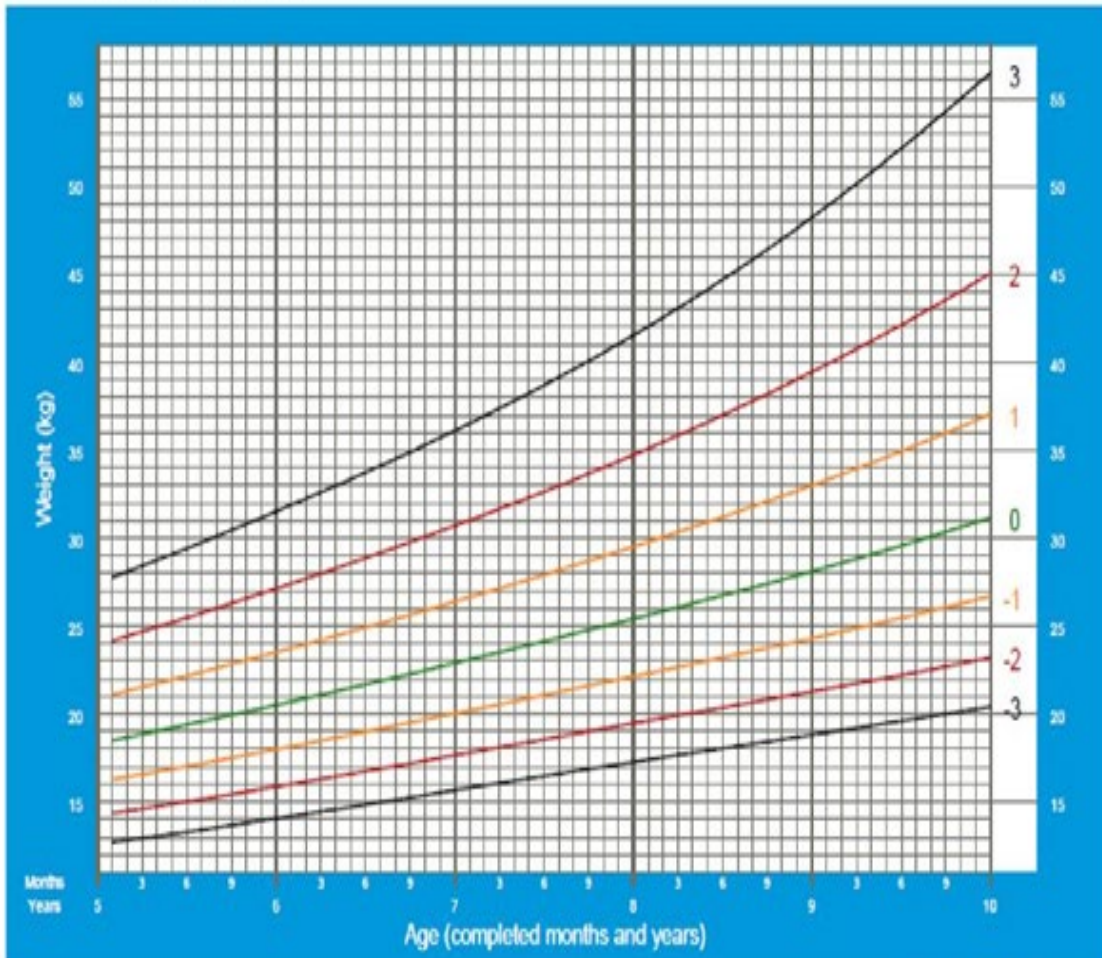
Birth to 5 years (z-scores)



WHO Child Growth Standards

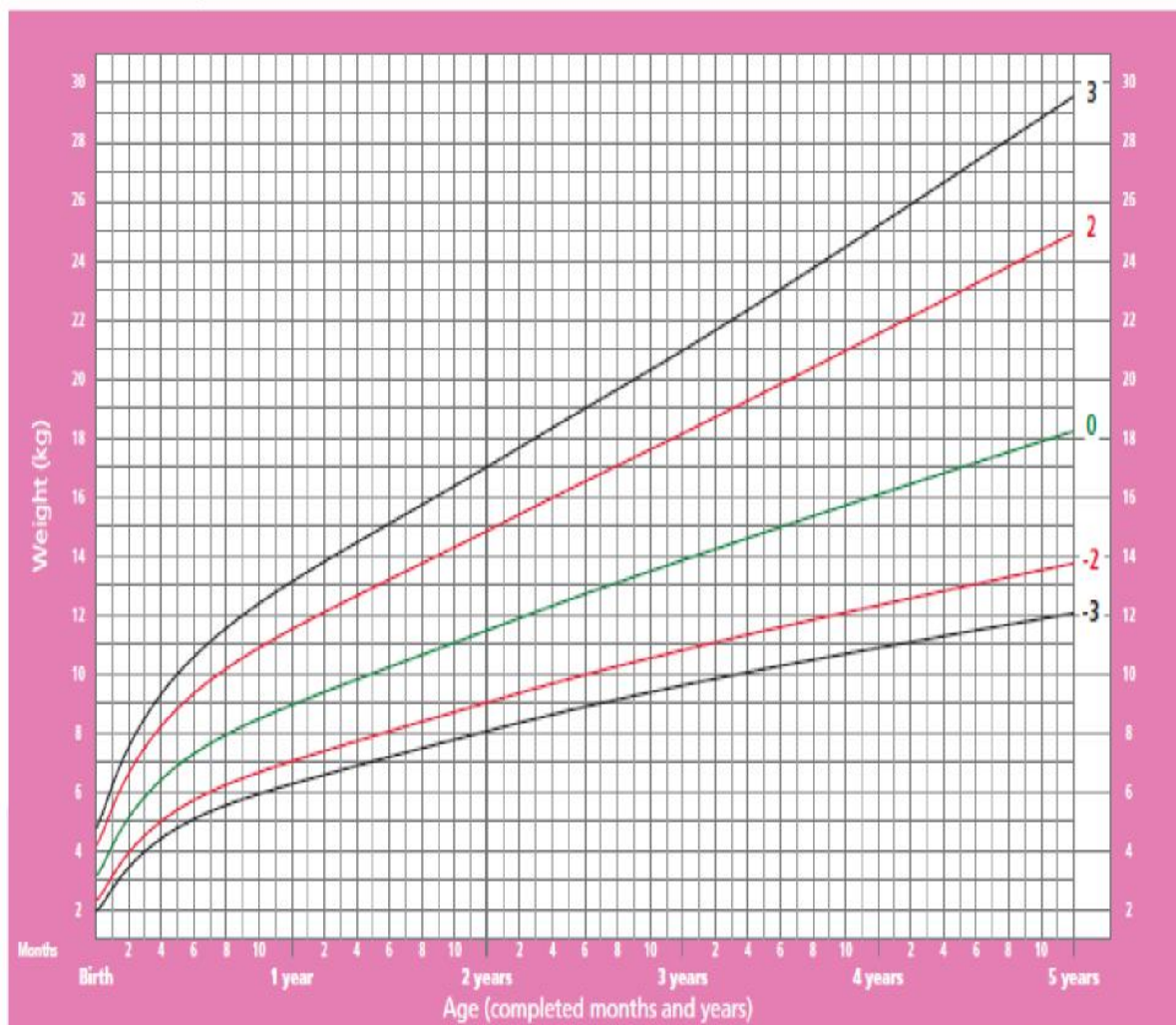
Weight-for-age BOYS

5 to 10 years (z-scores)



Weight-for-age GIRLS

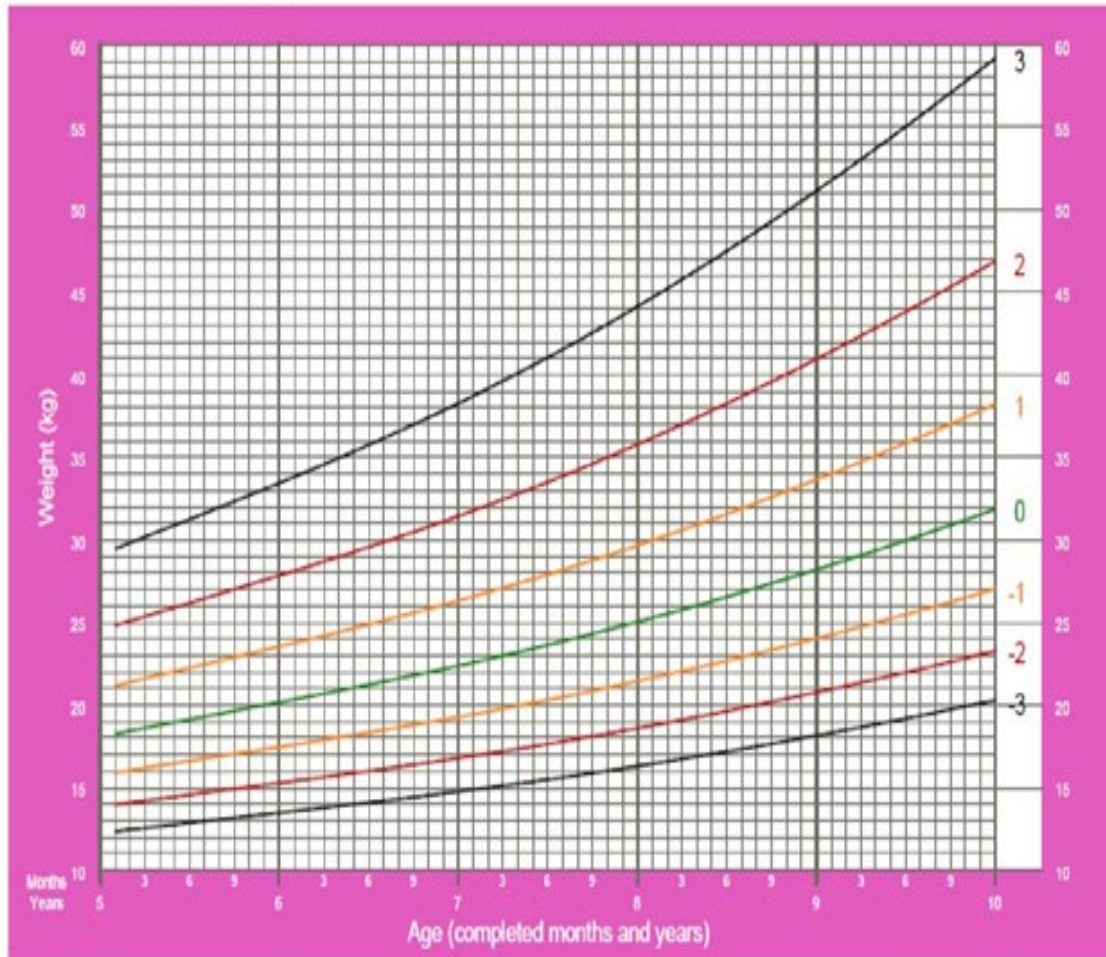
Birth to 5 years (z-scores)



WHO Child Growth Standards

Weight-for-age GIRLS

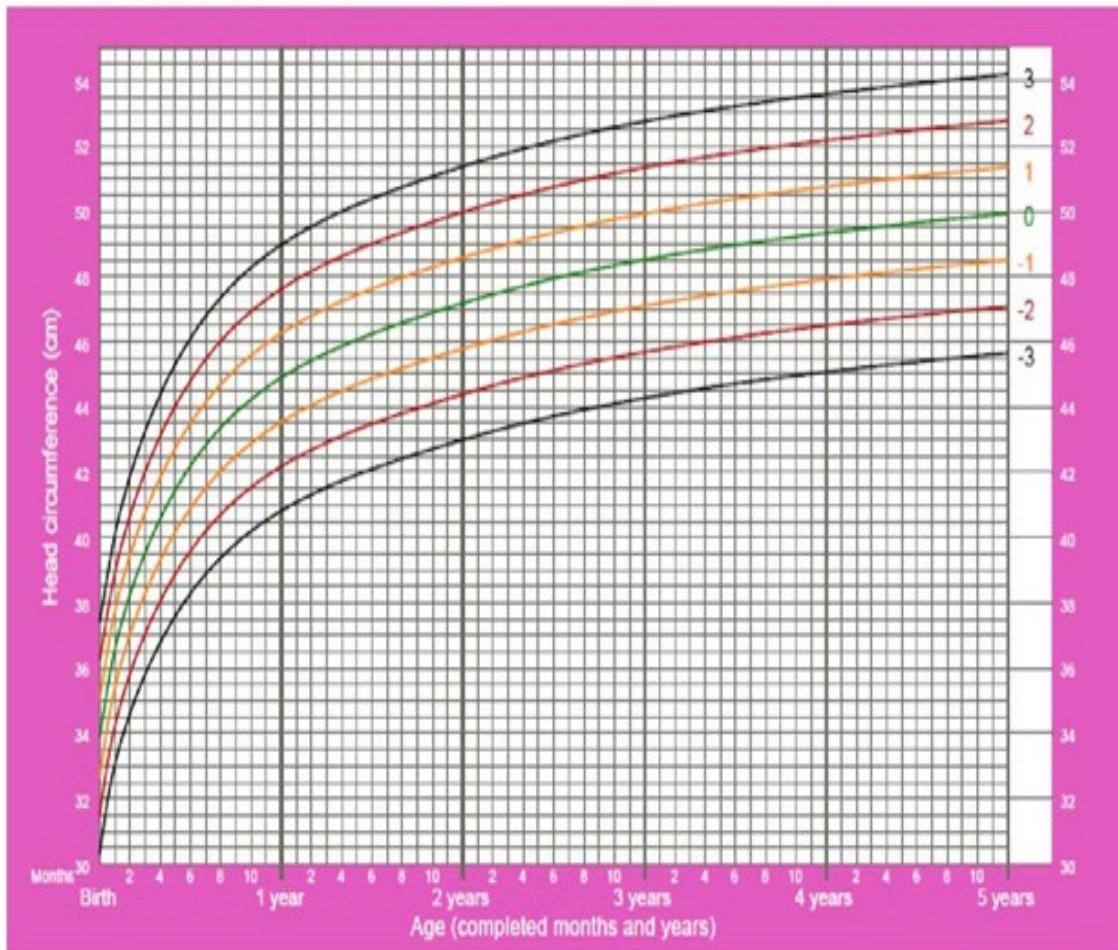
5 to 10 years (z-scores)



2007 WHO Reference

Head circumference-for-age GIRLS

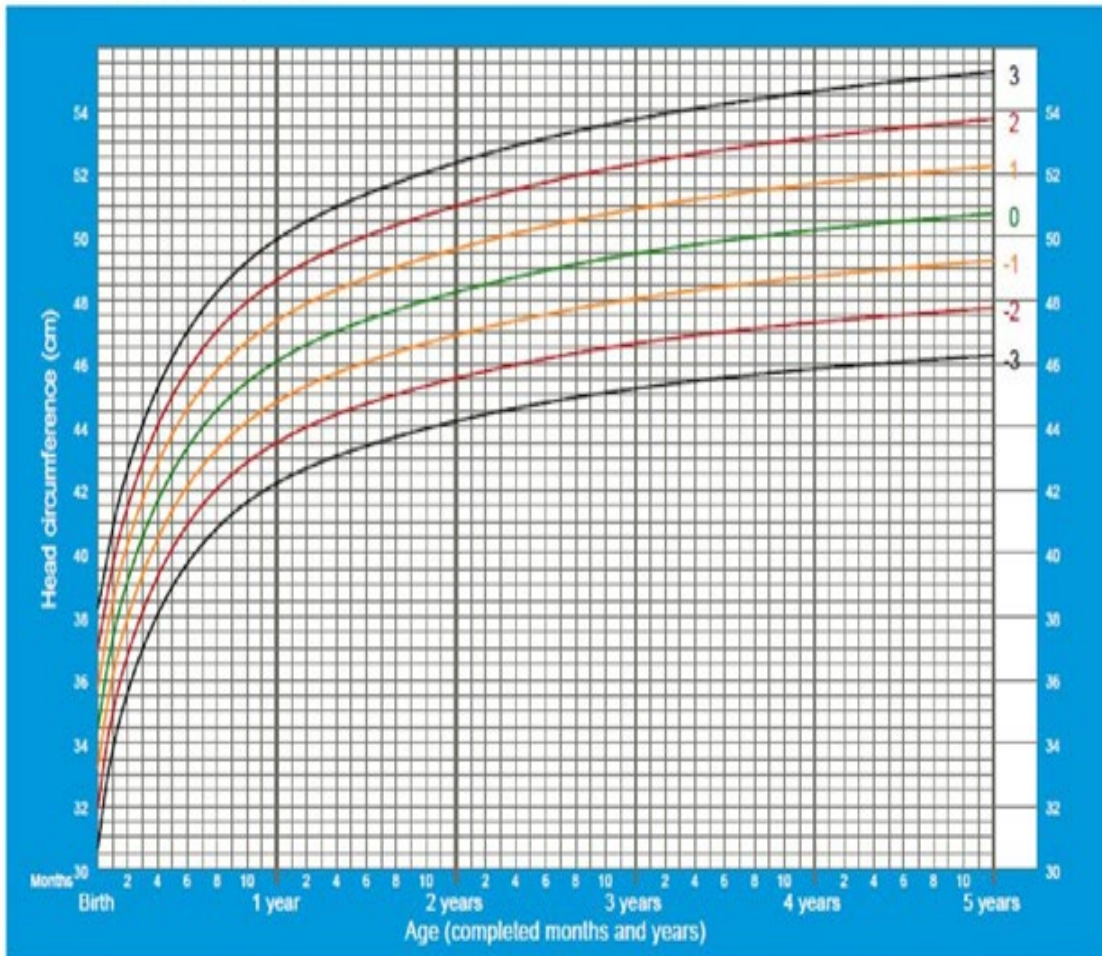
Birth to 5 years (z-scores)



WHO Child Growth Standards

Head circumference-for-age BOYS

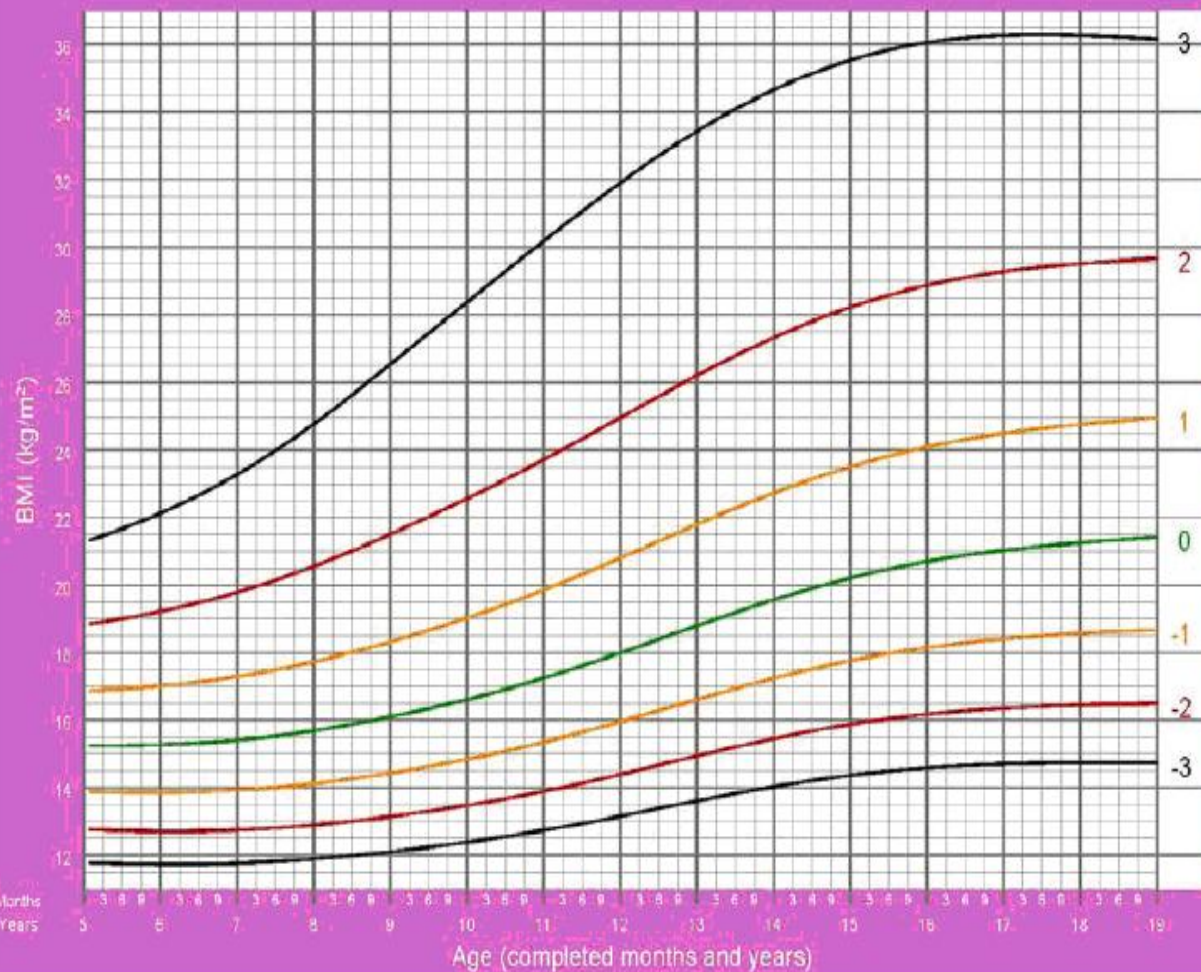
Birth to 5 years (z-scores)



WHO Child Growth Standards

BMI-for-age GIRLS

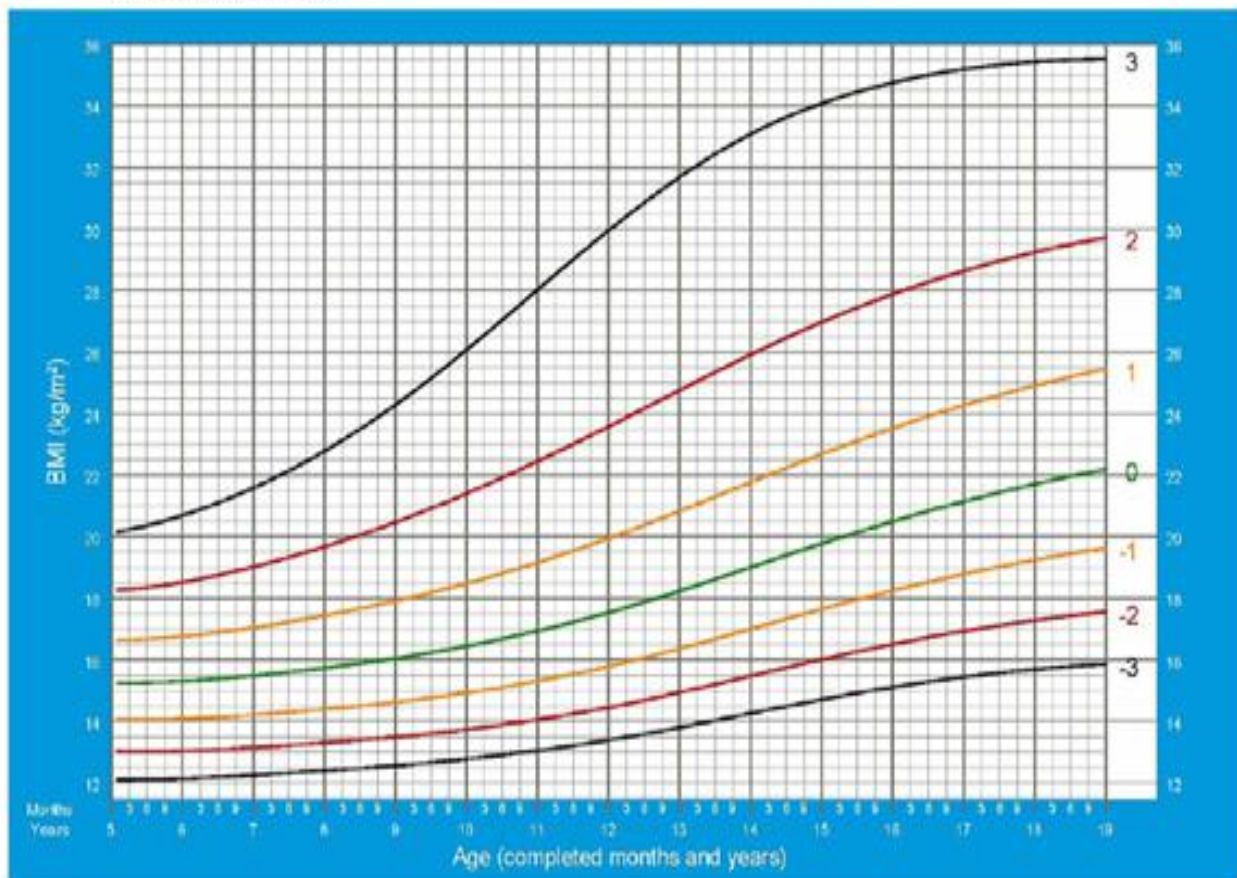
5 to 19 years (z-scores)



2007 WHO Reference

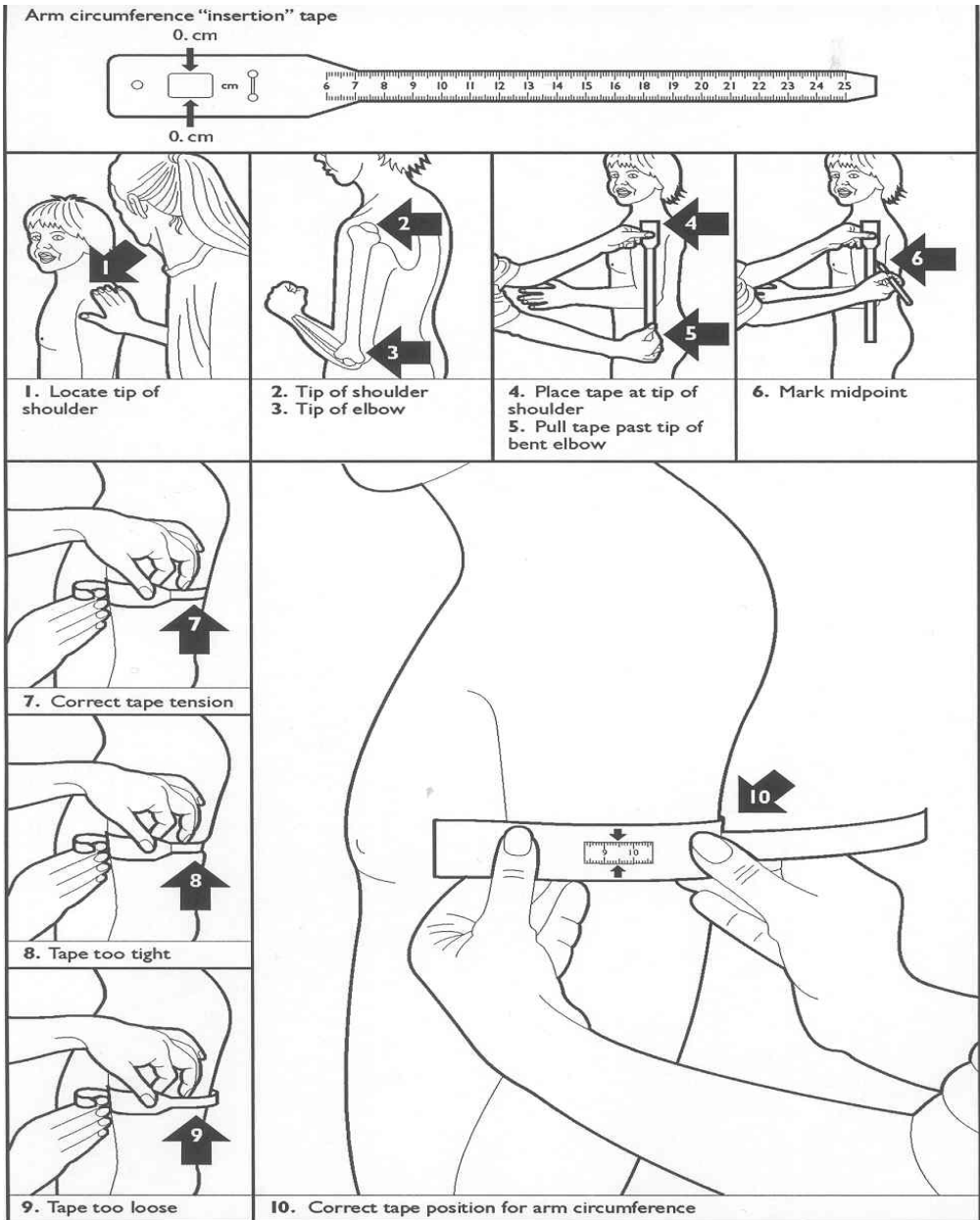
BMI-for-age BOYS

5 to 19 years (z-scores)



2007 WHO Reference

Annex 2. Instruction for MUAC measurement



Source: How to Weigh and Measure Children: Assessing the Nutritional Status of Young Children, United Nations, 1986.

Annex 3: Common Ethiopian meals, each providing 700 – 850 kilocalories

One Enjera with...			
	1 sauce	2 sauces	3 sauces
Shiro	5 small ladles shiro	4 small ladles shiro and one small ladle vegetable	3 small ladles shiro, 1small ladle vegetable alicha, and 1small ladle tomato salad
Miser	3 big ladles miser	2 big ladles miser sauce and 1 big ladle gommen	1 big ladle miser sauce , 1small ladle gommen, and 1 small ladle tomato
Bozena shiro	5 small ladles bozena Shiro	4 small ladles bozena shiro and 1 small ladle tomato salad	3 small ladles bozena shiro and 1 small ladle vegetable alicha, 1 small ladle tomato salad
Pumpkin	5 small ladles pumpkin	4 small ladles pumpkin and 1 small ladle vegetable alicha	3 small ladles pumpkin sauce, 1 small ladle vegetable alicha, and 1 small ladle tomato salad
Potato	5 small ladles potato	4 small ladles potato and 1 small ladle vegetable Alicha	3 small ladles potato sauce , 1 small ladle vegetable alicha, and 1small ladle ater kick
Ater kick	5 small ladles ater kick	4 small ladles ater kick and 1 small ladle vegetable alicha	3 small ladles ater kick , 1 small ladle vegetable alicha, and 1small ladle tomato salad
Gommen	3 big ladles gommen	2 big ladles gommen with 2 small ladles shiro	1 big ladle gommen, 1 small ladle shiro, and 1small ladle vegetable alicha
Siga	4-5 small ladles meat	3 small ladles meat sauce 1 small ladle vegetable	2 ladles meat sauce, 1 small ladle vegetable alicha and 1 small ladle tomato salad

N.B. Small ladle = 50 g

Medium ladle = 70 g

Large ladle =100 g

Annex 4: Energy Values of Locally Available Snacks (Mekses)

Snack (ingredients)	Amount in grams/ml	Calories	Protein
Kolo (roasted barley, wheat)	50 grams (1 small ladle)	195	5.1
Nifro (boiled wheat and chickpeas)	70 grams (1 medium ladle)	125	301
Kitta/ambasha	100 grams (1 slice)	222	6.8
Beso juice (beso and sugar)	55 grams (5 medium tablespoons beso and 1 teaspoon sugar)	205	5.05
Beso firfir (beso and oil)	65 grams (6 medium tablespoons beso and 1 teaspoon oil)	267	6.06
Sweat potato	100 grams (1 average-size sweet potato)	134	0.5
Boiled milk	140 ml (2 large coffee cups)	103	4.7
Tea with sugar	10 grams sugar (2 teaspoons)	40	0
Ashuk (roasted and boiled beans)	70 grams (1 large coffee cup)	192	11.48
Mango	100 grams (1 average-size mango)	44	0.30
Banana	100 grams (1 average-size banana)	88	0.8
Fried bread unleavened (wheat flour,spiced pepper, oil, salt, water)	300 grams	668	13.7
Thick porridge (wheat flour, oil, spiced pepper, salt, water)	350 grams	591	13.9
Chopped enjera with meat sauce (enjera, meat sauce)	300 grams	466	22.3
Chopped enjera without meat sauce (onion, pepper, oil, salt, water)	265 grams	456	7.6
Split wheat (kinche) (wheat, butter, salt)	160 grams	626	13.7

Annex 5: Dosage of antiretroviral drugs for adults and adolescents

Nucleoside reverse-transcriptase inhibitors (NRTIs)	
Abacavir (ABC)	300 mg twice daily or 600 mg once daily
Emtricitabine (FTC)	200 mg once daily
Lamivudine (3TC)	150 mg twice daily or 300 mg once daily
Zidovudine (AZT)	250–300 mg twice daily
Nucleotide reverse-transcriptase inhibitors (NtRTIs)	
Tenofovir (TDF)	300 mg once daily
Non-nucleoside reverse-transcriptase inhibitors (NNRTIs)	
Efavirenz (EFV)	400–600 mg once daily
Etravirine (ETV)	200 mg twice daily
Nevirapine (NVP)	200 mg once daily for 14 days, followed by 200 mg twice daily
Proteases inhibitors (PIs)	
Atazanavir + ritonavir (ATV/r)	300 mg + 100 mg once daily
Darunavir + ritonavir (DRV/r)	800 mg + 100 mg once daily ^a or 600 mg + 100 mg twice daily ^b
Lopinavir/ritonavir (LPV/r)	400 mg/100 mg twice daily
	Considerations for individuals receiving TB therapy
	In the presence of rifabutin, no dose adjustment required. In the presence of rifampicin, adjusted dose of LPV/r: LPV 800 mg + RTV 200 mg twice daily or LPV 400 mg + RTV 400 mg twice daily). or, SQV/r (SQV 400 mg + RTV 400 mg twice daily), with close monitoring.
Integrase strand transfer inhibitors (INSTIs)	
Dolutegravir (DTG)	50 mg once daily
Raltegravir (RAL)	400 mg twice daily

a For individuals with no previous use of protease inhibitors.

b For individuals with previous use of protease inhibitors.

Annex 6: Dosage of antiretroviral drugs in children twice daily basis

Simplified dosing of solid and oral liquid formulations **for twice-daily dosing** for infants and children 4 weeks of age and older

Drug	Formulations (tablet, capsule or oral liquid) and strength (mg/tab. or mg/ml for liquids)	Dose as number of tablets or ml by weight band, morning(AM) and evening (PM)										Adult tablets and their strength in mg	Dose as number of tablets by weight band			
		3 – 5.9Kg		6-9.9kg		10-13.9kg		14-19.9kg		20-24.9kg			25-29.9 kg		30-34.9	
		AM	PM	AM	PM	AM	PM	AM	PM	AM	PM			AM	PM	AM
Solid formulations																
ABC/3TC	120mg/60mg (scored, dispersible tab)	0.5	0.5	0.5	1	1	1	1	1.5	1.5	1.5	600mg/300mg tab	0.5	0.5	0.5	0.5
ABC/3TC	60mg/30mg (scored, dispersible tab)	1	1	1.5	1.5	2	2	2.5	2.5	3	3	600mg/300mg tab	0.5	0.5	0.5	0.5
AZT/3TC	60mg/30mg (scored, dispersible tab)	1	1	1.5	1.5	2	2	2.5	2.5	3	3	300mg/150mg tab	1	1	1	1
AZT/3TC/NVP	60mg/30mg/50mg (scored, dispersible tab)	1	1	1.5	1.5	2	2	2.5	2.5	3	3	300mg/150mg/200mg tab	1	1	1	1
AZT	60mg (dispersible tablet)	1	1	1.5	1.5	2	2	2.5	2.5	3	3	300mg tab	1	1	1	1
DRV ^a	75mg tablet	-	-	-	-	-	-	5	5	5	5	600mg				
LPV/r ^{b, c}	40mg/10mg oral pellets in capsules	2	2	3	3	4	4	5	5	6	6	200mg/50mg tab	2	1	2	1
	100mg/25mg tablet	-	-	-	-	2	1	2	2	2	2		2	1	2	1
NVP ^d	50mg (scored, dispersible	1	1	1.5	1.5	2	2	2.5	2.5	3	3	200mg tab	1	1	1	1
RAL	100 mg, chewable tablet	-	-	-	-	-	-	1	1	1.5	1.5	400mg tab	1	1	1	1
	25mg, chewable tablet	-	-	-	-	3	3	4	4	6	6		1	1	1	1
	100 mg granules per sachet	0.25	0.25	0.5	0.5	-	-	-	-	-	-		1	1	1	1
RTV ^f	25mg tablet	-	-	-	-	2	2	2	2	2	2					

Liquid formulations															
AZT	10mg/ml	6ml	6ml	9ml	9ml	12ml	12ml	-	-	-	-		-	-	-
NVP ^d	10mg/ml	5ml	5ml	8ml	8ml	10ml	10ml	-	-	-	-		-	-	-
LPV/r ^e	80mg/20mg/ml	1ml	1ml	1.5ml	1.5ml	2ml	2ml	2.5ml	2.5ml	3ml	3ml		-	-	-
DRV ^a	100mg/ml	-	-	-	-	2.5ml	2.5ml	3.5ml	3.5ml	-	-		-	-	-
RTV ^f	25mg tablet	-	-	-	-	2	2	2	2	2	2		2	2	2
	80mg/ml	0.5ml	0.5ml	0.5ml	0.5ml	0.5ml	0.5ml	-	-	-	-		-	-	-

^aDRV must be administered with 0.5ml of RTV 80mg/mL oral suspension if the child weighs less than 15kg and with RTV 50mg solid formulation for children weighing 15–30kg. DRV/r should not be used in children younger than 3 years of age.

^bLPV/r heat-stable oral pellets (presented in a capsule) must be administered by **opening the capsule and pouring the pellets over a small soft food** at room temperature and swallowed without chewing. The pellets **MUST NOT** be stirred, crushed, dissolved/dispersed in food. The capsules containing LPV/r oral pellets **must not** be swallowed whole.

^cThe LPV/r heat-stable tablet must be swallowed whole and should not be split, chewed, dissolved or crushed.

^d NVP dose escalation with half dose for 2 weeks when initiating ART is still recommended to avoid toxicity from high initial NVP levels.

^e LPV/r liquid requires a cold chain during transport and storage.

^f RTV is a booster for other protease inhibitors such as LPV, ATV and DRV.

ABC=Abacavir; AZT=Zidovudine; 3TC= Lamivudine; DRV=Darunavir; LPV/r= Lopinavir combined with ritonavir; NVP=Nevirapine; RAL=Raltegravir; RTV=Ritonavir; ATV=Atazanavir.

Simplified dosing of solid and oral liquid formulations for **once-daily dosing** for infants and children 4 weeks of age and older

Drug	Formulations (tablet, capsule or oral liquid) and strength (mg/tab. or mg/ml for liquids)	Dose as number of tablets or ml by weight band					Adult tablets and their strength in mg	Dose as number of tablets by weight band	
		3 – 5.9Kg	6-9.9kg	10-13.9kg	14-19.9kg	20-24.9kg		25-29.9kg	30-34.9kg
DTG	50 mg tab					1	50mg	1	1*
EFV ^a	200mg tablet	-	-	1	1.5	1.5	200 tab	2	2
	50mg capsule	-	-	4	6	6	600mg		
ABC/3TC	120mg/60mg (scored, dispersible tablet)	1	1.5	2	2.5	3	600mg/300mg	1	1
ABC/3TC	60mg/30mg (scored, dispersible tablet)	2	3	4	5	6	600mg/300mg	1	1
ATV ^b	100mg capsule	-	-	1	2	2		2 capsules of 100mg	1 tablet of 300mg
DRV ^c	150mg tablet				4	4	600mg	1	1
	600mg tablet				1	1			
TDF/3TC	300mg/300mg tablet	-	-	One third	One half	Two third		1	1
TDF/3TC/EFV	300mg/300mg/600mg tablet	-	-	One third	One half	Two third		1	1

^aTwo EFV 50mg capsules is administered in combination with EFV 200mg tablet for children weighing 14-24.9KG.

^b ATV is only approved for use for children 3 months and older. ATV single-strength capsules should be administered with RTV 100 mg for all weight bands. The ATV powder formulation enables administration of ATV to infants and children as young as 3 months. Infants and children weighing 5–10 kg should be administered 200 mg of ATV with 80 mg of RTV oral solution (5 ml).

^c DRV in combination with RTV should be used, in children older than 3 years, once daily when this is used without previous exposure to PI. While approved dosing for 30-35 kg is 675 mg, preliminary data from adult studies suggest that even lower DRV doses may be effective, therefore use of 600 mg dose was extended to the entire 25-35 kg weight band.

^d TDF is only approved for use for children 2 years and older. Target dose: 8 mg/kg or 200mg/m²(maximum 300mg). A child would not receive more than 25% above the maximum target dose or more than 5% below the minimum target dose.

EFV=Efavirenz; TDF=Tenofovir disoproxil fumarate.

Annex 7: Pediatric ARV Drug Formulations, Side Effects and Special considerations in Children

Drug / Formulation	Dosing recommendation and instructions	Side Effects
Nucleoside reverse transcriptase		
Lamivudine (3TC) : Formulations:Fixed-Dose Combination Tablets <i>With Abacavir :</i> • 3TC60mg+120mgABC, scored and dispersible tablet <i>With Zidovudine :</i> • 3TC30mg+AZT60mg, dispersible tablet <i>With Zidovudine and Nevirapine:</i> • 3TC30mg+AZT60mg+NVP50mg, dispersible tablet	Instruction for dosing/administration: • Can be given with food. • For those who can't swallow whole tablet, <u>dispersible tablets</u> can be <u>dissolved</u> with 2 teaspoons (10ml) water in a small and clean container and taken immediately. • <u>Scored tablets</u> can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: • Adult formulations are to be used for children older than 10 years with body weight > 35kg. • <i>Please see dosing Chart for dose recommendation.</i> Storage: • Can be stored at room temperature	Side effects of 3TC: Common: head ache, nausea, abdominal pain. Less common: pancreatitis, neutropenia, increased LFTs. Hepatitis may exacerbate after discontinuation of lamivudine in the setting of chronic Hepatitis B virus infection. For those patients with HBV and HIV co-infection please consult “the national guideline on management of HBV infection-2016.”
Abacavir (ABC) : Formulations:Fixed-Dose Combination Tablets <i>With Lamivudine :</i> • ABC120mg+3TC60mg, scored and dispersible tablet	Instruction for dosing/administration: • Can be given with food. • For those who can't swallow whole tablet, <u>dispersible tablets</u> can be <u>dissolved</u> with 2 teaspoons (10ml) water in a small and clean container and taken immediately. • <u>Scored tablets</u> can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: • Adult formulations are to be used for children older than 10 years with body weight > 35kg. • <i>Please see dosing Chart for dose recommendation.</i> Storage: • Can be stored at room temperature	Side effects of ABC: Common: head ache, GI upset and rash. Less common: lactic acidosis, hepatomegaly with steatosis. Life threatening: potentially fatal hypersensitivity reaction (fatigue, rash, nausea and vomiting, sore throat, joint and muscle pain, cough, and dyspnea). Occurrence of hypersensitivity reactions requires immediate and permanent discontinuation of ABC.

		Advise patients/care givers what to do if hypersensitivity reactions occur at home. DO NOT re-challenge after hypersensitivity reaction.
Zidovudine (AZT or ZDV) Formulations: Oral solution, 50mg/5ml, 100 ml Fixed-Dose Combination Tablets: <i>With Lamivudine :</i> • AZT60mg + 3TC30mg, dispersible tablet <i>With Lamivudine and Nevirapine:</i> AZT60mg+3TC30mg+NVP50mg, dispersible tablet	Instruction for dosing/administration: • Can be given with food. • The syrup is for infant prophylaxis. • <u>For those who can't swallow whole tablet, dispersible tablets can be dissolved</u> with 2 teaspoons (10ml) water in a small and clean container and taken immediately. • <u>Scored tablets</u> can be broken to administer lower doses for younger children as recommended in the national ART guidelines. Note: • <i>Adult formulations are to be used for children older than 10 years with body weight > 35kg.</i> • <i>Please see dosing Chart for dose recommendation.</i> Storage: • Can be stored at room temperature • Syrup is light sensitive, store in a brown glass jar and protect from sun light.	Side effects of AZT: Common: neutropenia, anemia, granulocytopenia, macrocytosis, and head ache. Less common: myositis, myopathy, mitochondrial disease.

Non-nucleoside reverse transcriptase:		
Efavirenz (EFV) Formulations: Efavirenz 200 mg, scored tablet. Efavirenz 50 mg, tablet or capsule	Instructions for dosing/administration: • Used only for children ≥ 3 yrs. • Can be given with food but avoid administration with a high-fat meal because of potential for increased absorption that leads to CNS toxicity. • Bedtime dosing is recommended, particularly during the first 2 to 4 weeks of therapy, to improve tolerability of central nervous system side effects. • <u>Scored tablets</u> can be broken to administer lower doses for younger children as recommended in the national ART guidelines. • Efavirenz can be swallowed as a whole capsule or tablet. For children who can't swallow the whole capsule/tablet, the capsule content can be administered by sprinkling on food to avoid/mask peppery taste of EFV. For sprinkling: • Hold capsule horizontally over a small container and carefully twist to open to avoid spillage. • Gently mix capsule contents with 1–2 teaspoons of an age-appropriate soft food (e.g. yogurt or banana or infant formula milk) at room temperature. • Administer the mixture that contains the sprinkle using a 10-mL syringe. • After administration, an additional 2 teaspoons of food or infant formula must be added to the container, stirred, and dispensed to the patient. Note: • <i>Adult formulations are to be used for children older than 10 years with body weight > 35kg.</i> • <i>Please see dosing Chart for dose recommendation.</i> Storage: • Can be stored at room temperature.	Side effects of EFV: Common: skin rash, dizziness, somnolence, insomnia, abnormal dreams, confusion, hallucinations, impaired concentration, psychosis, seizures, suicidality. Less common: increased LFTs.

Nevirapine (NVP) Formulations: Nevirapine oral syrup, 50mg/5ml (10mg/ml) Fixed-Dose Combination Tablets: <i>With Lamivudine and zidovudine :</i> AZT60mg+3TC30mg+NVP50mg, dispersible tablet	Instructions for administration: • Can be given with food. • The syrup is for infant prophylaxis • <u>Scored tablets</u> can be broken to administer lower doses for younger children as recommended in the national ART guidelines. • <u>For those who can't swallow whole tablet, dispersible tablets can be dissolved</u> with 2 teaspoons (10ml) water in a small and clean container and taken immediately. Caution: Multiple drug interactions. Note: • <i>Adult formulations are to be used for children older than 10 years with body weight > 35kg.</i> • <i>Please see dosing Chart for dose recommendation.</i> Storage: Store at room temperature.	Side effects of NVP: Common: skin rash, head ache, nausea, diarrhea. Patients should be warned of rash. Do not escalate dose if rash occurs. Less common: increased LFTs. Life threatening: Steven Johnsons syndrome, TEN, fatal hepatitis. • For SJS and TEN discontinue drug and do not re-challenge.
Protease Inhibitors		
Lopinavir (LPV) Formulations: Combined with Ritonavir: Oral syrup, (LPV 80mg+RTVLPV20mg)/ml Oral pellets, (LPV40mg+Ritonavir10mg)/capsule Tablet, (LPV100mg+Ritonavir25mg)	Instructions for administration: • Do not administer to neonates before a post-menstrual age of 42 weeks and a postnatal age of at least 14 days. • Should be taken with food. • For the capsules: open the capsules, add the pellets from the capsule content onto small soft or milk (not more than 2ml) and administer immediately. • Tablets should not be opened or crushed, swallow whole. Tablets are for older children. • Liquid has low volume but bitter taste. Caution: Multiple drug interactions. Note: • <i>Adult formulations are to be used for children older than 10</i>	Side effects of LPV Common: diarrhea, head ache, nausea, vomiting,, increase in blood lipids Less common: pancreatitis, diabetes, hyperglycemia, hepatic toxicity, fat redistribution.

	<i>years with body weight > 35kg.</i> • <i>Please see dosing Chart for dose recommendation.</i> Storage: • For the syrup, store it in a refrigerator at least until dispensed. Can be stored at room temperature (25⁰C) for 60 days. • For tablets and oral pellets, store at room temperature.	
Ritonavir (RTV) Formulations: Combined with Lopinavir or Atazanavir:	Take with food to increase absorption and reduce GI side effects. Oral solution must be refrigerated. Can be kept at room temperature (25⁰C) if used within 30 days. Bitter taste, coat mouth with peanut butter or chocolate milk.	Side effects of RTV: Common: N/V, diarrhea, headache, abdominal pain, anorexia Less Common: circumoral paraesthesia, increased LFTs, lipodystrophy, elevated cholesterol and triglycerides, hyperglycemia.

Annex 8: Grading of Adverse Events in children

Parameter	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Severe Life-threatening
Diarrhoea ≥1 year of age <1 year of age	Transient or intermittent episodes of unformed stools OR increase of ≤3 stools over baseline per day. Liquid stools (more unformed than usual) but usual in number.	Persistent episodes of unformed to watery stools or increase of 4-6 stools over baseline per day. Liquid stools with increased number of stools or mild dehydration.	Grossly bloody diarrhoea or increase of ≥7 stools per day or IV fluid replacement indicated. Liquid stools with moderate dehydration.	Life-threatening consequences (e.g. hypotensive shock). Liquid stools resulting in severe dehydration with aggressive rehydration indicated or hypotensive shock.
Nausea	Transient (<24 hours) or intermittent nausea with no or minimal interference with oral intake.	Persistent nausea resulting in decreased oral intake for 24-48 hours.	Persistent nausea resulting in minimal oral intake for >48 hours or aggressive rehydration indicated (e.g. IV fluids).	Persistent nausea with no or minimal oral intake resulting in dehydration with aggressive rehydration indicated.
Vomiting	Transient or intermittent vomiting with no or minimal interference with oral intake.	Frequent episodes of vomiting with no or mild dehydration.	Persistent vomiting resulting in orthostatic hypotension or aggressive rehydration indicated (e.g. IV fluids).	Life threatening consequences (e.g. hypotensive shock).
Acute systemic allergic reaction	Localized urticaria (wheals) lasting a few hours.	Localized urticaria with medical intervention indicated or mild angio oedema.	Generalized urticaria or angio oedema with medical intervention indicated or symptomatic mild bronchospasm.	Acute anaphylaxis or life-threatening bronchospasm or laryngeal oedema.
Pancreatitis	NA	Symptomatic and hospitalization not indicated (other than emergency treatment).	Symptomatic and hospitalization not indicated (other than emergency treatment).	Life-threatening consequences (e.g. Circulatory failure, haemorrhage, sepsis).
			Diffuse macular, maculopapular, or	Extensive or generalized bullous

Parameter	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Severe Life-threatening
Rash	Localized macular rash	Diffuse macular, maculopapular, or morbilliform rash or target lesions.	morbilliform rash with vesicles or limited number of bullae or superficial ulcerations of mucous membrane limited to one site.	lesions or Stevens-Johnson syndrome or ulceration of mucous membrane involving two or more distinct mucosal sites or Toxic Epidermal Necrolysis.
Alteration in personality-behaviour or mood	Alteration causing no or minimal interference with usual social and functional activities	Alteration causing greater than minimal interference with usual social and functional activities.	Alteration causing inability to perform usual social and functional activities and intervention indicated.	Behaviour potentially harmful to self or others or with life-threatening consequences.
Altered Mental Status	Changes causing no or minimal interference with usual social and functional activities	Mild lethargy or somnolence causing greater than minimal interference with usual social and functional activities.	Onset of confusion, memory impairment, lethargy, or somnolence causing inability to perform usual social and functional activities.	Onset of delirium, obtundation, or coma.
<i>Source: Antiretroviral therapy of HIV infection in infants and children in resource-limited settings. WHO 2006</i>				

Annex 9: Grading toxicities in children by selected laboratory findings

Parameter	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 Severe Life threatening
Haemoglobin (g/dL)	8.5 – 10.0	7.5 – <8.5	6.5 – <7.5	<6.5
ANC (mm ³)	750 – <1,000	500 – 749	250 – 500	<250
Platelets (mm ³)	100,000 – <125,000	50,000 – <100,000	25,000 – <50,000	<25,000 or bleeding
ALT (SGPT)	1.25 – 2.5 x ULN	2.6 – 5.0 x ULN	5.1 – 10.0 x ULN	>10.0 x ULN
AST (SGOT)	1.25 – 2.5 x ULN	2.6 – 5.0 x ULN	5.1 – 10.0 x ULN	>10.0 x ULN
Bilirubin (>2 weeks of age)	1.1 – 1.5 x ULN	1.6 – 2.5 x ULN	2.6 – 5.0 x ULN	>5.0 x ULN
Lipase	1.1 – 1.5 x ULN	1.6 – 3.0 x ULN	3.1 – 5.0 x ULN	>5.0 x ULN
Pancreatic amylase	1.1 – 1.5 x ULN	1.6 – 2.0 x ULN	2.1 – 5.0 x ULN	>5.0 x ULN
Cholesterol (fasting, <18 years old) mg/dL	170 – <200	200 – 300	>300	NA
Glucose, serum, Nonfasting (mg/dL)	116 – <161	161 – <251	251 – 500	>500
Glucose, serum, high: Fasting (mg/dL)	110 – <126	126 – <251	251 – 500	>500
Lactate	<2.0 x ULN without acidosis	≥2.0 x ULN without acidosis	Increased lactate with pH <7.3 without life threatening consequences or related condition present	Increased lactate with pH <7.3 with life threatening consequences (e.g. neurological findings, coma) or related condition present
Triglycerides: Fasting (mg/dL)	NA	500 – <751	751 – 1,200	>1,200
<i>Source: Antiretroviral therapy of HIV infection in infants and children in resource-limited settings. WHO 2006</i>				

Annex 10: Adverse Drug Event Reporting form

Patient Name (abbreviation)	Card No	Age, Date of birth	Sex	Weight	Height	
-----	-----	-----	-----	-----	-----	
Ethnic group-----		Substance of abuse-----				
Information on suspected drug/vaccine S=suspected drug C=concomitantly used drugs						
Drug name(write all information including brand name batch no and manufacturer)	S/C	Dose/dosage form, route, frequency	Date drug taking was started (D/M/Y)	Date drug reaction started (D/M/Y)	Date drug taking was stopped (D/M/Y)	Indication (Reason for drug use)
Adverse drug event description(include all available laboratory test results)						
Reaction necessitated Discontinuation of drug/s ☐ YES ☐ No Hospitalization prolonged ☐ YES ☐ No			Reaction subside after D/C of suspected drug ☐ YES ☐ No ☐ Information not available Reaction reappear after restart of suspected drug ☐ YES ☐ No ☐ Information not available			
Treatment of reaction						
Outcome: ☐ Died due to the adverse event ☐ Died, drug may be contributory ☐ Not yet recovered ☐ Recovered without sequelae ☐ Recovered with sequelae ☐ Unknown						
Sequelae						
Relevant medical conditions such as allergies, renal disease, liver disease, other chronic diseases, pregnancy etc						
Reported by: Name	Profession:	Email address:	Telephone			
-----	-----	-----	-----			
Name of health institution				Date		
Product quality problem: Color change, separating of components, powdering, crumbling, caking, molding, change of odor, incomplete pack, suspected contamination, poor packaging/poor labeling, etc (Write if anything different than given above)						
Drug trade name	Batch No	Registration no	Dosage form and strength	Size /type of package		
For office use only						
Received on:			Registration no:			
Key: D/M/Y ; Date /Month/Year D/C; Discontinue treatment Y;YES N;NO						

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All suspected reactions to drugs	
Unknown or unexpected reactions	
Serious adverse drug reactions	
Unexpected therapeutic effects	
All suspected drug interactions	
Product quality problems	
Treatment failures	
Medication errors	
NB. Drugs includes	
<i>This ADE reporting form was prepared</i>	Conventional drugs
<i>by EFDA in collaboration with MSH/SPS</i>	Herbal drugs
<i>and the financial support from USAID</i>	Traditional medicines
	Biologicals
	Medical supplies
	Medicated cosmetics
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Food, Medicine and Health Care Administration and Control Authority
Regulatory Information Development and Dissemination Team
P.O.Box 5681-Tel.0115-523142
Addis Ababa, Ethiopia

Annex 11. Family Matrix/ Family HIV care and treatment enrollment form

Date: ____/____/____

MRN number _____

ART client: Unique ART number of the index client _____

Pre ART Client: Pre ART serial number _____

Mother, Age _____ 0 1 2 3 4 5 6 7 _____ IIAN: _____	Father, Age _____ 0 1 2 3 4 5 6 7 _____ IIAN: _____		
1st child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	2nd child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	3rd child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	4th child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____
5th child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	6th child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	7th child, Age _____ 0 1 2 3 4 5 6 7 _____ UAN: _____	

Legend

0 = Not counseled	1 = Counseled not tested	2 = HIV exposed infant	3 = HIV negative
4 = HIV positive	5 = Pre-ART	6 = ART	7 = Dead

Remark _____

NB:

- For clients who are on ART please write the unique ART number in the box
- Identify the index case by putting asterisk (*) or by marking the box with highlighter
- A child is defined here less than 14 years of age
- HIV exposed infant is an infant less than 18 months of age who is HIV antibody positive or born to HIV positive mother

Annex 12: Algorithm for Treatment of HIV/Leishmaniasis Co-infection



Annex 13. Appointment Spacing Model of HIV Care, Client Classification Tool

A. Client's Current data

1. Client Name _____	Fathers' Name _____	Garand Fathers' _____	
For child only: - Mother's /Fathers Name: _____			
2. Age: _____ years/months for < 1years	Sex: ☐ M ☐ F		
3. Marital Status: ☐ Never Married ☐ Married ☐ Divorced ☐ Widowed			
4. Level of education: ☐ No education ☐ Primary ☐ Secondary ☐ Tertiary ☐ Other/specify _____			
5. Occupation: _____			
6. Address: Region _____ Zone/Sub city _____ Woreda _____ Kebele _____ House No. _____ Telephone _____			
7. Client reside within the catchments area ☐ Yes ☐ No, If no; Counsel and encourage for referral to a nearby facility to his/her home			
8. Date ART started(EC) _____		Month on ART _____	
9. Unique ART ID: _____ / _____ / _____ / _____		MRN _____	
10. Current ART regimen			
☐ First Line Regimen: _____		☐ Date (ET) Initiated ____/____/____ (dd/mm/yyyy)	
☐ Second Line Regimen: _____		☐ Date (ET) Initiated ____/____/____ (dd/mm/yyyy)	
11. Current ART Adherence ☐ Good >95% ☐ Fair (85 - 94%) ☐ Poor <85%			
12. Is the client pregnant ☐ Yes ☐ No		Breastfeeding ☐ Yes ☐ No	
13. CD4 count (CD4% for <5 years): Most recent result _____ cells/ul		Date (ET) ____/____/____	
14. If VL test done: Most recent result _____ copies/ml			
15. Current Clinical observations/symptoms: WHO (Treatment) Staging : ☐ I ☐ II ☐ III ☐ IV			

B. Classification based on their current data and care needs

S.N.	Category	Criteria or Description
	Category 1a	People, who present when well, potentially with higher CD4 cell(>200cells/ml) counts but either not yet initiated on ART or on ART for < 12 months
	Category 1b	Pregnant/Breast feeding women with no other complications, rapidly growing children (0–5 years old) and adolescents with no advanced disease or other complications.
	Category 2	People with advanced disease are defined as those presenting to care with a CD4 count below 200cells/mm3 or WHO disease stages 3 and 4.
	Category 3	Those who are already on ART but need careful monitoring to ensure timely action as required (Treatment failure suspects, patients with other chronic comorbidities, patients with identified adherence barriers etc).
	Category 4	Stable individuals who have received ART for at least one year and have no adverse drug reactions that require regular monitoring and have good understanding of lifelong adherence and evidence of treatment success. ^a

a: Two consecutive viral load measurements below 1000 copies/mL, with no current illnesses, excluding children, adolescents, pregnant and lactating women; In the absence of viral load monitoring, rising CD4 cell counts or CD4 counts above 200cells/mm3, an objective adherence measure either from the clients self-report or by doing pill count; can be used to indicate treatment success.

C. If there is change in category specify

	New category	Reason for category change
Change one		
Change Two		
Change Three		

Annex 14. ASMof HIV Care Client Enrolment Register

Year _____

Registration										ARV Reg.	Follow Up Decision						
S. No	Date of enrolment	MRN	UAN	Full Name	Age	Sex	ART start date	Month on ART	First line regimen	Current category	Counseled (✓)	If they are couples write UAN of the other partner	New category	If category changed reason (write code) and date changed	Date and Reason of Termination	Remark	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	

Reasons for Termination

1. Category change
2. Lost/ Drop
3. TO
4. Dead

Reasons for change in client category

1. Pregnant/breast feed
2. Developed advanced disease
3. Other

INSTRUCTIONS FOR APPOINTMENT SPACING MODEL OF HIV CARE ENROLMENT REGISTER

Register (HC/Hospital-Appointment spacing register) kept in ART room. The Appointment spacing register is completed by the Data Clerk.

Location information to be completed at front of register:

Region	Write the region where the Health Facility is located
Woreda /Sub-City	Write the woreda/sub-city where the Health Facility is located
Name of Health Facility	Write the name of the health facility where the Health Facility is located.
Register begin date	Enter the date of the first entry in the register, written as (EC) Day/Month /Year (DD/MM/YY)
Register end date	Enter the date of the last entry in the register, written as (EC) Day/Month /Year (DD/MM/YY)

SN	Datum	Comments
1.	Serial Number	Sequential serial number in registration book; to be entered on client's registration card for later identification in register
2.	Date of enrollment	Write date of enrollment to appointment spacing model, written as (EC) Day / Month / Year (DD/ MM/YY)
3.	MRN	Unique individual identifier used on medical information folder, for HC and hospital.
4.	UAN	Patients should be assigned a unique ART number when they leave pre-ART and begin ART. This will be: region number / facility type code / specific facility code / patient assigned number. Region number: the following code numbers are used: 1. Tigray 2. Afar 3. Amhara 4. Oromia 5. Somali 6. Benishangul Gummuz (BG) 7. SNNPR (SN) 8. Gambella 9 Harari 10 Addis Ababa 11. Dire Dawa Facility type code: 08 = Hospital 09 = Health Center Specific facility code: Each HC / hospital in each region is coded with three digits starting from 001. These specific facility codes are assumed to be given by regions together with federal, which means it is pre-coded and given to each facility centrally. Patient assigned number: A 5-digit number unique within the facility; the first patient to start ART in the clinic will be given 00001
5.	Full Name	Write the patient's first name, father's and grandfather's name.
6.	Age	Enter the age of patient's age in years – yr. For example, a 17-year-old child is entered as 17 yr
7.	Sex	M=Male; F=Female
8.	ART Start Date	Enter date patient started ART, written as (EC) Day / Month / Year (DD/ MM/YY)
9.	Months on ART	Write months on ART in number
10.	First line regimen	Write the code for the 1st line regimen that that patient has being taken
11.	Current category	Write the patients current category as Category 1a: - People, who present when well, potentially with higher CD4 cell(>200cells/ml) counts but either not yet initiated on ART or on ART for < 12 months Category 1b: - Pregnant/Breast feeding women with no other complications, rapidly growing children (0–5 years old) and adolescents with no advanced disease or other complications. Category 2: - People with advanced disease are defined as those presenting to care with a CD4 count below 200cells/mm3 or WHO disease stages 3 and 4. Category 3: - Those who are already on ART but need careful monitoring to ensure timely action as required (Treatment failure suspects, patients with other chronic comorbidities, patients with identified adherence barriers etc). Category 4: - Stable individuals who have received ART for at least one year and have no adverse drug reactions that require regular monitoring and have good understanding of lifelong adherence and evidence of treatment success. a
12.	Counselled	Tick if the clients offered counselling on the new appointment spacing
13.	If they are couples write UAN of the other partner	If the client has couples write UAN of the other partner
14.	New category	If client category changed from one category to other write the new category
15.	If category changed, specify reason and date	Write the code for reasons change in client category on the upper space and 1. pregnant/breast feed 2. developed advanced disease 3. other Write the date category changed on the lower space
16.	Date and Reason for Termination	Write the code of reason for termination on the upper space and write the date of termination on the lower space 1. category change 2. Lost from follow up 3. Dead 4. TO
17.	Remark	Write any additional suggestions, comments...follow up appointment

Annex 15. Viral Load Request Form

1. Health Facility Information Facility Name: _____ Facility Code: _____ Tel.No. _____ Region: _____ Zone: _____ District: _____ Requested by: Name _____ Signature: _____ Date (ET) ___/___/___ (dd/mm/yyyy)	
2. Client Information Unique ART ID: ___/___/___/___/___/___/___/___/___/___ MRN _____ ☐ Sex: ☐ M ☐ F Age (years) _____ <1 year (in months) ___/___/___	
3. Current ART regimen ☐ Adult First Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy) ☐ If Pediatric, First Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy) ☐ Adult Second Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy) ☐ If Pediatric, Second Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy) ☐ Adult Third Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy) ☐ If Pediatric, third Line Regimen: _____ Date (ET) Initiated ___/___/___ (dd/mm/yyyy)	4. ART Adherence ☐ Good >95% ☐ Fair (85-94%) ☐ Poor <85%
5. Is the client pregnant ☐ Yes or ☐ No Breastfeeding ☐ Yes or ☐ No	
6. CD4 count history (CD4% for <5 years) Most recent result _____ cells/ul Date (ET) ___/___/___ (dd/mm/yyyy) Baseline result (pre ART or at the time of ART initiation) _____ cells/ul Date (ET) ___/___/___ (dd/mm/yyyy)	
7. Current Clinical Status: WHO Clinical (Treatment) Staging : ☐ I ☐ II ☐ III ☐ IV	
8. Reason for Test Routine viral load: ☐ First Viral Load (VL) test (6 mons. post ART) ☐ Second VL test (12 mons. post ART) ☐ Annual VL Test Suspected ART Failure: ☐ Initial Viral load >1000 copies/ml (repeat) ☐ Immunological ☐ Clinical	
9. To be filled by referring laboratory Date (ET) Specimen Collected: ___/___/___ (dd/mm/yyyy) Time (ET): _____ Specimen type ☐ Whole Blood ☐ DBS ☐ Plasma Date specimen sent to Reference laboratory ___/___/___ (dd/mm/yyyy) Time (ET): _____	
10. For Testing Laboratory use only LAB ID: _____ Date Received (ET) ___/___/___ (dd/mm/yyyy) Specimen quality ☐ Acceptable ☐ Unacceptable Reason _____	Test results: Test Date (ET): ___/___/___ (dd/mm/yyyy) Test result: _____ copies/ml Tested by: _____ Signature _____ Reviewed by: _____ Signature _____ Dispatch date (ET) ___/___/___ (dd/mm/yyyy)
Comments: _____ _____	

Annex 16 Re-testing register

1 Ministry of Health

Retesting Register for HIV positive Clients before ART Initiation (Retesting for Verification)

Ser. No	Date of Initial test	Clients MRN/Code No	Age	Sex	HIV positive Clients Entry Point (OPD, IPD, Pre ART pool, VCT, Other units within facility (specify)), Other facility (Specify)	Date of retesting for verification	Retesting for verification result				If the initial and the verification tests are discrepant (Initial +ve, retesting –ve or Inconclusive) Refer sample for testing at reference laboratories or at referral hospitals	Action Taken/Final result
							Test 1	Test 2	Test 3	Final result		
							Kit Name: _____	Kit Name: _____	Kit Name: _____			
							Lot No.: _____	Lot No.: _____	Lot No.: _____			
							Exp. Date: _____	Exp. Date: _____	Exp. Date: _____			
1	2	3	4	5	6	7	8	9	10	11	12	13

Annex 17. High Viral Load Client Follow-up Form

Name of Health Facility:				
1. Patient Information: Name: _____ Age: _____ Sex: ☐ M ☐ F UAN: _____				
2. ARV Information		Viral Load Results		
ARV Regimen Date of initiation: _____ / _____ / _____ _____ / _____ / _____ _____ / _____ / _____		1st VL result: _____ c/ml Date: ____ / ____ / ____ Previous VL: _____ c/ml Date: ____ / ____ / ____ Date 1st VL result & 1st EAC Session given: ____ / ____ / ____		
3. Give the viral load test result and explain the following to the client				
• Viral Load is the number of HIV copies in the blood • High VL result can be due to poor adherence to medication or can be due to primary resistance. • When VL result is high in the blood, the CD4 count decreases, OIs are flare up and disease progresses. • High VL can be reduced as a result of good adherence to medication within three months.				
4. Assess current adherence to treatment and document →		Adherence Rate at EAC Session 1		
		>95%	85-94%	<85%
		Good	Fair	Poor
4.1	How many ARV dose/s do you take/day? ☐ Once ☐ Twice	≤2 dose ≤3 doses	3-4 doses 4-9 doses	≥5 doses >9 doses
4.2	Did you miss ARV doses in the past 1 month? Yes No			
4.3	If Yes, # dose/s you missed? ☐ ≤2 ☐ 3-4 ☐ ≥5 ☐ ≤3 ☐ 4-9 ☐ >9			
5. Explore Medical and Psychosocial Reasons for High VL			EAC	
5.1 Identify Medical Reasons for High Viral Load			Session 1	
• Did you take other drugs than ARVs without consulting your physician? (Yes, No), • If Yes, identify the drug/s, review interaction with ARVs, counsel the client & discontinue • Have you ever developed recurrent OIs including cough, fever, weight loss, night sweat, diarrhoea, and vomiting in the past? (Yes, No). If yes, investigate for TB and chronic diarrhea and manage • Have you ever developed severe ARV drugs side effects in the past? (Yes, No), If Yes, investigate and manage for ARVs side effects. • Have ever taken ART/PMTCT prophylaxis in the past prior to ART initiation? • (Yes, No), If Yes, suspect primary resistance and consult ART physician • Did you discontinue your ARV in the past? (Yes, No), If Yes, identify reasons and develop treatment plan with the client. • If the client is a child, check proper dosing for weight by reviewing chart and readjust ARV dosing for weight.				
5.2 Identify Psychosocial Reasons for High Viral Load				
5.2.1 Cognitive barriers: understanding and expectation -counsel & explain expected outcomes				

What were the reason/s for missing your ARV dose/s in the past? Identify the reason/s, counsel and motivate the client to develop medication taking plan.	
5.2.2. Socio-economic barriers: lack of social support, disclosure, stigma, & poor living condition - Have you disclosed your HIV status to anyone? If No, provide disclosure counseling and encourage - Do you have treatment supporter? (Yes, No) If No, counsel to designate treatment supporter	
5.2.3 Behavioural barriers: attitude, motivation, confidence & skills – educate, motivate, and empower client to manage medication taking and develop reminder - What do you do to remind yourself to take drugs on time? - Are you confident to take your ARV openly at home?	
5.2.4 Psychological/Emotional barriers: common mental illness: depression, PTSD, substance abuse, and psychosis – link to psychiatric clinic for psychotherapy & treatment Have you ever felt sad for more than 2 weeks? Have you ever lost interest in activities that usually give you pleasure for more than 2 weeks? Have you regularly taken alcohol or chew Khat?	

4. Assess current adherence to treatment and document		Rate at Session 2			Rate at Session 3		
		>95%	85-94%	<85%	>95%	85-94%	<85%
		Good	Fair	Poor	Good	Fair	Poor
4.1	How many ARV dose/s do you take/day? ☐ Once ☐ Twice	≤2 ≤3	3-4 4-9	≤5 >9	≤2 ≤3	3-4 4-9	≥5 >9
4.2	Did you miss ARV doses in the past one month? Yes No						
4.3	If Yes, dose/s missed? ☐ ≤2 ☐ 3-4 ☐ ≥5 ☐ ≤3 ☐ 4-9 ☐ ≥9						

5. Explore Medical and Psychosocial Reasons for High VL	Date: __/__/__	Date: __/__/__
5.1 Follow-up status of identified medical reason/s for High Viral Load	EAC Session 2	EAC Session 3
5.2 Follow-up status of identified psychosocial reasons for High Viral Load		
5.2.1 Cognitive barriers:		
5.2.2 Socio-economic barriers:		
5.2.3 Behavioural barriers:		
5.2.4 Psychological/Emotional barriers:		

Outcome of EAC Sessions:

- Did the patient attend all the appointments? (Yes No) If no, any reason? _____
- **Major remaining barriers identified after EAC sessions:** ☐ Behavioral ☐ Cognitive ☐ Socio-economic ☐ Emotional. _____
- If patient is defaulted EAC session/s or adherence barriers were not resolved, second Viral Load should be deferred and EAC extended.
- Share decision with the team.
- Date of collection of 2nd Viral Load: __/__/__, **2nd Viral Load result:** ____ c/ml Date: __/__/__
- Your next plan? ☐ continue current regimen ☐ referring to ART Physician

ART Nurse: _____ **Date of final assessment:** __/__/__

Decision by ART physician

What is the plan for this patient? ☐ Shifted to Second-line Regimen. ☐ continue current regimen

Comment: _____

ART Physician: _____ **Date:** __/__/__

Annex 18. High viral load register

SN	UAN	Client Name	Age	Sex	Date 1 st VL taken	Date 1 st VLR Received	1 st VL Result	Date 1 st EAC given	Date 2 nd EAC given	Date 3 rd EAC given	Date 2 nd VL taken	Date 2 nd VLR Received	2 nd VL Result	Date 2 nd VL given to client	Measures Taken	Date of 2 nd line regimen started
															1. Switched to 2 nd line 2. Remain on 1st line	

Annex 19. Screening tool for ICT & High Viral Load

SCREENING TOOL for ICT & VL testing									
1. Partner & Family based Index case testing service									
Index Name:		Age:		Sex:		MRN:		UAN:	
If index case is adult, check (Circle) Marital status: 1. Single 2. Married 3. Divorced 4. Separated 5. Widowed 6. Cohabiting									
If index case is a child, check (Circle) maternal HIV status: 1. Known Positive 2. Negative 3. Unknown status & alive 4. * Dead									
Initial or Follow up Date (DD/MM/YY)	Check and Circle-Y (Yes), N (No) or NA (Not applicable) at visit time							Circle E (Eligible): If ≥ 1 Y (YES) NE (Not Eligible): otherwise	
	Newly enrolled	Has High Viral Load (HVL)?	Drop out & Return ed?	**Partner(s)/ Parent NOT Tested?	Child (ren)/ Sibling(s) NOT Tested?	Has STI?	KP_FSW		
	Y N	Y N	Y N	Y N NA	Y N NA	Y N NA	Y N NA	E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
		Y N	Y N	Y N	Y N NA	Y N		E	NE
N.B., Write here the Serial number if the index is Eligible and registered on the Partner and Family based ICT Register: _____									
2. Screening for Viral Load testing									
	Follow up 1	Follow up 2	Follow up 3	Follow up 4	Follow up 5	Follow up 6	Follow up 7		
Initial OR Follow up date (dd/mm/yy)	/ /	/ /	/ /	/ /	/ /	/ /	/ /	/ /	/ /
Date Recent VL sample taken (dd/mm/yy)	/ /	/ /	/ /	/ /	/ /	/ /	/ /	/ /	/ /
Eligible for VL test?	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N
If Yes, is VL test requested?	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N
If NO, is Appointment card issued with due date?	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N	Y N
* If maternal status is Dead, check HIV status for the father and other siblings									
** Partner (s) shall be considered tested if testing date was only with in the last 12 months or less									

Annex 20: ART regimen coding

ADULT FIRST LINE	CHILD FIRST LINE
1c=AZT-3TC-NVP 1d= AZT - 3TC - EFV 1e= TDF - 3TC - EFV 1f= TDF + 3TC+ NVP 1g= ABC + 3TC + EFV 1h= ABC + 3TC +NVP 1j= TDF+3TC+DTG 1k=AZT+3TC+DTG 1i= Other specify	4c=AZT+ 3TC+NVP 4d=AZT+3TC+EFV 4e= TDF+3TC+EFV 4f= AZT +3TC+LPV/r 4g= ABC+3TC+LPV/r 4i= TDF+3TC+DTG 4j=ABC+3TC+DTG 4k=AZT+3TC+DTG 4L=ABC+3TC+EFV 4h= Other specify
ADULT SECOND LINE	CHILD SECOND LINE
2e= AZT +3TC +LPV/r 2f=AZT+3TC +ATV/r 2g=TDF + 3TC+LPV/r 2h= TDF + 3TC + ATV/r 2i= ABC + 3TC+ LPV/r 2j= Other specify	5e=ABC+3TC+LPV/r 5f=AZT + 3TC + LPV/r 5g=TDF + 3TC + EFV 5h=ABC + 3TC + EFV 5i=TDF + 3TC+LPV/r 5k = RAL + AZT + 3TC 5L = RAL + ABC + 3TC 5m = ABC + 3TC + DTG 5n = AZT + 3TC + DTG 5j = Other specify
ADULT THIRD LINE	CHILD THIRD LINE
3a=DRV/r+DTG+AZT+3TC 3b=DRV/r+DTG+TDF+3TC 3D = DRV/r+ABC+EFV+3TC 3c= Other specify	6a=DRV/r + RAL + AZT +3TC 6b=DRV/r + RAL + TDF +3TC 6c=DRV/r + DTG + AZT +3TC 6d=DRV/r + DTG +TDF+3TC 6F =DRV/r+DTG+ABC+3TC 6e= other specify

Annex 21: Case Investigation Form, index

1. Partner and Family based Index Case Testing Case Investigation Form (CIF) (Index Client form)							
A. Facility Information							
Region		Town		Facility Name		Name of Service provider	
B. Index Client Information							
Name	Father name	Grandfather	Sex	1. Male 2. Female	Age	MRN	UAN
						—	—
Address:	Sub city/Zone		Woreda		Kebele		House No
	Phone # (primary):		Alternative phone #1:		Alternative phone #2:		
Marital status: 1. Single 2. Married 3. Divorced 4. Separated 5. Widowed 6. Co-habiting							
Population group: 1. FSW 2. Long distance drivers 3. Mobile Workers 4. Pregnant Mothers 5. Refugee 6. Prisoner 7. Military 8. Vulnerable Adolescents 9. Others Specify (general population, etc)							
C. Reporting Information (Eligibility)							
1. HIV new diagnosis Date HIV tested (DD/MM/YYYY) _____			2. PLHIV (known HIV+) with: 2.1 Viral Load >1000 copies/mm3 2.2 Drop out and returned 2.3 Not yet disclosed to partner 2.4 New partner 2.5 Disclosed but not tested yet 2.6 STI 2.7 Key population (Female sex workers)				
D. Interview Information							
If offered, (DD/MM/YYYY) Date 1 st Offered _____ Date 2 nd Offered _____ Date 3 rd Offered _____			If interviewed, (DD/MM/YYYY) Date 1 st interview _____ Date 2 nd interview _____ Date 3 rd Interview _____			If NOT interviewed, Reason: 1. Refused 2. Dead 3. Other (specify): _____	
Index client is willing to provide name(s) of partner(s). 1. Yes 2. No after 3 rd interview, what are the reasons for not willing to provide the name(s)? If yes, proceed to Page 2							

Annex 22: Case Investigation Form, sexual partner

II. Partner and Family based Index Case Testing Case Investigation Form (CIF)						
Sexual Partner Form (If index is adult)/ Parents (If index is child)						
Name	Father Name	Grandfather Name	Sex	1. Male 2. Female	Age	MRN No of Index client
Address	Sub city	Woreda	Kebele		House No	
	Phone 1	Phone 2	Phone 3 (relative)			
Currently living with index client 1. Yes 2. No				Relation with Index client: 1. Spouse 2. Non spouse		
Date of last exposure (DD/MM/YYYY) 1. < 3 months 2. 3 to 12 months 3. >12 months				*Priority category: 1 ☐ 2 ☐ 3 ☐ 4 ☐		
Intimate partner violence assessment						
1. Has [partner's name] ever hit, kicked, slapped, or otherwise physically hurt you A. Yes B. No		2. Has [partner's name] ever threatened to hurt you? A. Yes B. No		3. Has [partner's name] ever forced you to do something sexually that made you feel uncomfortable? A. Yes B. No		
Notification plan: 1. No notification (IPV+) 2. Client Referral 3. Provider Referral 4. Dual Referral 5. Contract Referral						
Status: 1. Date Contacted (DD/MM/YYYY): _____ 2. Not contacted 2.1 Reason for not contacted: 2.1.1 Unable to locate 2.1.2 Other _____ 3. Notified, Date (DD/MM/YYYY): _____ 4. Not notified 4.1 Reason for not notified: 4.1.1 Refused 4.1.2 Out of jurisdiction 4.1.3 Other _____						
Tested for HIV 1. Yes 2. No		If tested for HIV, date (DD/MM/YYYY) _____		If not tested for HIV, reason: 1. Known HIV+ 1.1. On ART UAN: _____ 1.2. Not on ART 2. Refused		
If tested for HIV, HIV test result: 1. HIV +ve 2. HIV -ve 3. Indeterminate		If HIV +ve, Linked to HIV care, 1. Yes; ART# _____ 2. No		If Tested –ve tested after 3 month 1. Yes 1.1. HIV +ve 1.2. HIV -ve 2. No		
		If HIV +ve, Linked to PS 1. Yes 2. No				
Priority category: 1. Index case reports high risk sexual behavior (Recent unprotected sex with partner, Large number of sex partners (e.g. FSW and their partner(s)), adolescent girls with Large age difference partners) and ART with STI. 2. Newly diagnosed HIV positive index client & Index client who drop from care & returned(both have High Viral load) and ART patients who are in care but with high viral load(≥ 1000 c/ml) 3. Index case or partner is pregnant or breast feeding and Index case or partner is an adolescent girl or young woman (due to their high incident rates) 4. All others(Index client who is Non disclosed , disclosed but not tested, new partner and priority population)						

Annex 23: Case Investigation form, Siblings

Case investigation form, Biological Children		
Name of Index: _____		
Unique ART No. of Index: _____		
Gender: ☐ Male ☐ Female Age: _____		
Date form completed (dd/mm/yyyy): ____/____/____		
Child 1	Child 2	Child 3
Name: _____	Name: _____	Name: _____
Gender: ☐ Male ☐ Female	Gender: ☐ Male ☐ Female	Gender: ☐ Male ☐ Female
Age: _____	Age: _____	Age: _____
Type of Family Testing:	Type of Family Testing:	Type of Family Testing:
☐ Facility	☐ Facility	☐ Facility
☐ Community	☐ Community	☐ Community
☐ Contract	☐ Contract	☐ Contract
Child's HIV Status:	Child's HIV Status:	Child's HIV Status:
☐ HIV-positive	☐ HIV-positive	☐ HIV-positive
☐ HIV-negative	☐ HIV-negative	☐ HIV-negative
☐ Unknown	☐ Unknown	☐ Unknown
If tested HIV+, is ART started? ☐ Yes ☐ No	If tested HIV+, is ART started? ☐ Yes ☐ No	If tested HIV+, is ART started? ☐ Yes ☐ No
ART Start Date: ____/____/____	ART Start Date: ____/____/____	ART Start Date: ____/____/____
Unique ART No. _____	Unique ART No. _____	Unique ART No. _____
Instructions ☐ Complete this testing form for all biologic children of the index client. If the index patient has more than 3 children, complete additional forms as needed so that all children are recorded. · If the index HIV patient is a child, complete the form for all the child's siblings and biological parents · Children of male index clients do not need HIV testing unless their biological mother is HIV-positive, deceased, or her HIV status is unknown/ not documented.		

Annex 24: Mental Health Screening Tool

Brief Mental Health Disorder Symptom Screening Tool for PLHIV and Referral Tool

MRN: _____ UniqueART No: _____ Date: _____ Patient's Name: _____

This checklist is to assist you in assessing and making a timely referral of the client to the treatment team. All behaviors listed below are important and should be taken seriously; they are also designed to help you decide if you should refer the client to the treatment team for further assistance. An answer of "yes" to any one of the following questions should prompt further referral and evaluation by the treatment team or mental health professional. Please put a (✓) to indicate a yes answer.

Questions to Identify Depression: In the past 3 months;

() Was there ever a time when you felt sad, or depressed for more than 2 weeks in a row?

() Was there ever a time lasting more than 2 weeks when you lost interest in most things like hobbies, work, or activities that usually give you pleasure?

Questions to Identify Anxiety: In the past 3 months;

() Did you ever have a period lasting more than 1 month when most of the time you felt worried and anxious?

() Did you have a spell or an attack when all of a sudden you felt frightened, anxious, or very uneasy when most people would not be afraid or anxious?

() Did you ever have a spell or an attack when for no reason your heart suddenly started to race, you felt faint, or you couldn't catch your breath?

Questions to Identify Mania: In the past 3 months;

() When not high or intoxicated, did you ever feel extremely energetic, elated or irritable and more talkative than usual that stayed for at least a week?

Questions to Identify Substance Abuse

() Have you ever felt the need to cut down on your use of alcohol or drugs?

() Has anyone annoyed you by criticizing your use of alcohol or drugs?

() Have you ever felt guilty because of something you've done while drinking or using drugs?

() Have you ever taken a drink or used drugs to steady your nerves or get over a hangover (eye-opener)? *A total of ≥ 2 may be suggestive of a problem.*

Questions to identify suicidal ideation: Since your last visit [or in the last 2 months];

() Have you wished you were dead, or wished you could go to sleep and not wake up?

() Have you had actual thoughts of killing yourself? () Have you ever attempted to harm/kill yourself?

Questions to Identify Psychosis: Observe or ask families whether the patient (in the last 3 months);

() Talking & acting strangely or becoming very quiet and avoid talking.

() Claiming to hear voices or see things that other people don't.

() Being very suspicious, perhaps claiming that other people are trying to harm him/her.

Questions to Identify Dementia: Interview the patient or families whether the patient (in the last 3 months);

() Has trouble with memory () Has diminished orientation to time, place and person () Has diminished executive function.

Questions to Identify Epilepsy:

() Did you ever have partial or generalized fits [sharp, shaky movements], frothing accompanied by loss of control of bowel or bladder function, sudden loss of consciousness, and stiff limbs?

Referred by: _____ Date: _____

Feedback (confirm the assessment)

The patient has: () Mental Health Disorder (specify) _____ () Non Mental Health Disorder

Name of clinician: _____ Date: _____